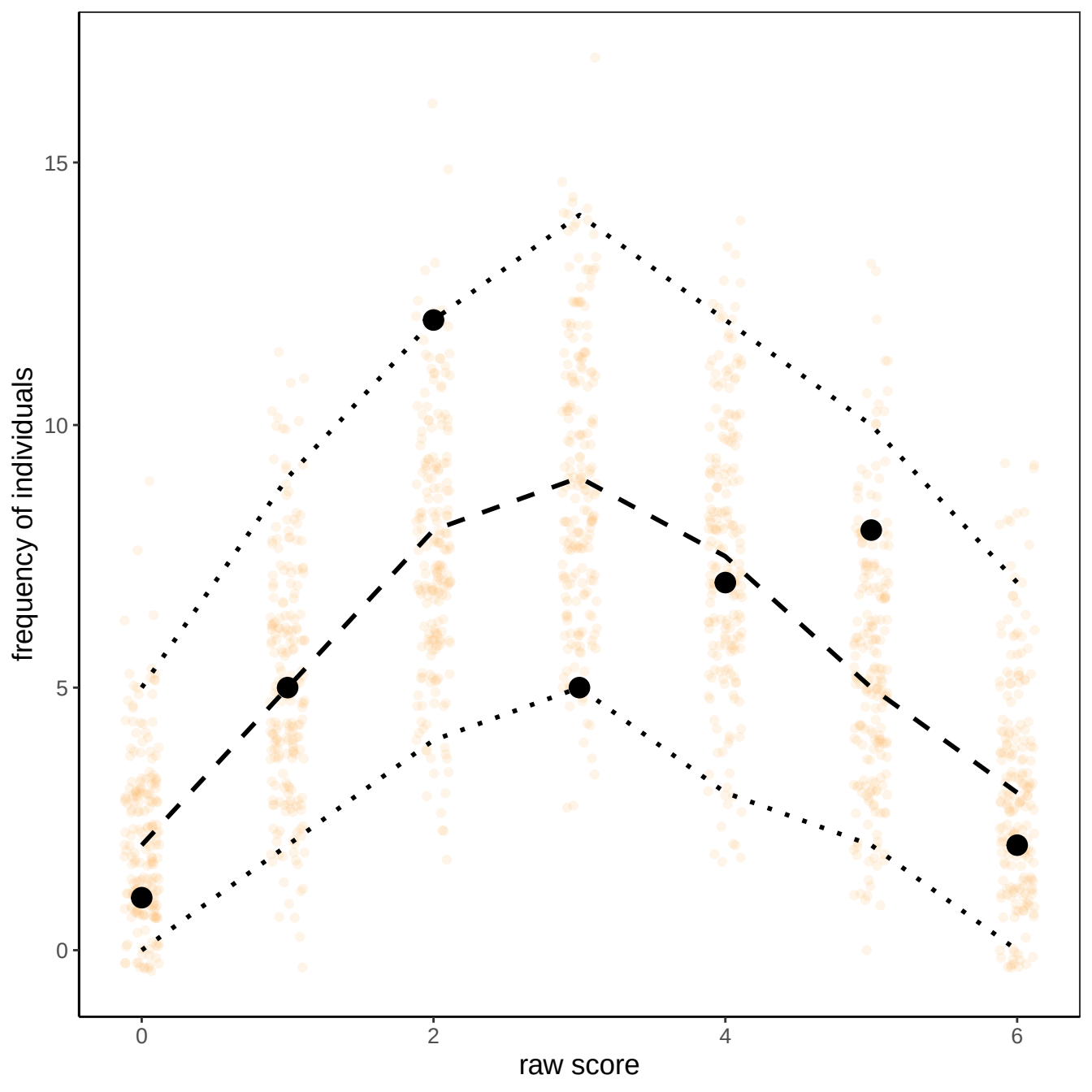
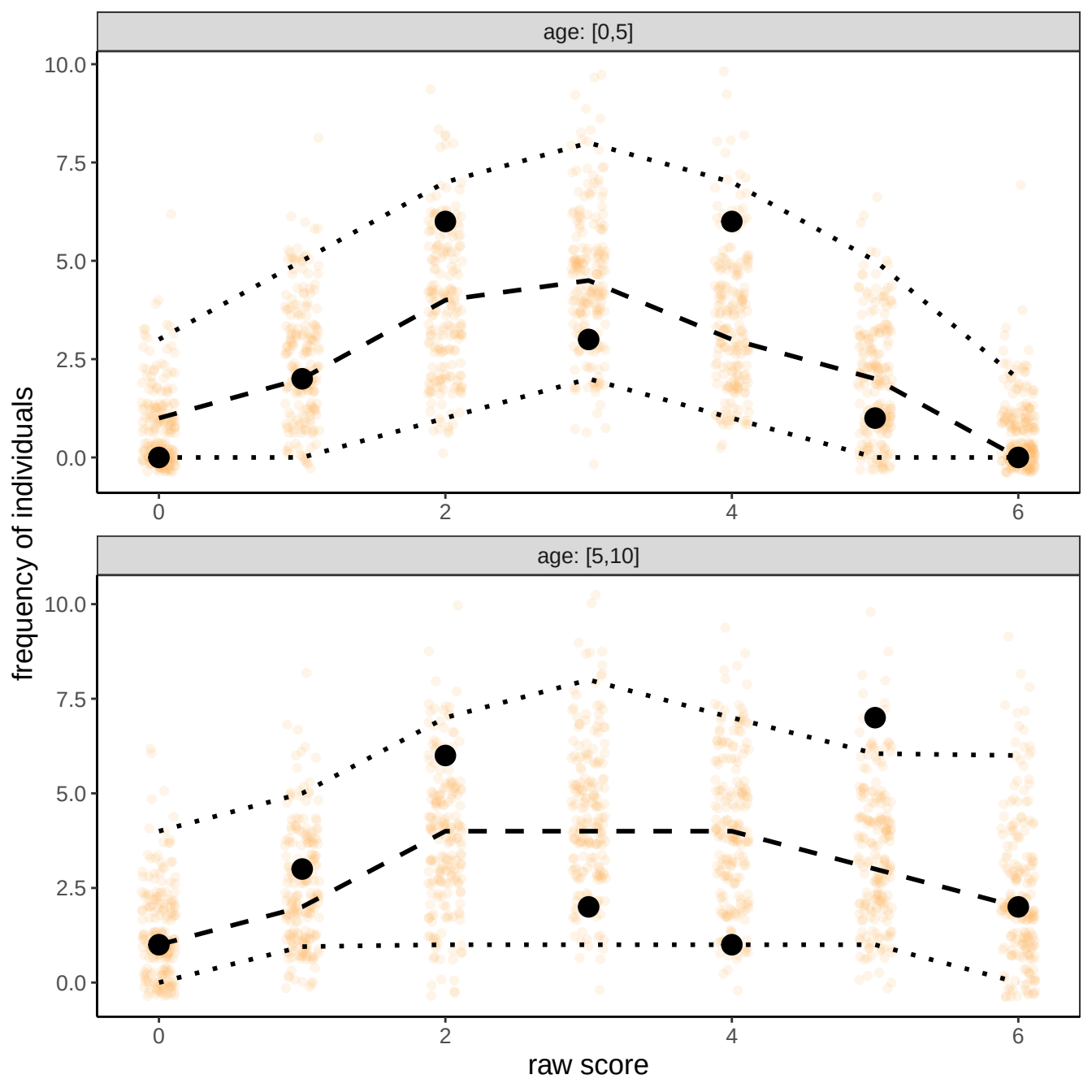
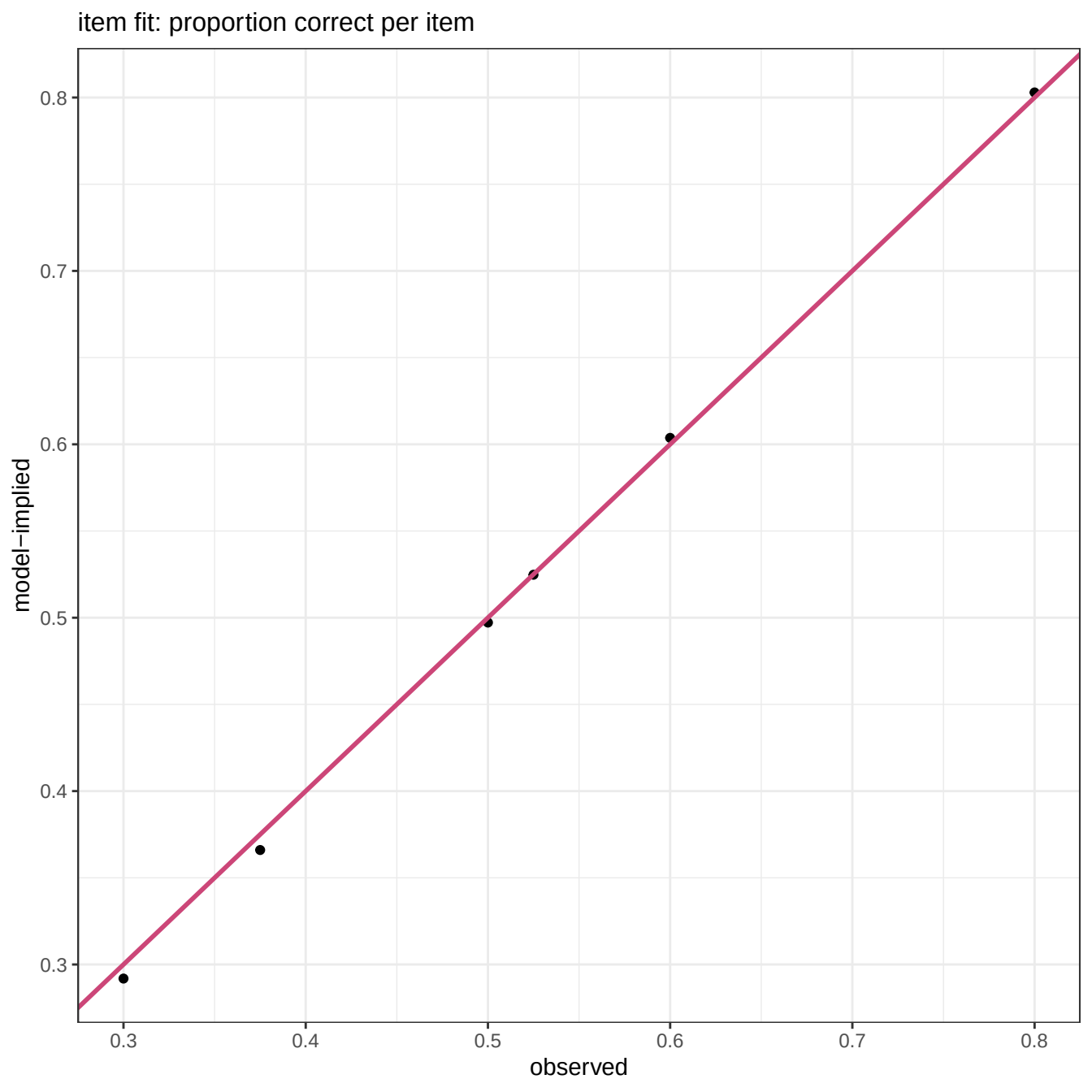
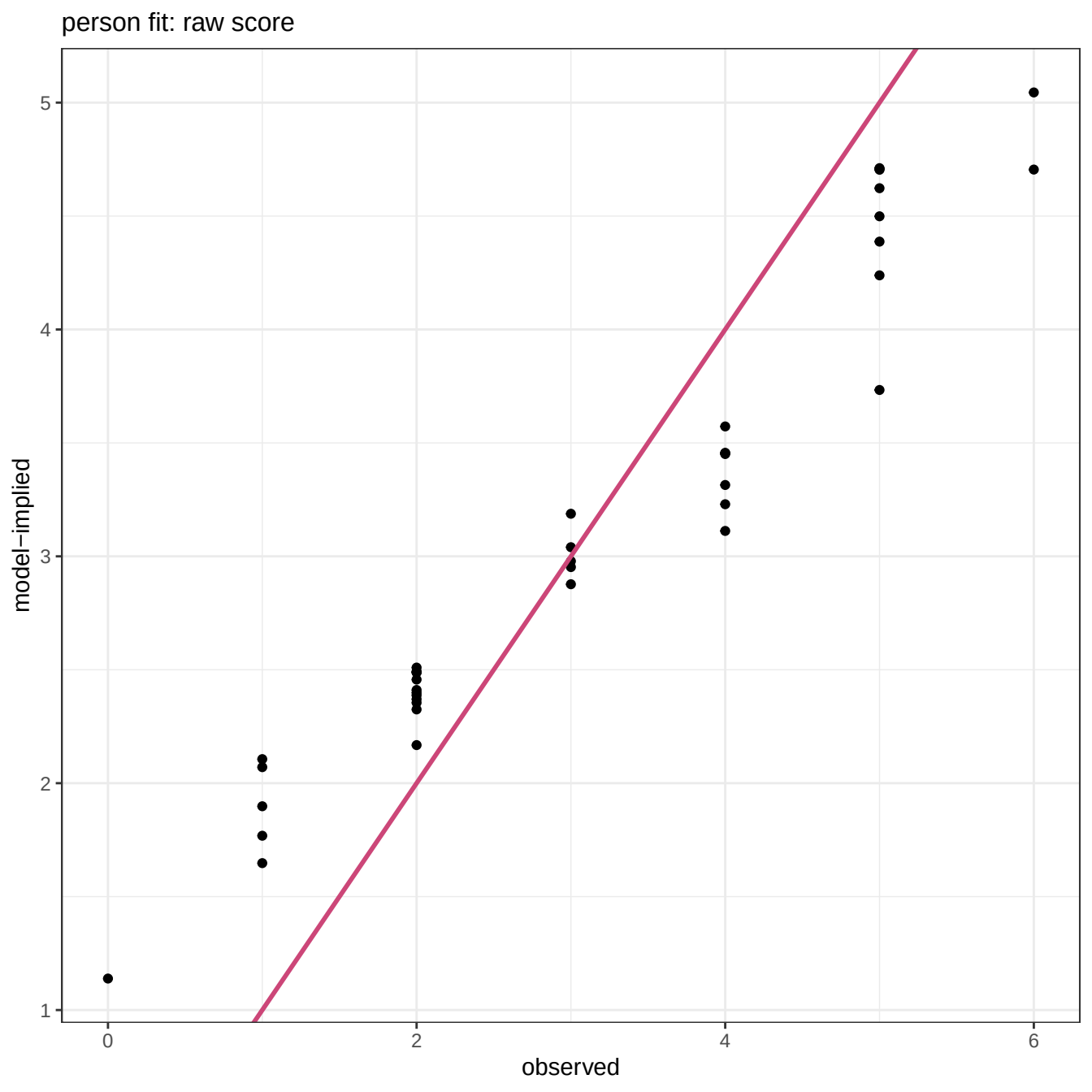
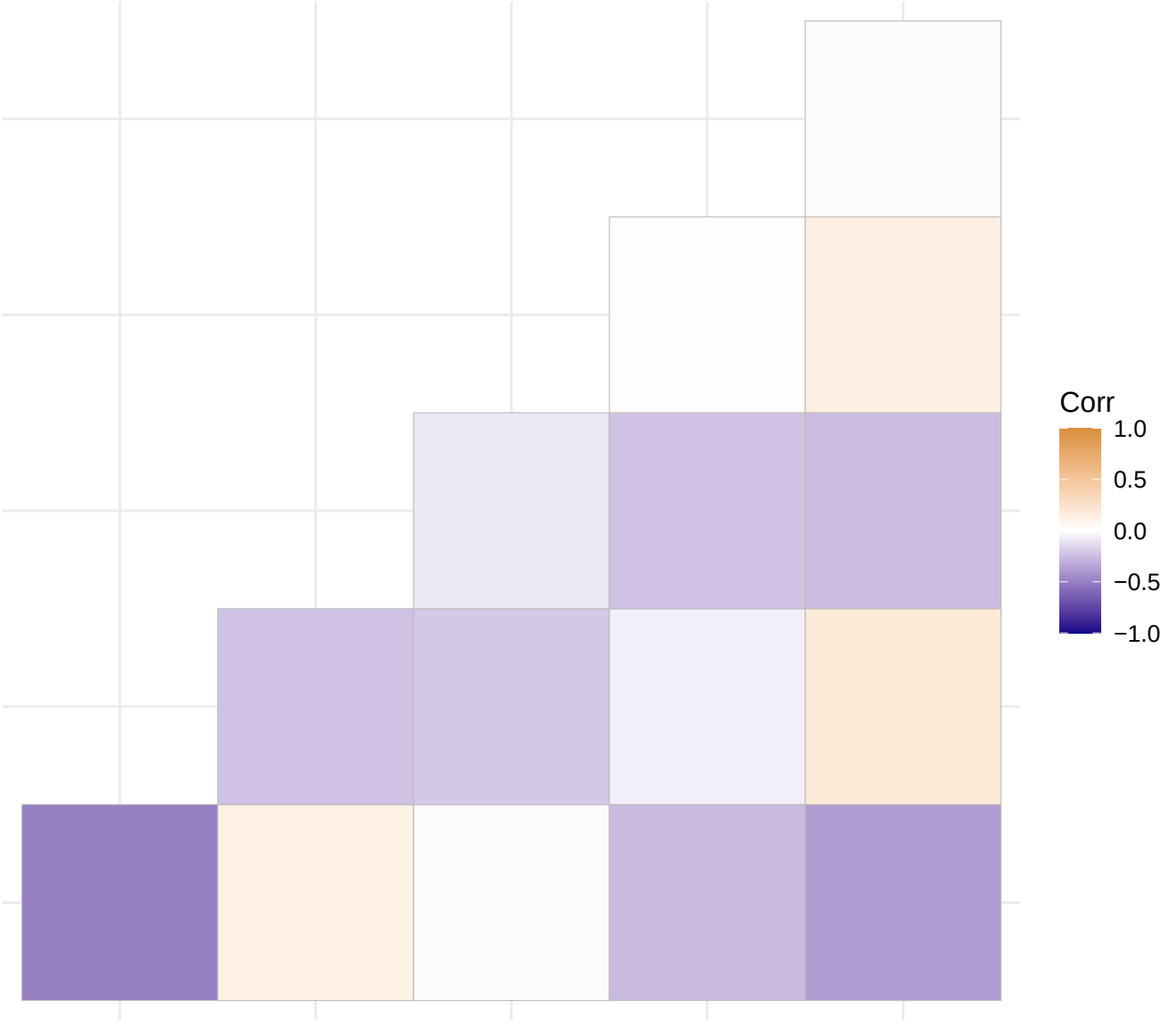
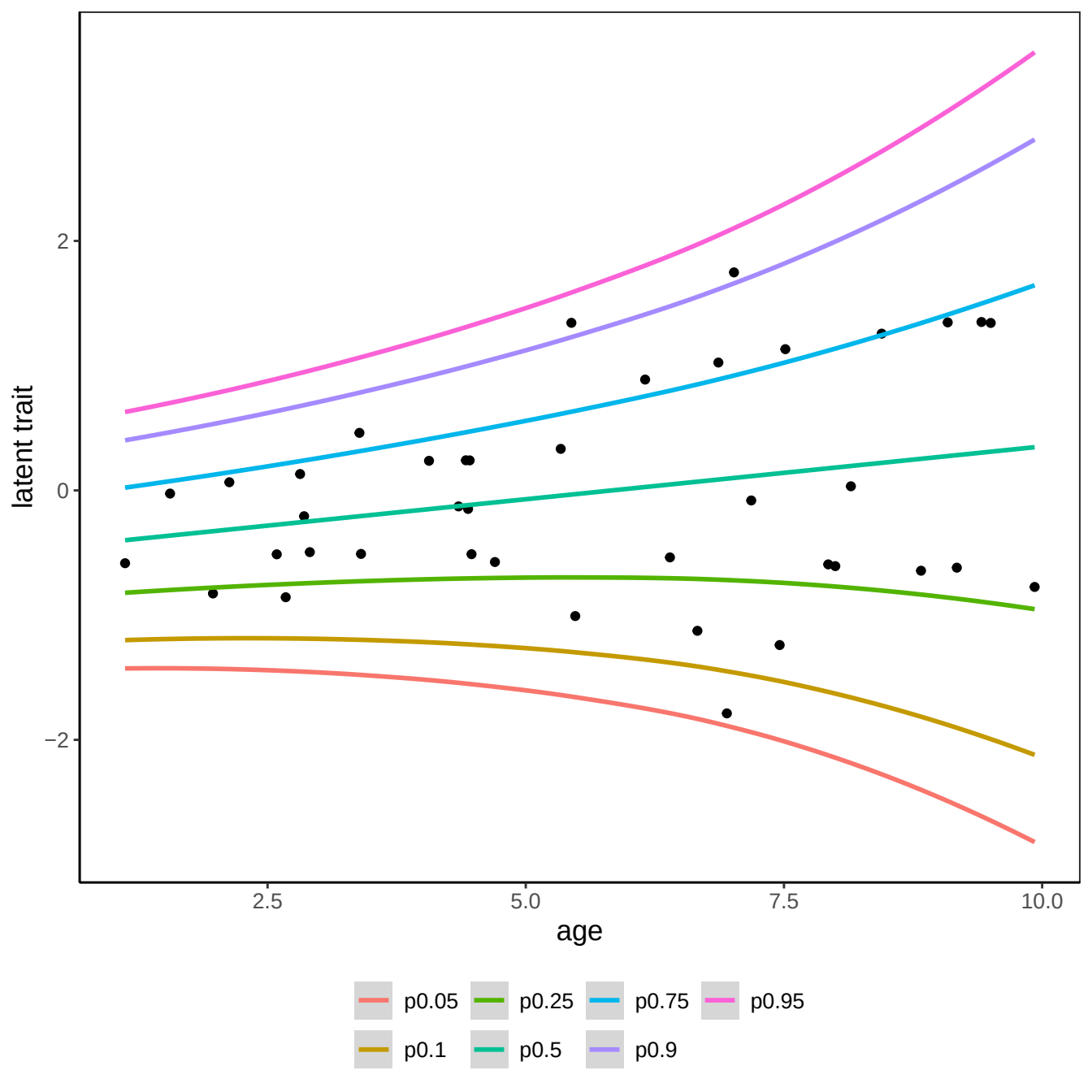

15
10
5
0
frequency of individuals
0
2
4
6
raw score

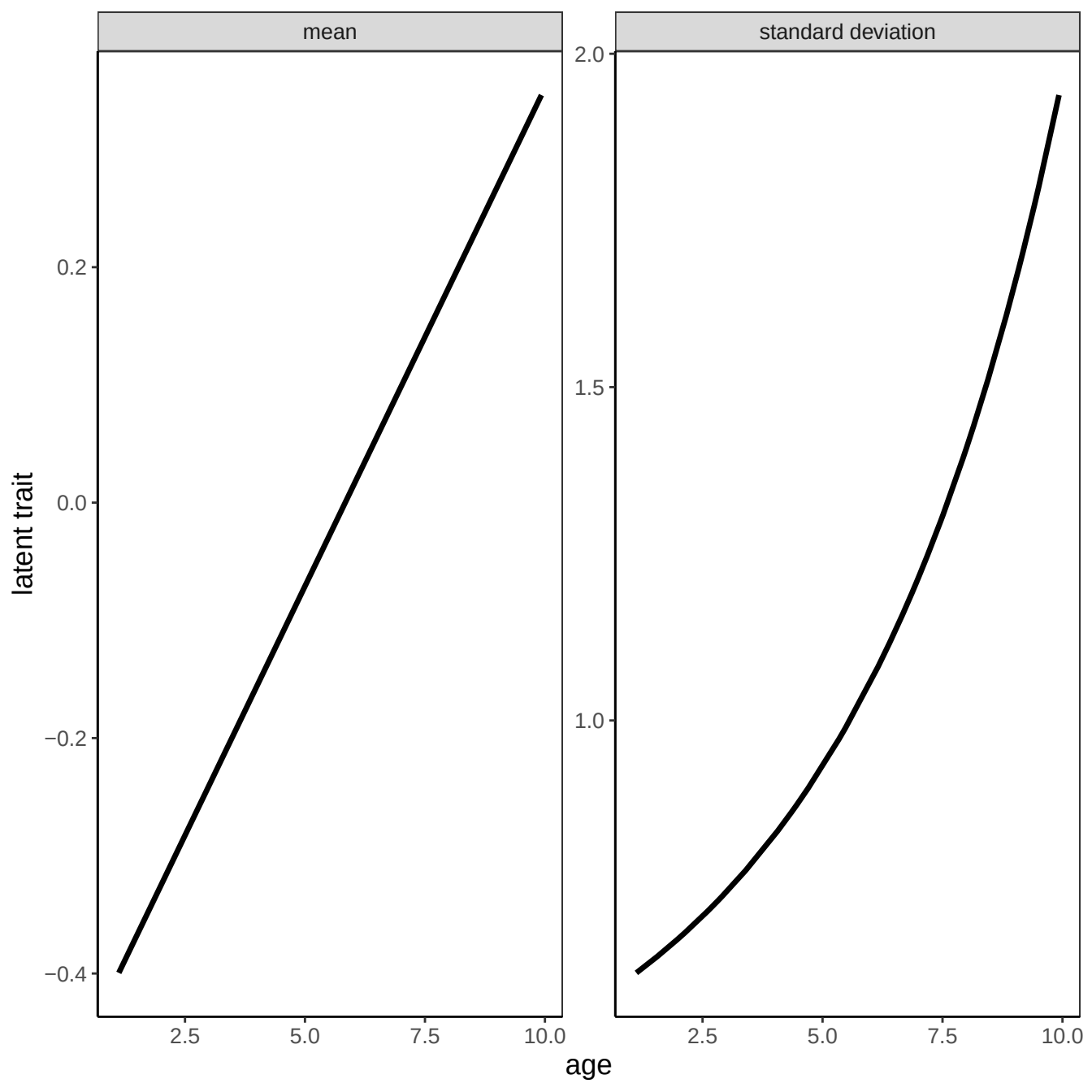

age: [0,5]
age: [5,10]
frequency of individuals
raw score
10.0
7.5
5.0
2.5
0.0
0
2
4
6

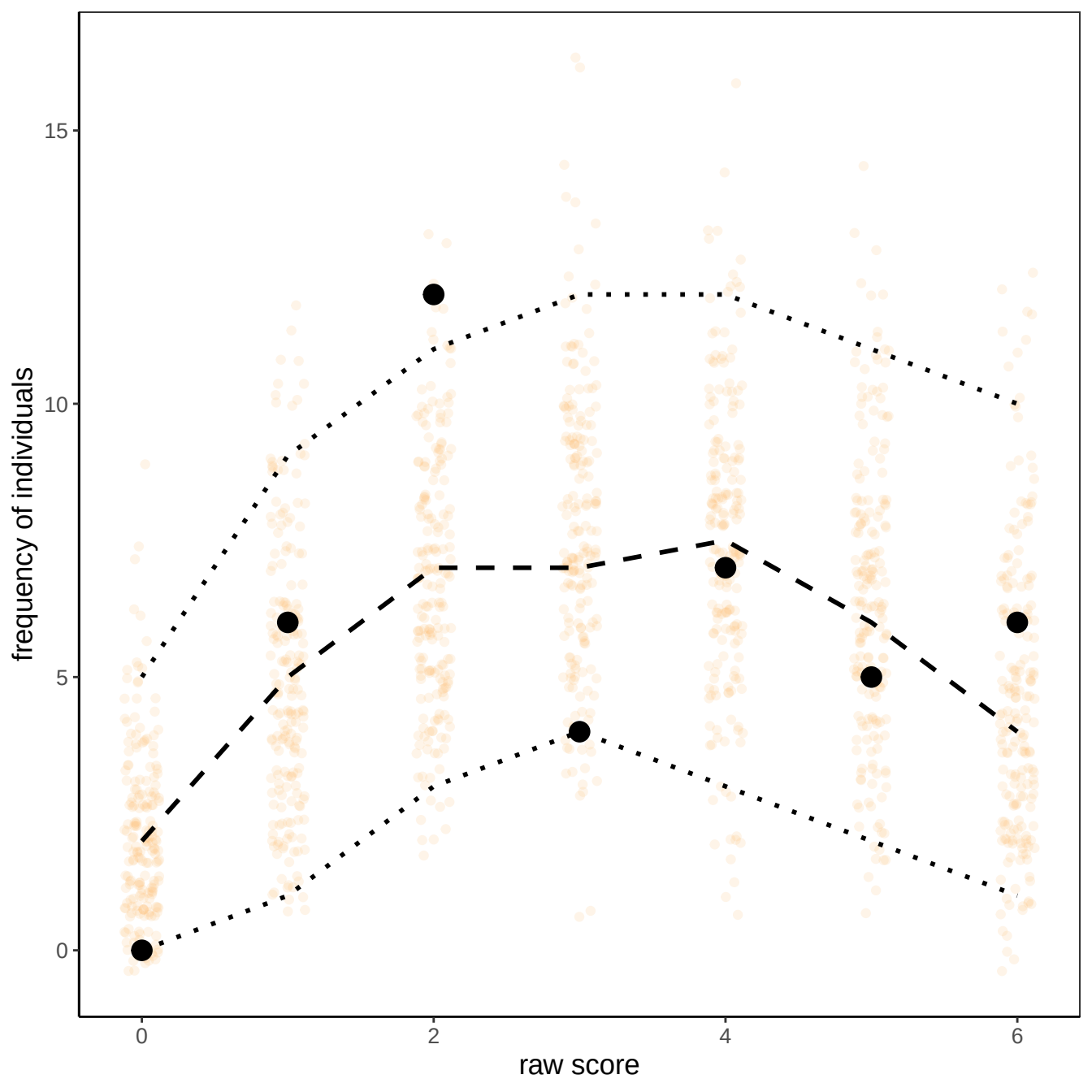

item fit: proportion correct per item
model−implied
observed
0.8
0.7
0.6
0.5
0.4
0.3
0.3
0.4
0.5
0.6
0.7
0.8

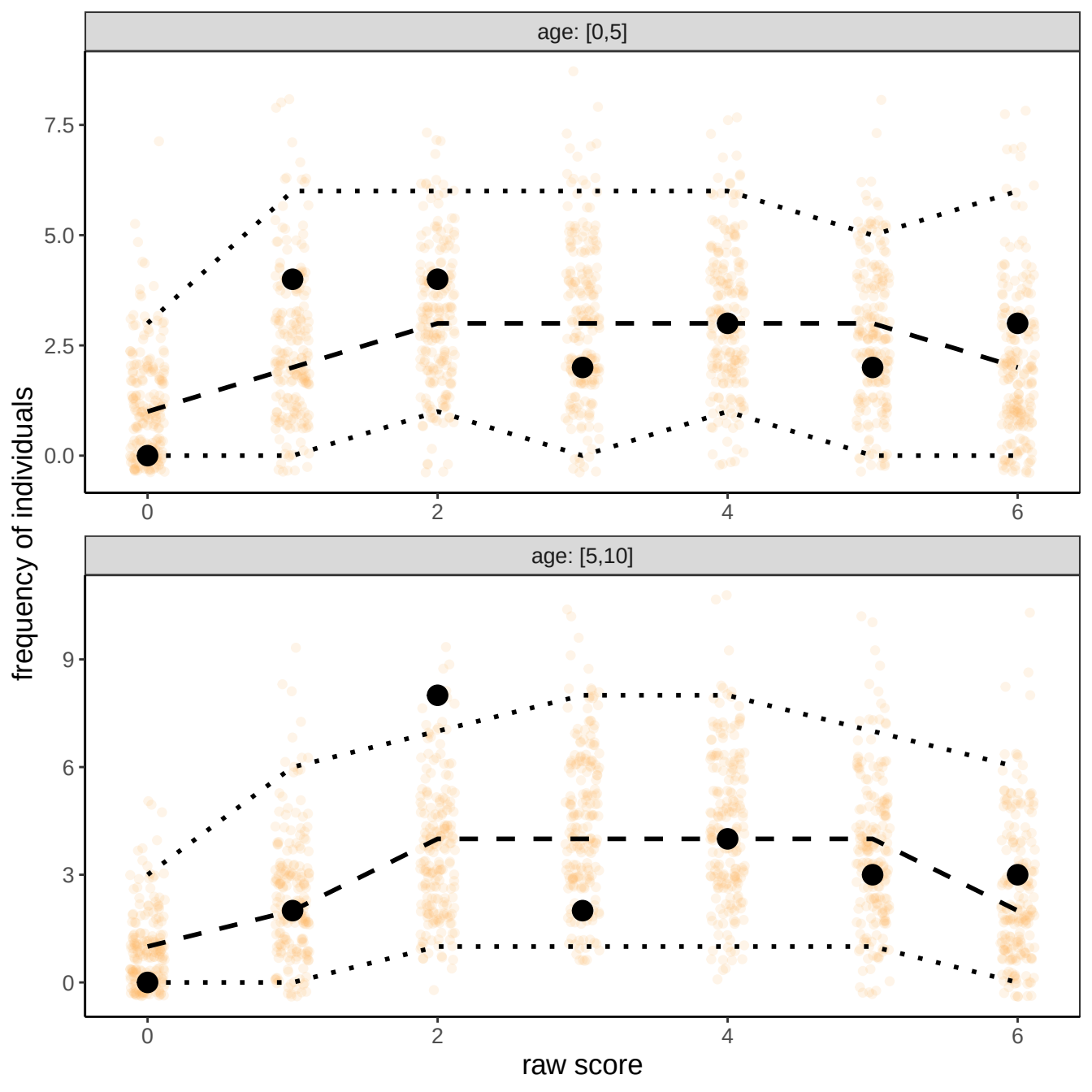

person fit: raw score
model−implied
observed
1
2
3
4
5
0
2
4
6

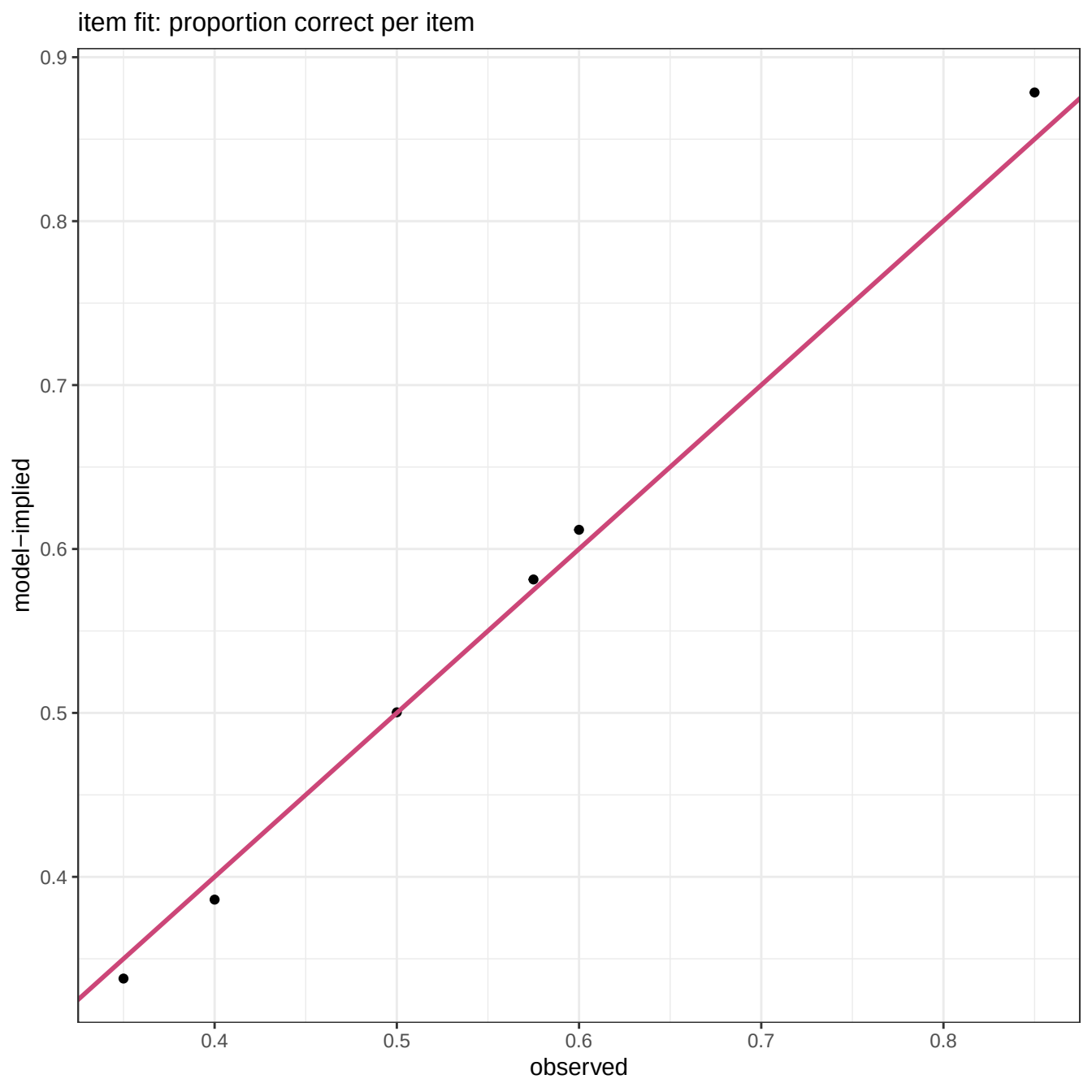

Corr
1.0
0.5
0.0
−0.5
−1.0

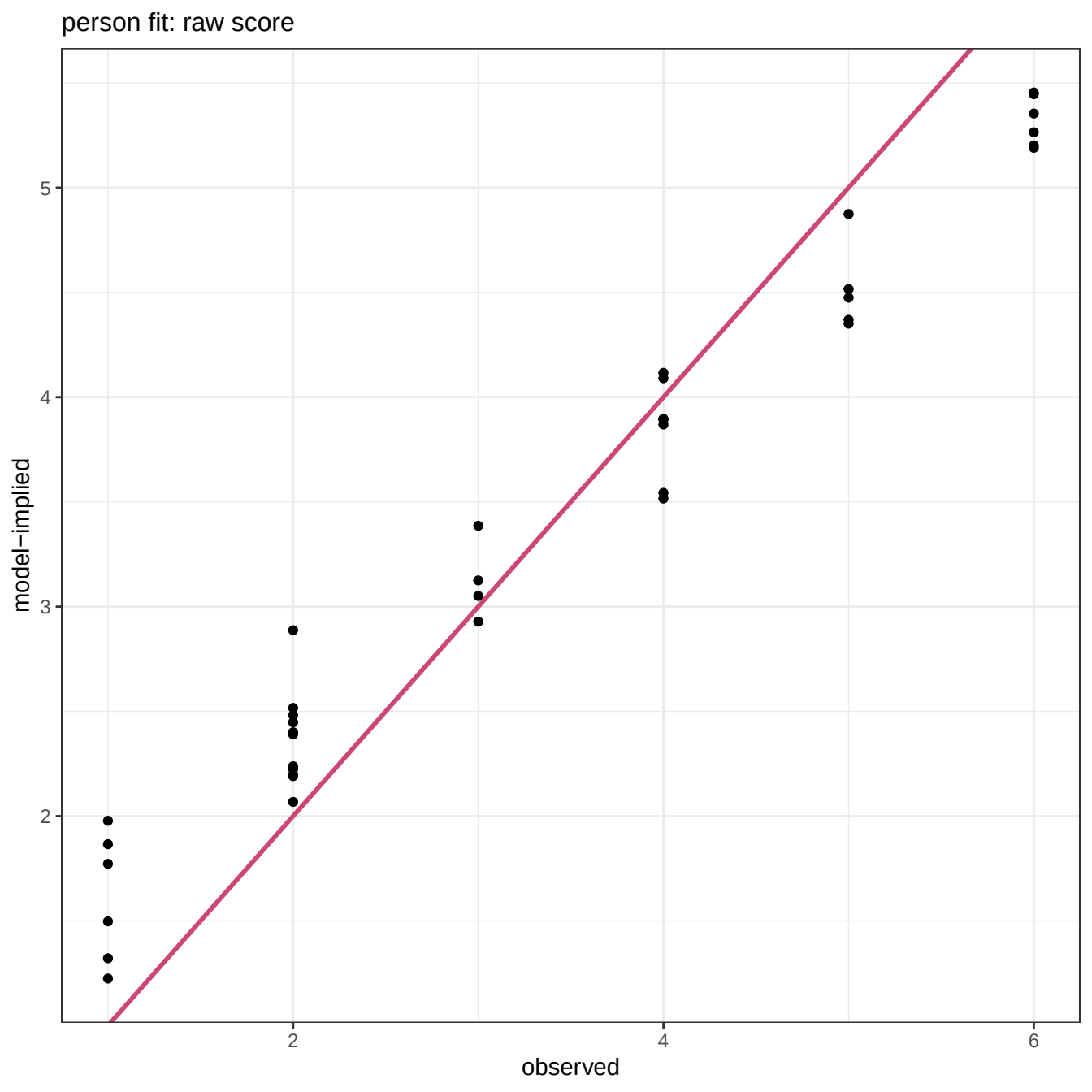

latent trait
2
0
−2
2.5
5.0
7.5
10.0
age
p0.05
p0.25
p0.75
p0.95
p0.1
p0.5
p0.9

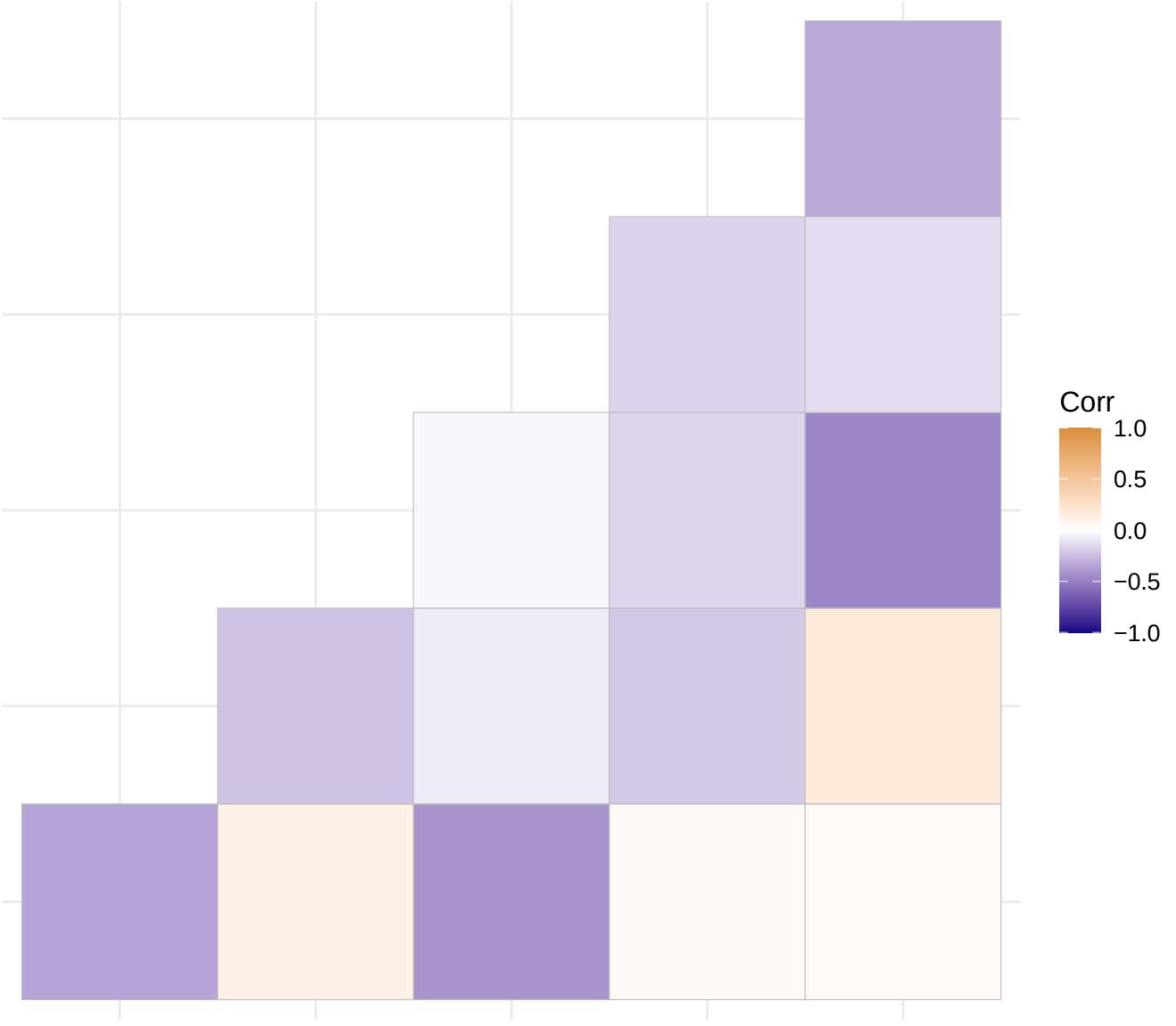

mean
standard deviation
latent trait
age
0.2
0.0
−0.2
−0.4
2.0
1.5
1.0
2.5
5.0
7.5
10.0

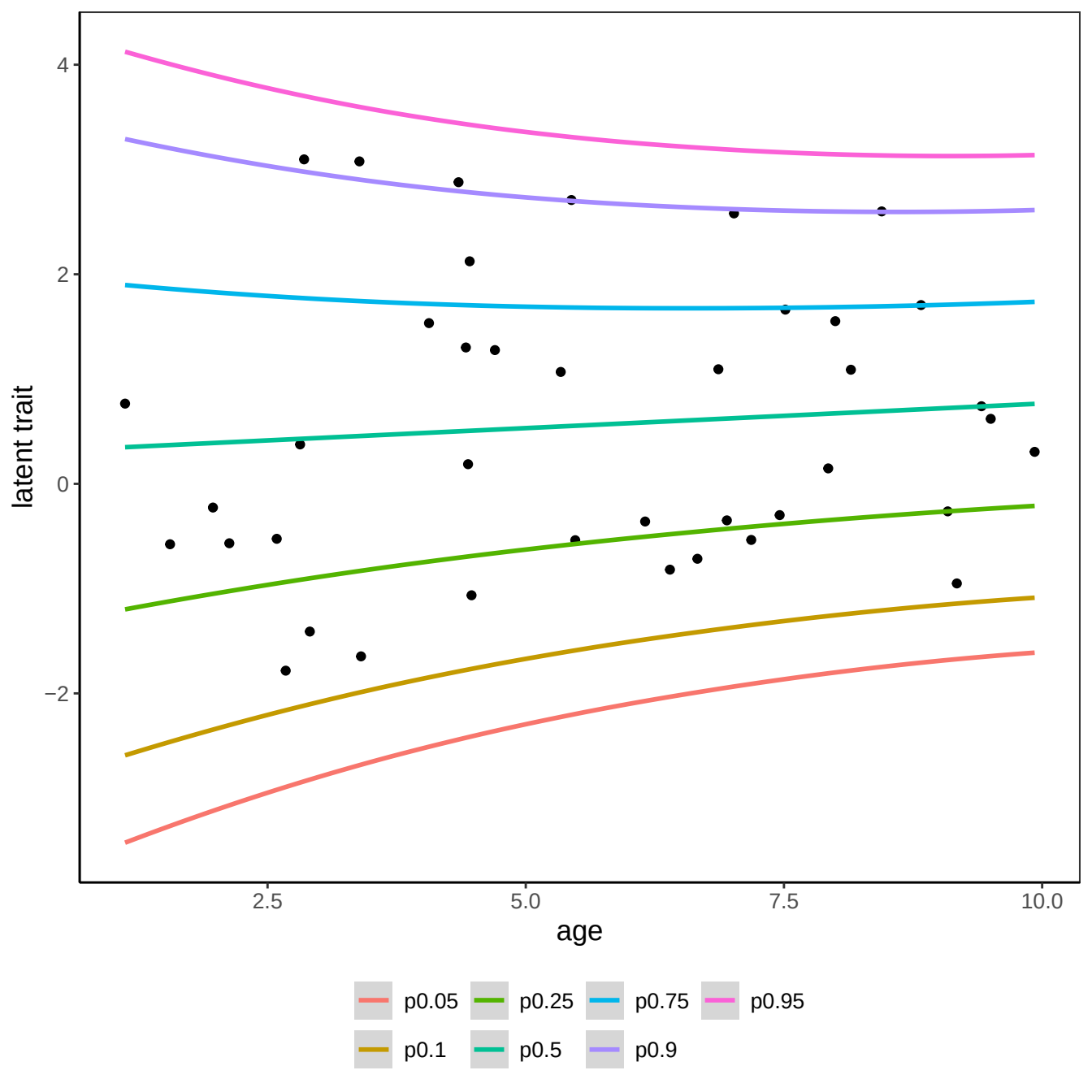

15
10
5
0
frequency of individuals
0
2
4
6
raw score

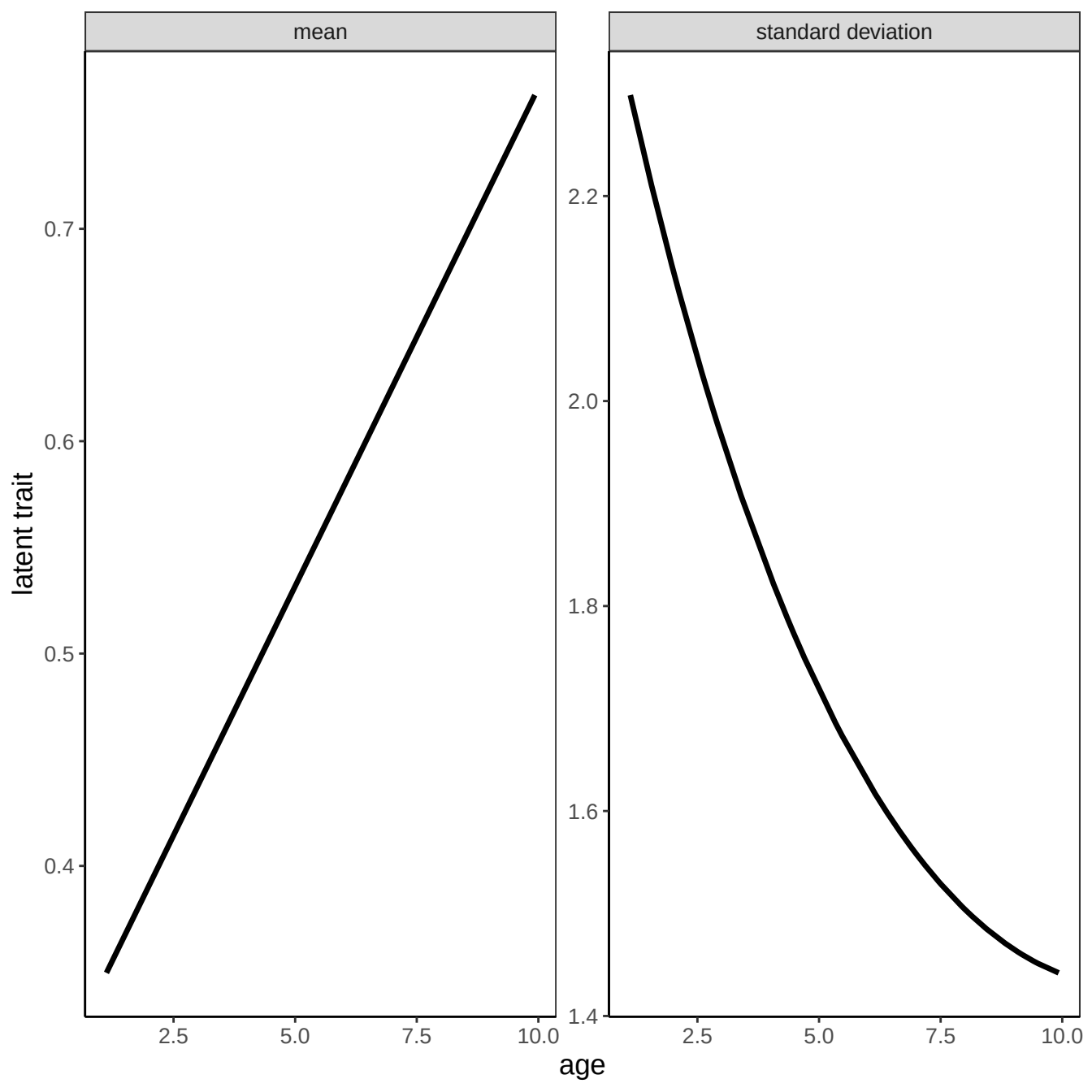

age: [0,5]
age: [5,10]
frequency of individuals
raw score
0
2
4
6
0.0
2.5
5.0
7.5
0
3
6
9

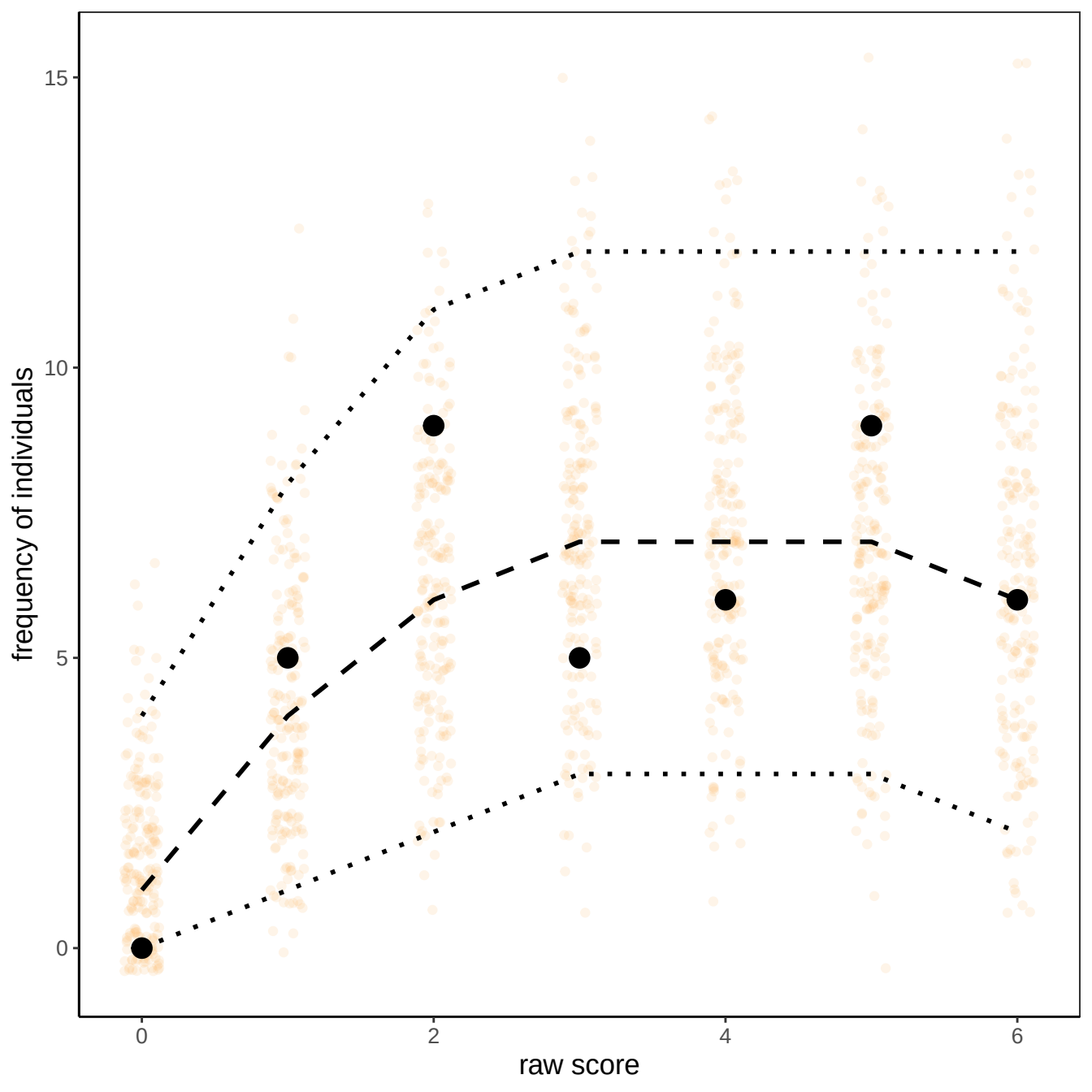

item fit: proportion correct per item
model−implied
observed
0.9
0.8
0.7
0.6
0.5
0.4
0.4
0.5
0.6
0.7
0.8

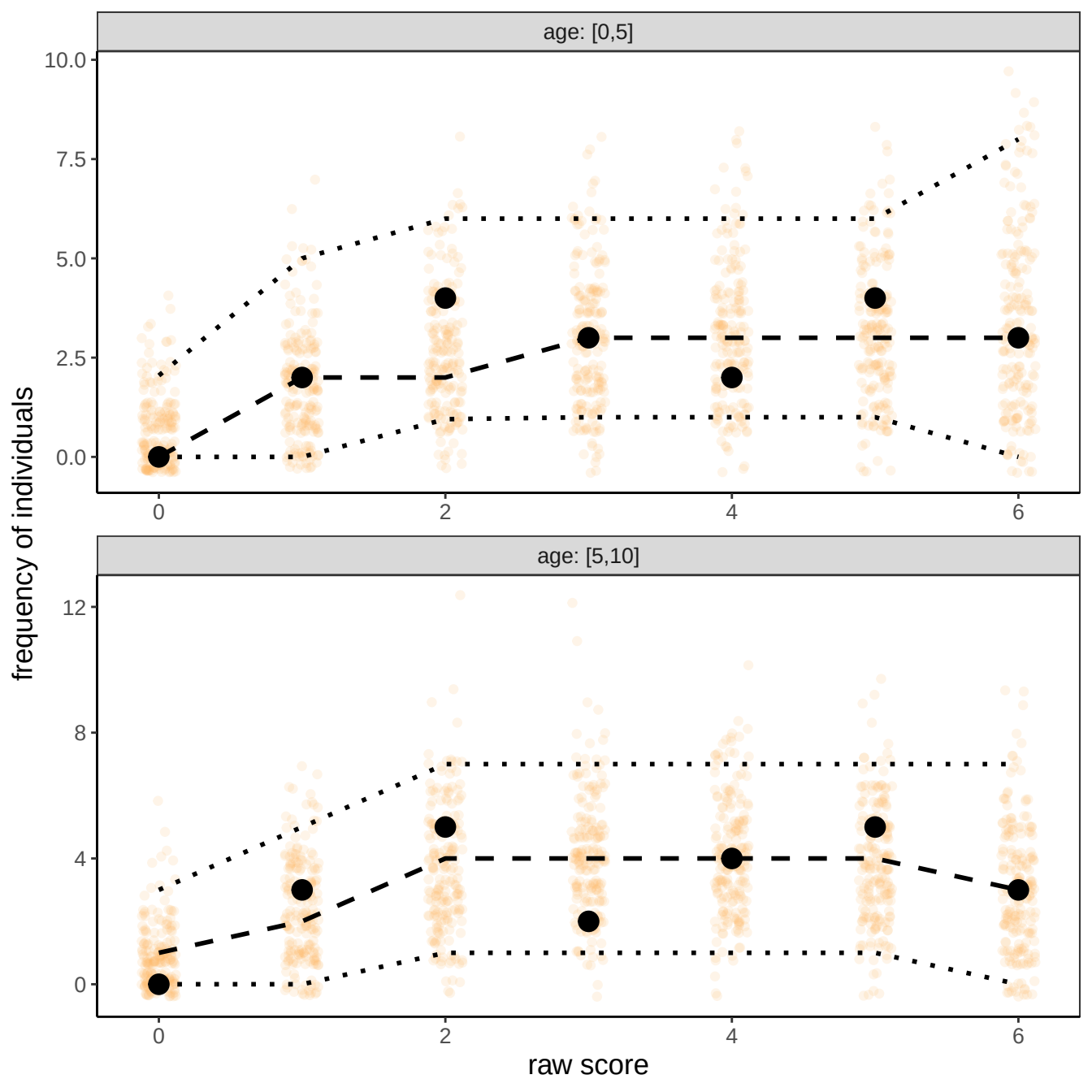

person fit: raw score
model−implied
observed
2
3
4
5
2
4
6

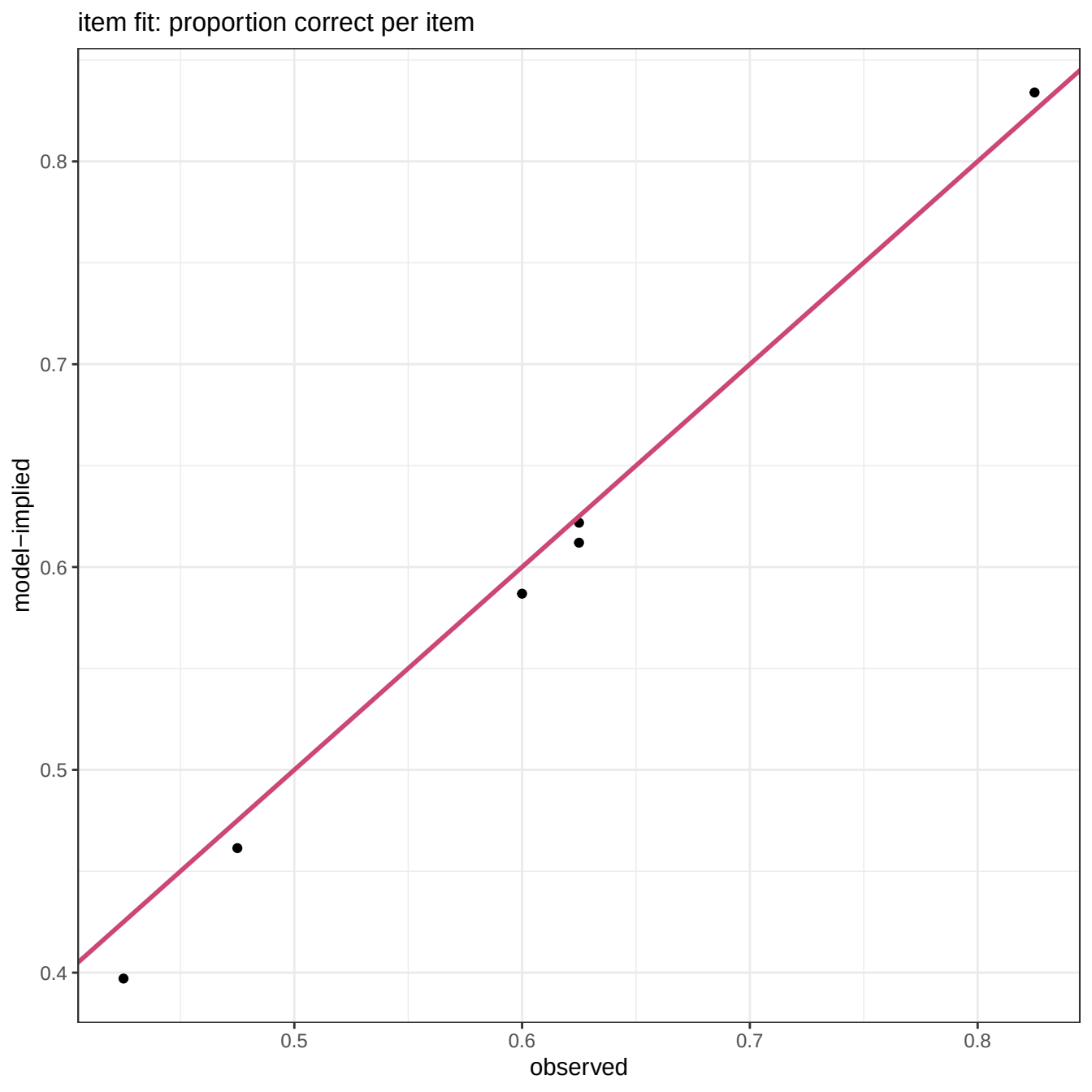

Corr
1.0
0.5
0.0
−0.5
−1.0

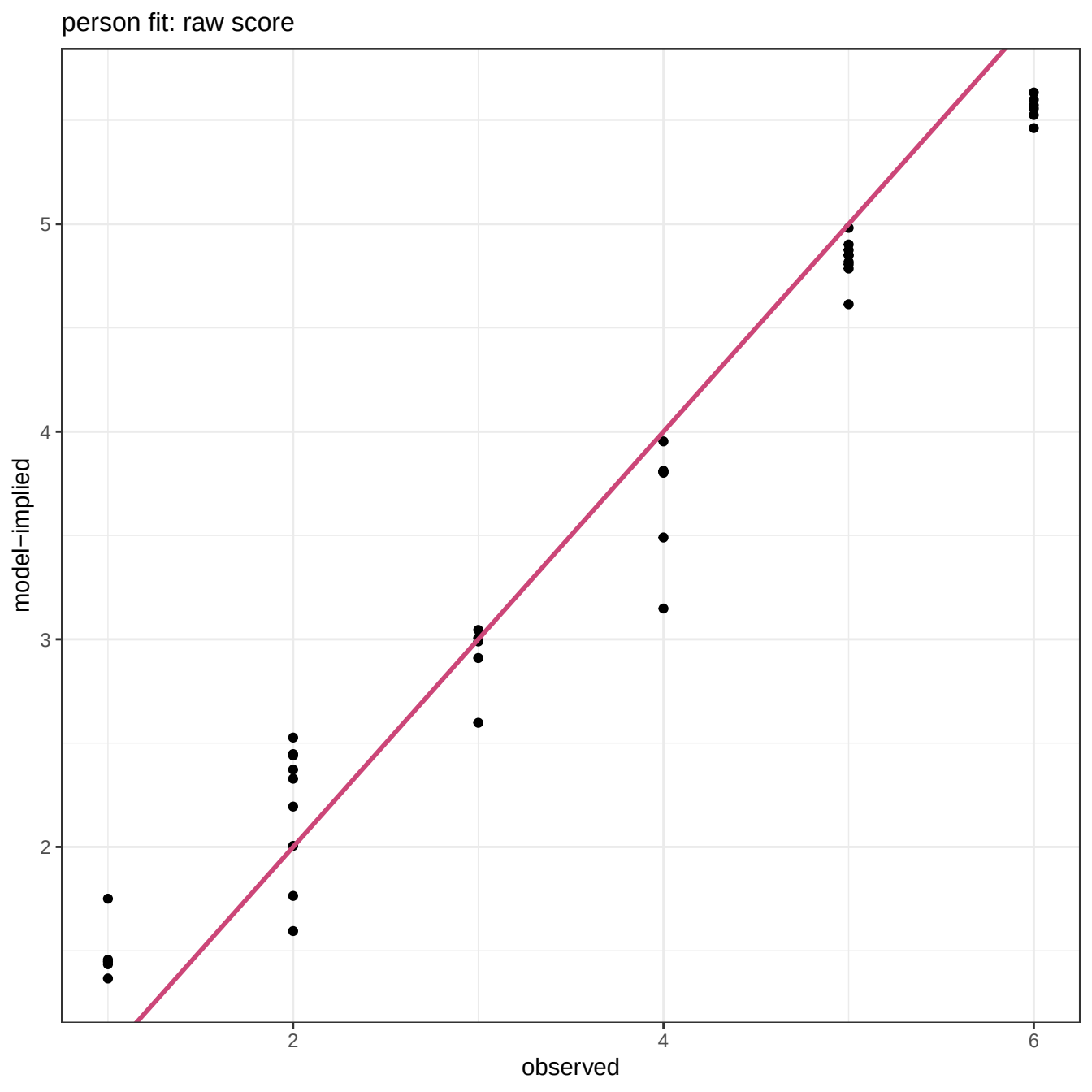

latent trait
4
2
0
−2
2.5
5.0
7.5
10.0
age
p0.05
p0.25
p0.75
p0.95
p0.1
p0.5
p0.9

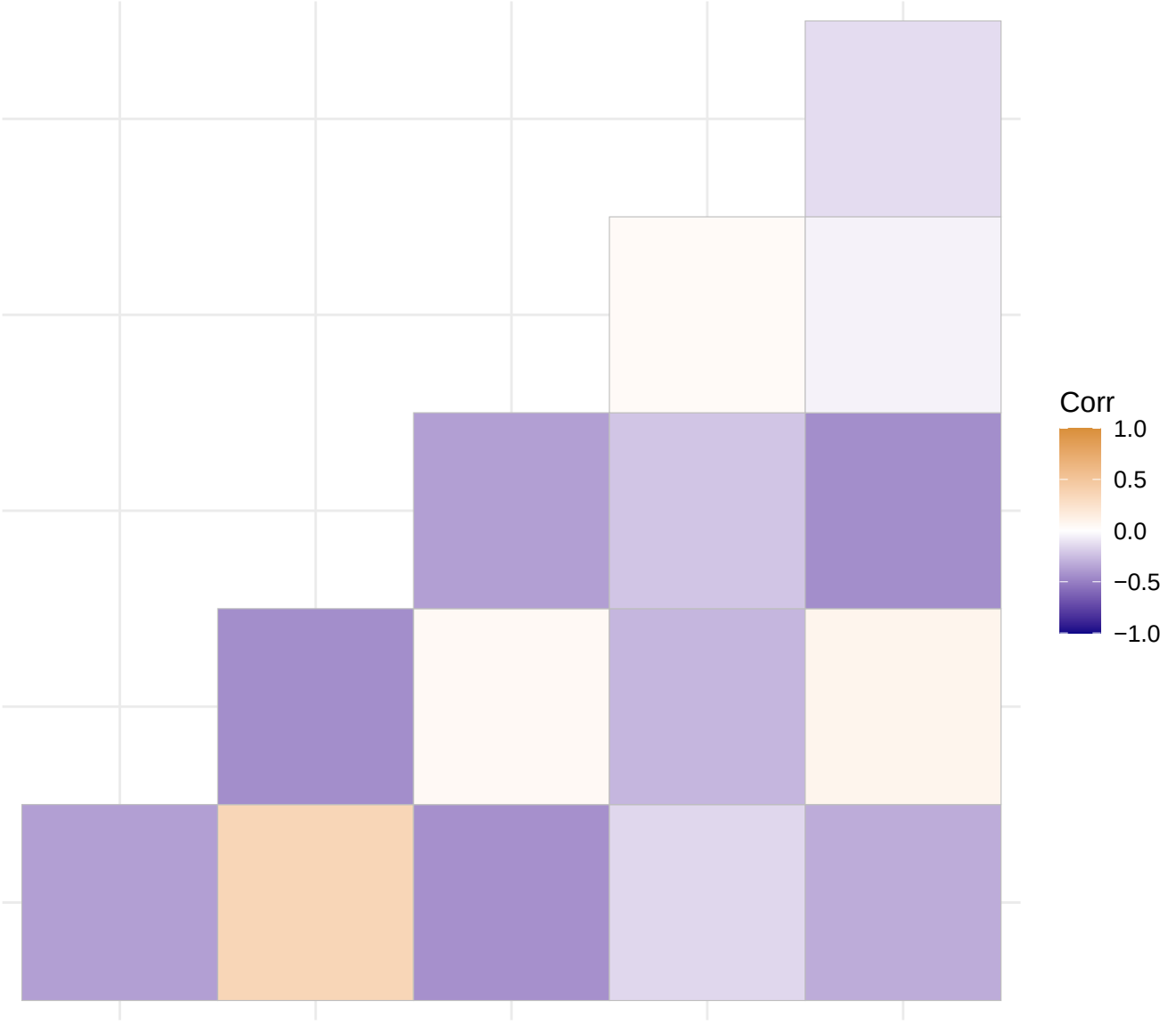

mean
standard deviation
latent trait
0.7
0.6
0.5
0.4
2.2
2.0
1.8
1.6
1.4
2.5
5.0
7.5
10.0
age

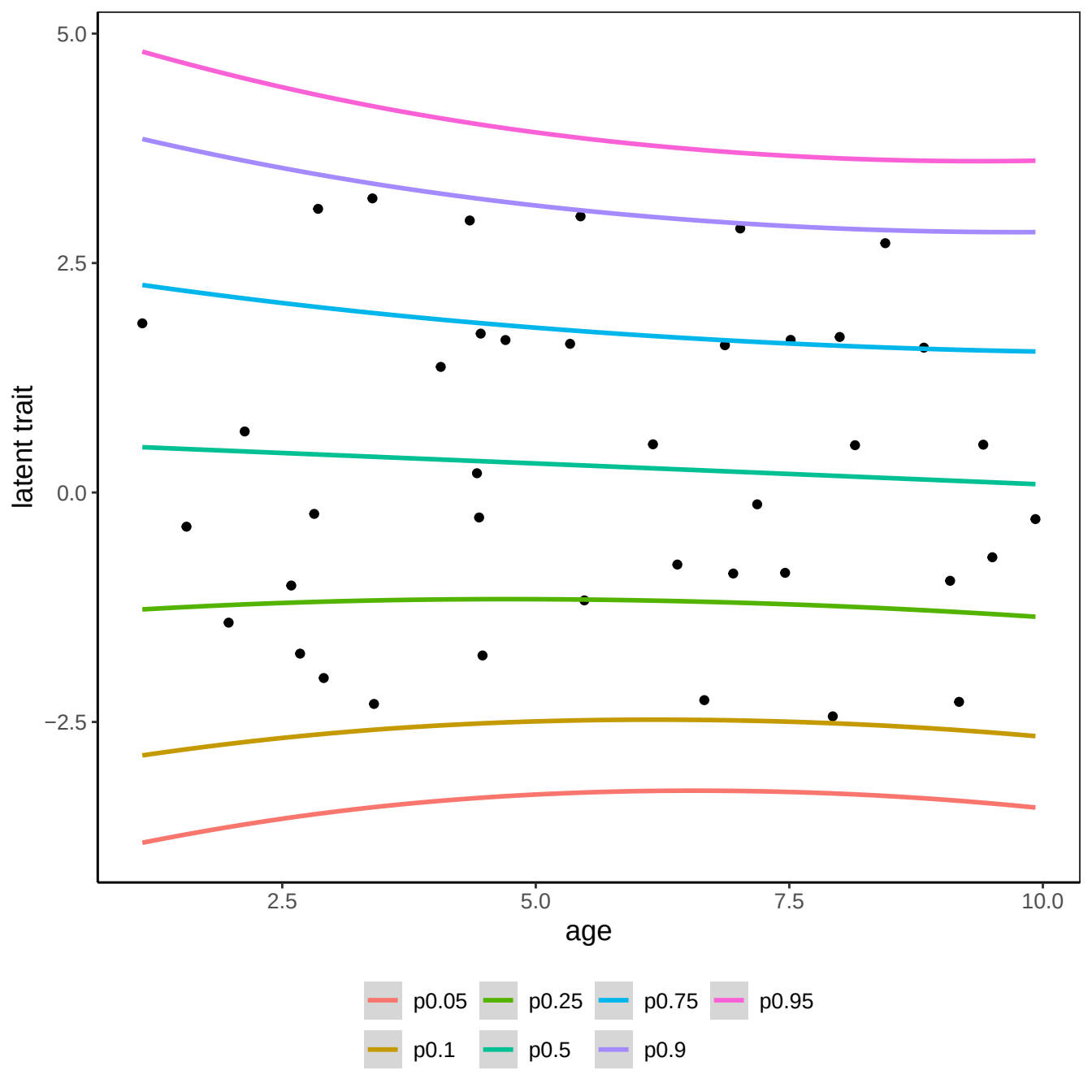

15
10
5
0
frequency of individuals
0
2
4
6
raw score

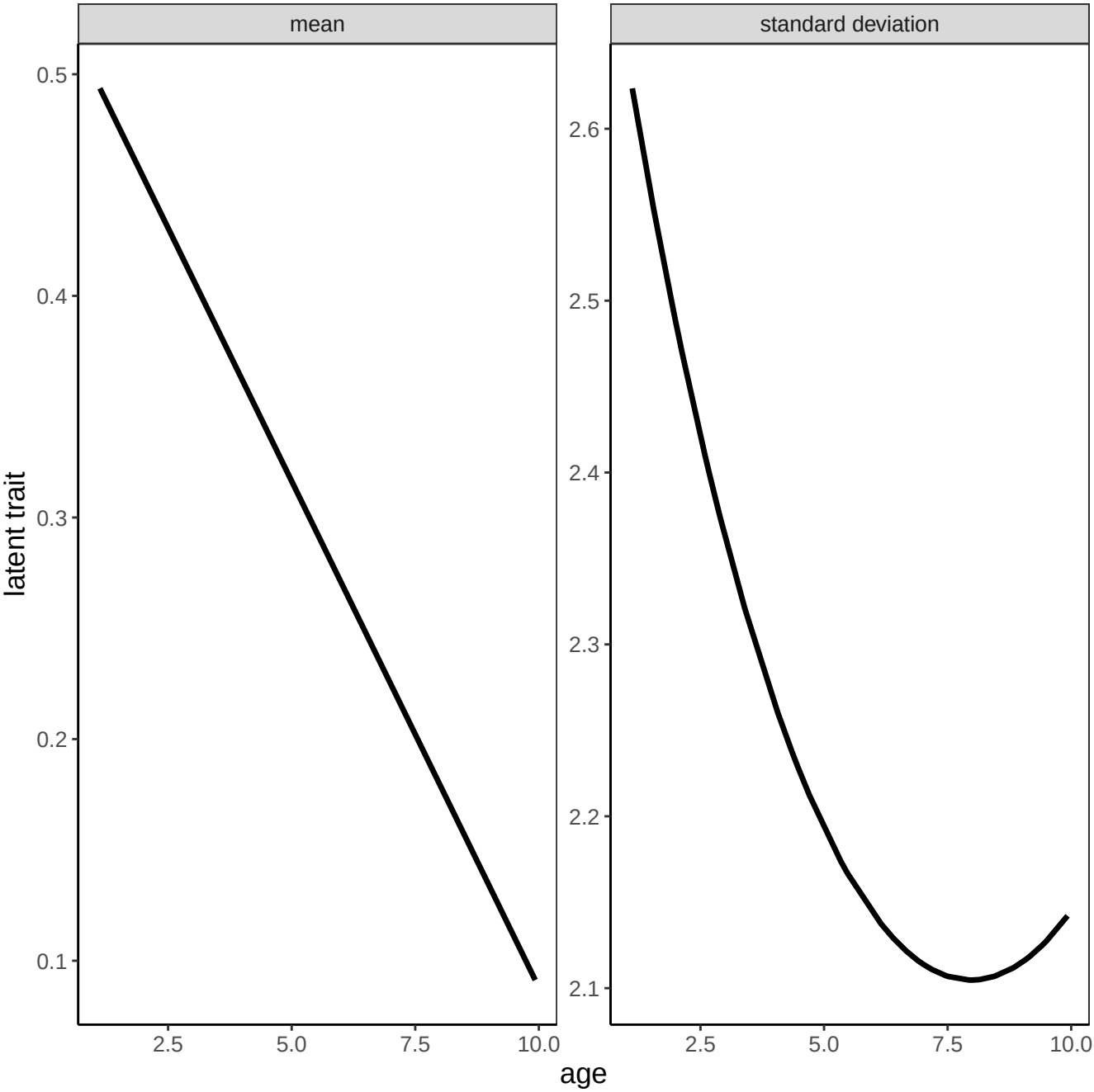

age: [0,5]
age: [5,10]
frequency of individuals
raw score
10.0
7.5
5.0
2.5
0.0
12
8
4
0
0
2
4
6

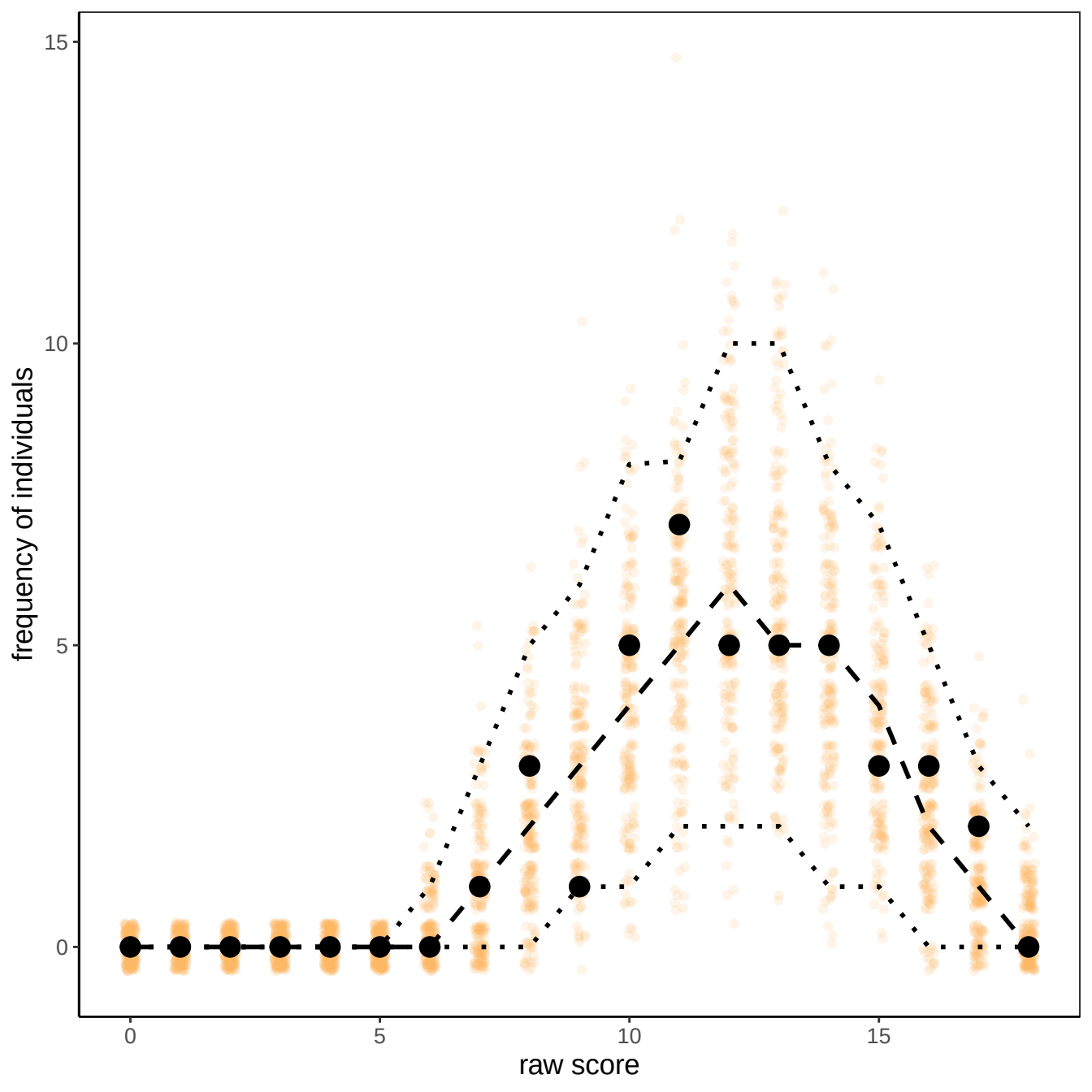

item fit: proportion correct per item
model−implied
observed
0.8
0.7
0.6
0.5
0.4
0.5
0.6
0.7
0.8

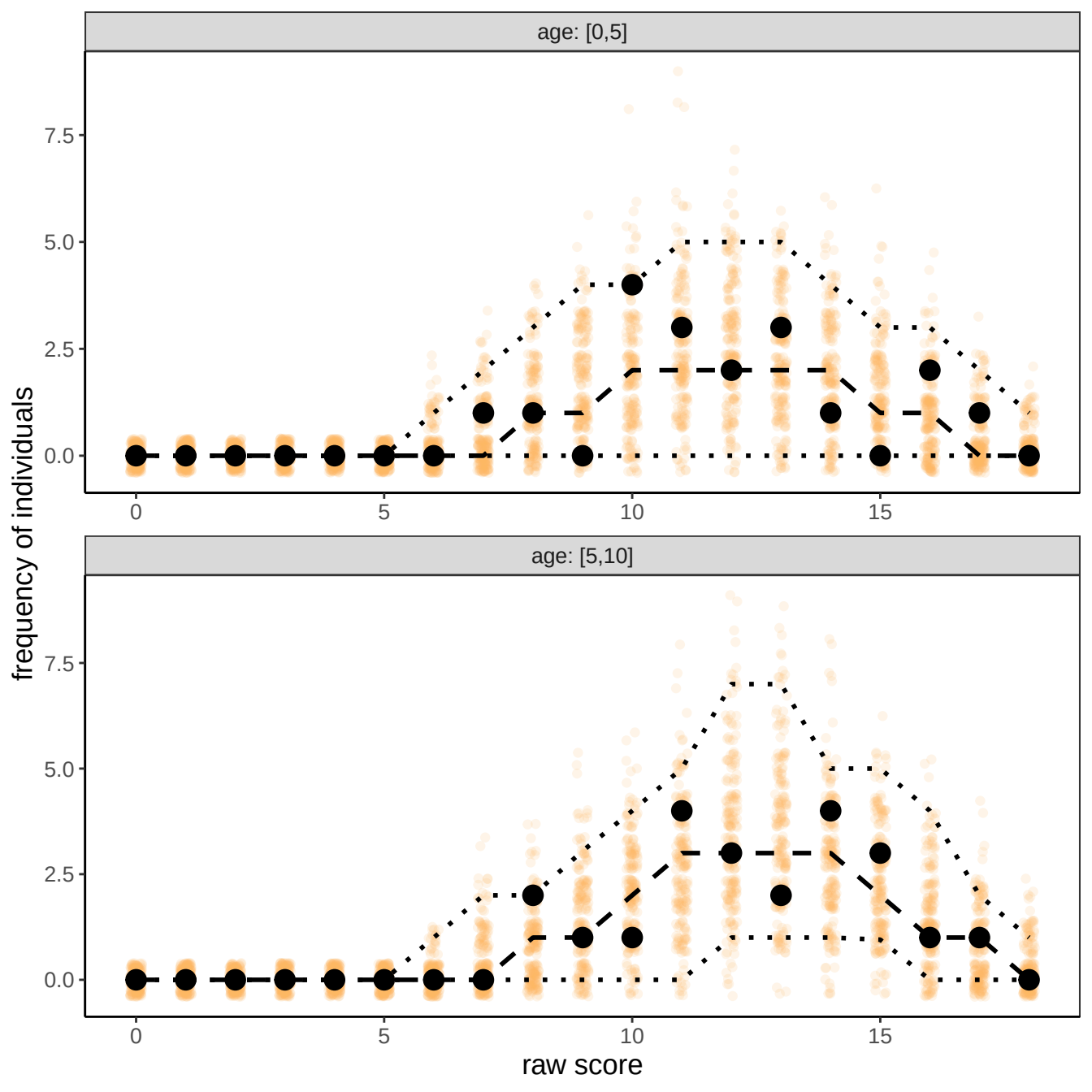

person fit: raw score
model−implied
observed
2
3
4
5
2
4
6

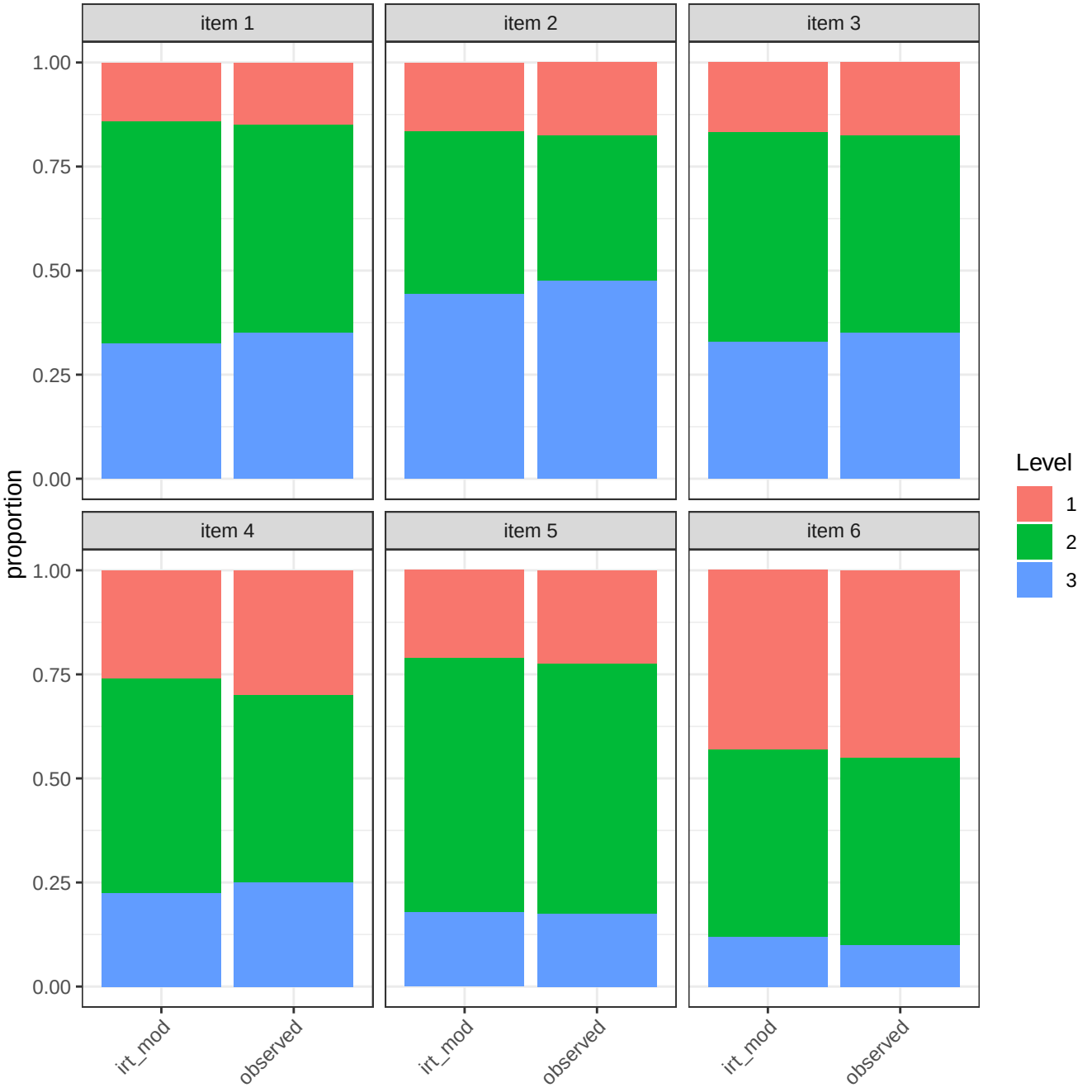

Corr
1.0
0.5
0.0
−0.5
−1.0

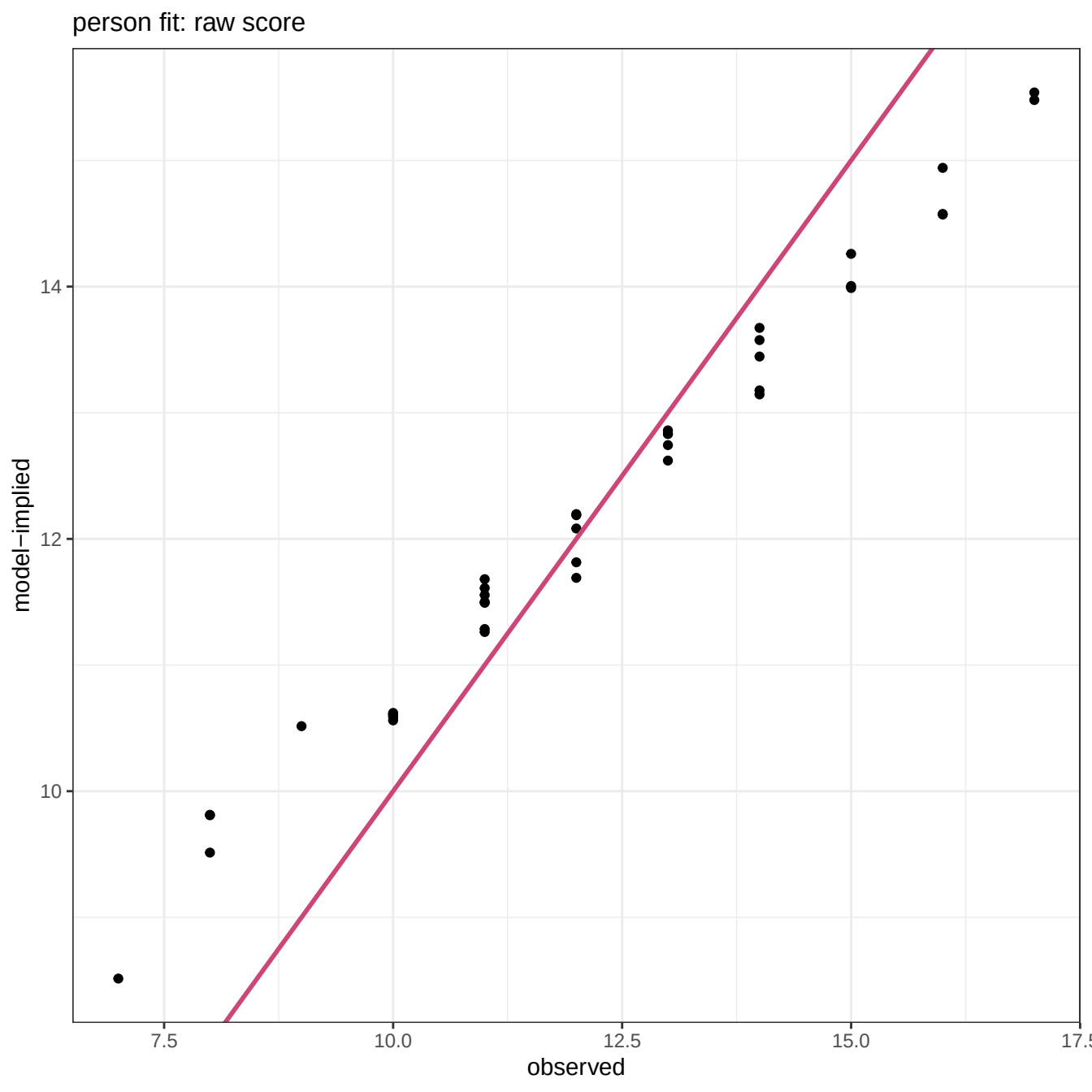

5.0
2.5
0.0
−2.5
latent trait
2.5
5.0
7.5
10.0
age
p0.05
p0.25
p0.75
p0.95
p0.1
p0.5
p0.9

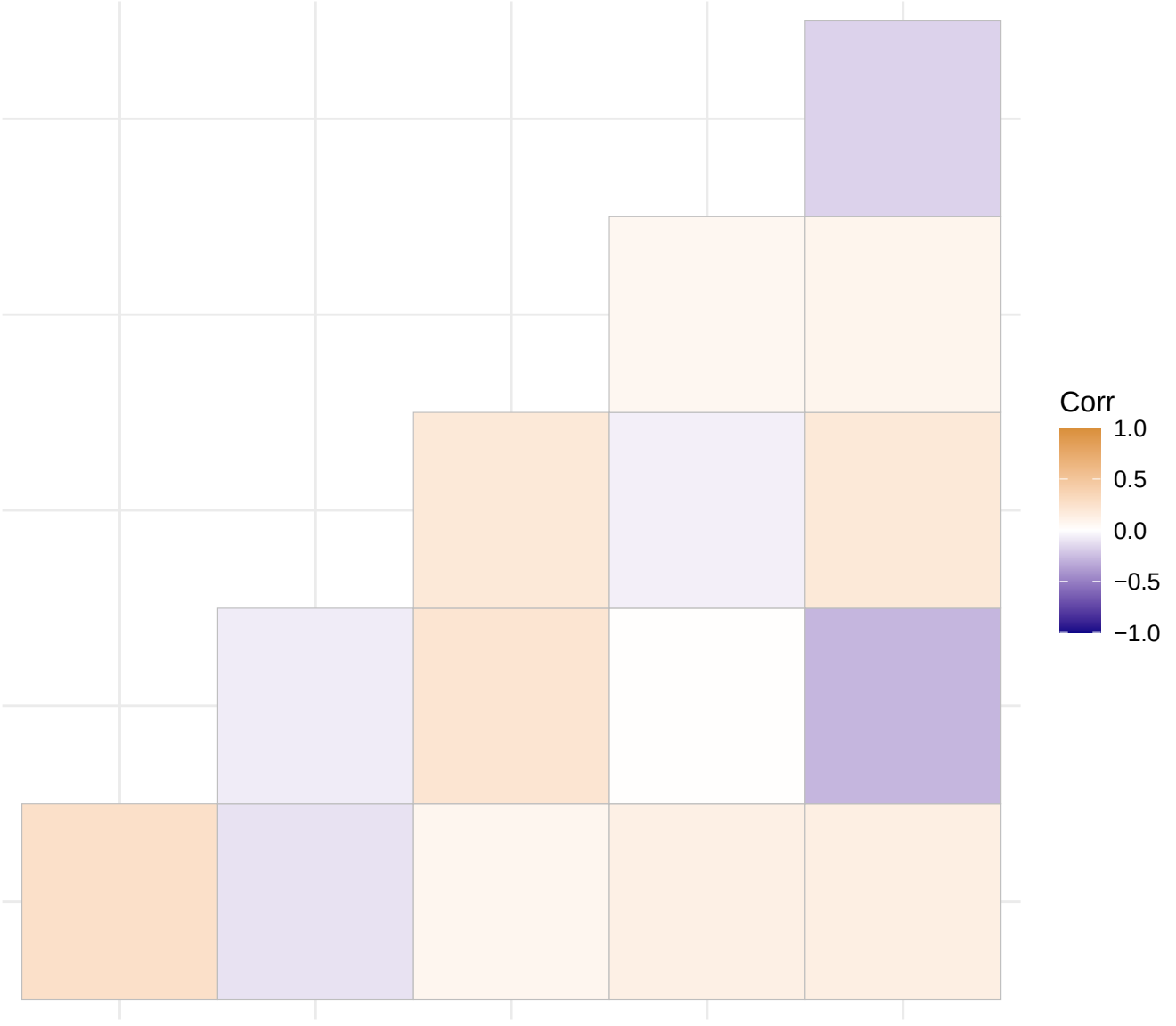

mean
standard deviation
latent trait
age
0.5
0.4
0.3
0.2
0.1
2.6
2.5
2.4
2.3
2.2
2.1
2.5
5.0
7.5
10.0

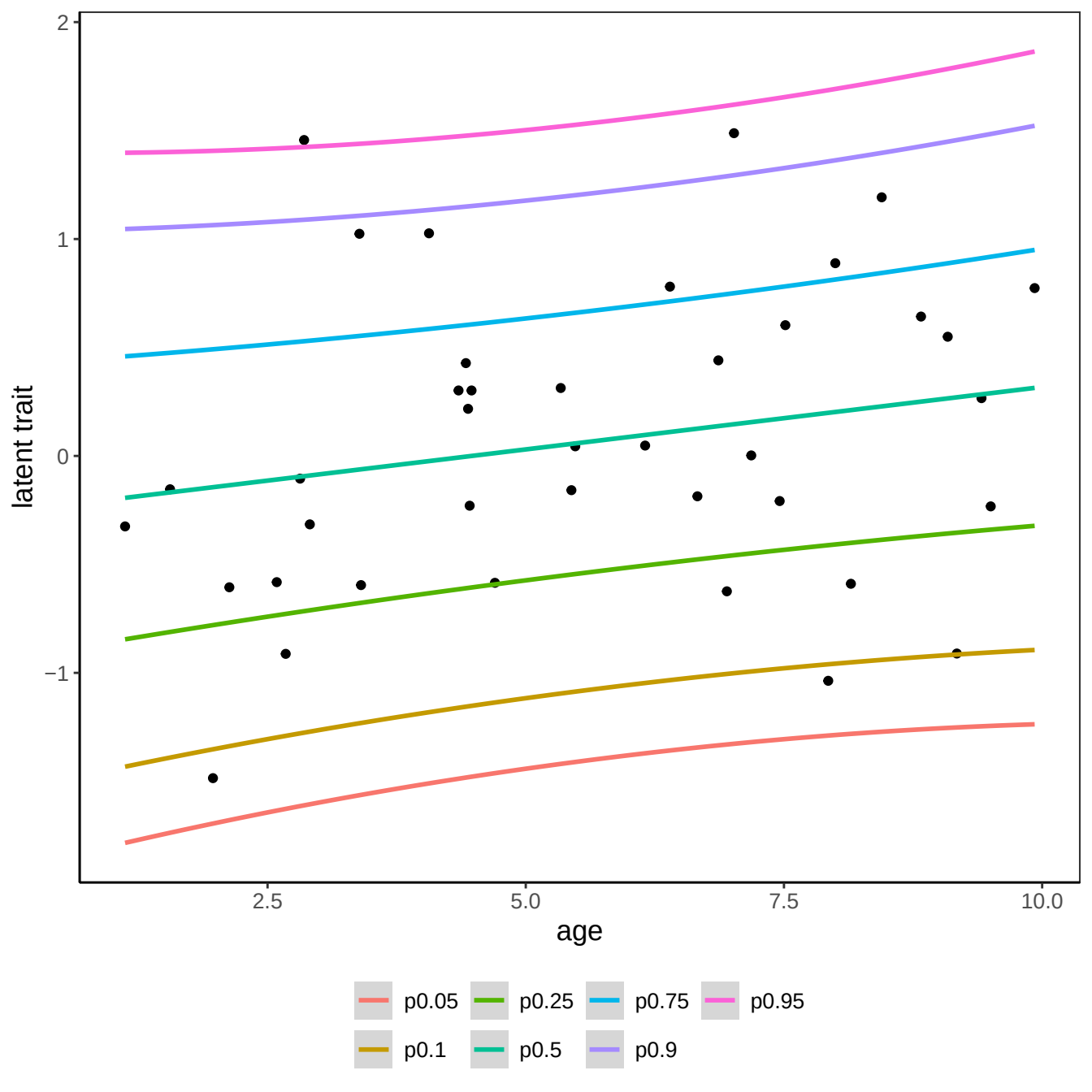

15
10
5
0
frequency of individuals
0
5
10
15
raw score

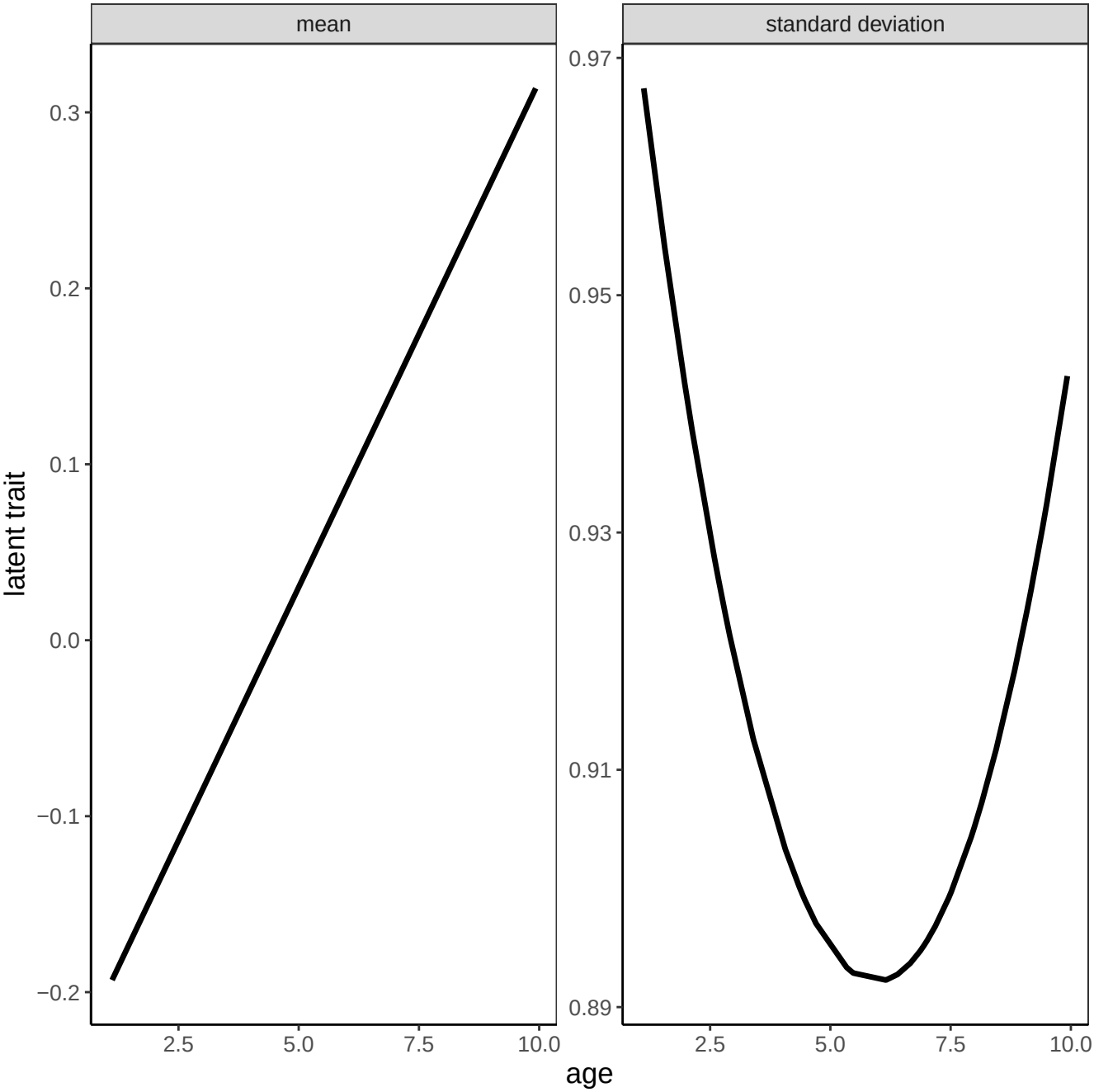

age: [0,5]
age: [5,10]
frequency of individuals
raw score
0.0
2.5
5.0
7.5
0
5
10
15

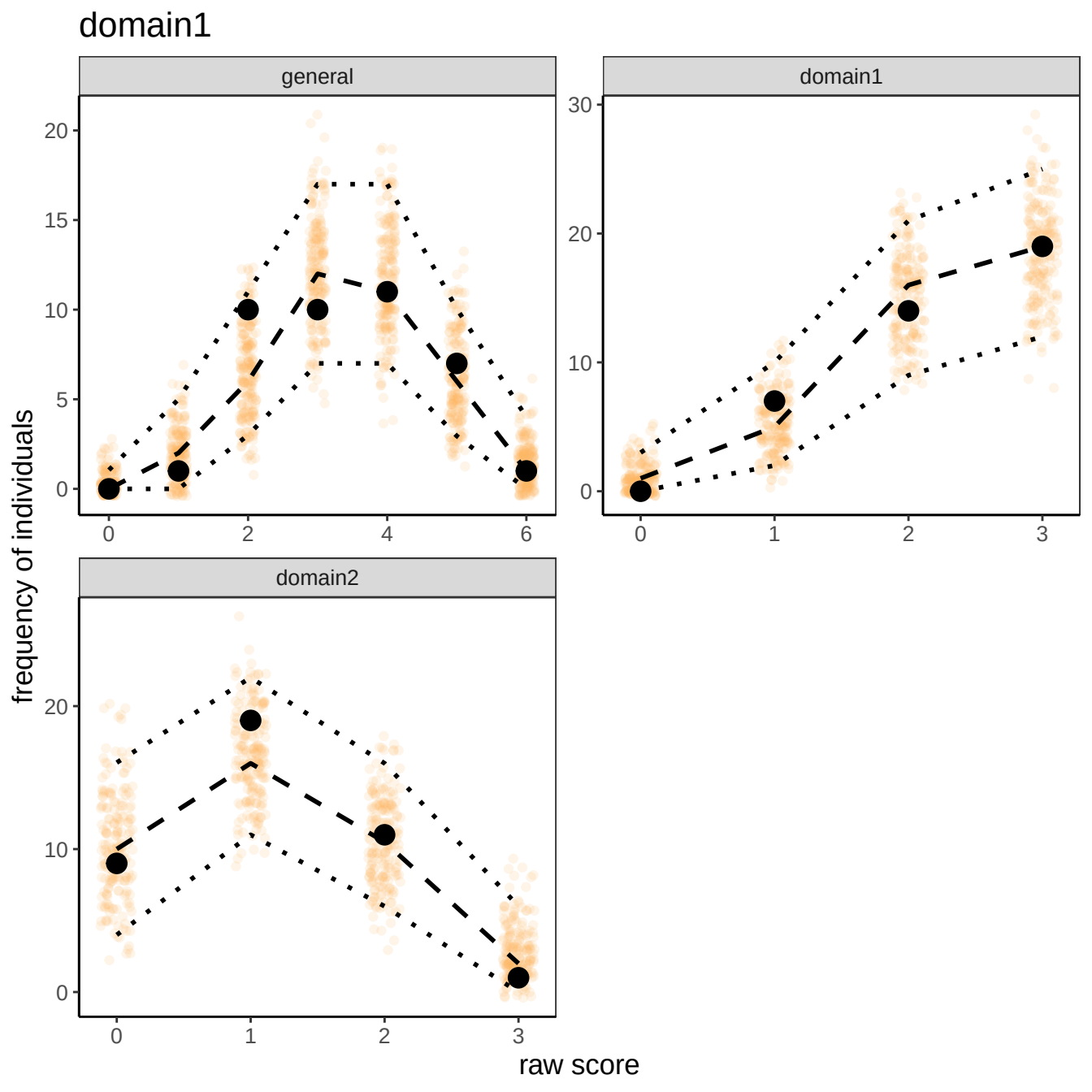

item 1
item 2
item 3
item 4
item 5
item 6
proportion
1.00
0.75
0.50
0.25
0.00
irt_mod
observed
Level
1
2
3

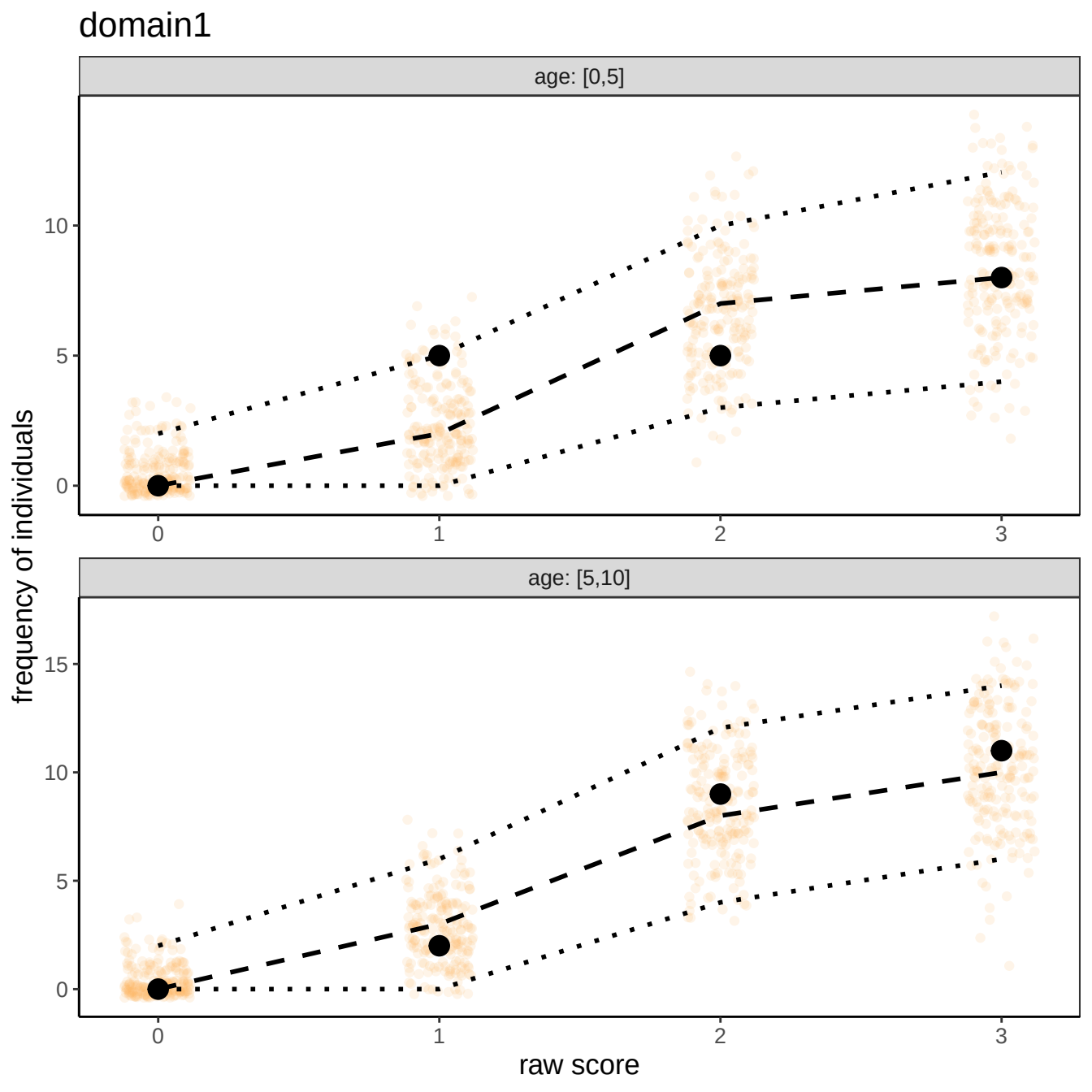

person fit: raw score
model−implied
observed
14
12
10
7.5
10.0
12.5
15.0
17.5

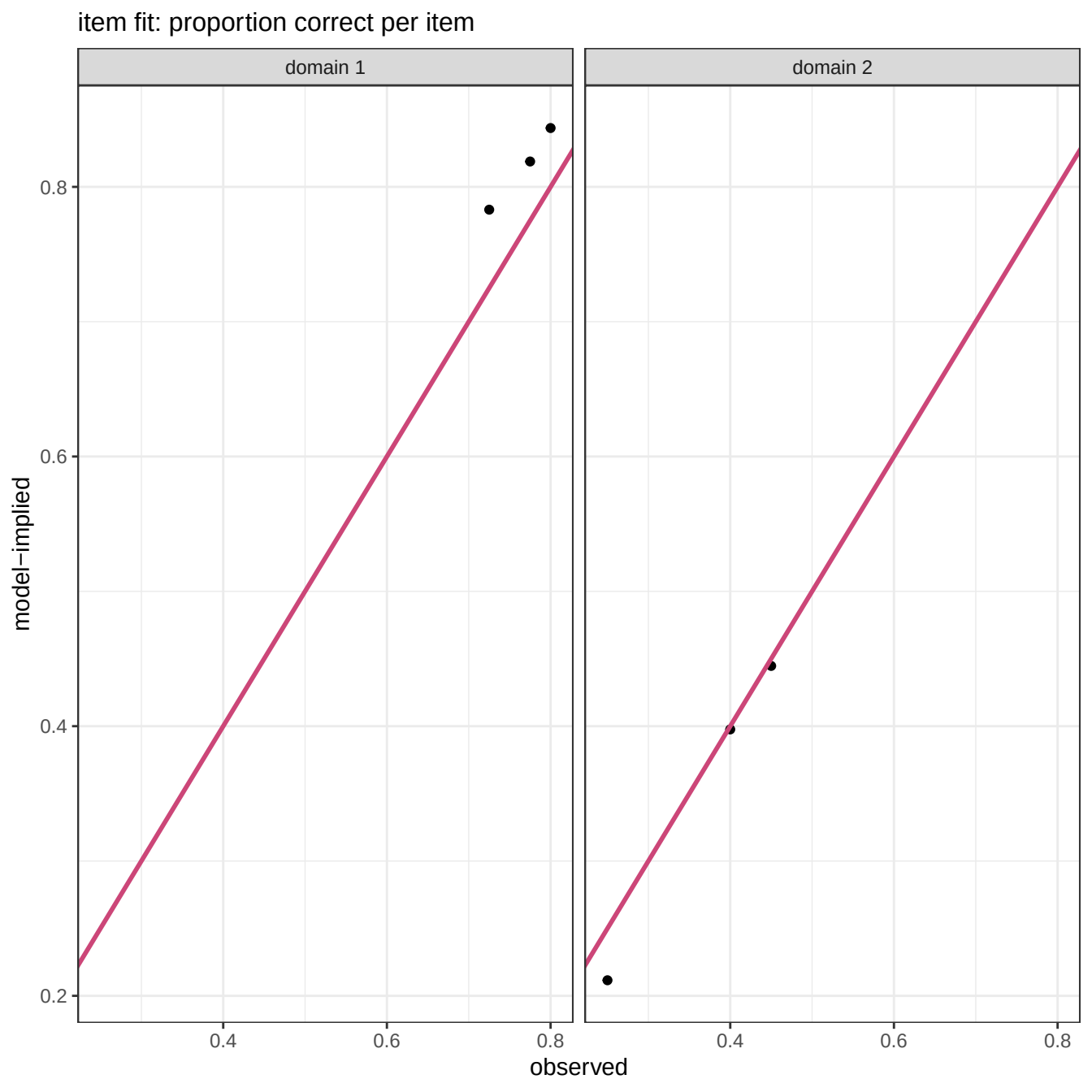

Corr
1.0
0.5
0.0
−0.5
−1.0

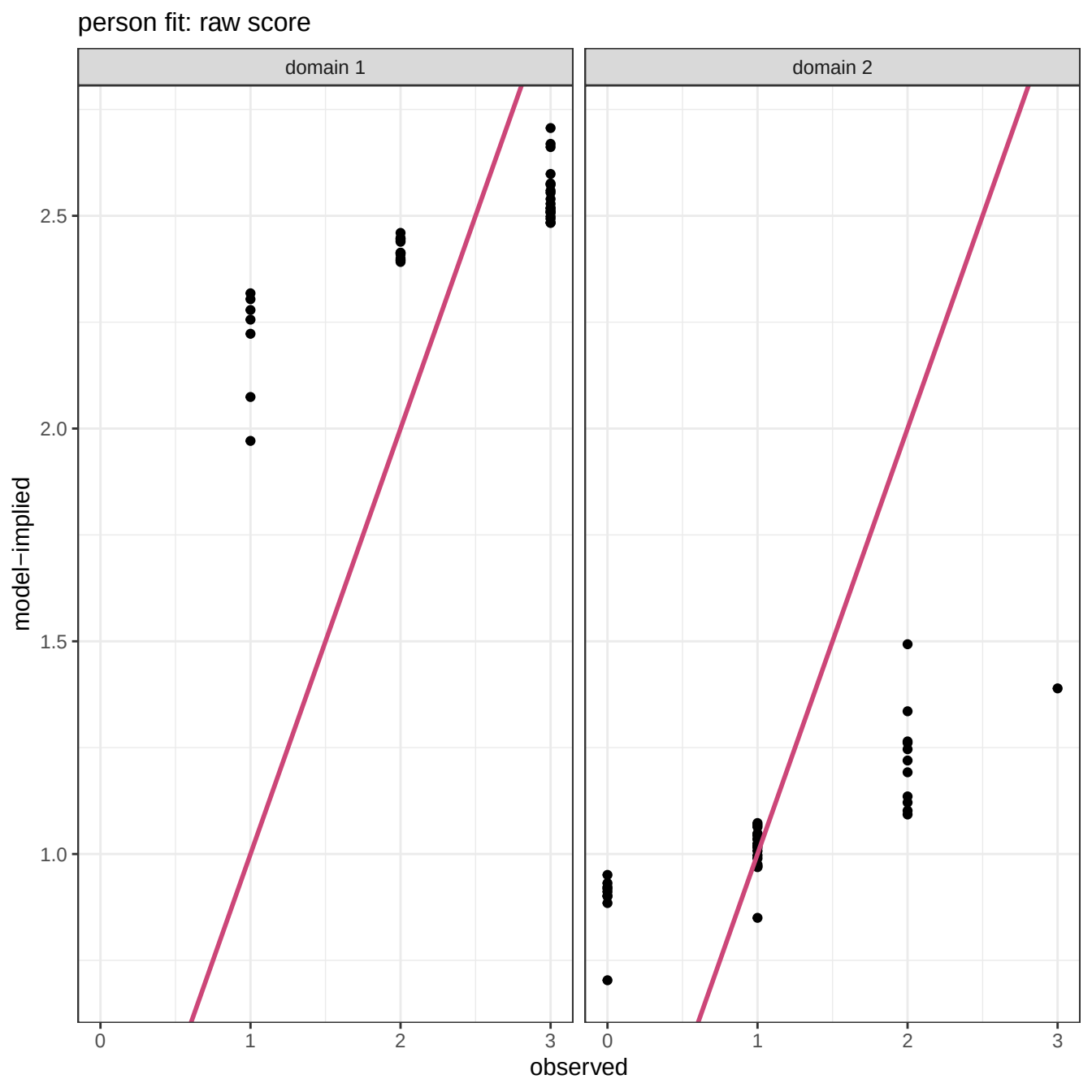

2
1
0
−1
latent trait
2.5
5.0
7.5
10.0
age
p0.05
p0.25
p0.75
p0.95
p0.1
p0.5
p0.9

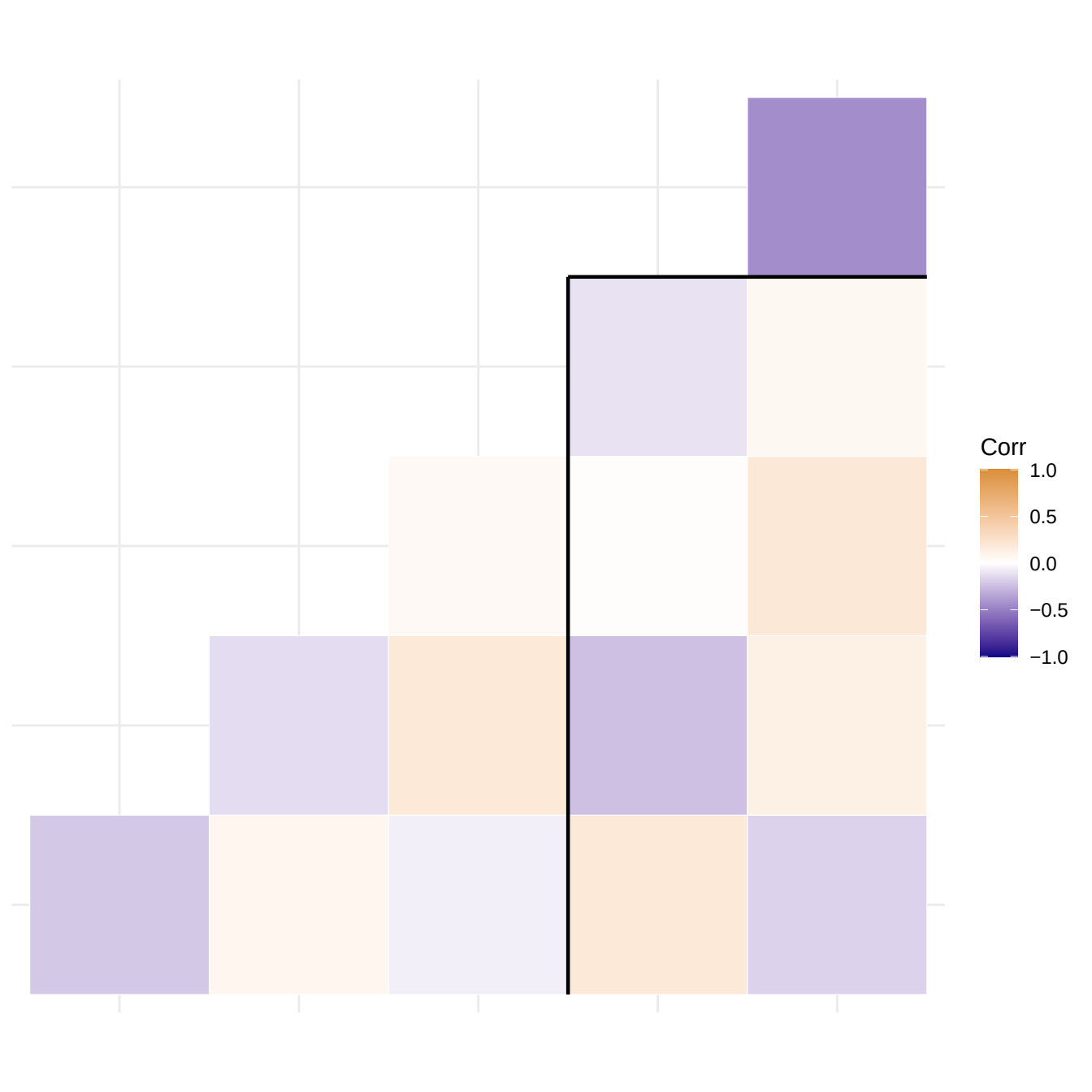

mean
standard deviation
latent trait
age
0.3
0.2
0.1
0.0
−0.1
−0.2
0.97
0.95
0.93
0.91
0.89
2.5
5.0
7.5
10.0

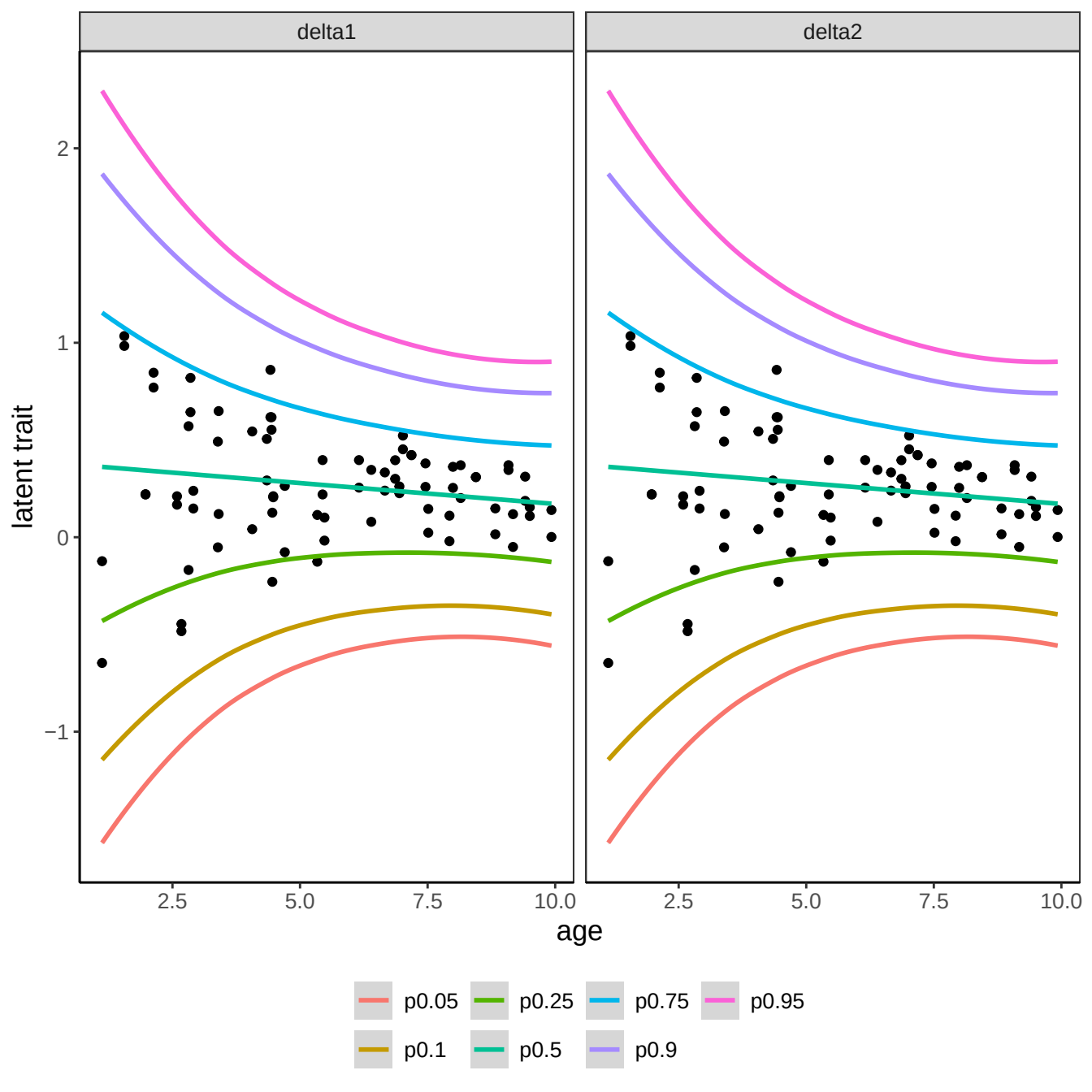

domain1
general
domain1
domain2
frequency of individuals
raw score
0
5
10
15
20
30
0
1
2
3
4
6

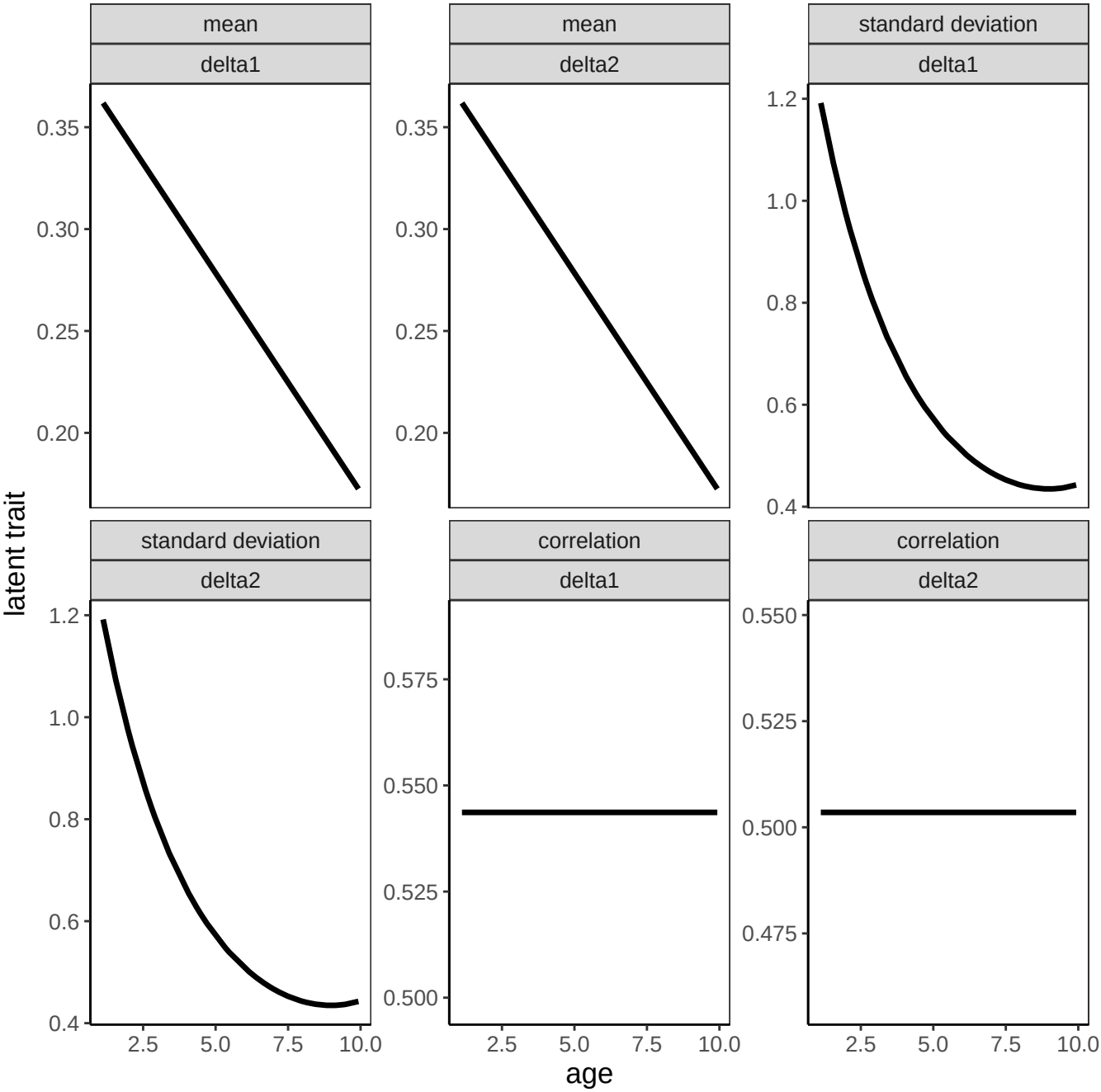

domain1
age: [0,5]
age: [5,10]
frequency of individuals
raw score
0
5
10
15
0
1
2
3

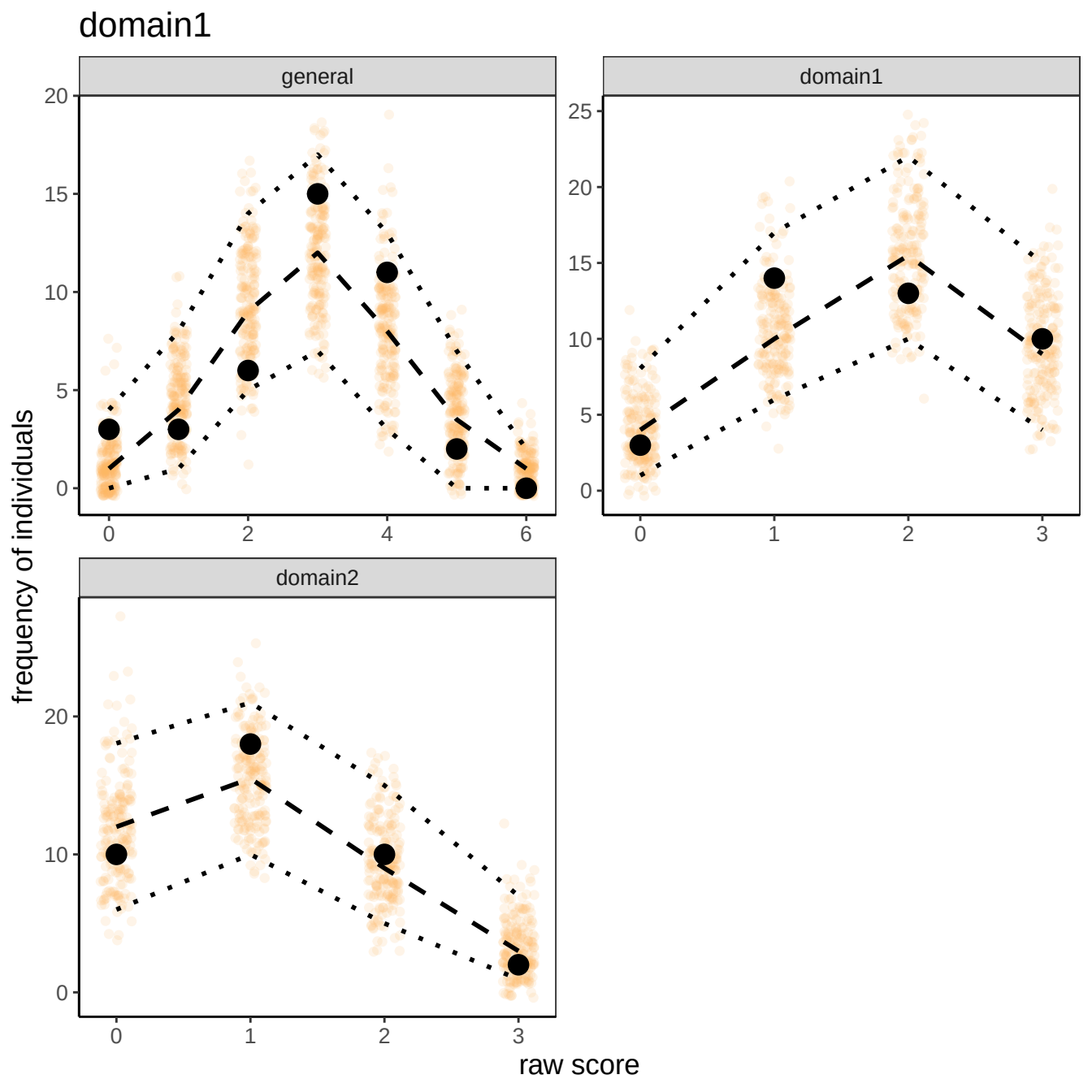

item fit: proportion correct per item
domain 1
domain 2
model−implied
observed
0.2
0.4
0.6
0.8
0.4
0.6
0.8

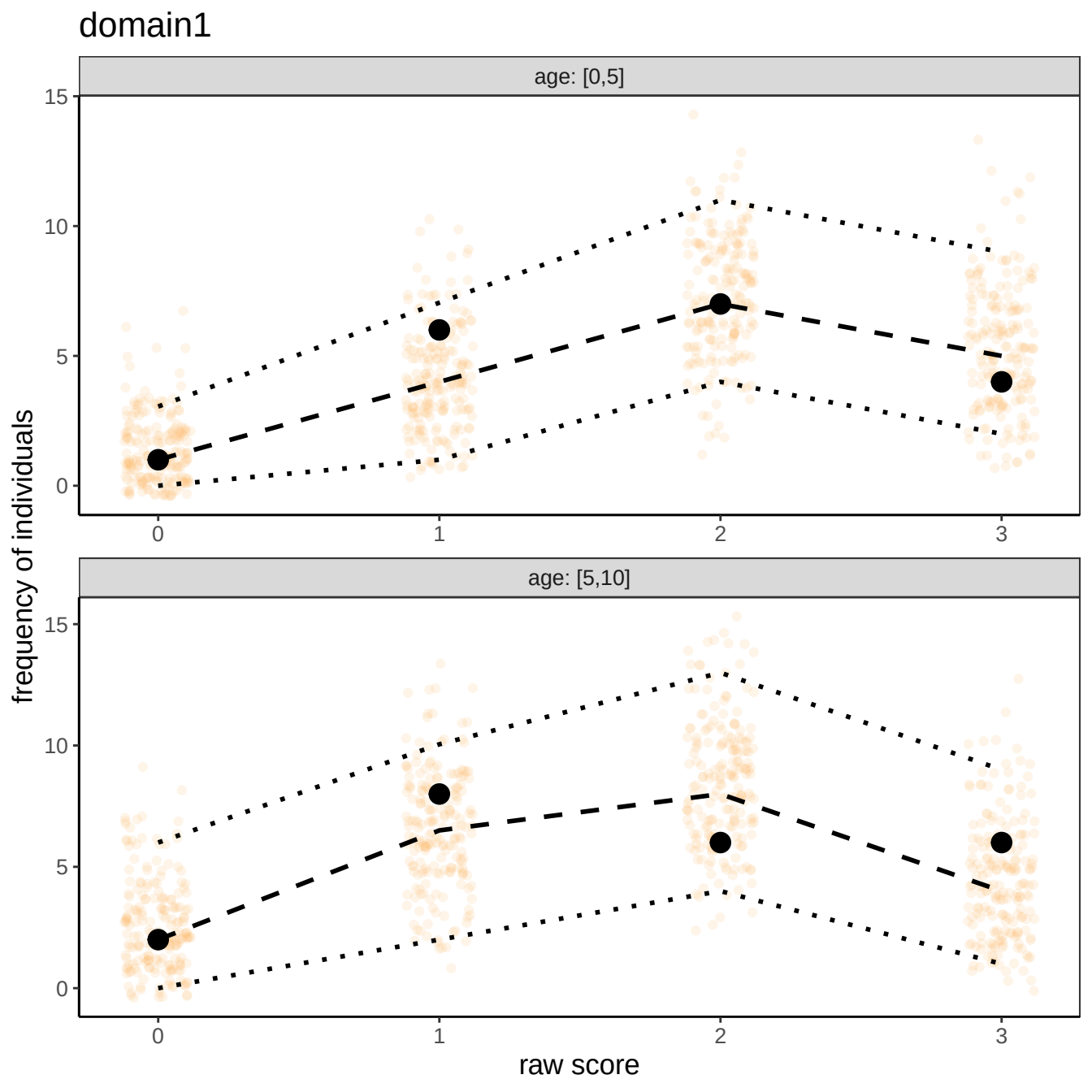

person fit: raw score
domain 1
domain 2
model−implied
observed
2.5
2.0
1.5
1.0
0
1
2
3

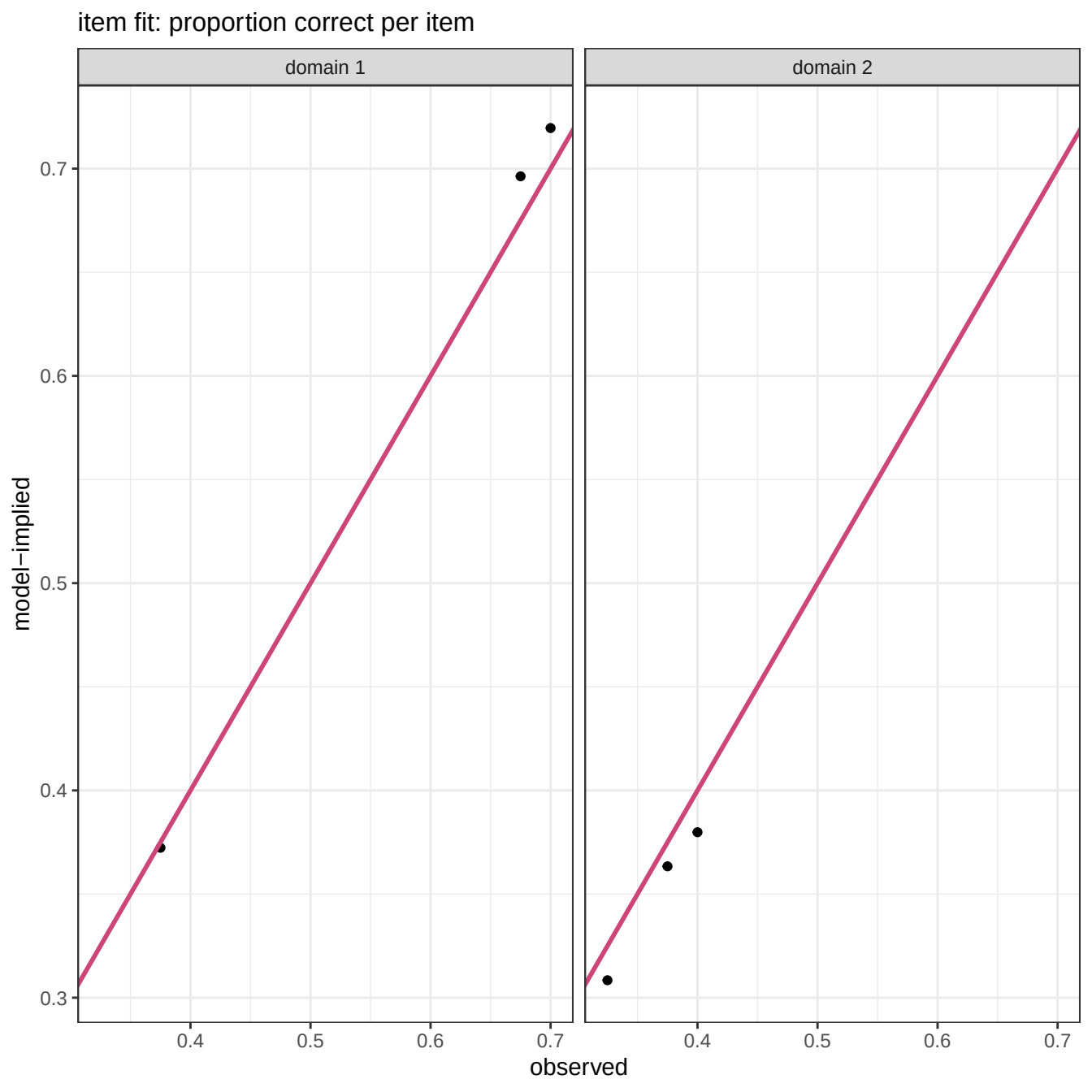

Corr
1.0
0.5
0.0
−0.5
−1.0

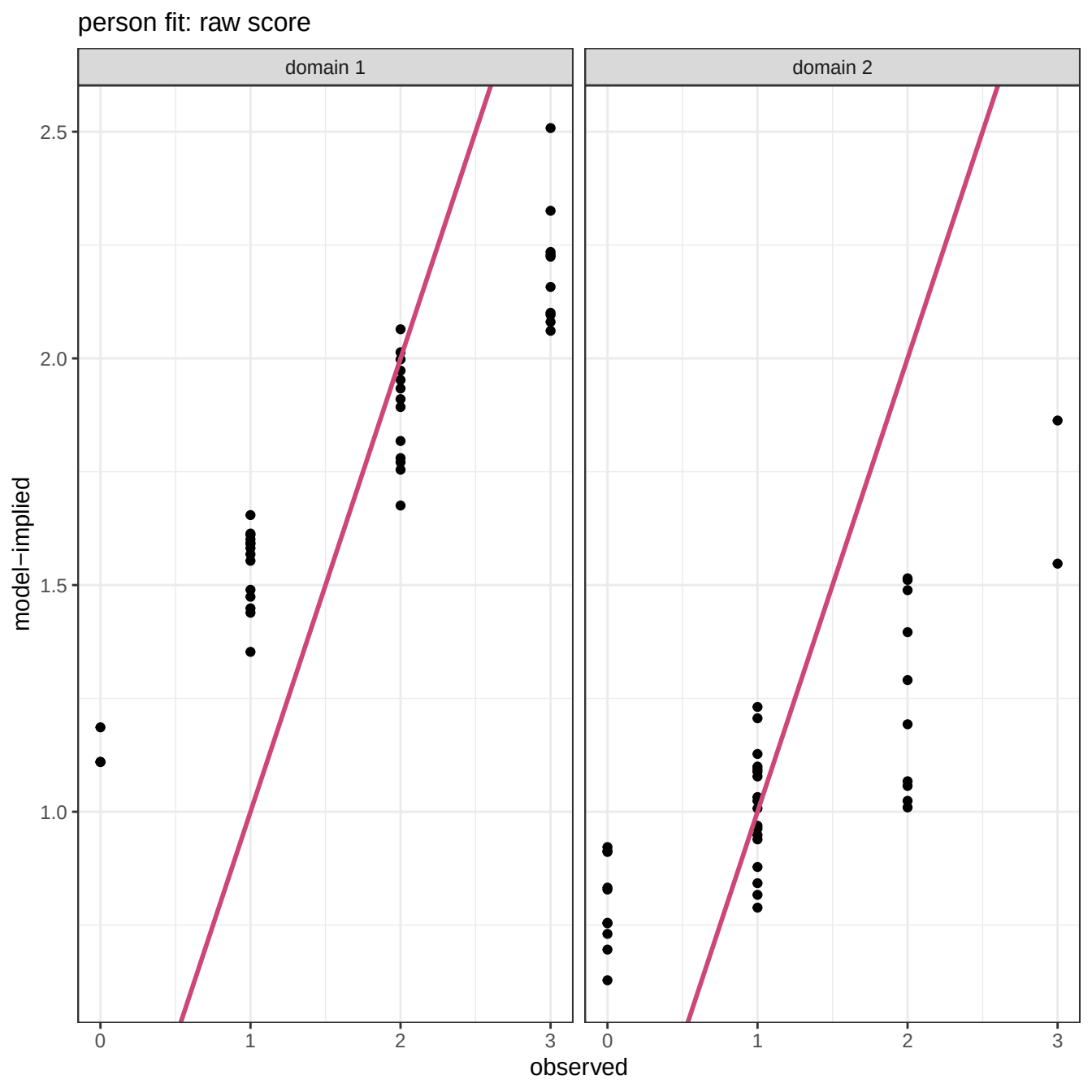

delta1
delta2
latent trait
2
1
0
−1
2.5
5.0
7.5
10.0
age
p0.05
p0.25
p0.75
p0.95
p0.1
p0.5
p0.9

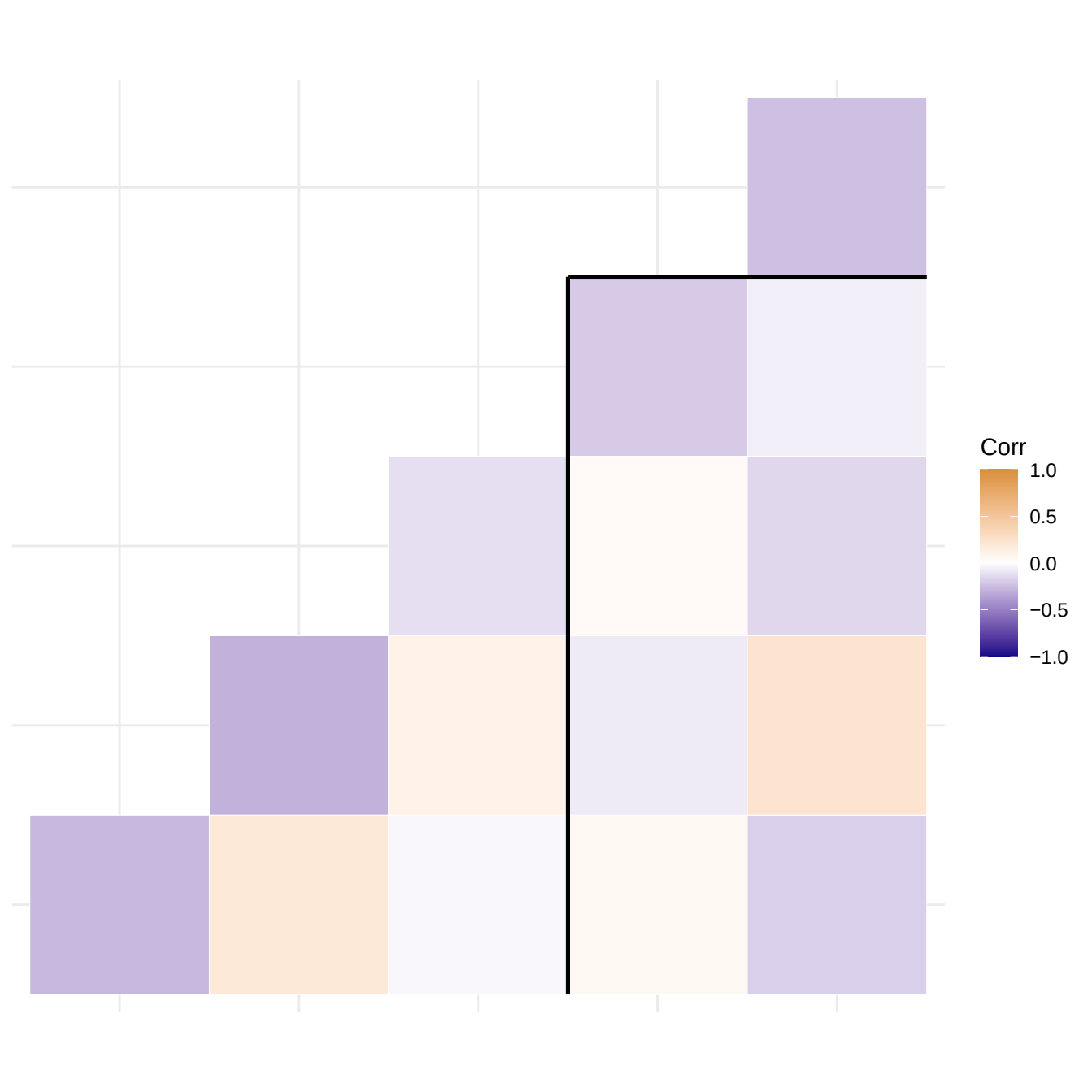

mean
delta1
mean
delta2
standard deviation
delta1
standard deviation
delta2
correlation
delta1
correlation
delta2
latent trait
age
0.35
0.30
0.25
0.20
1.2
1.0
0.8
0.6
0.4
0.575
0.550
0.525
0.500
0.475
2.5
5.0
7.5
10.0

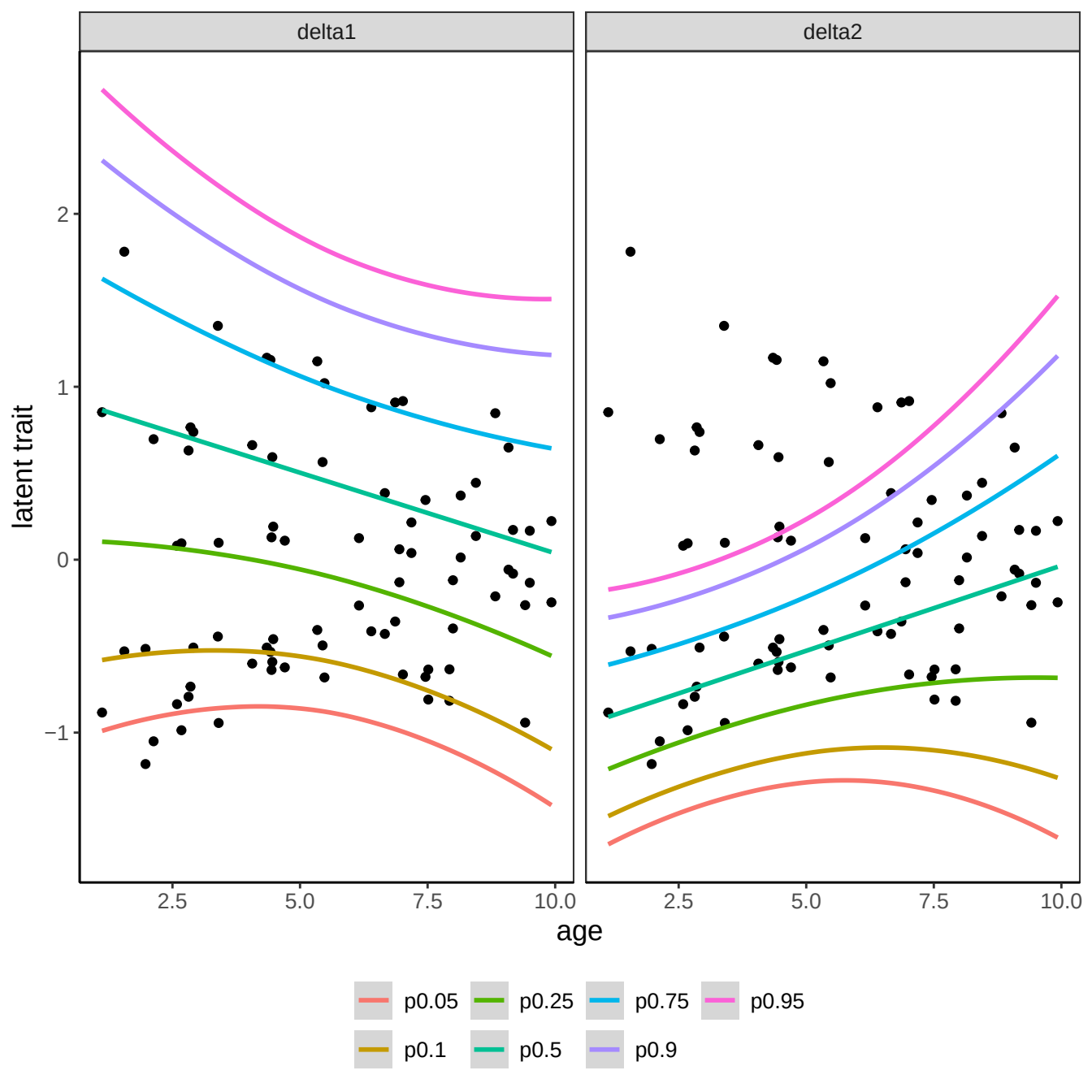

domain1
general
domain1
domain2
frequency of individuals
raw score
0
5
10
15
20
25
0
2
4
6
0
1
2
3

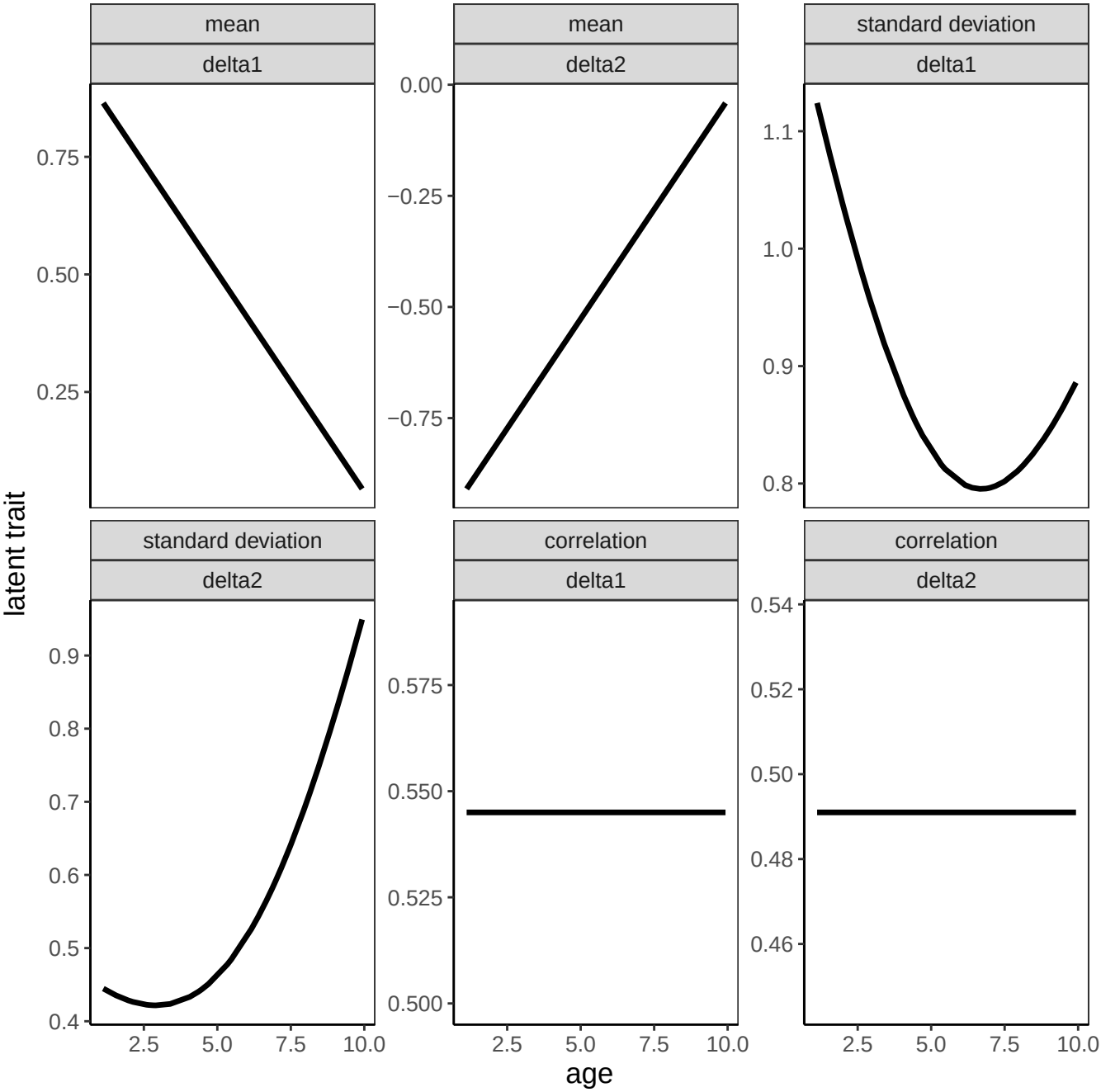

domain1
age: [0,5]
age: [5,10]
frequency of individuals
raw score
0
5
10
15
0
1
2
3

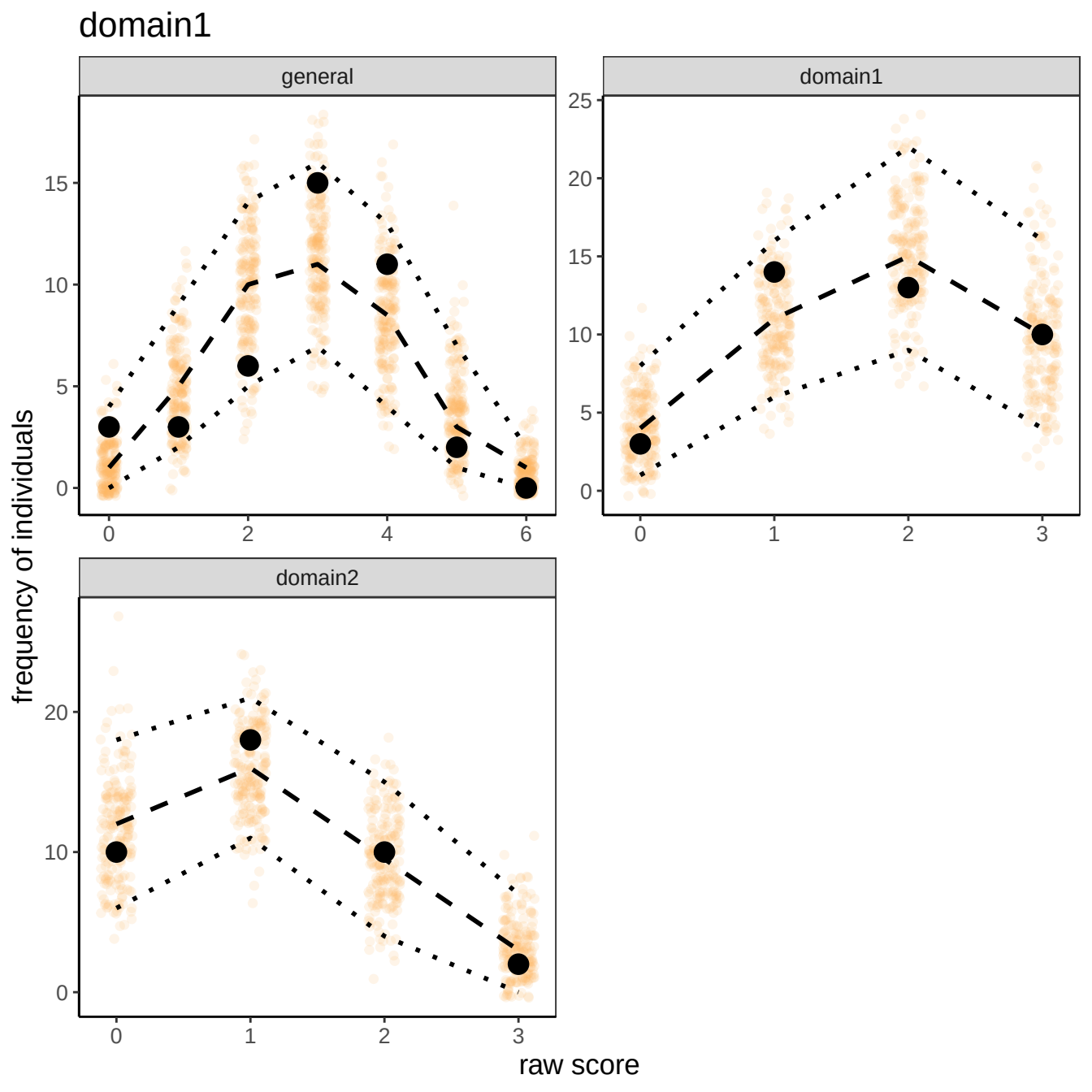

item fit: proportion correct per item
domain 1
domain 2
model−implied
observed
0.3
0.4
0.5
0.6
0.7
0.4
0.5
0.6
0.7

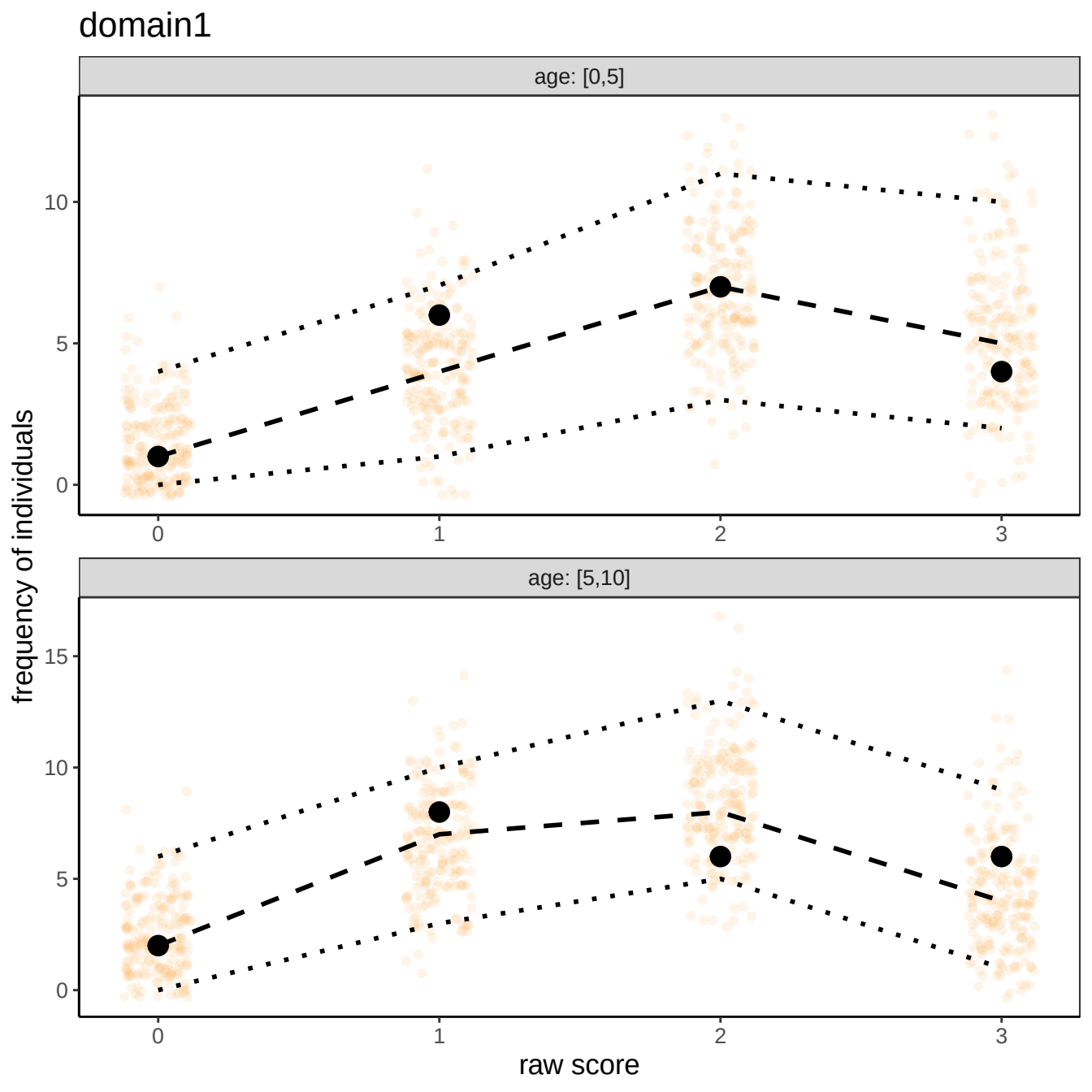

person fit: raw score
domain 1
domain 2
model−implied
observed
2.5
2.0
1.5
1.0
0
1
2
3

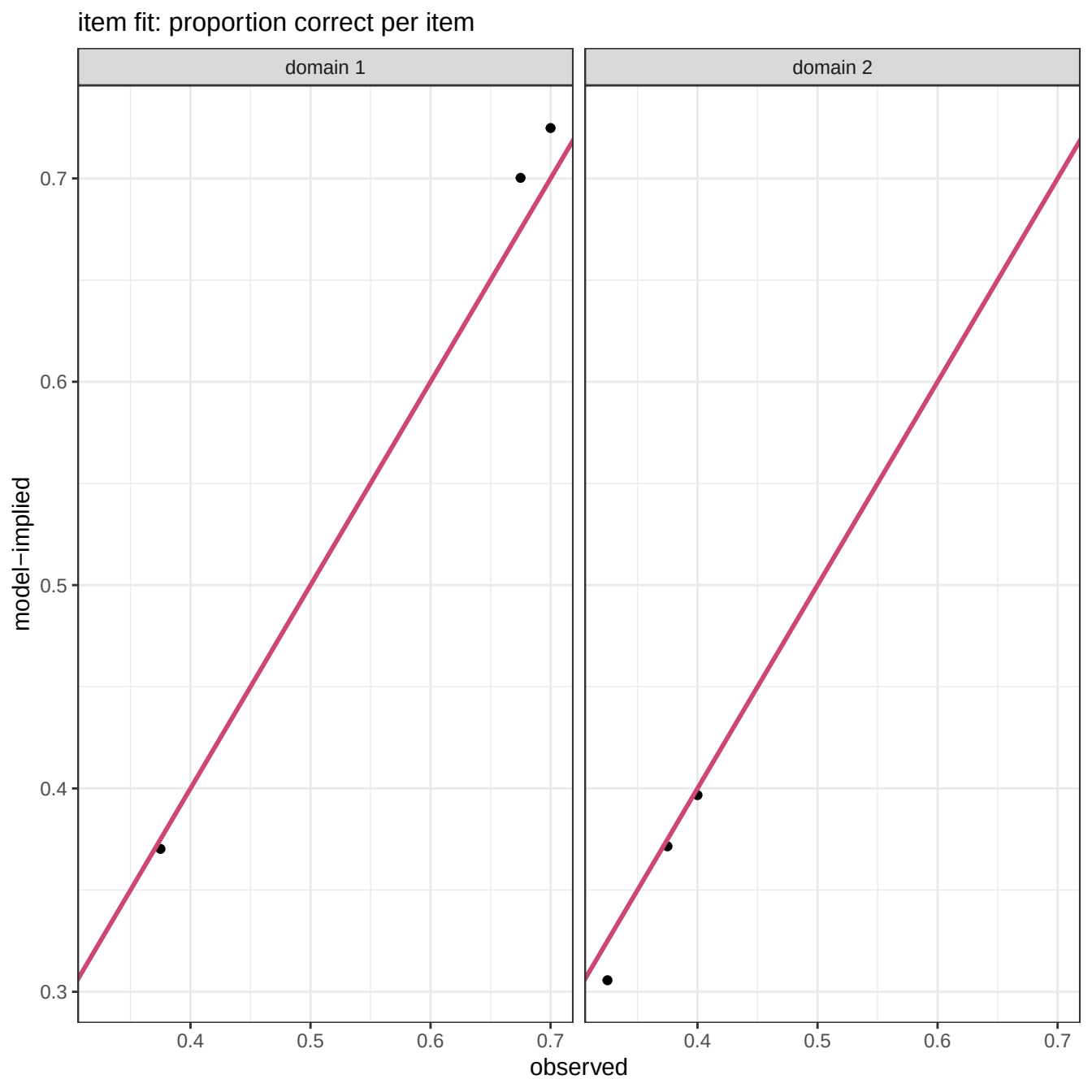

Corr
1.0
0.5
0.0
−0.5
−1.0

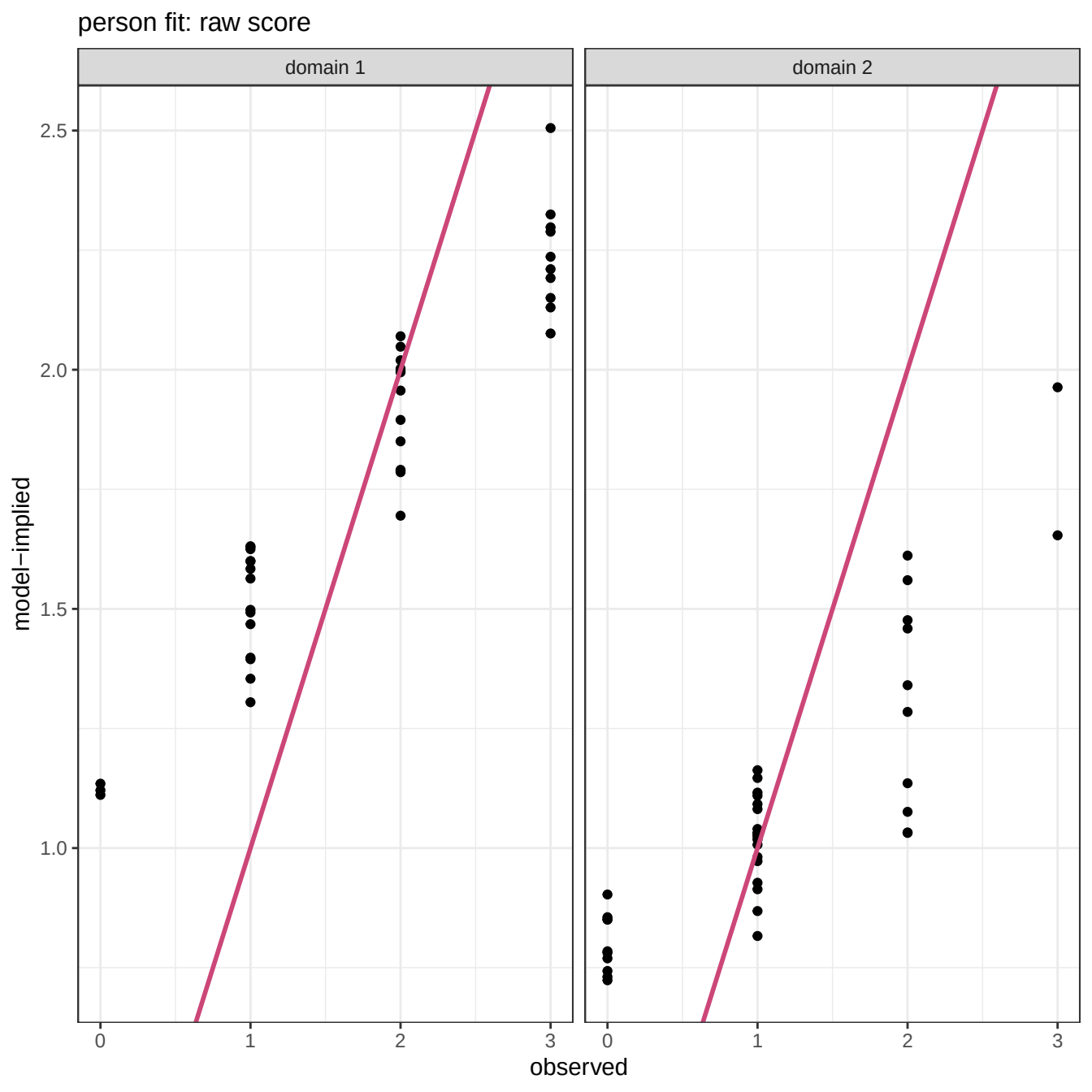

delta1
delta2
latent trait
2
1
0
−1
2.5
5.0
7.5
10.0
age
p0.05
p0.1
p0.25
p0.5
p0.75
p0.9
p0.95

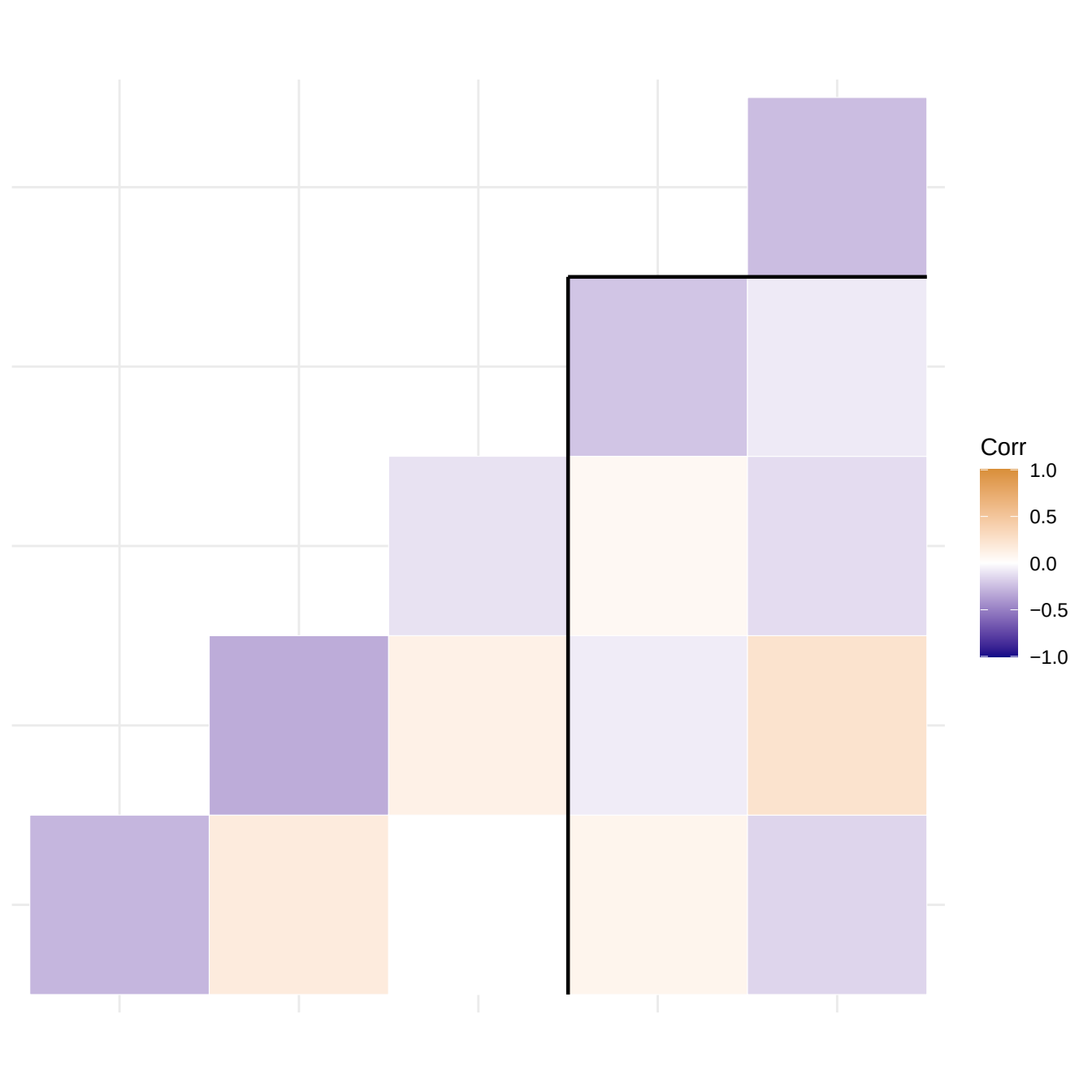

mean
delta1
mean
delta2
standard deviation
delta1
standard deviation
delta2
correlation
delta1
correlation
delta2
latent trait
age
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
1.1
1.0
0.9
0.8
0.9
0.8
0.7
0.6
0.5
0.4
0.575
0.550
0.525
0.500
0.54
0.52
0.50
0.48
0.46
2.5
5.0
7.5
10.0

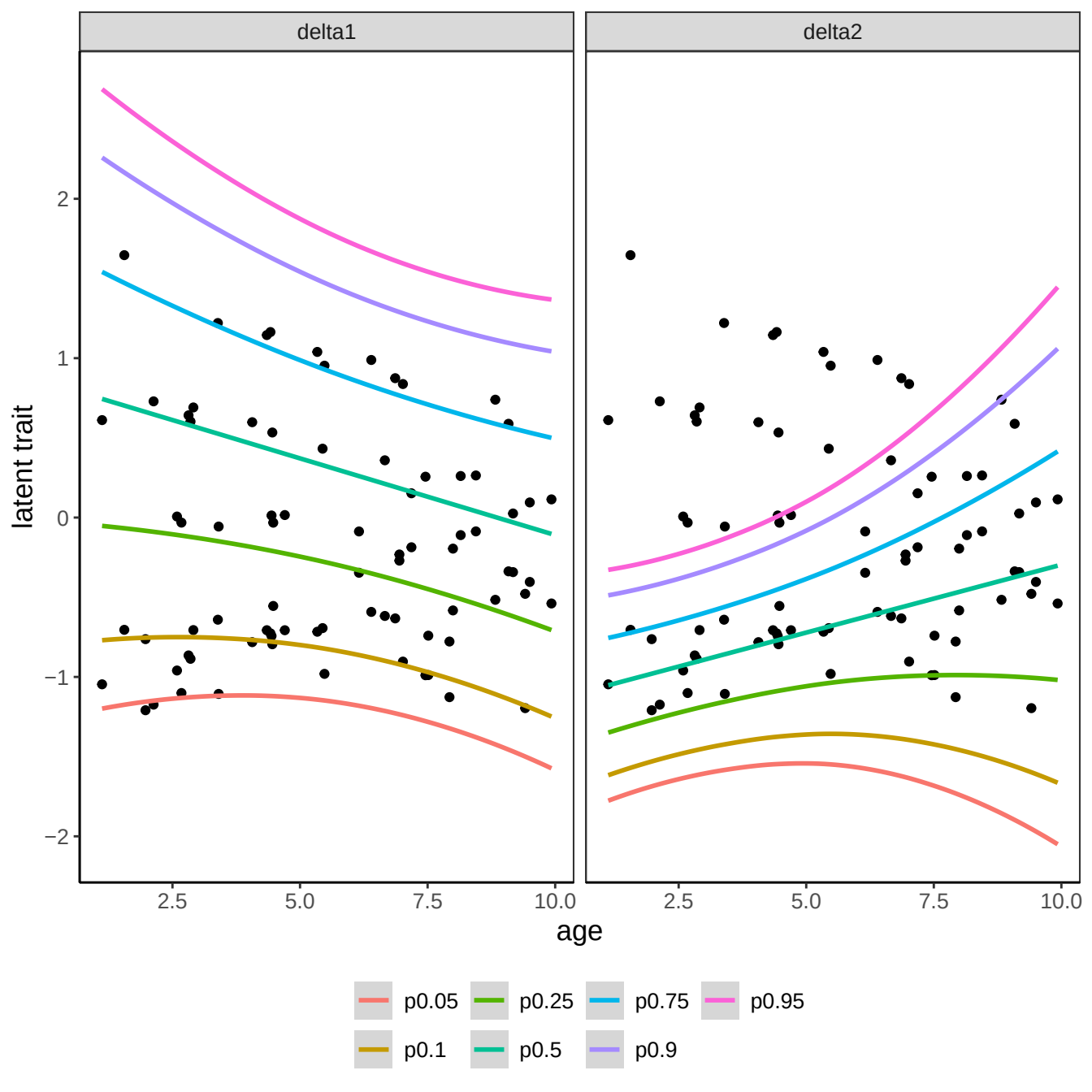

domain1
general
domain1
domain2
frequency of individuals
raw score
0
5
10
15
20
25
0
2
4
6
0
1
2
3

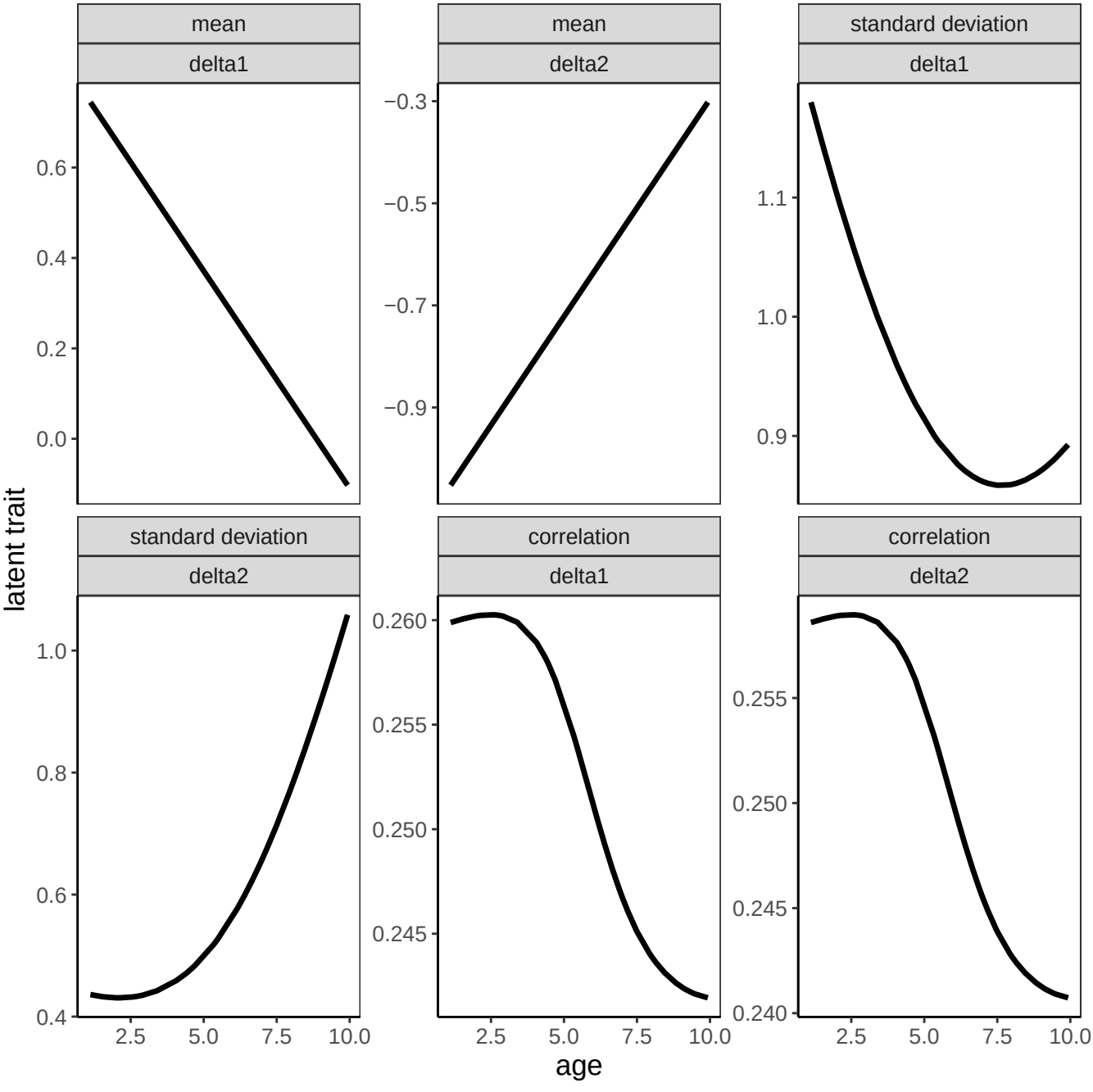

domain1
age: [0,5]
age: [5,10]
frequency of individuals
raw score
0
5
10
15
0
1
2
3

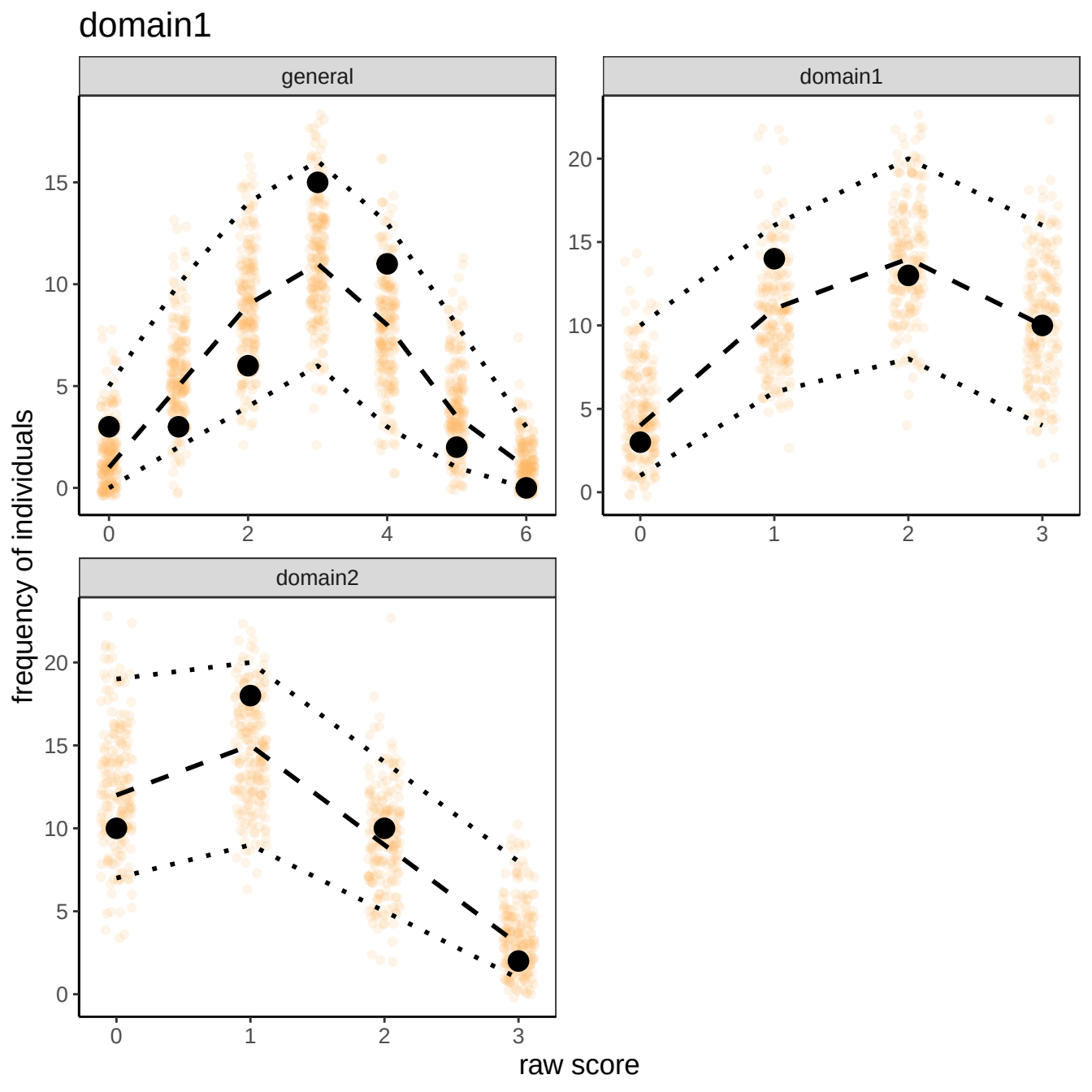

item fit: proportion correct per item
domain 1
domain 2
model−implied
observed
0.3
0.4
0.5
0.6
0.7
0.4
0.5
0.6
0.7

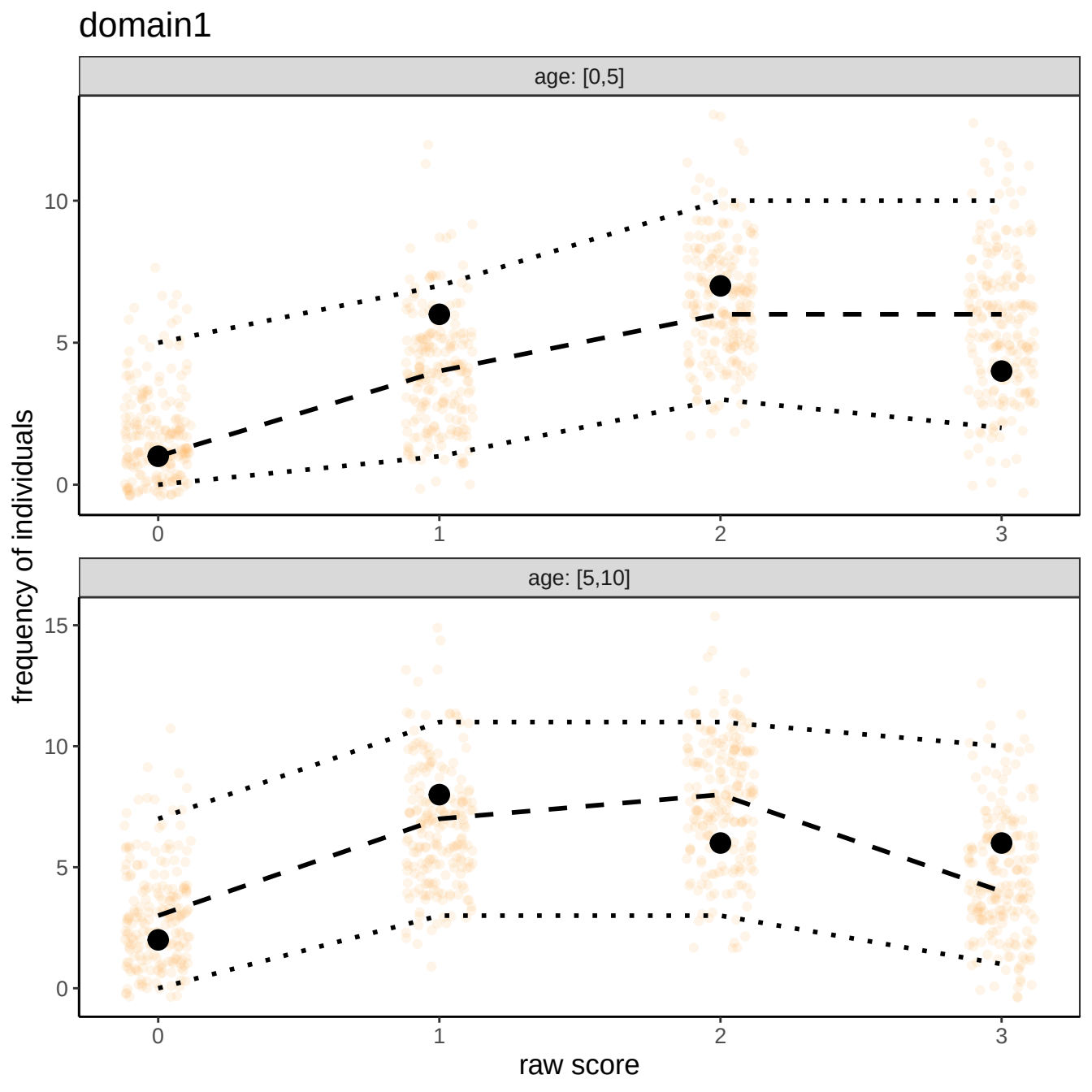

person fit: raw score
domain 1
domain 2
model−implied
observed
2.5
2.0
1.5
1.0
0
1
2
3

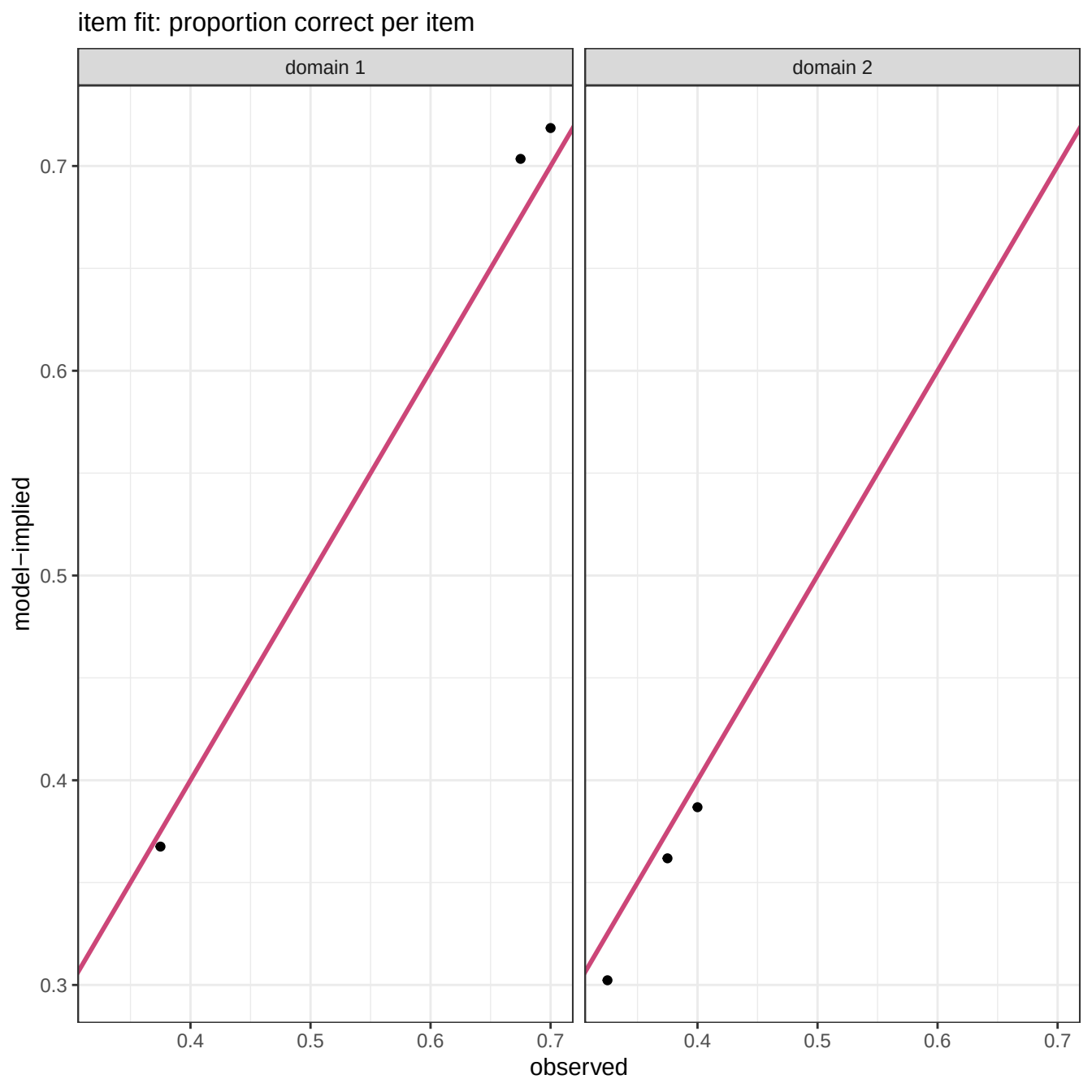

Corr
1.0
0.5
0.0
−0.5
−1.0

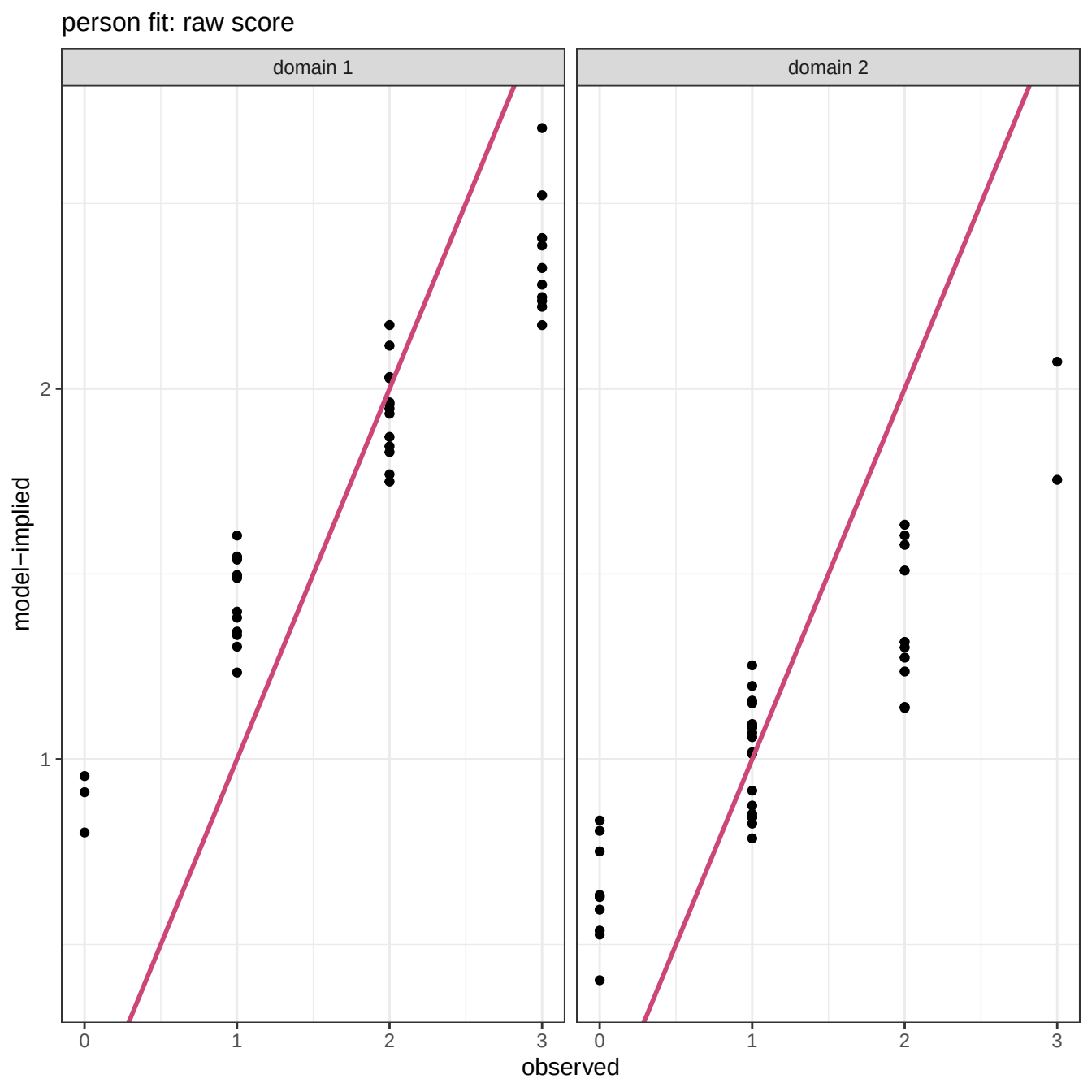

delta1
delta2
latent trait
2
1
0
−1
−2
2.5
5.0
7.5
10.0
age
p0.05
p0.25
p0.75
p0.95
p0.1
p0.5
p0.9

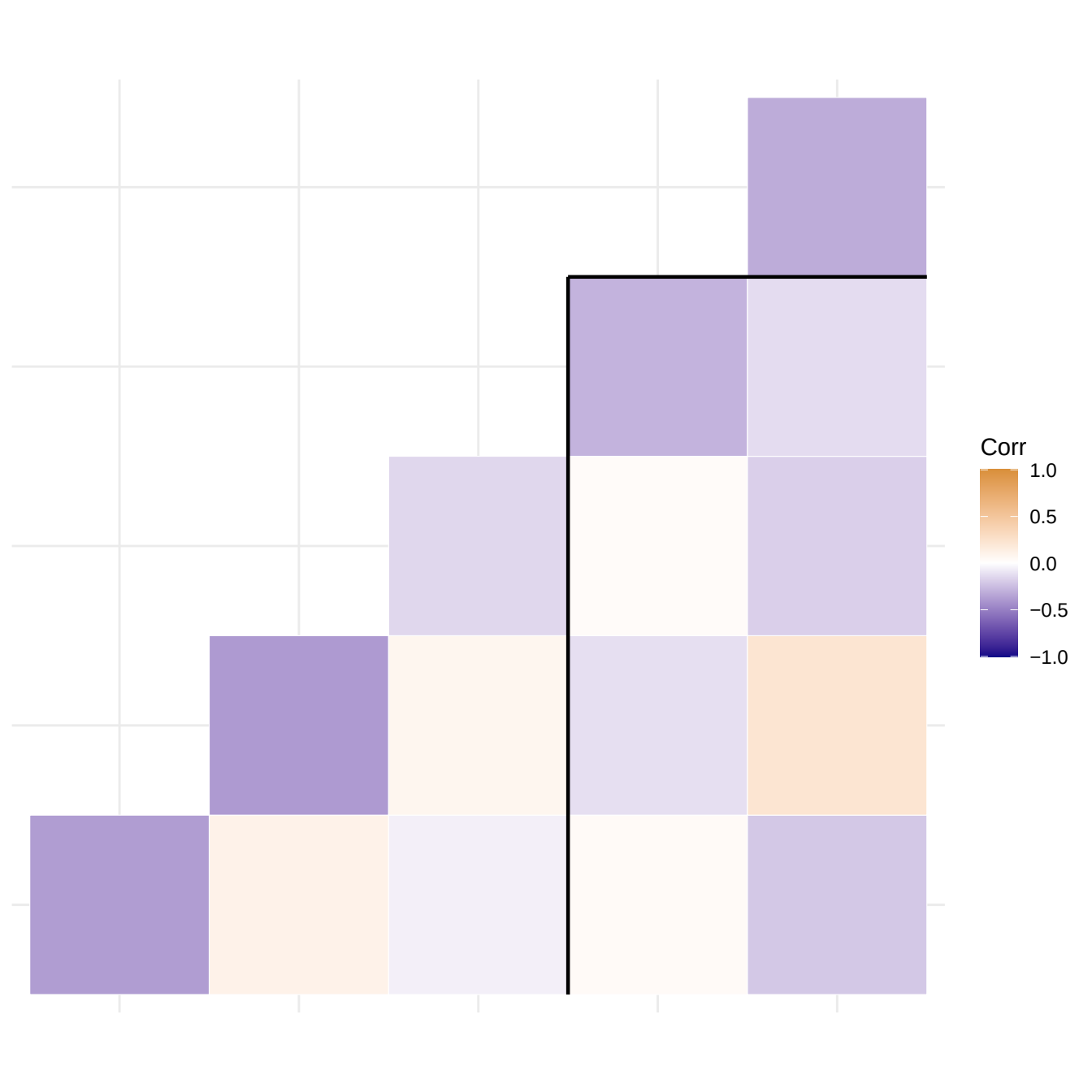

mean
delta1
mean
delta2
standard deviation
delta1
standard deviation
delta2
correlation
delta1
correlation
delta2
latent trait
age
0.6
0.4
0.2
0.0
−0.3
−0.5
−0.7
−0.9
1.1
1.0
0.9
1.0
0.8
0.6
0.4
0.260
0.255
0.250
0.245
0.255
0.250
0.245
0.240
2.5
5.0
7.5
10.0

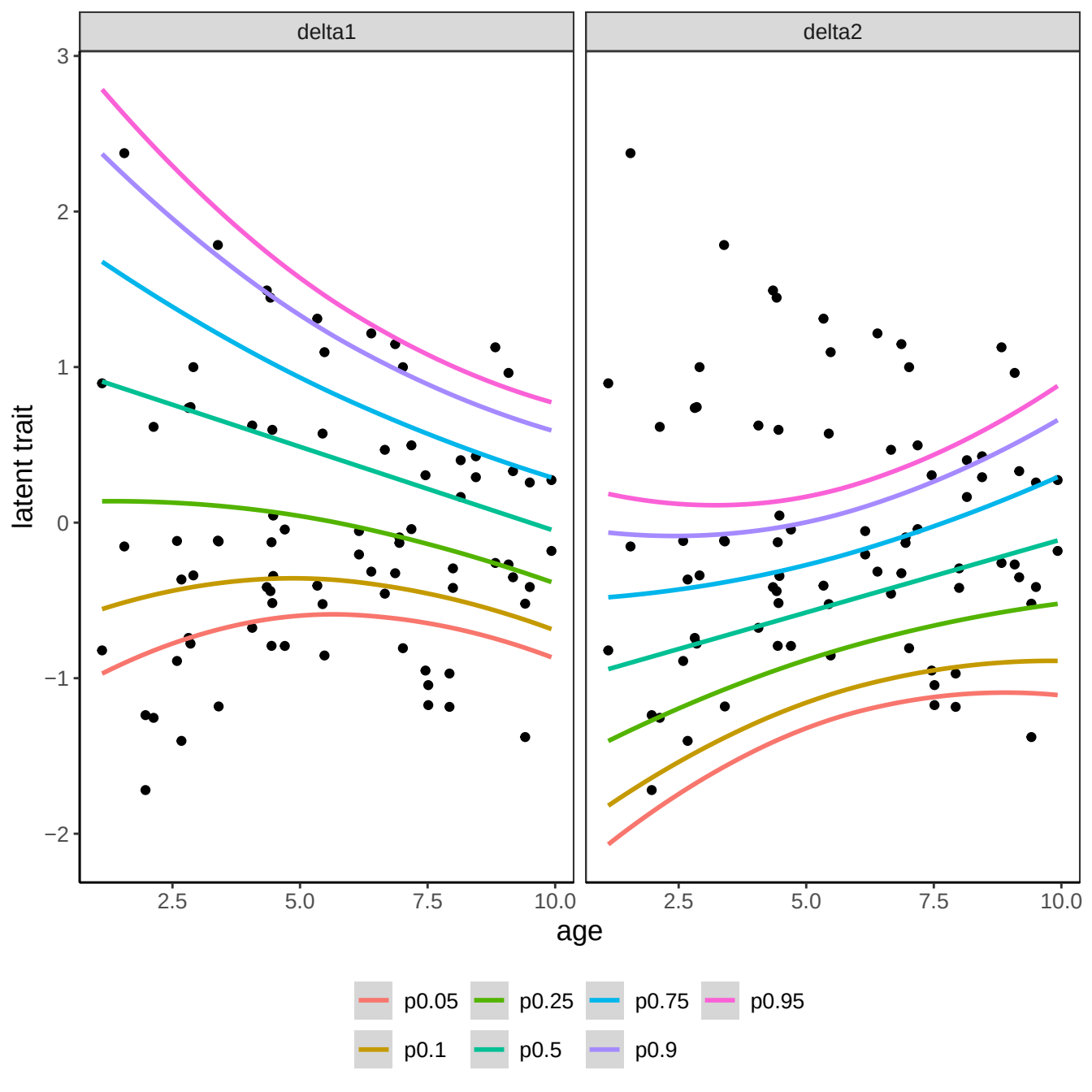

domain1
general
domain1
domain2
frequency of individuals
raw score
0
5
10
15
20
0
2
4
6
0
1
2
3

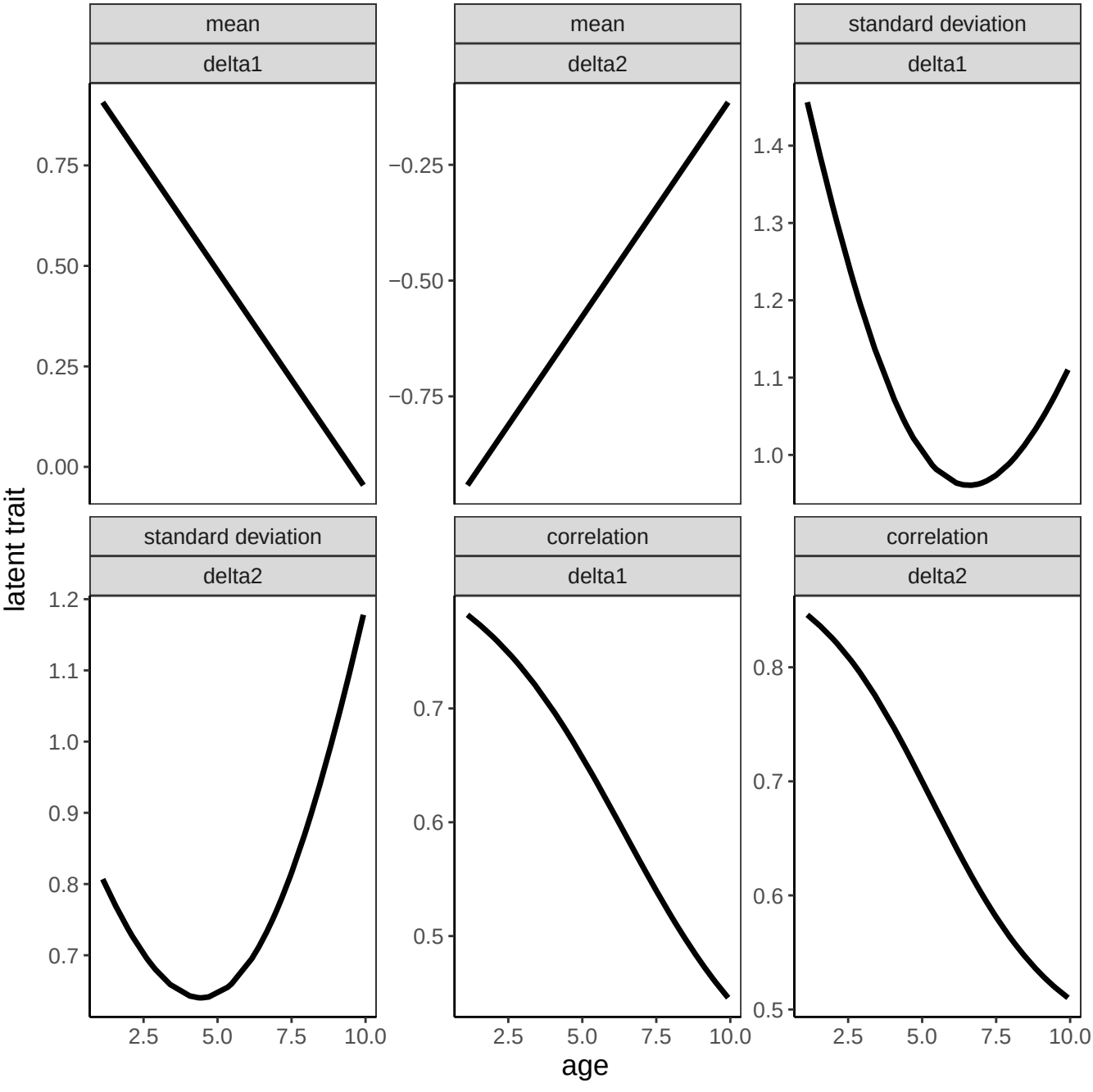

domain1
age: [0,5]
age: [5,10]
frequency of individuals
raw score
0
5
10
15
0
1
2
3

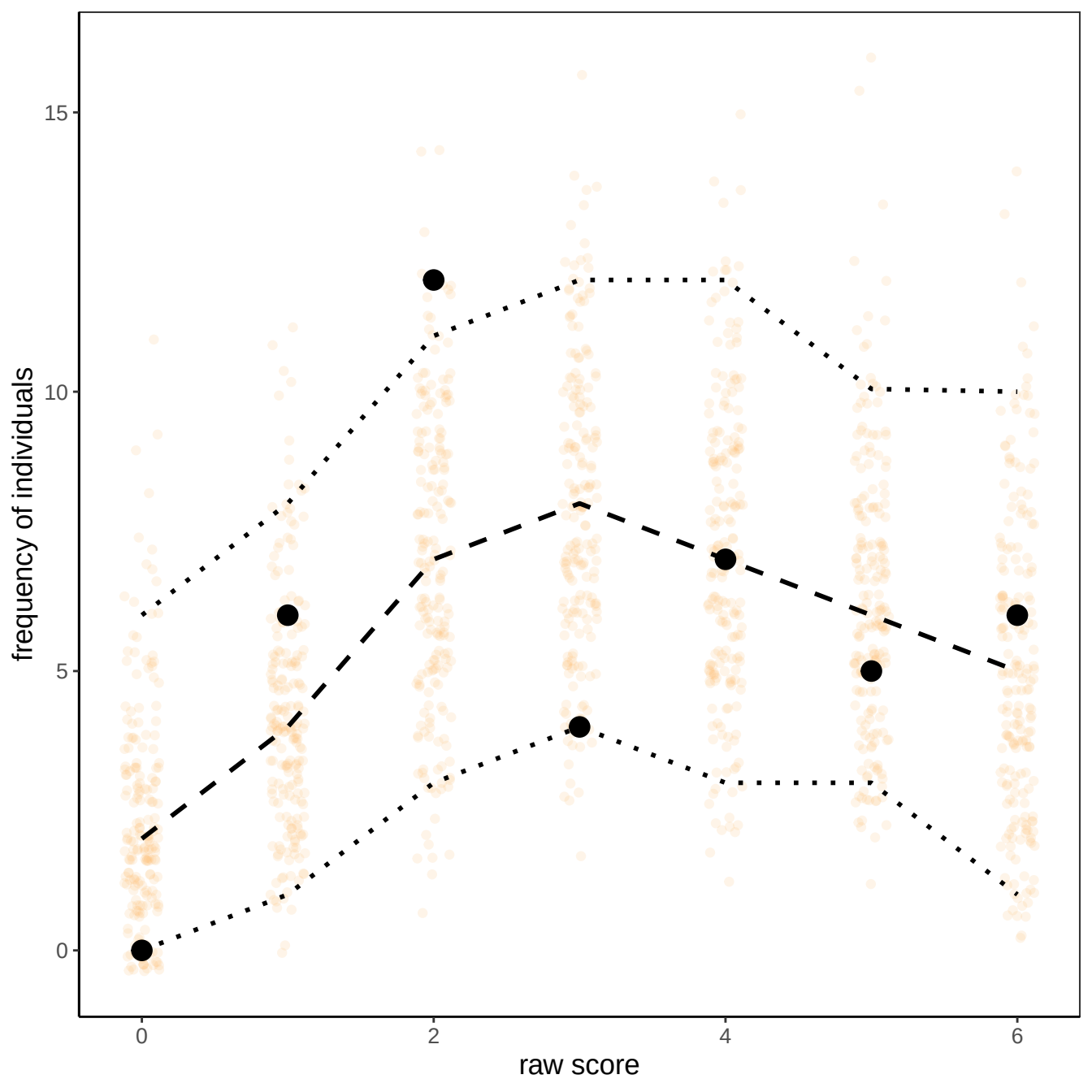

item fit: proportion correct per item
domain 1
domain 2
model−implied
observed
0.3
0.4
0.5
0.6
0.7
0.4
0.5
0.6
0.7

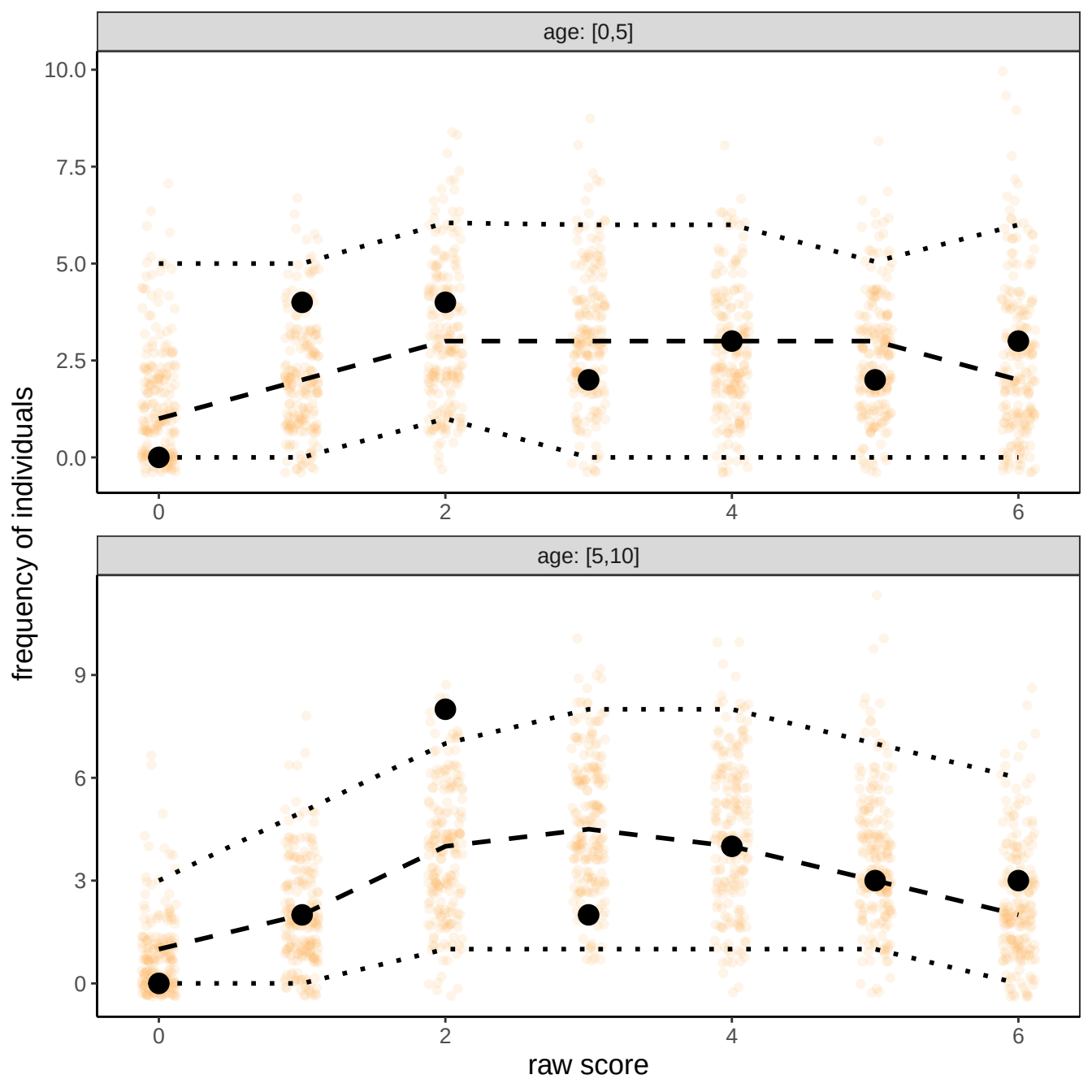

person fit: raw score
domain 1
domain 2
model−implied
observed
0
1
2
3

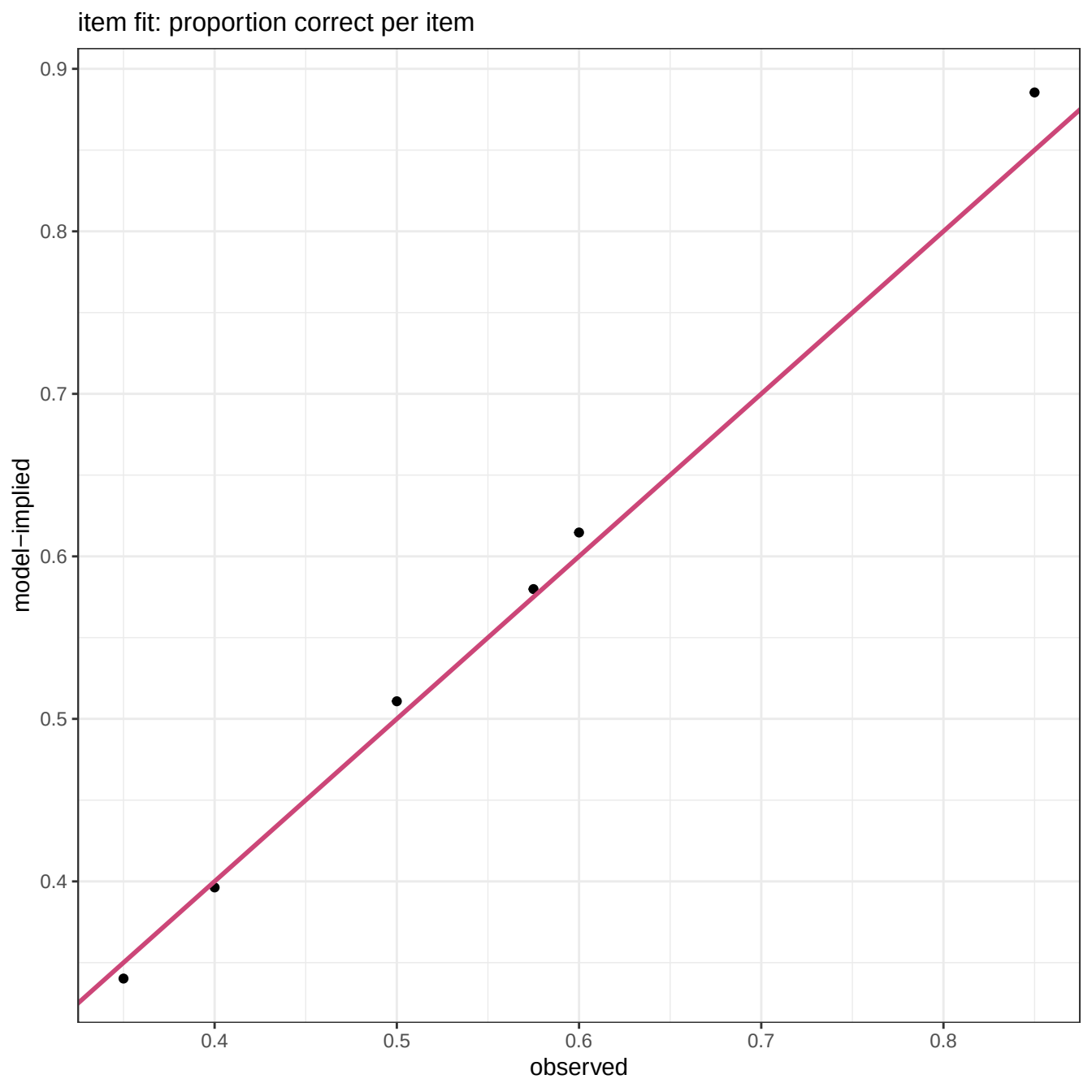

Corr
1.0
0.5
0.0
−0.5
−1.0

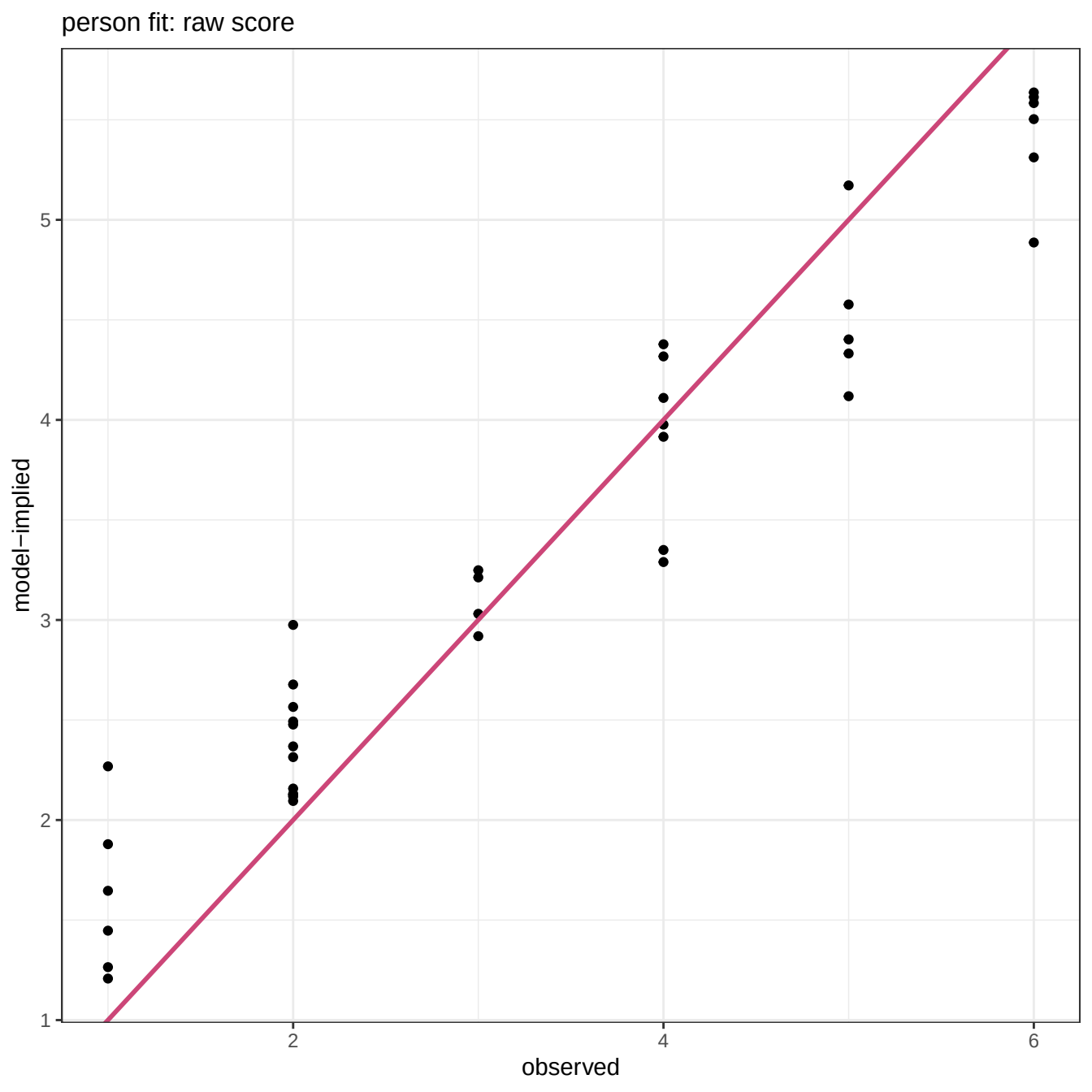

delta1
delta2
latent trait
age
3
2
1
0
−1
−2
2.5
5.0
7.5
10.0
p0.05
p0.1
p0.25
p0.5
p0.75
p0.9
p0.95

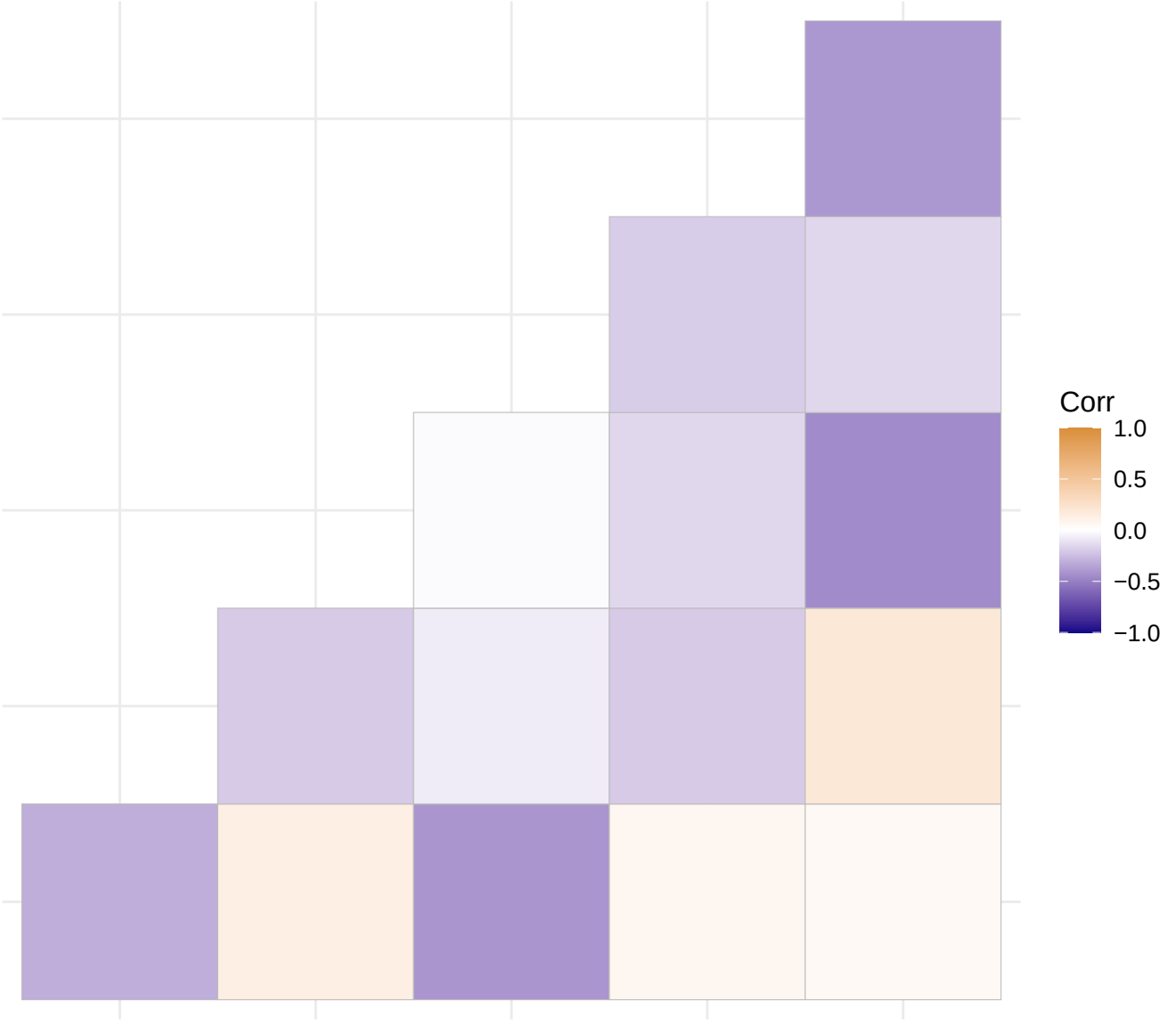

mean
delta1
mean
delta2
standard deviation
delta1
standard deviation
delta2
correlation
delta1
correlation
delta2
latent trait
age
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
1.4
1.3
1.2
1.1
1.0
1.2
1.1
1.0
0.9
0.8
0.7
0.7
0.6
0.5
0.8
0.7
0.6
0.5
2.5
5.0
7.5
10.0

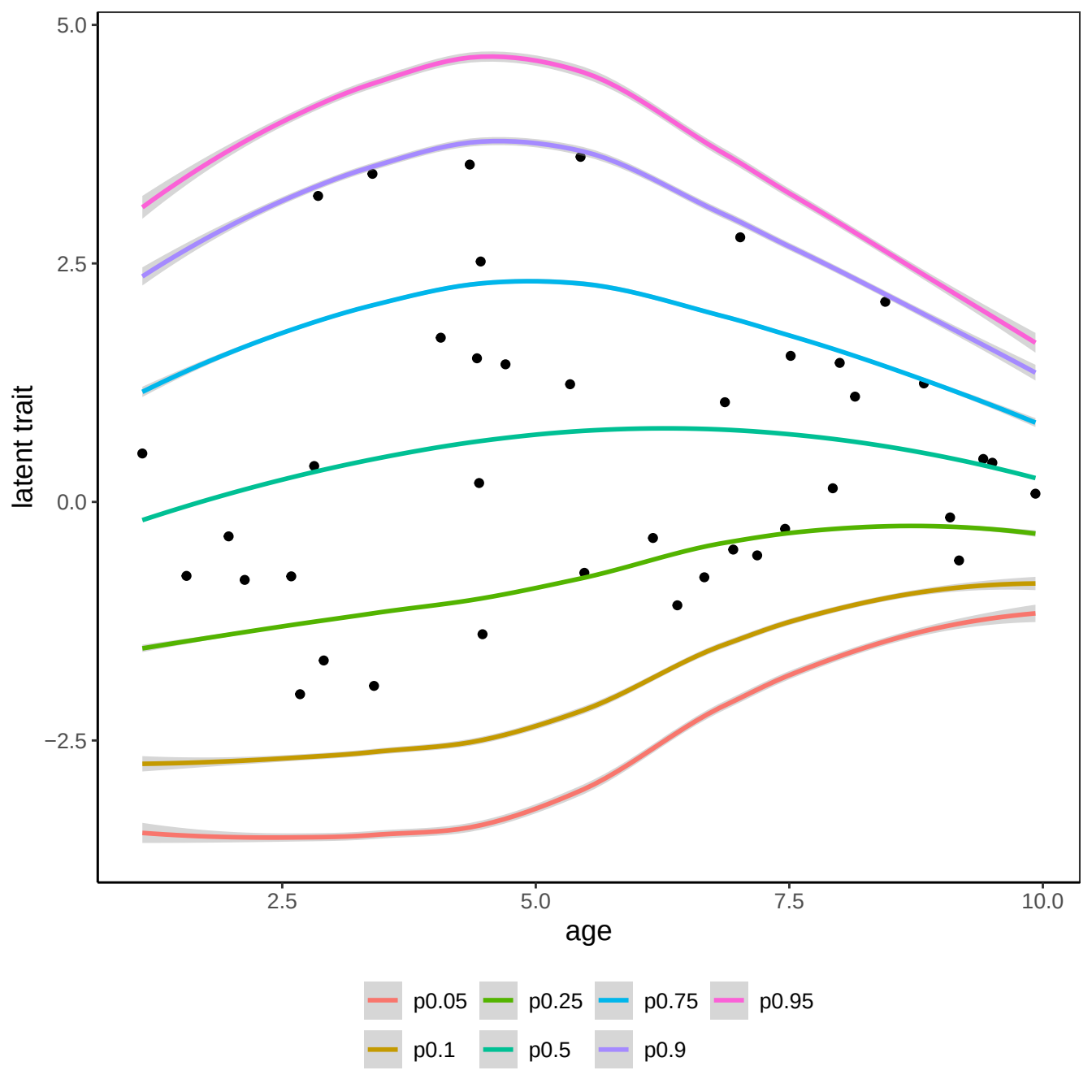

15
10
5
0
frequency of individuals
0
2
4
6
raw score

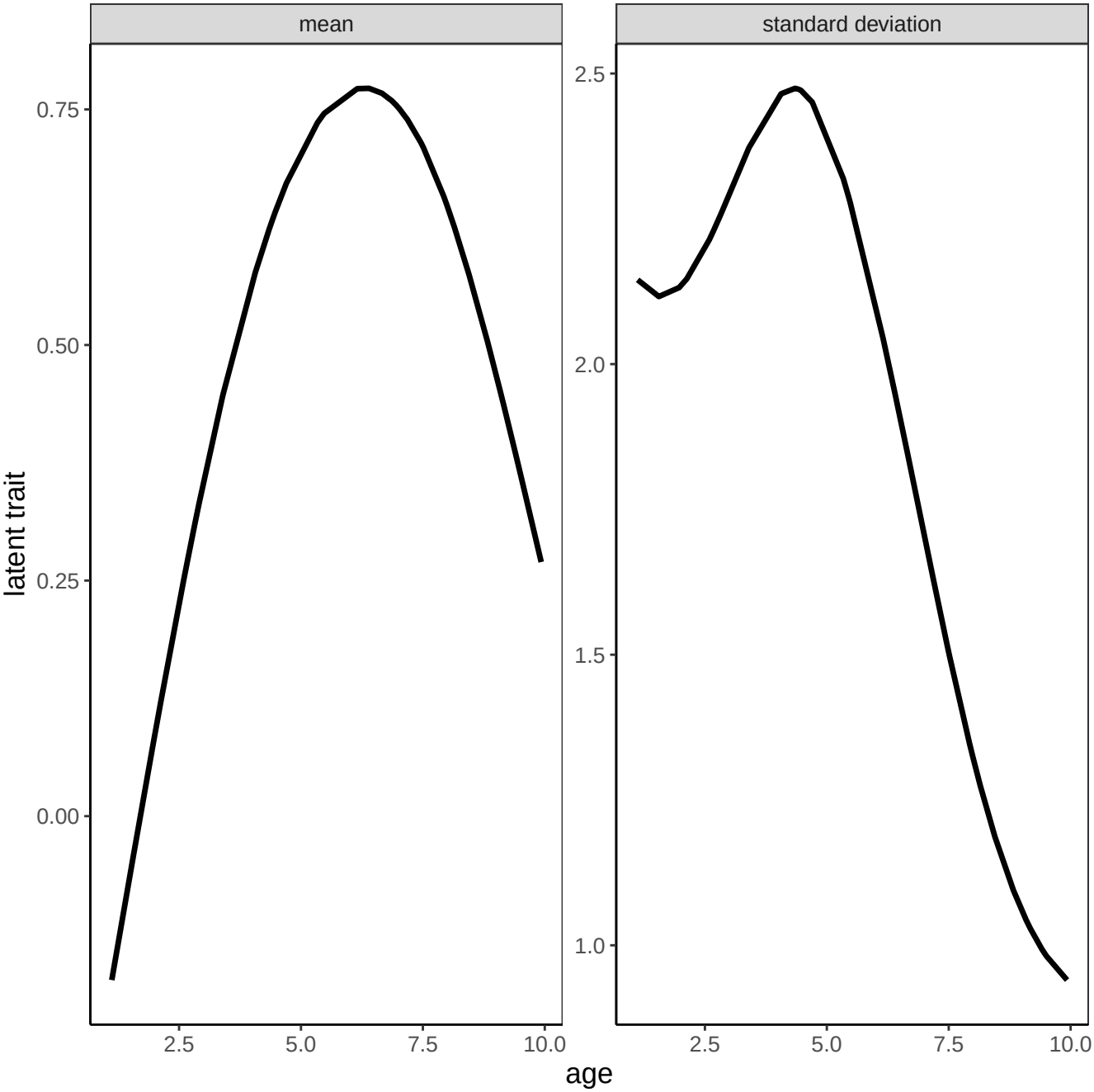

age: [0,5]
age: [5,10]
frequency of individuals
raw score
10.0
7.5
5.0
2.5
0.0
9
6
3
0
0
2
4
6

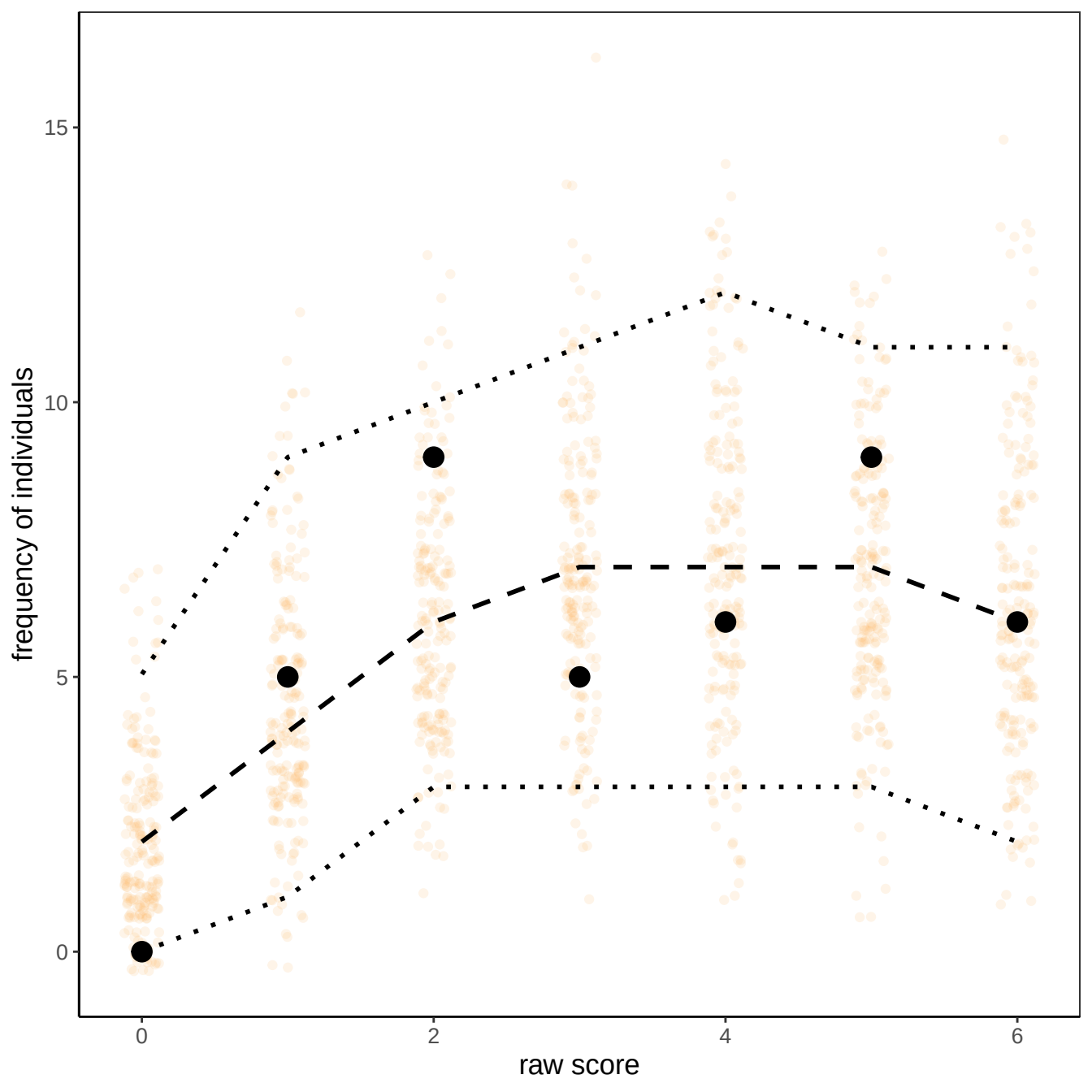

item fit: proportion correct per item
model−implied
observed
0.9
0.8
0.7
0.6
0.5
0.4
0.4
0.5
0.6
0.7
0.8

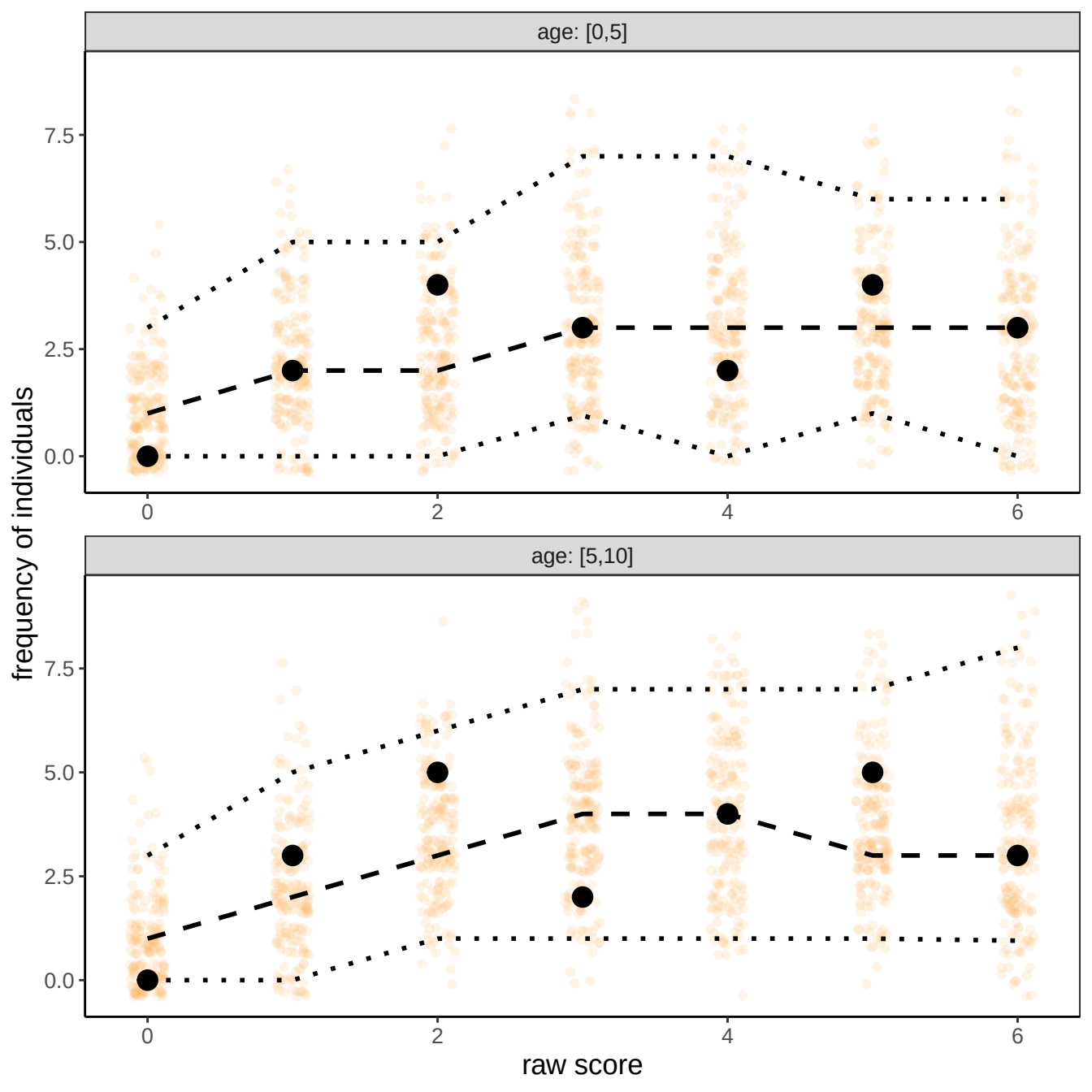

person fit: raw score
model−implied
observed
1
2
3
4
5
2
4
6

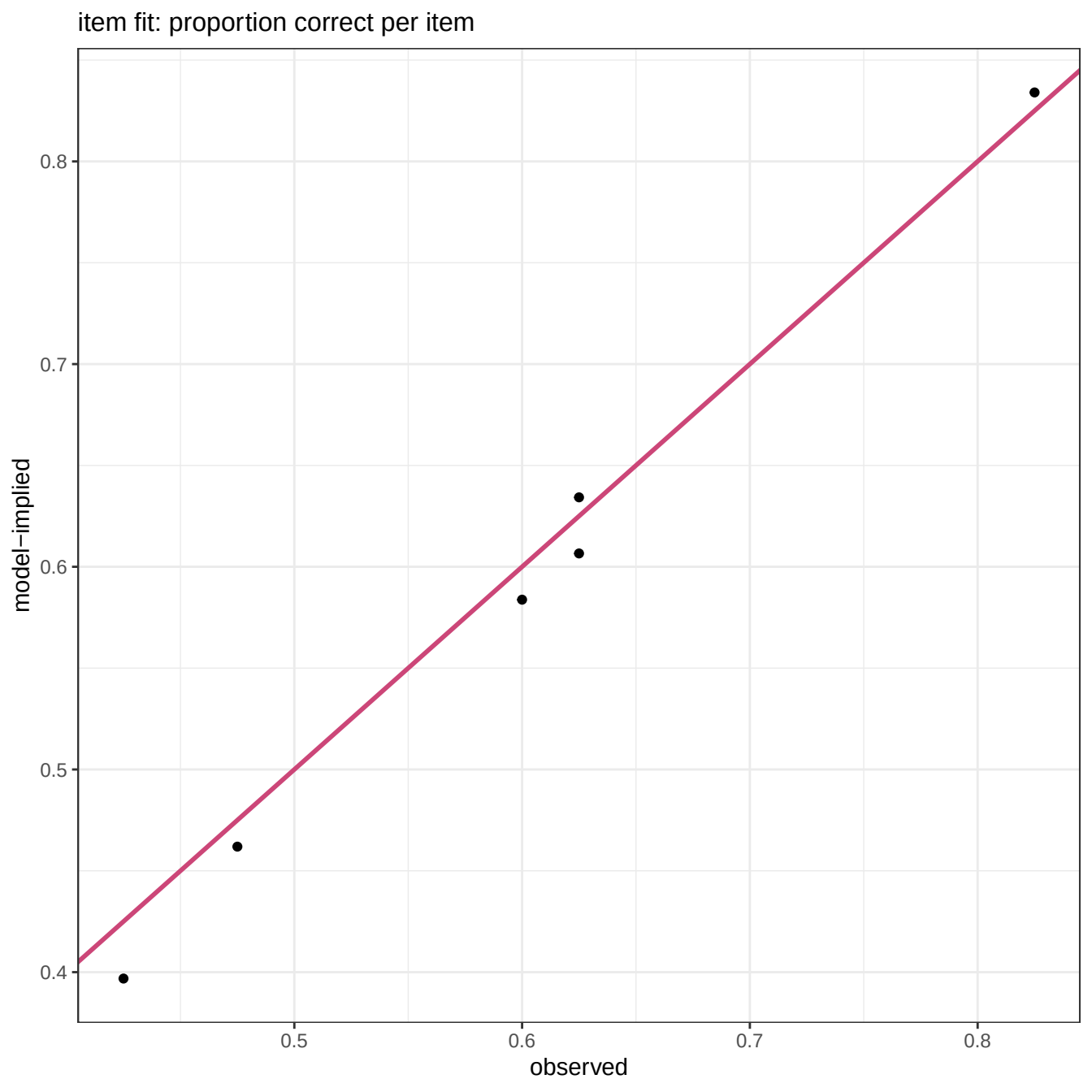

Corr
1.0
0.5
0.0
−0.5
−1.0

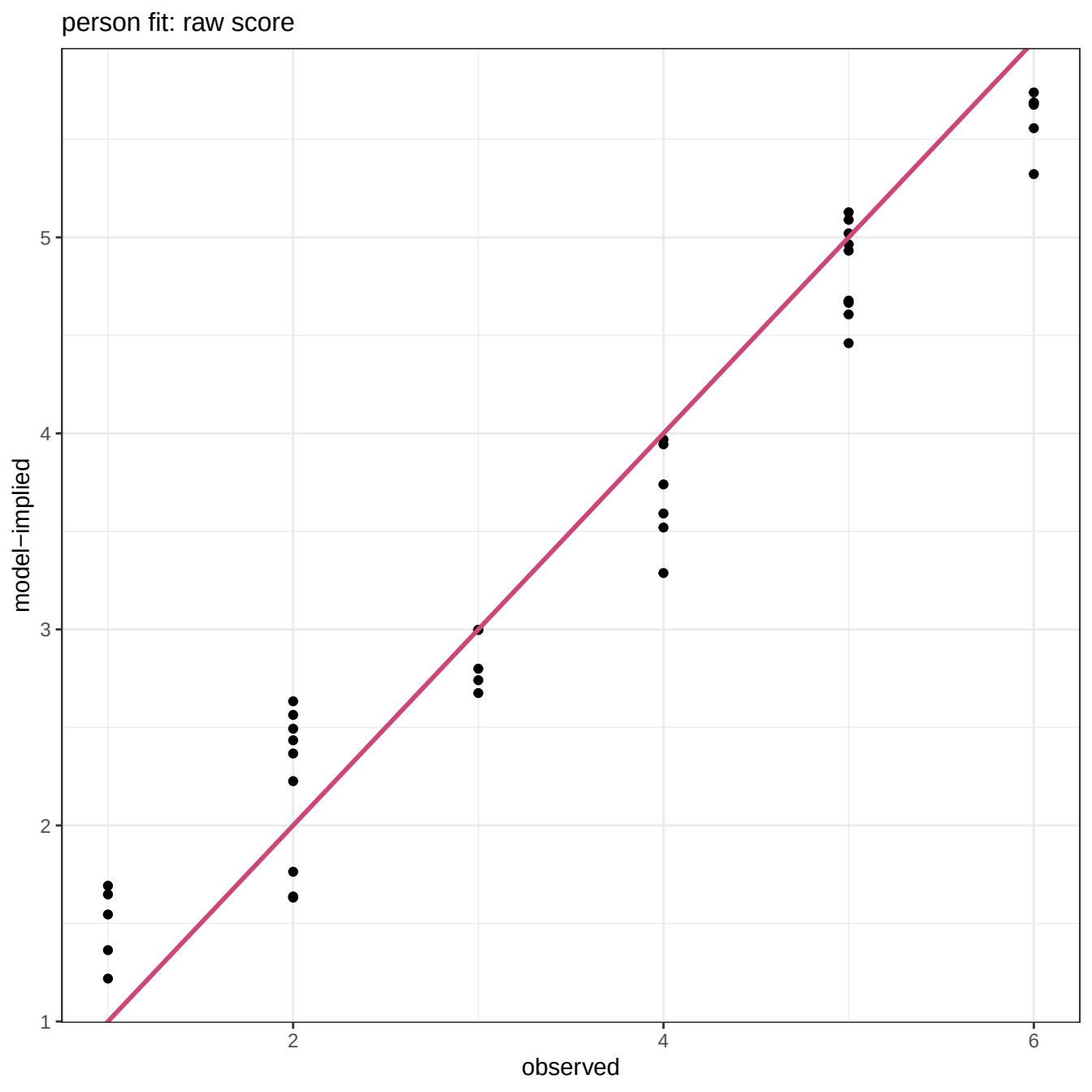

5.0
2.5
0.0
−2.5
latent trait
2.5
5.0
7.5
10.0
age
p0.05
p0.25
p0.75
p0.95
p0.1
p0.5
p0.9

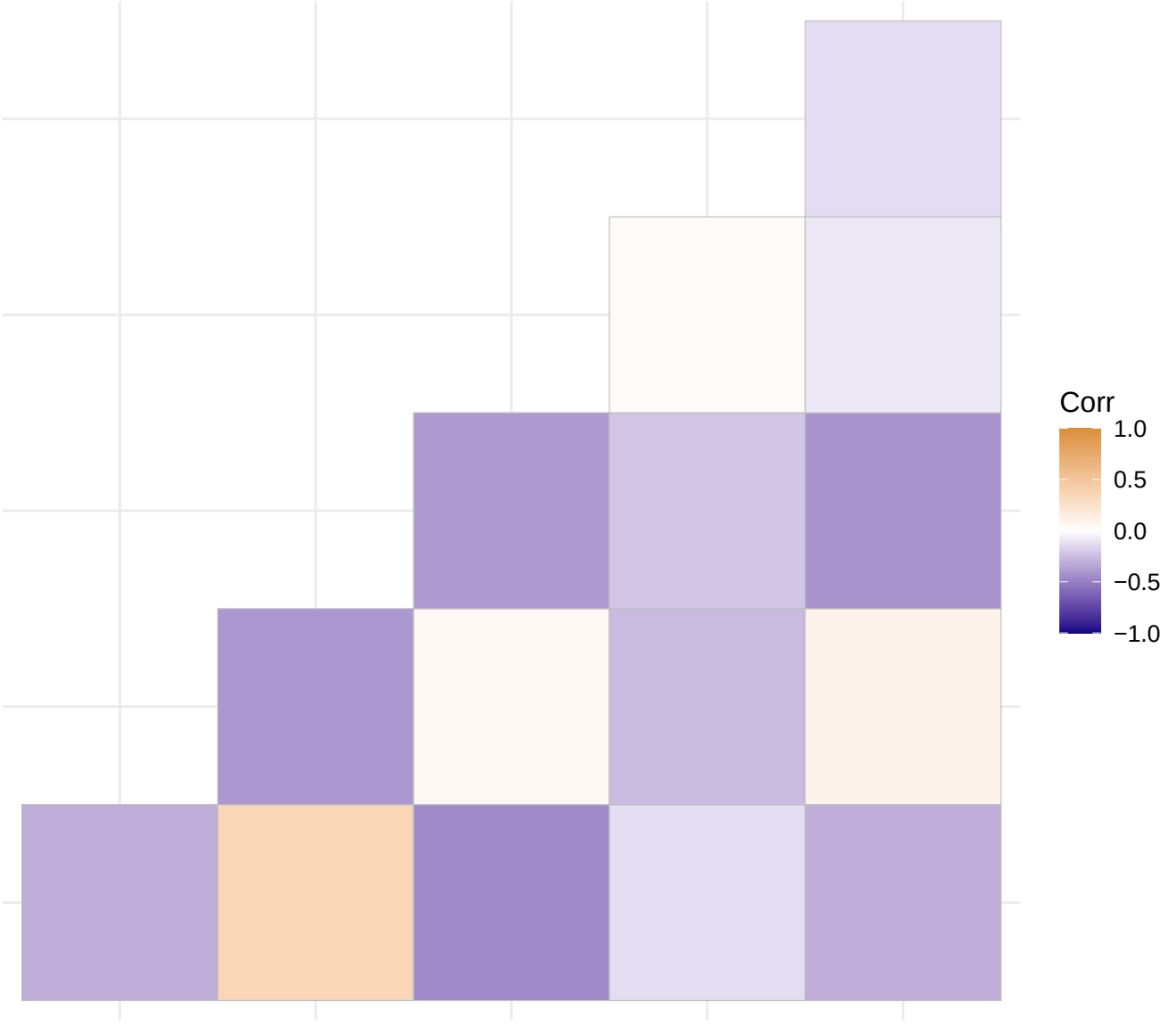

mean
standard deviation
latent trait
0.75
0.50
0.25
0.00
2.5
2.0
1.5
1.0
2.5
5.0
7.5
10.0
age

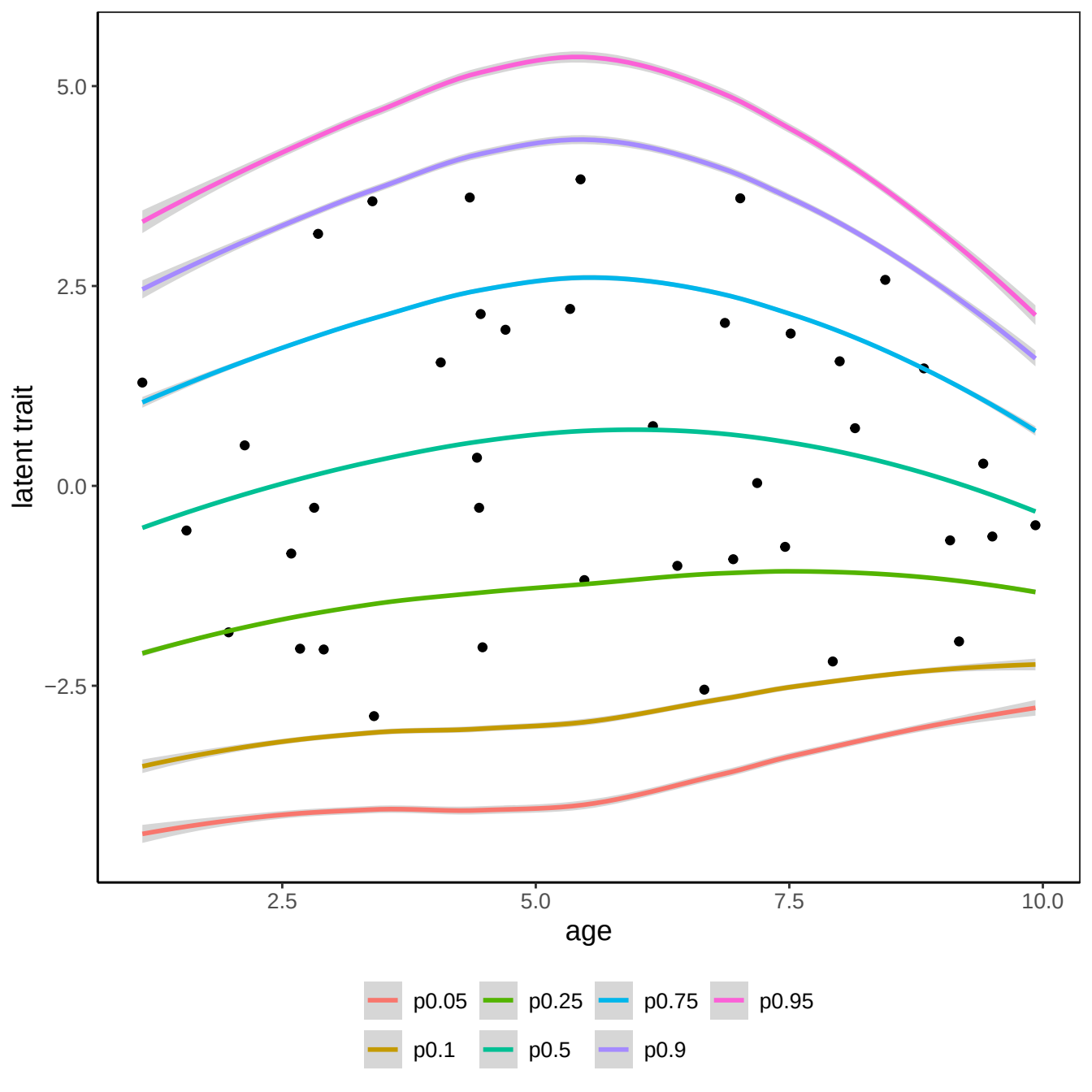

15
10
5
0
frequency of individuals
0
2
4
6
raw score

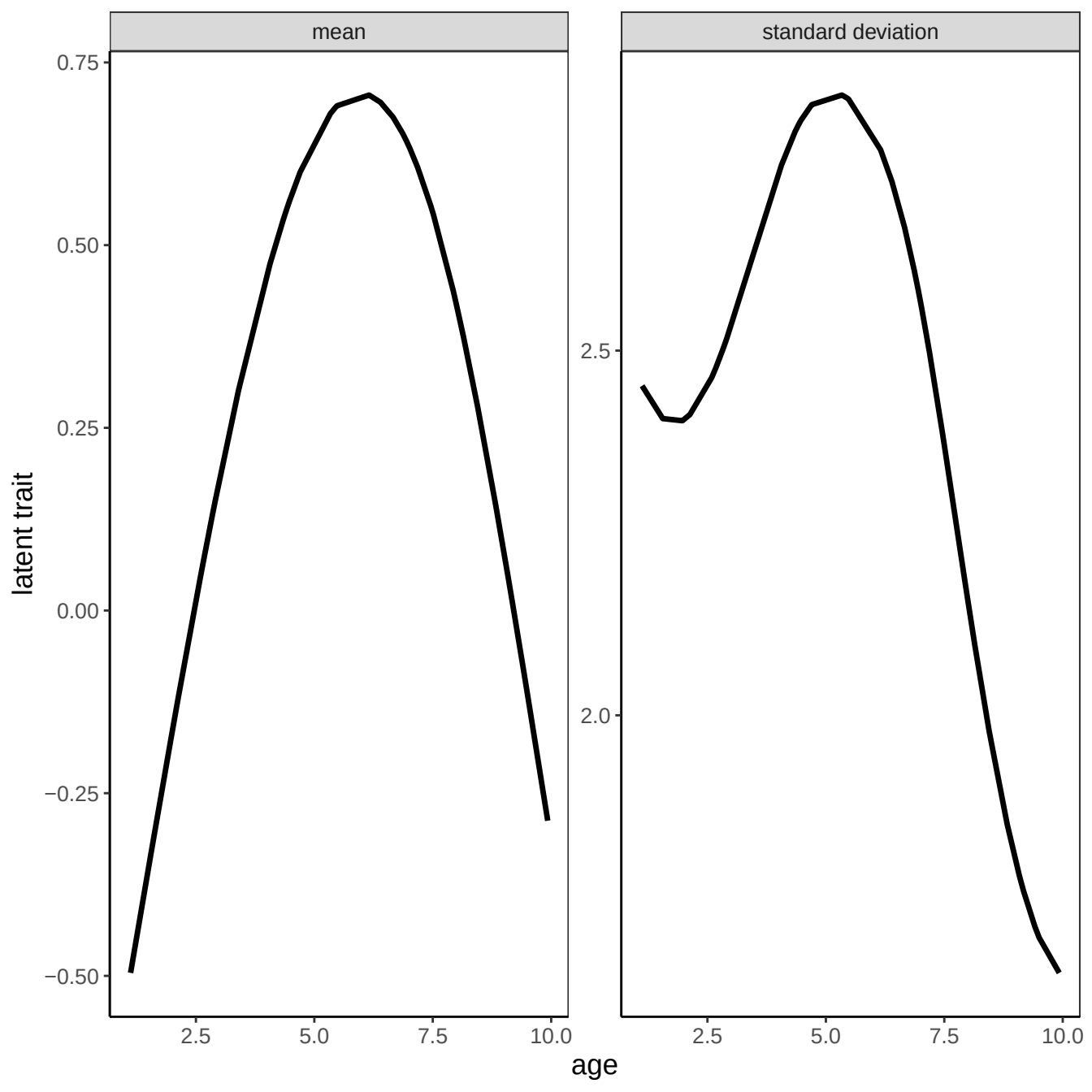

age: [0,5]
age: [5,10]
frequency of individuals
raw score
0.0
2.5
5.0
7.5
0
2
4
6

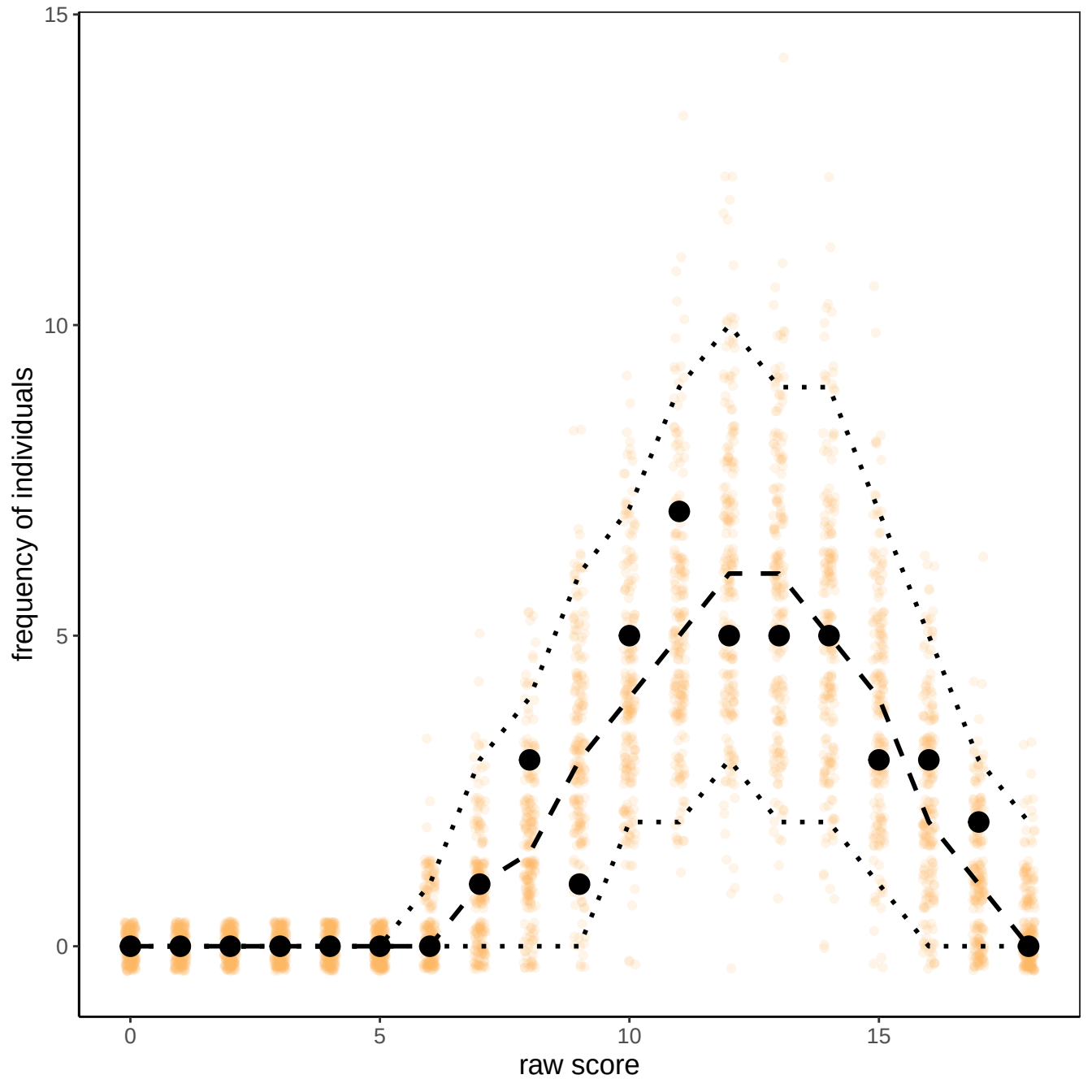

item fit: proportion correct per item
model−implied
observed
0.8
0.7
0.6
0.5
0.4
0.5
0.6
0.7
0.8

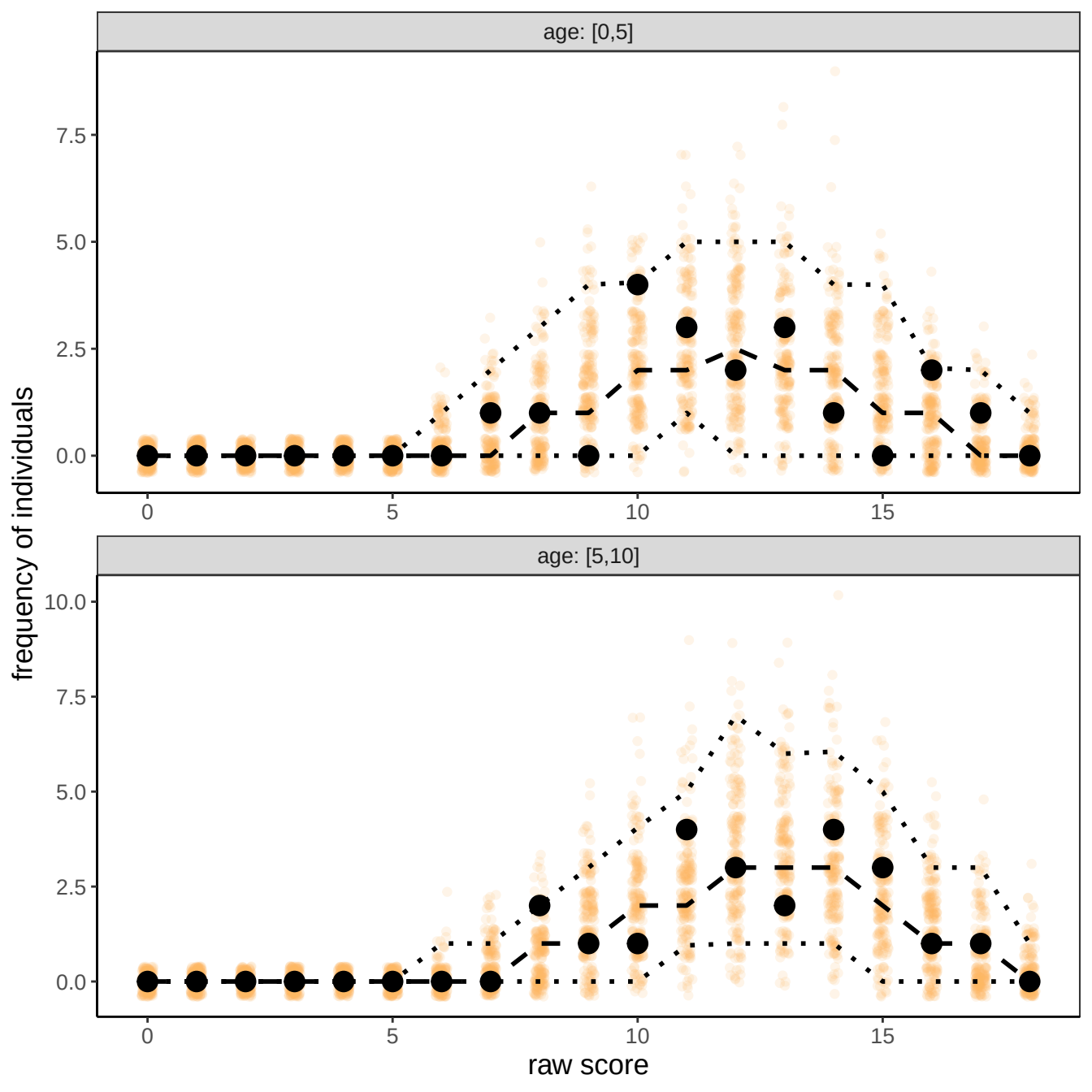

person fit: raw score
model−implied
observed
1
2
3
4
5
2
4
6

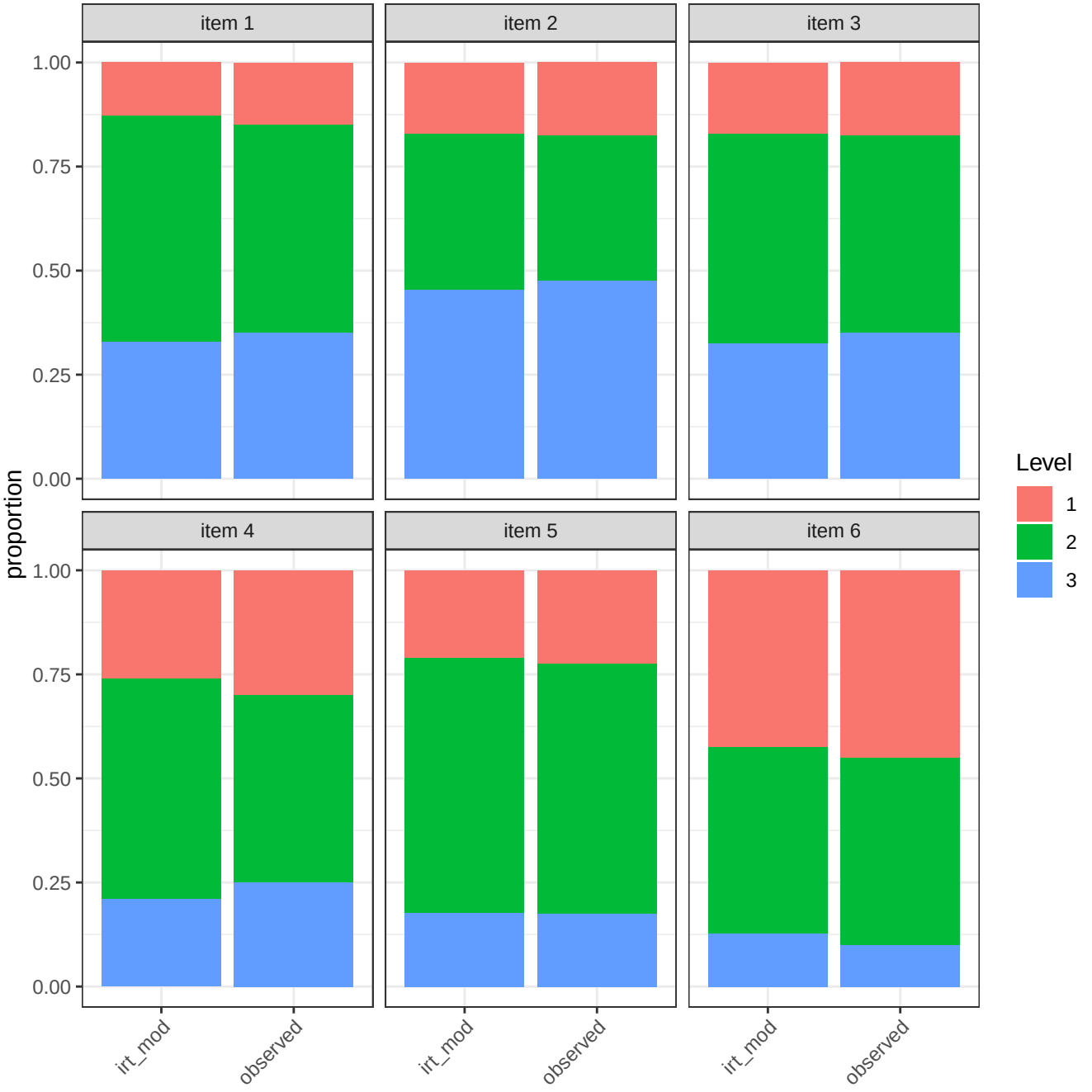

Corr
1.0
0.5
0.0
−0.5
−1.0

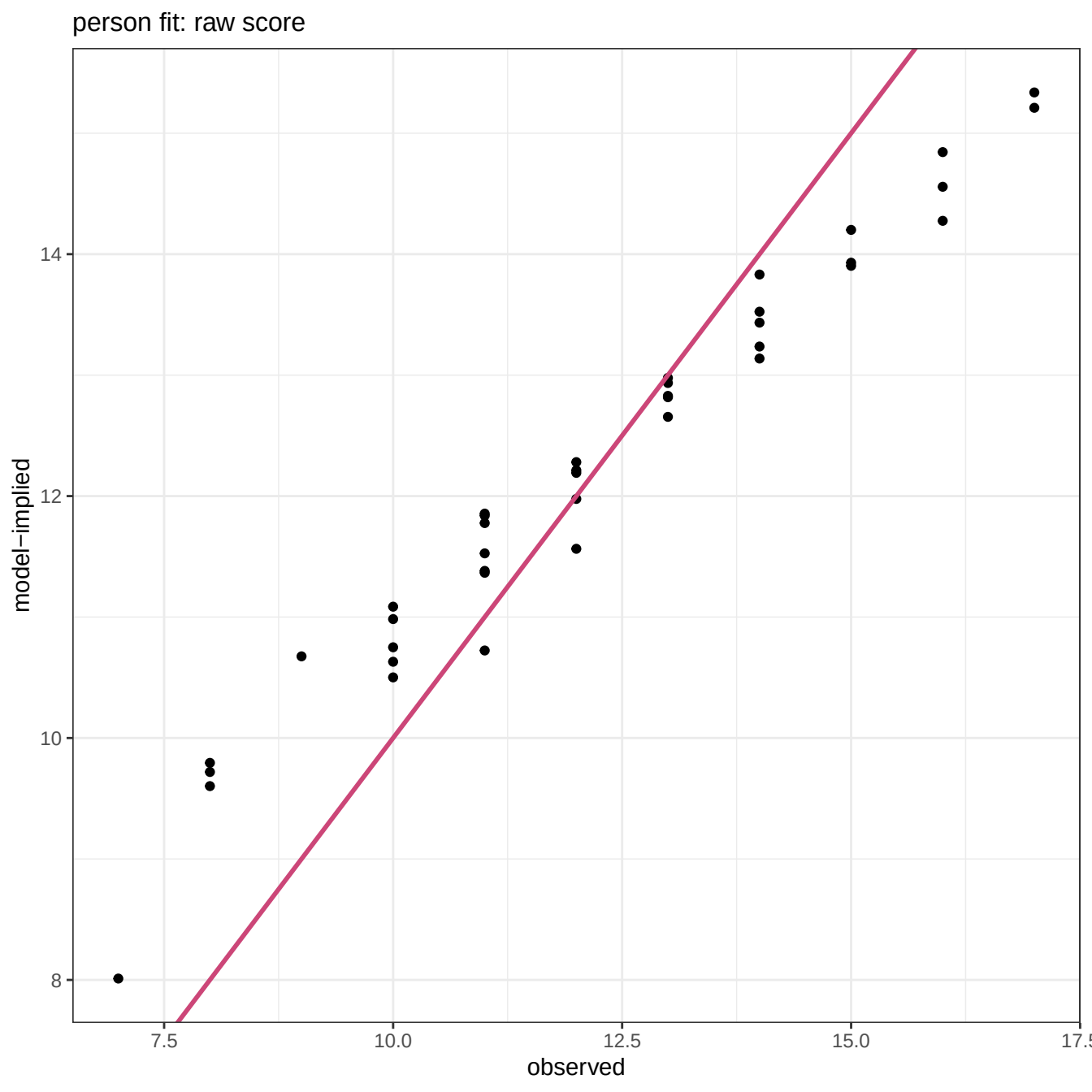

5.0
2.5
0.0
−2.5
latent trait
2.5
5.0
7.5
10.0
age
p0.05
p0.25
p0.75
p0.95
p0.1
p0.5
p0.9

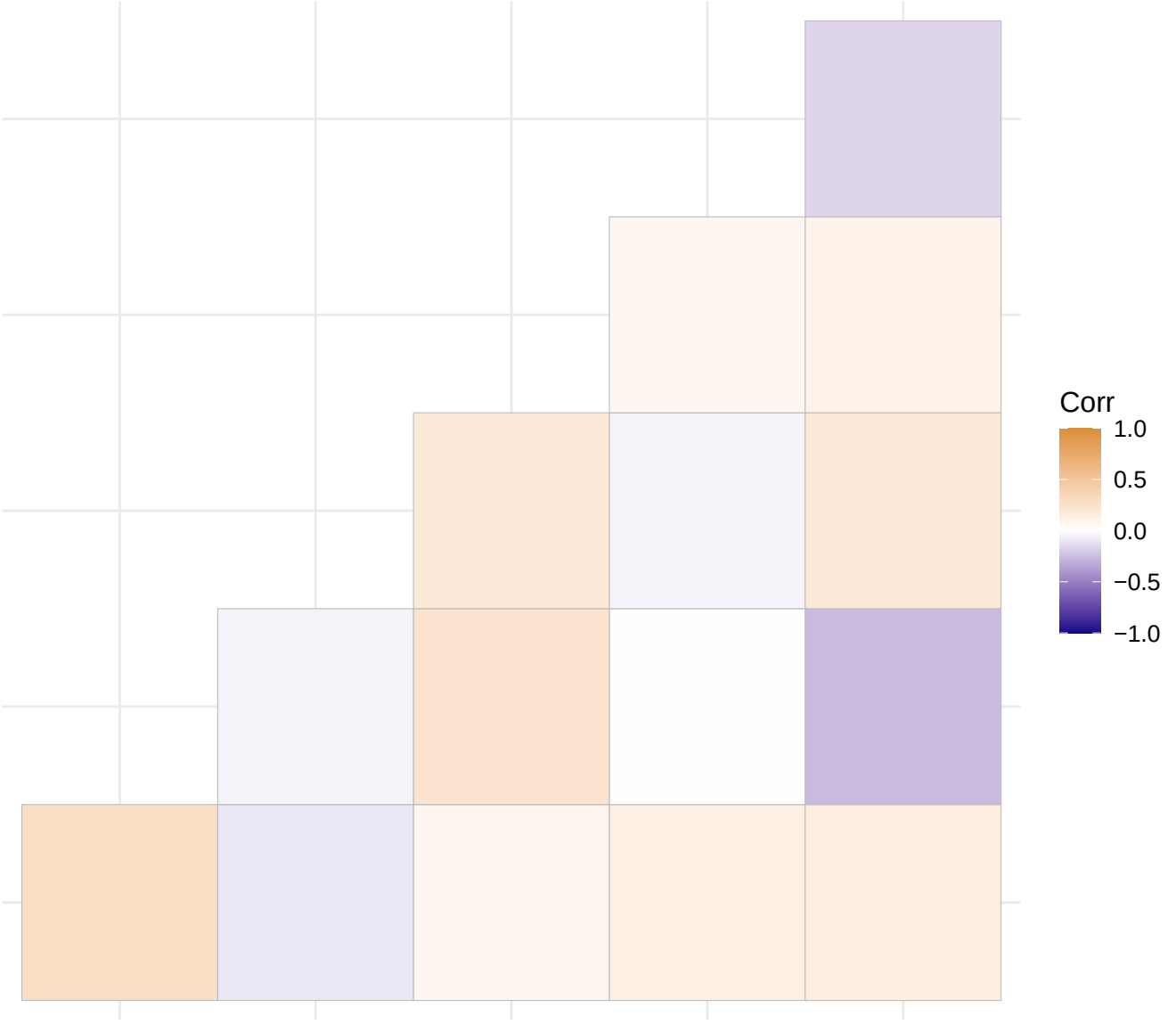

mean
standard deviation
latent trait
age
0.75
0.50
0.25
0.00
−0.25
−0.50
2.5
2.0
2.5
5.0
7.5
10.0

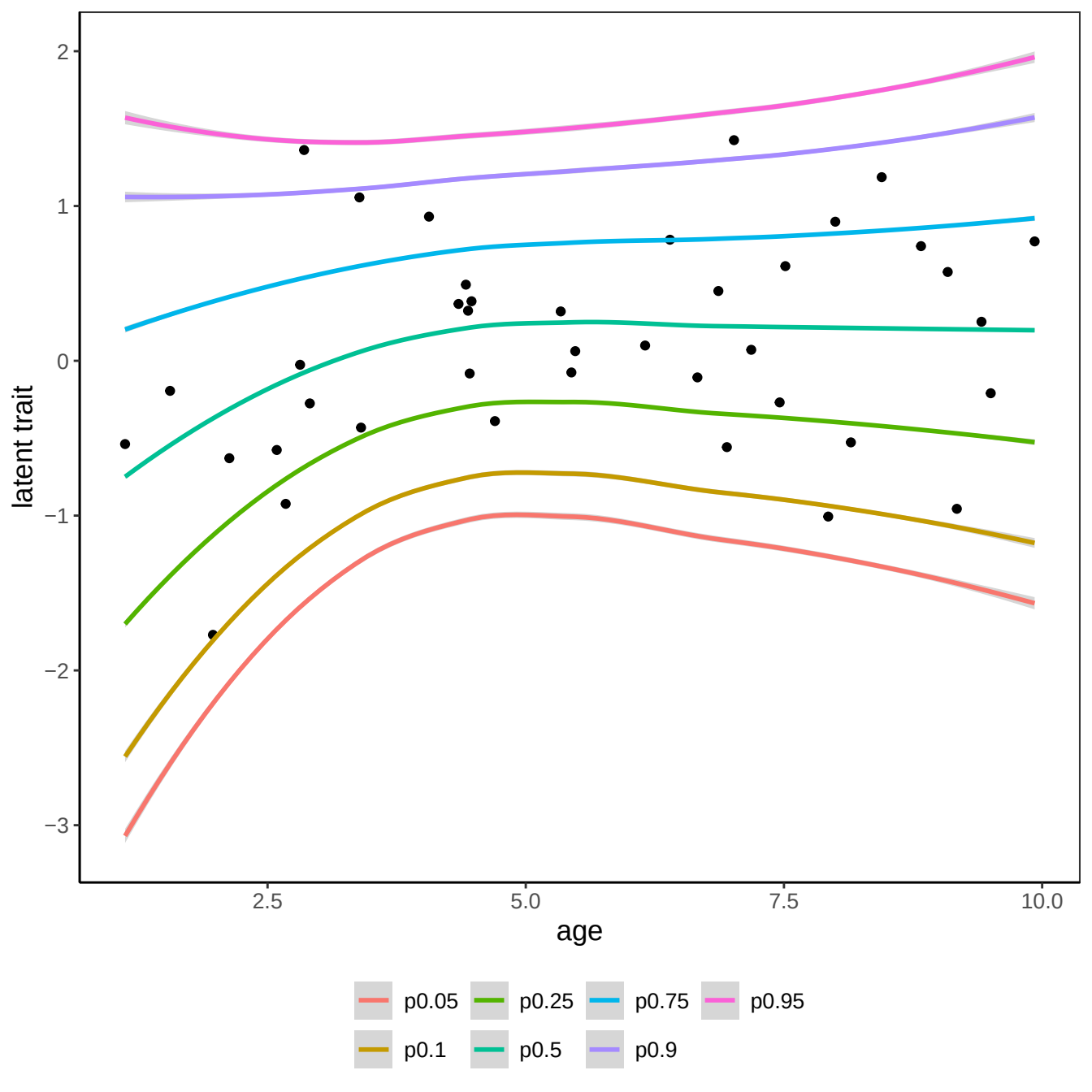

15
10
5
0
frequency of individuals
0
5
10
15
raw score

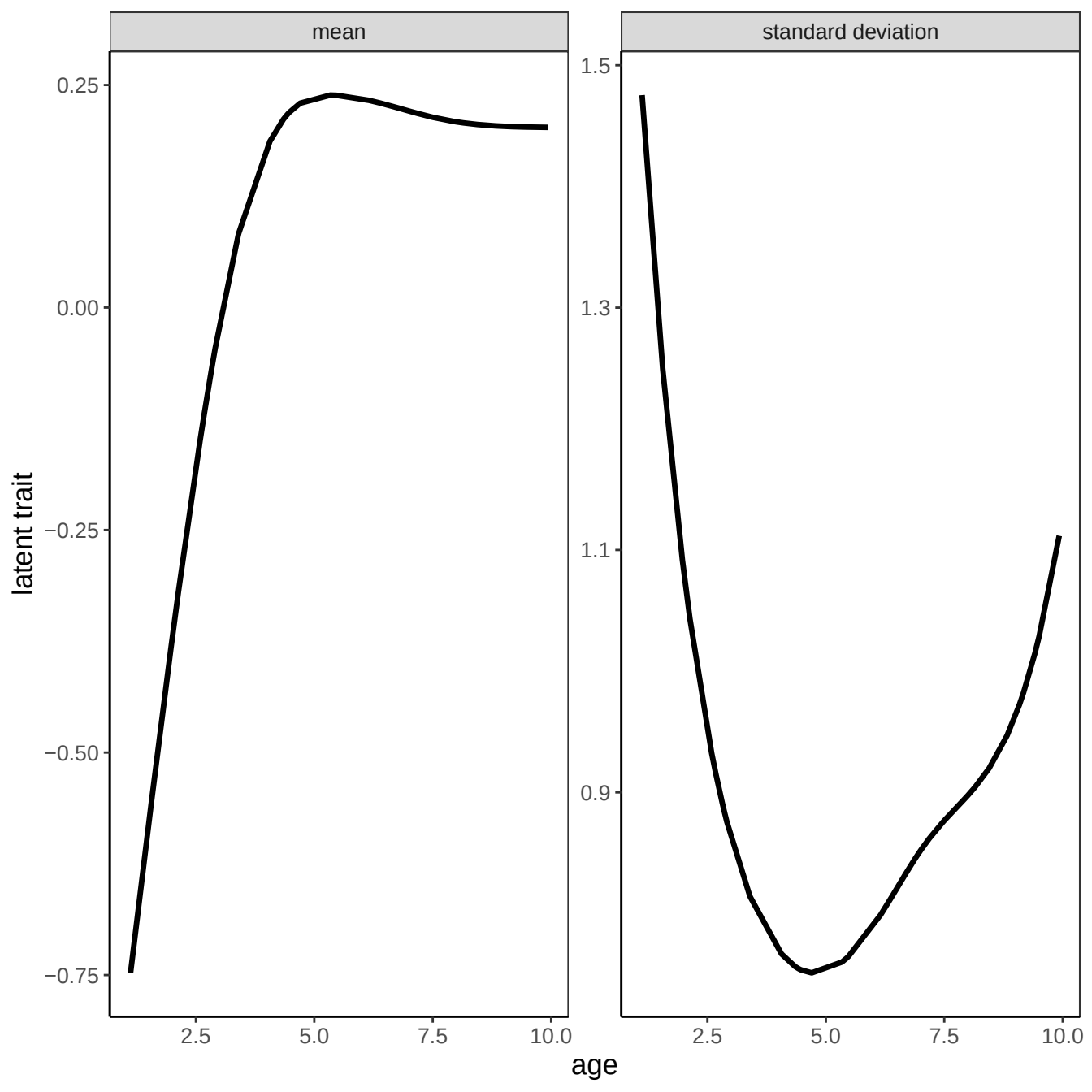

age: [0,5]
age: [5,10]
frequency of individuals
raw score
0.0
2.5
5.0
7.5
10.0
0
5
10
15

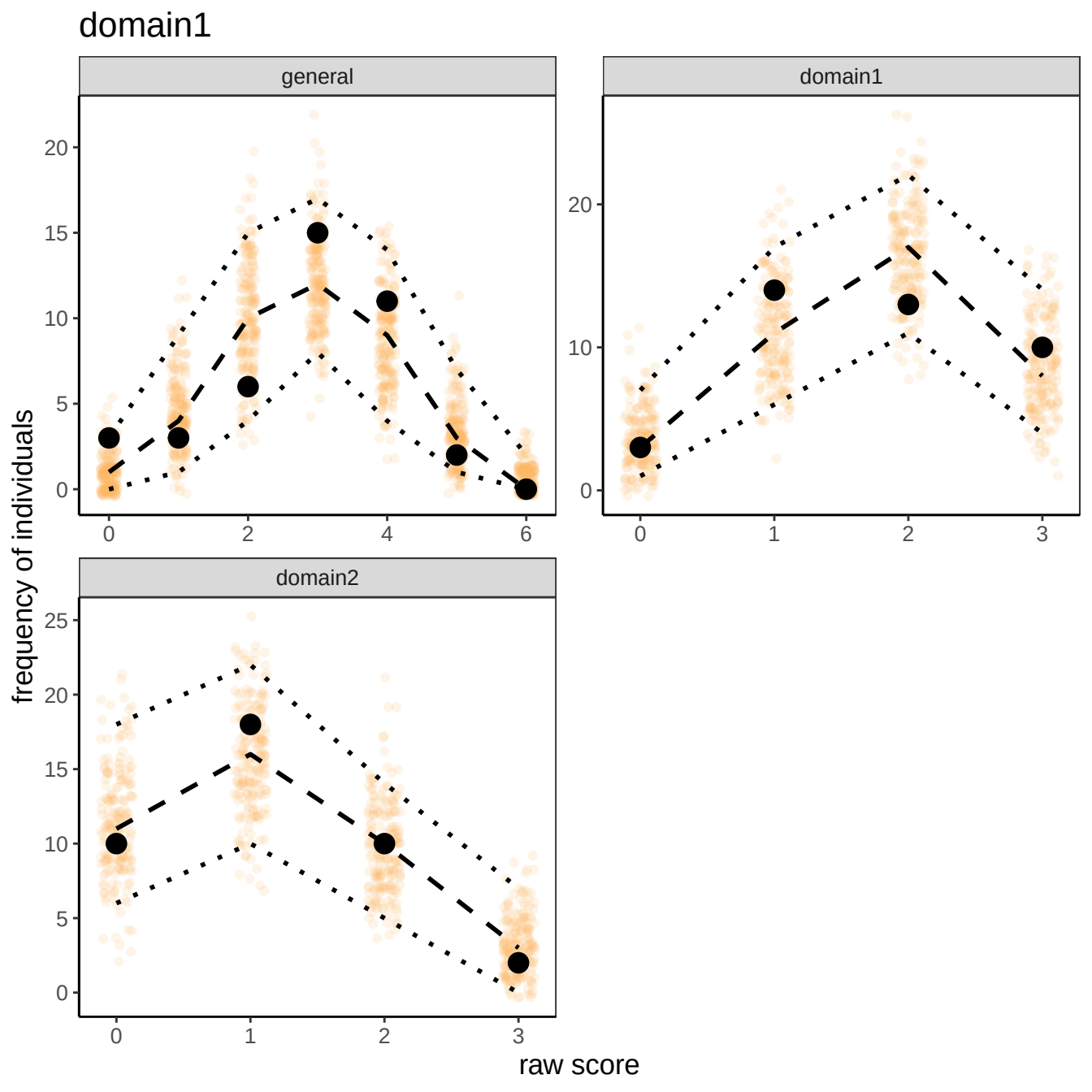

item 1
item 2
item 3
item 4
item 5
item 6
proportion
1.00
0.75
0.50
0.25
0.00
irt_mod
observed
Level
1
2
3

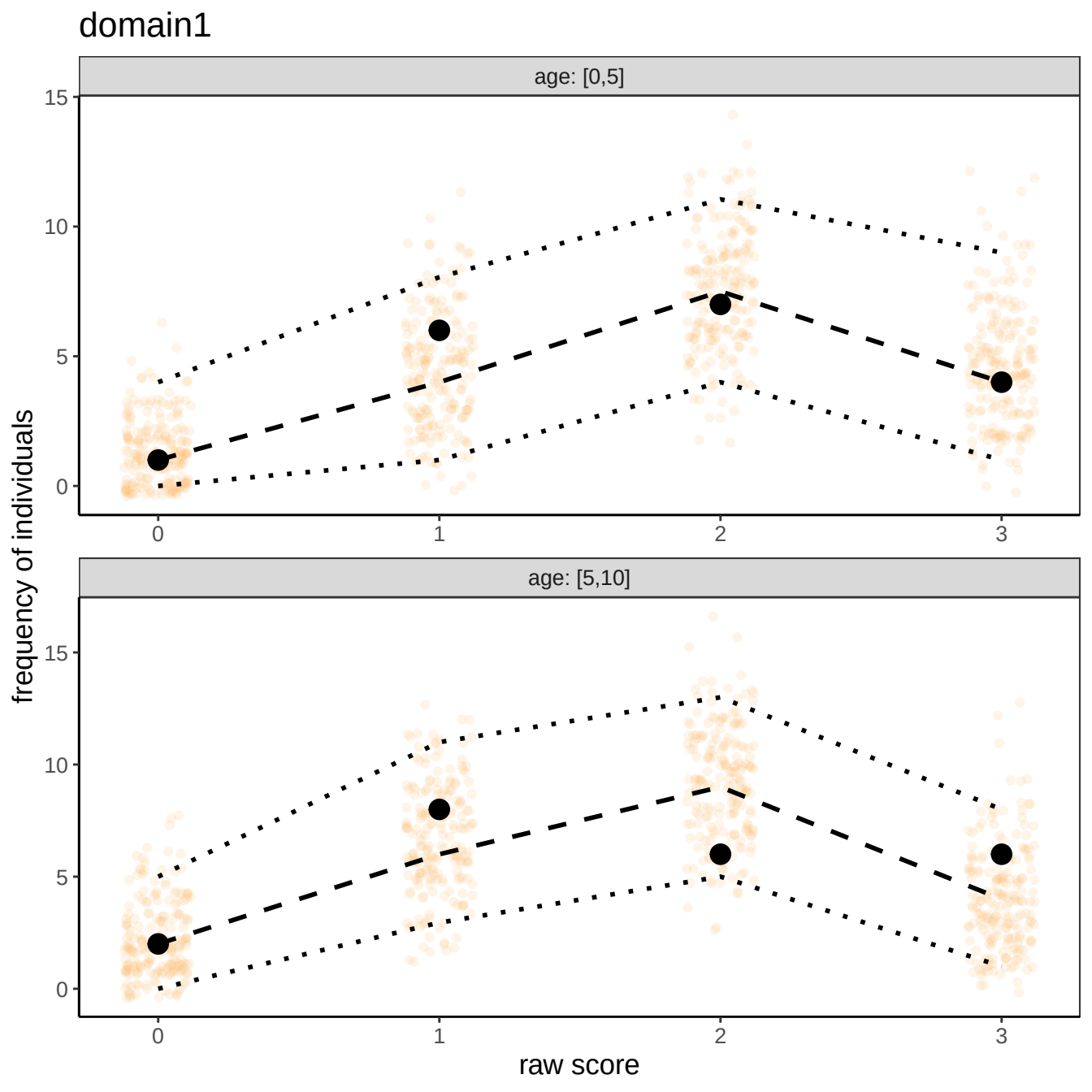

person fit: raw score
model−implied
observed
14
12
10
8
7.5
10.0
12.5
15.0
17.5

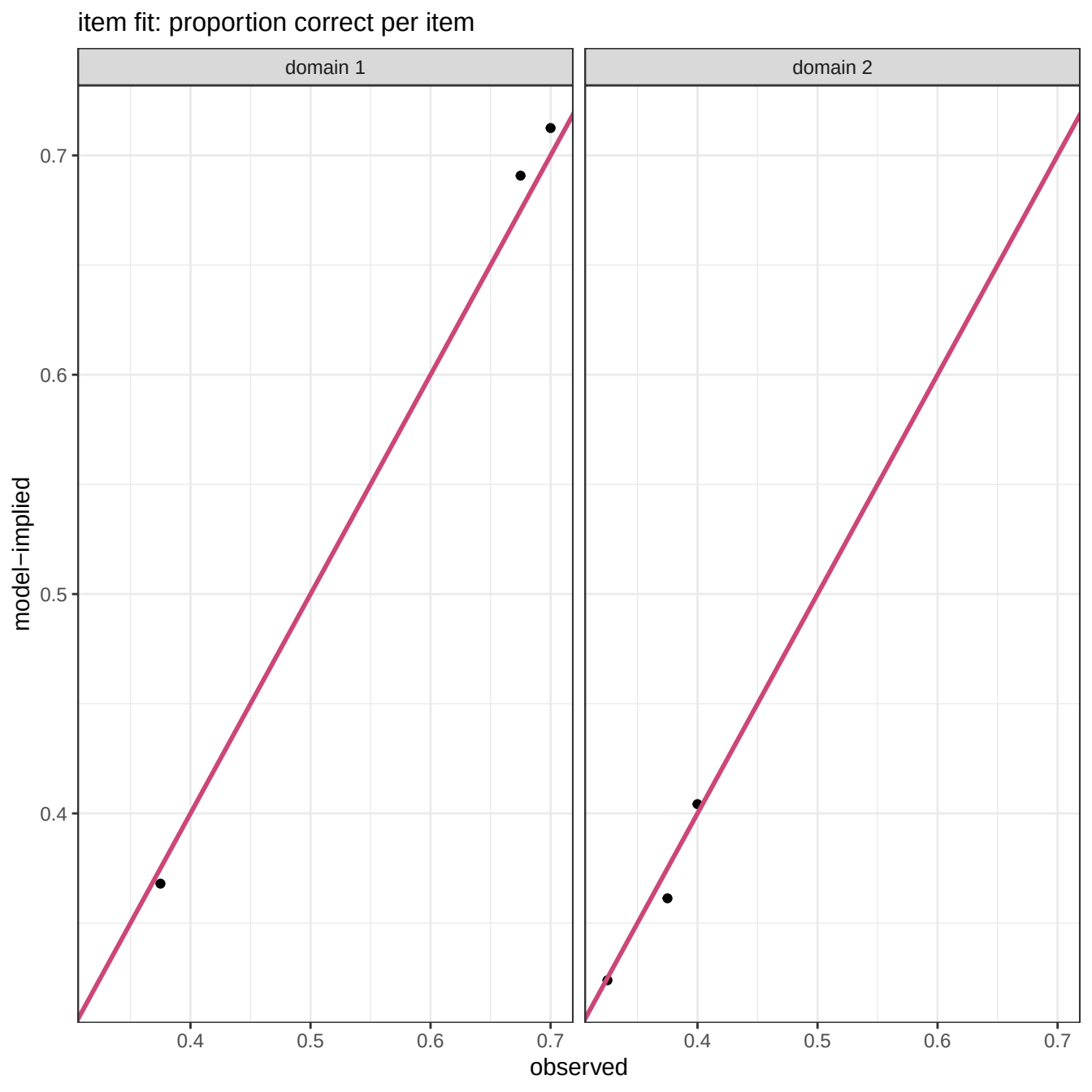

Corr
1.0
0.5
0.0
−0.5
−1.0

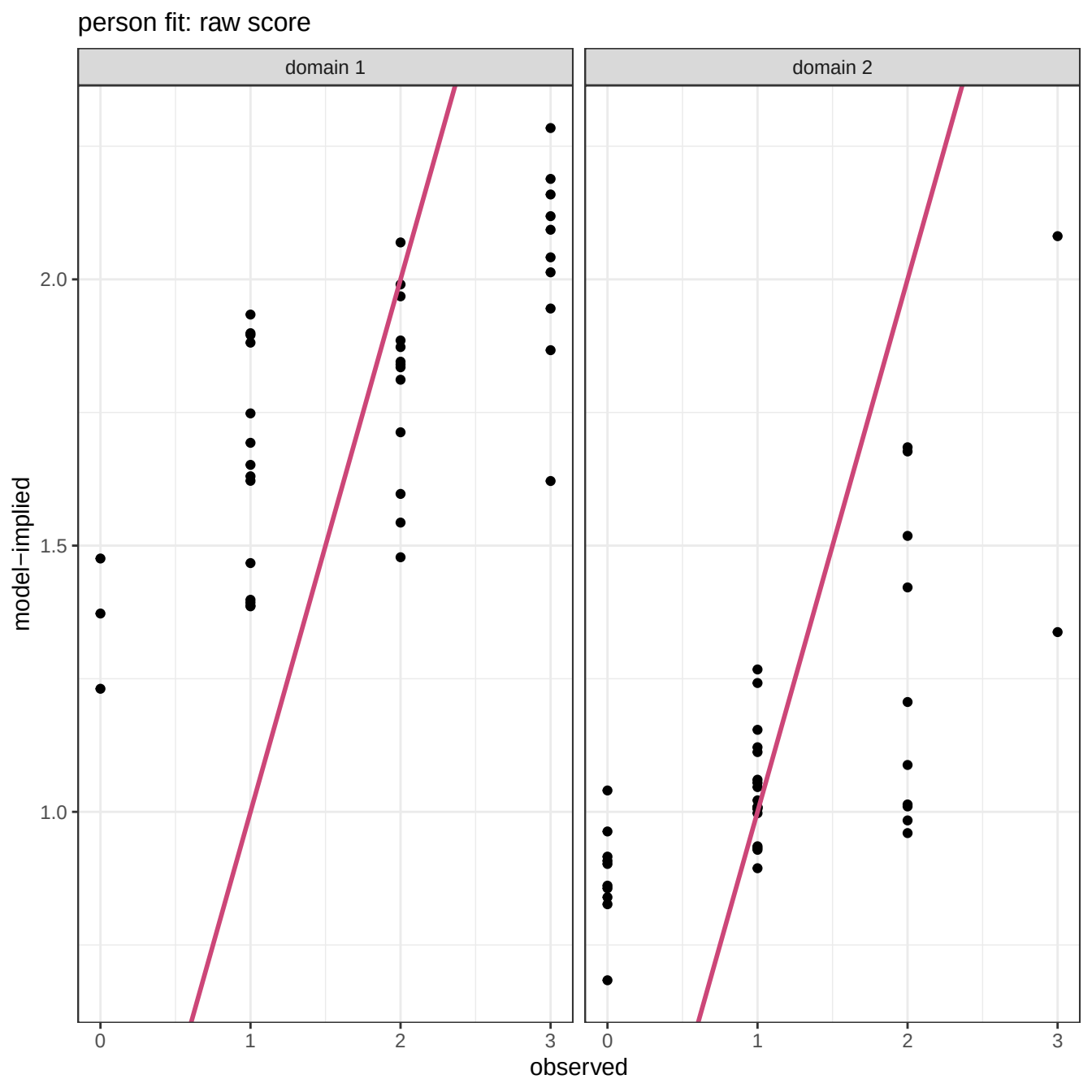

latent trait
2
1
0
−1
−2
−3
2.5
5.0
7.5
10.0
age
p0.05
p0.25
p0.75
p0.95
p0.1
p0.5
p0.9

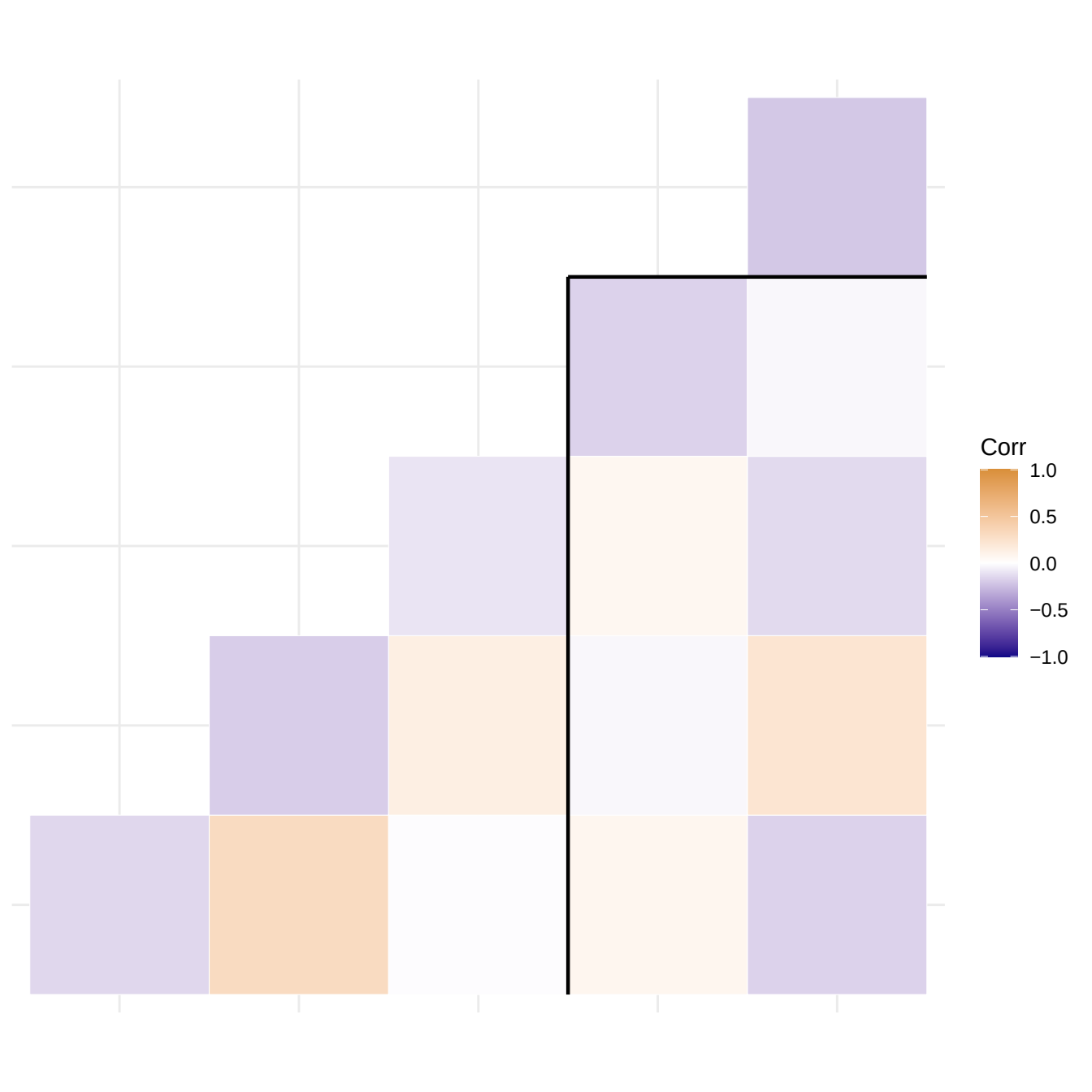

mean
standard deviation
latent trait
0.25
0.00
−0.25
−0.50
−0.75
1.5
1.3
1.1
0.9
2.5
5.0
7.5
10.0
age

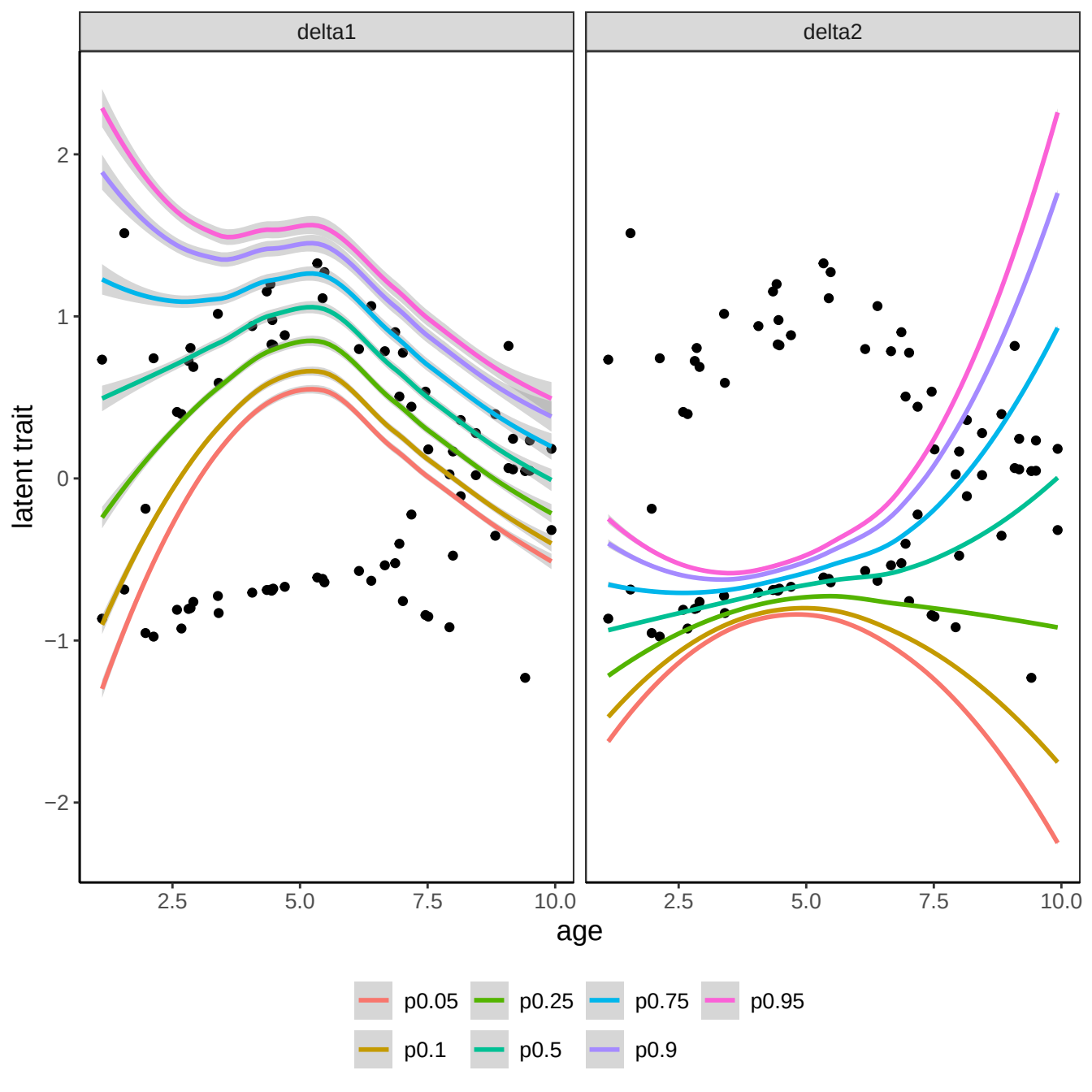

domain1
general
domain1
domain2
frequency of individuals
raw score
0
5
10
15
20
25
0
2
4
6
0
1
2
3

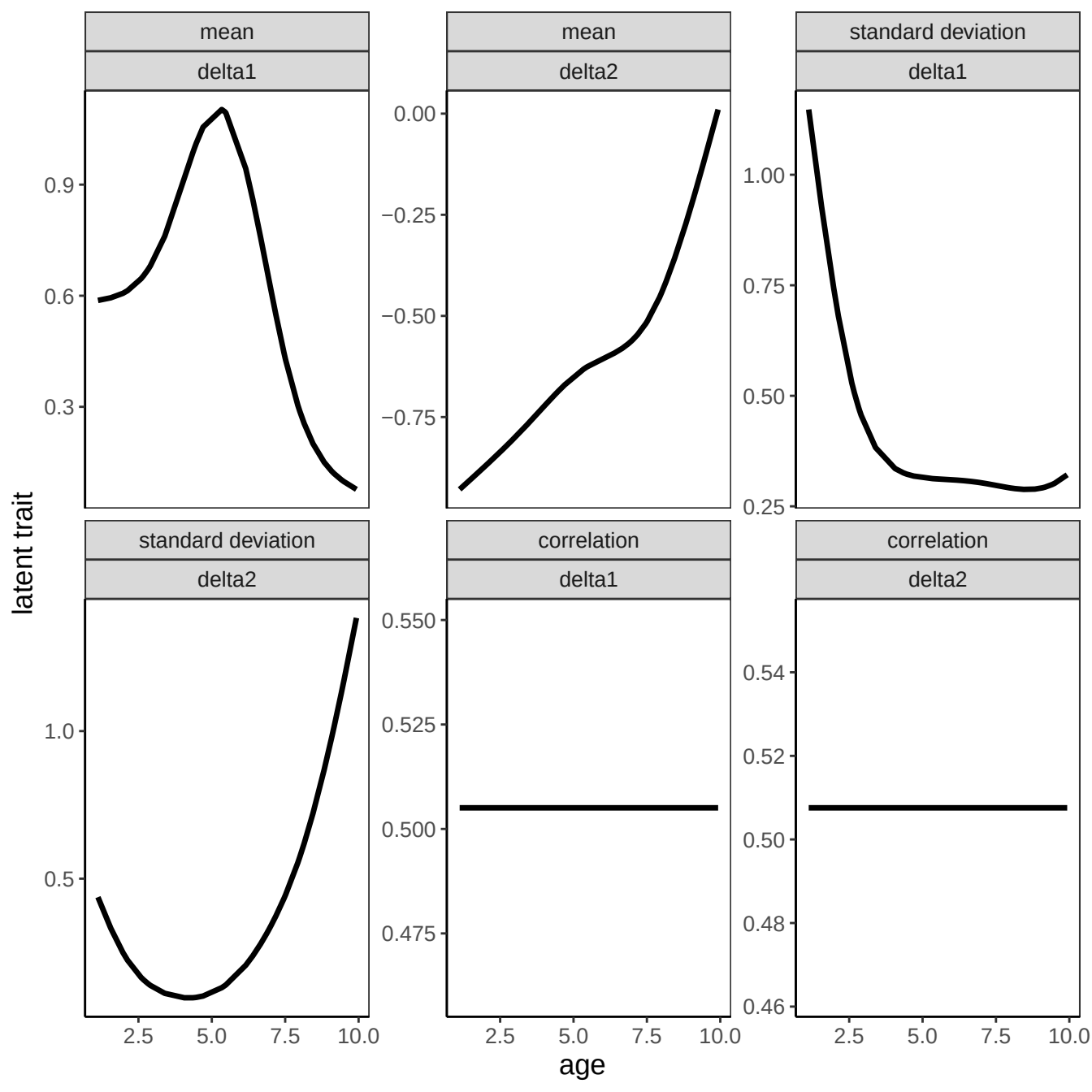

domain1
age: [0,5]
age: [5,10]
frequency of individuals
raw score
0
5
10
15
0
1
2
3

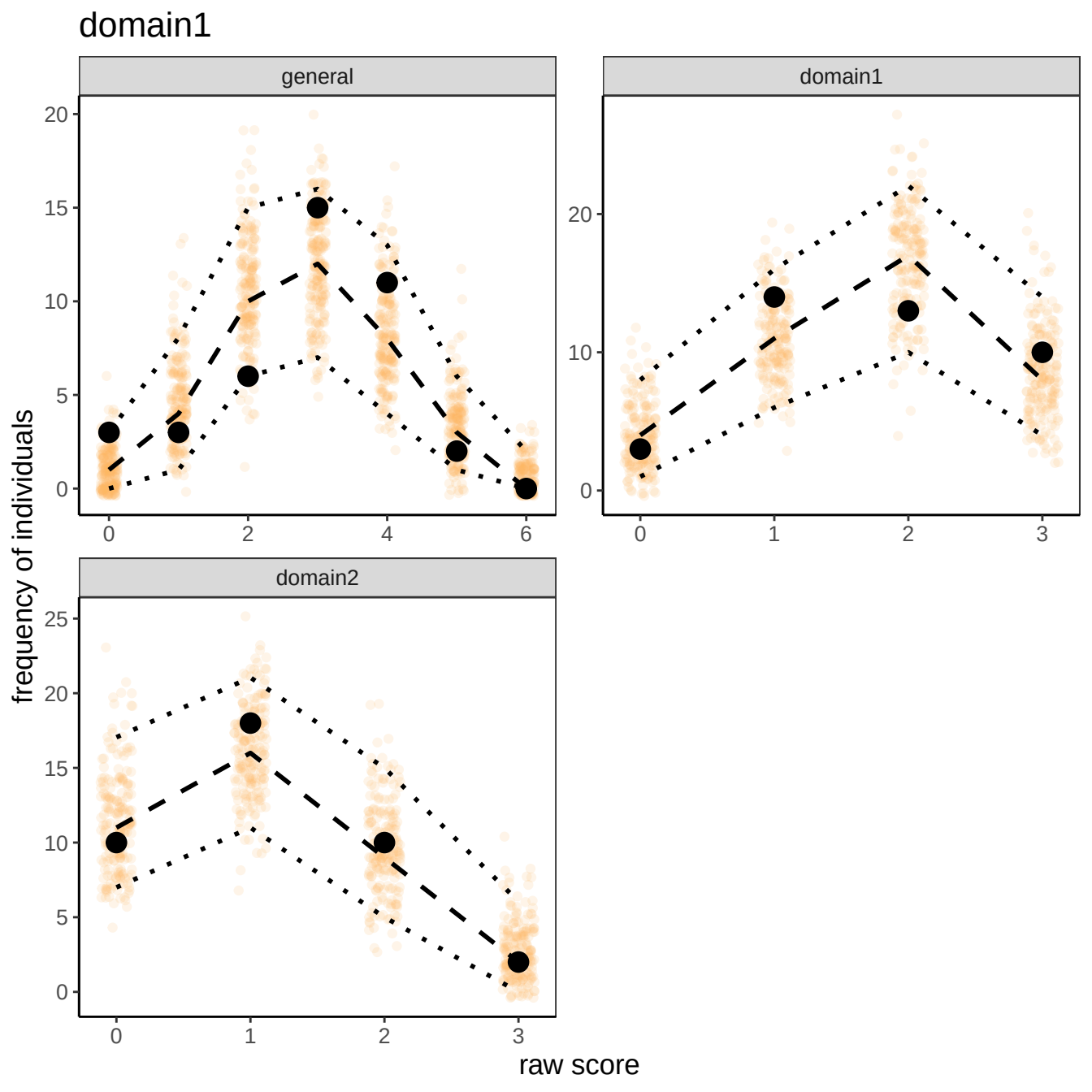

item fit: proportion correct per item
domain 1
domain 2
model−implied
observed
0.4
0.5
0.6
0.7

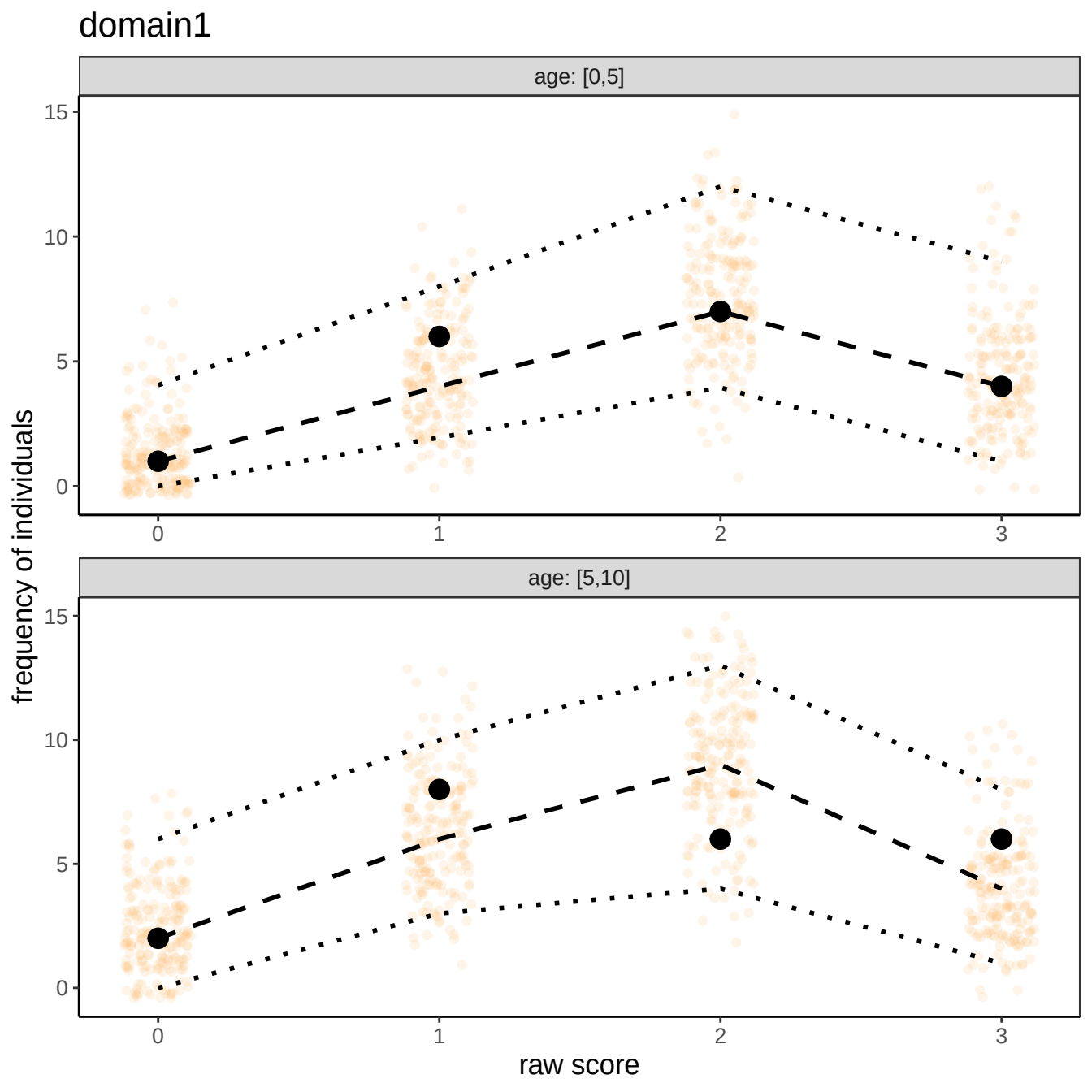

person fit: raw score
domain 1
domain 2
model−implied
observed
2.0
1.5
1.0
0
1
2
3

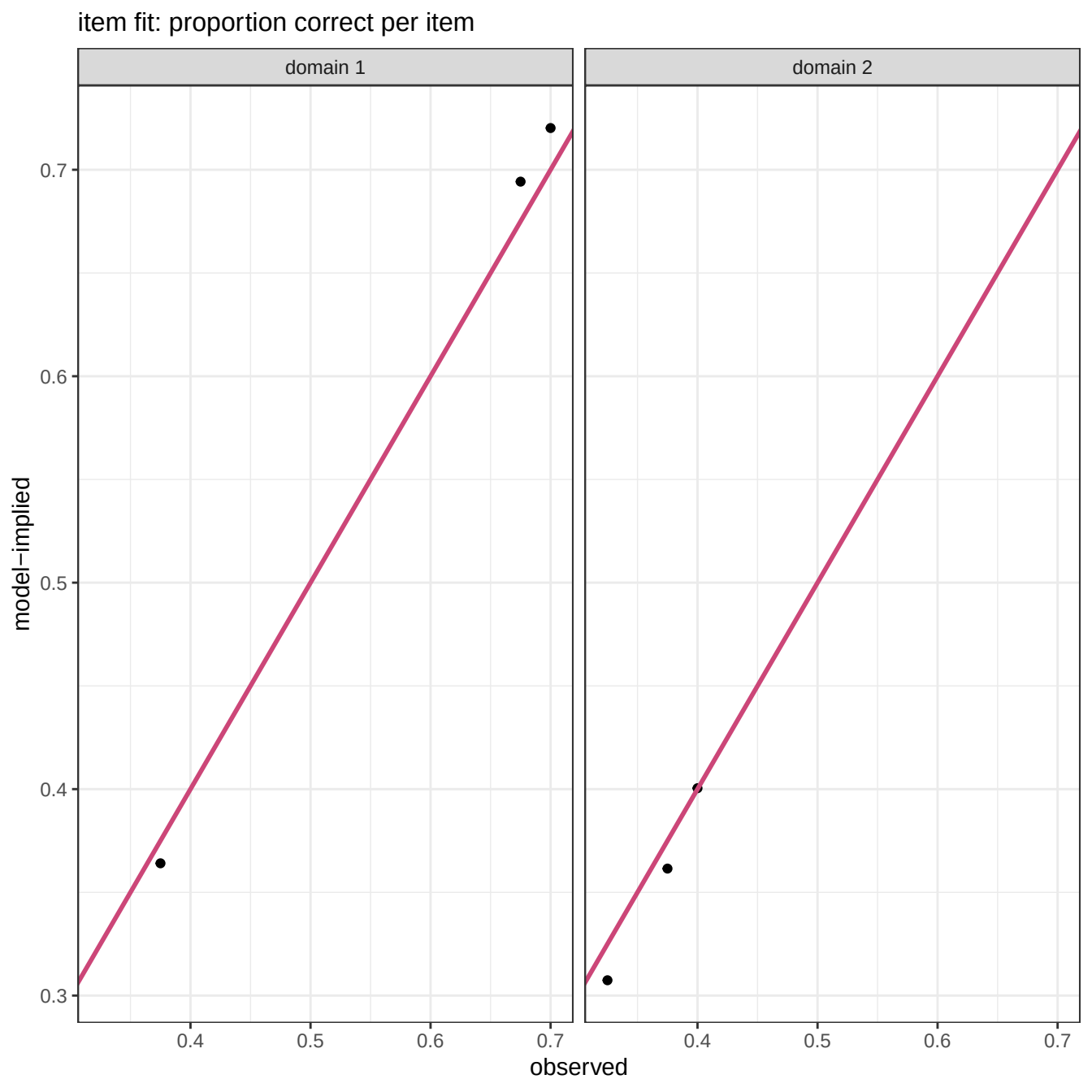

Corr
1.0
0.5
0.0
−0.5
−1.0

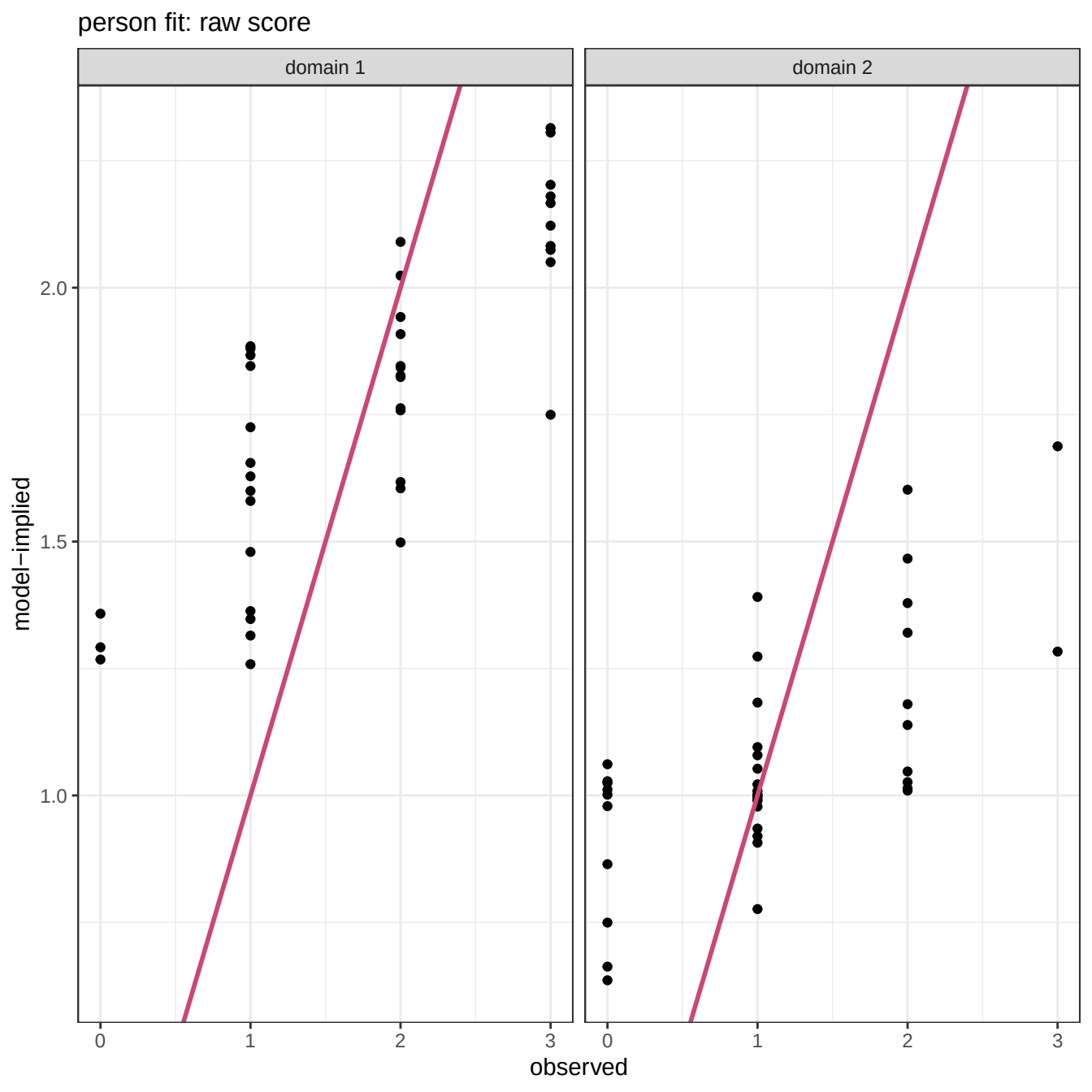

delta1
delta2
latent trait
2
1
0
−1
−2
2.5
5.0
7.5
10.0
age
p0.05
p0.1
p0.25
p0.5
p0.75
p0.9
p0.95

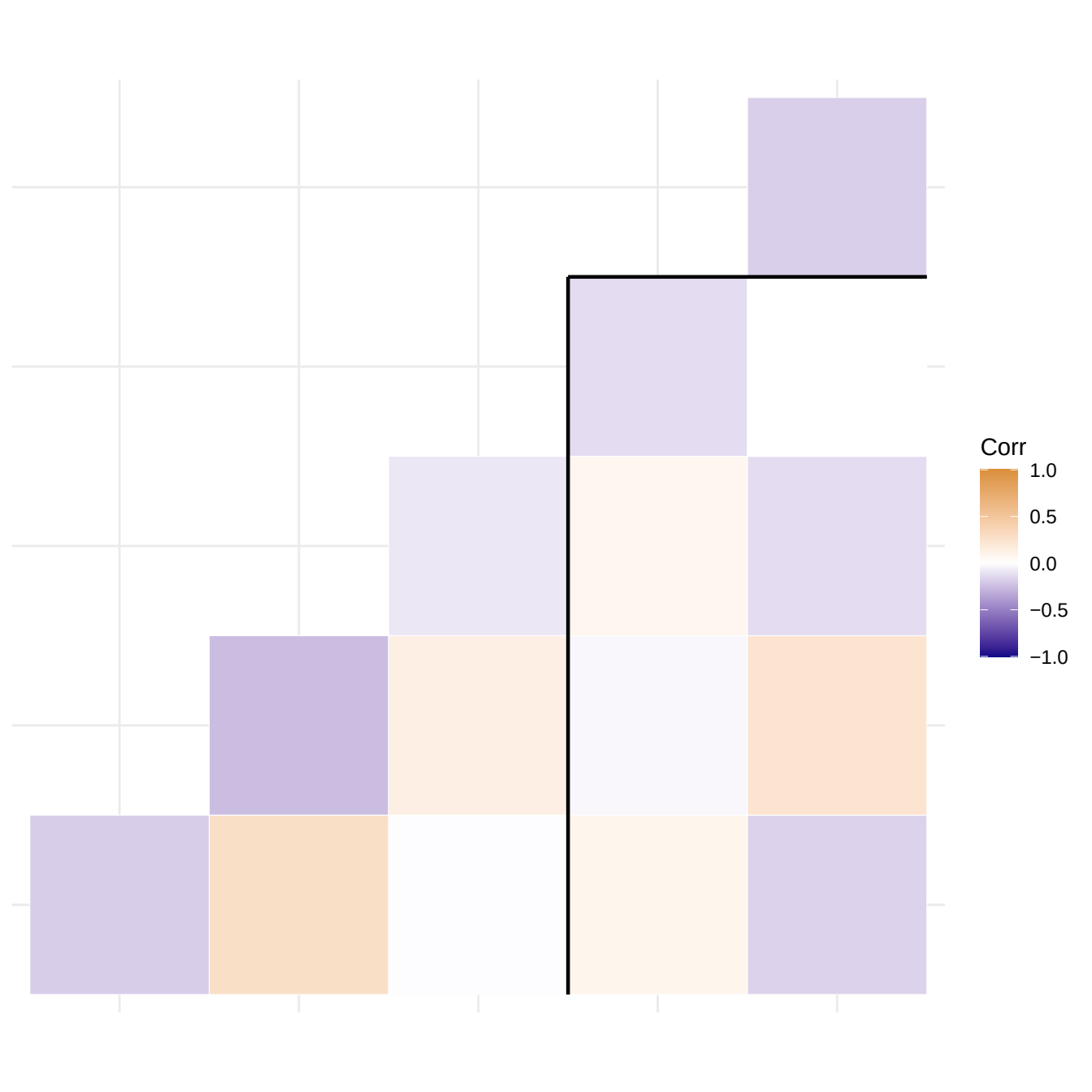

mean
delta1
mean
delta2
standard deviation
delta1
standard deviation
delta2
correlation
delta1
correlation
delta2
latent trait
age
0.9
0.6
0.3
0.00
−0.25
−0.50
−0.75
1.00
0.75
0.50
0.25
1.0
0.5
0.550
0.525
0.500
0.475
0.54
0.52
0.50
0.48
0.46
2.5
5.0
7.5
10.0

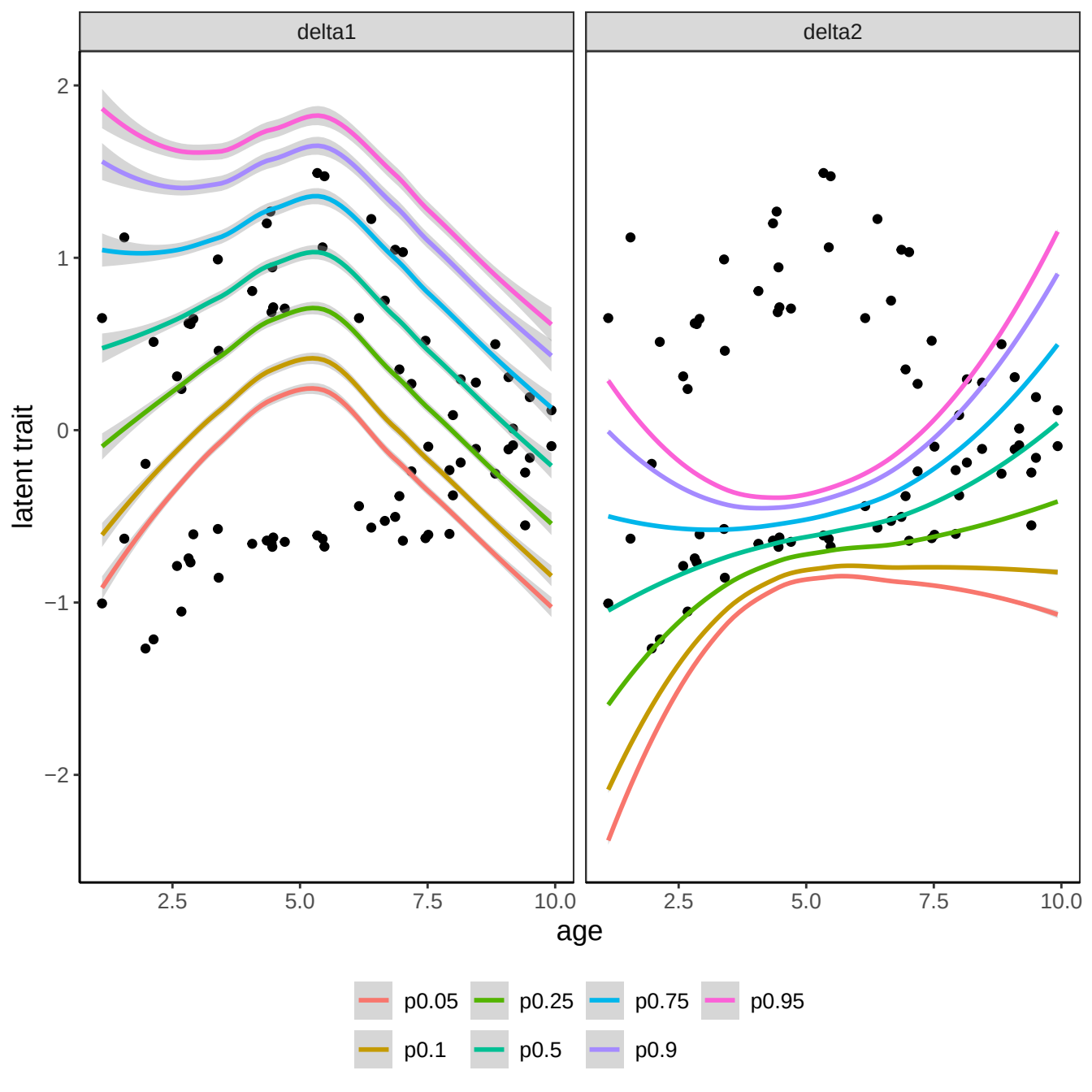

domain1
general
domain1
domain2
frequency of individuals
raw score
0
5
10
15
20
25
0
2
4
6
0
1
2
3

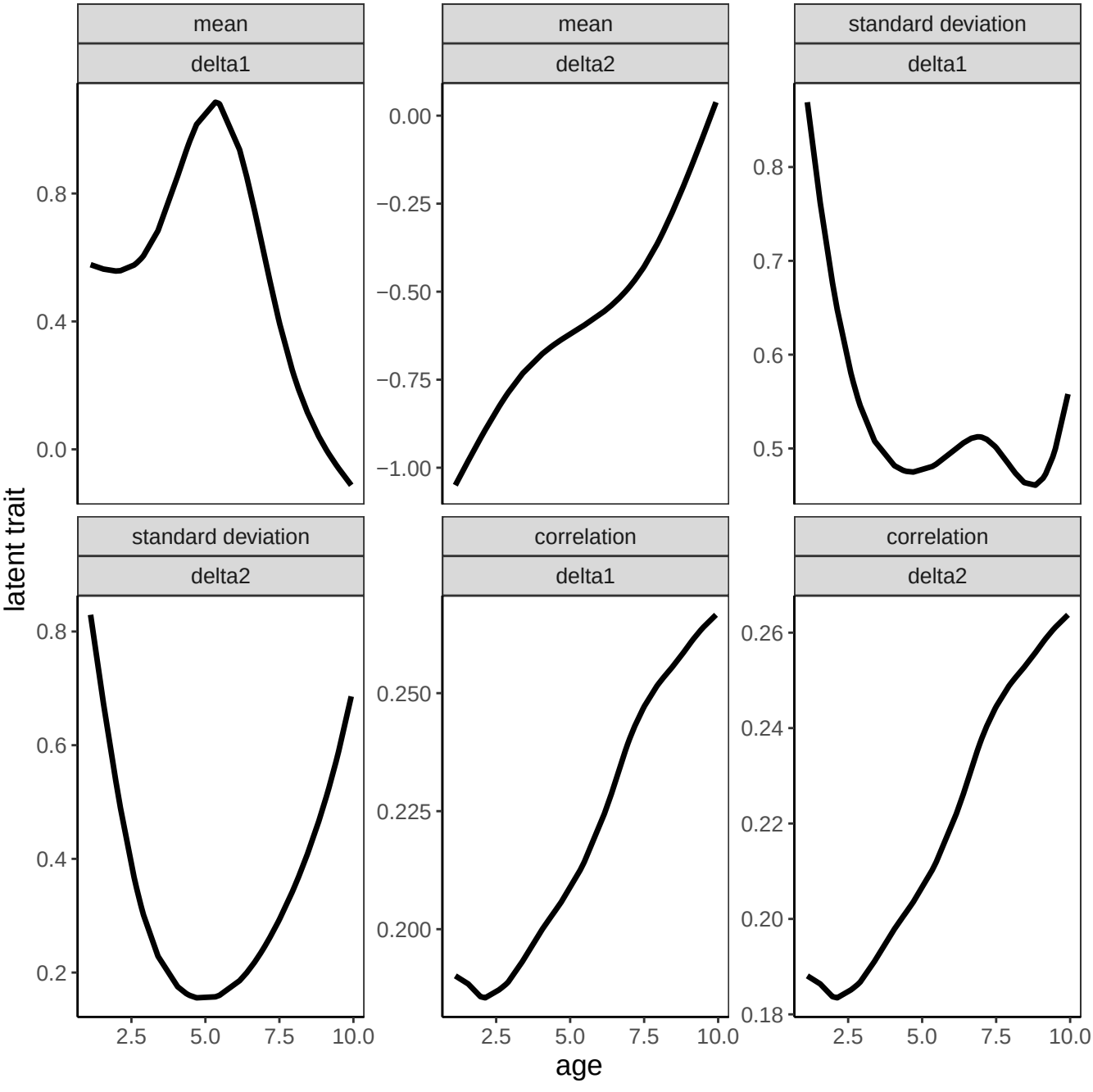

domain1
age: [0,5]
age: [5,10]
frequency of individuals
raw score
0
5
10
15
0
1
2
3

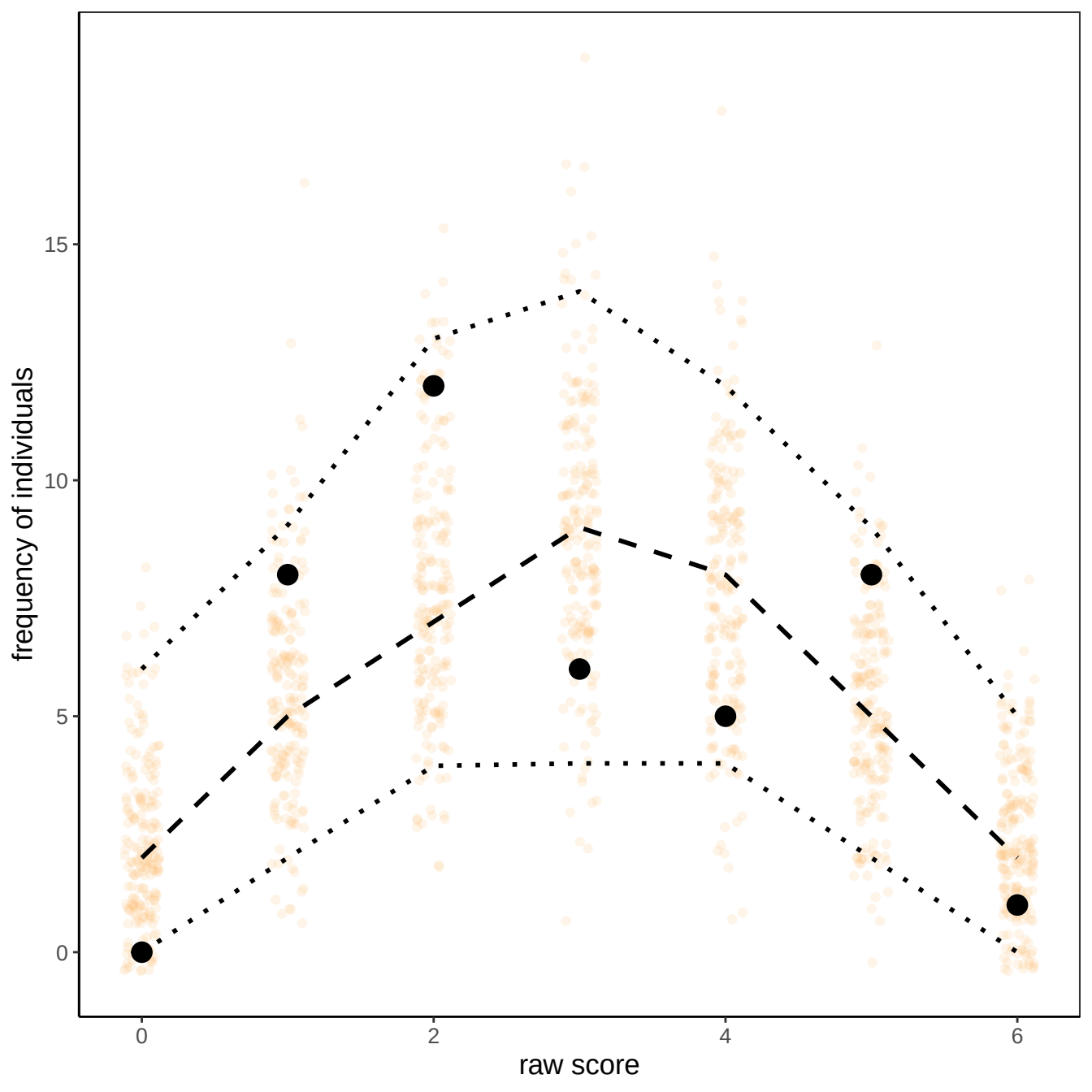

item fit: proportion correct per item
domain 1
domain 2
model−implied
observed
0.3
0.4
0.5
0.6
0.7
0.4
0.5
0.6
0.7

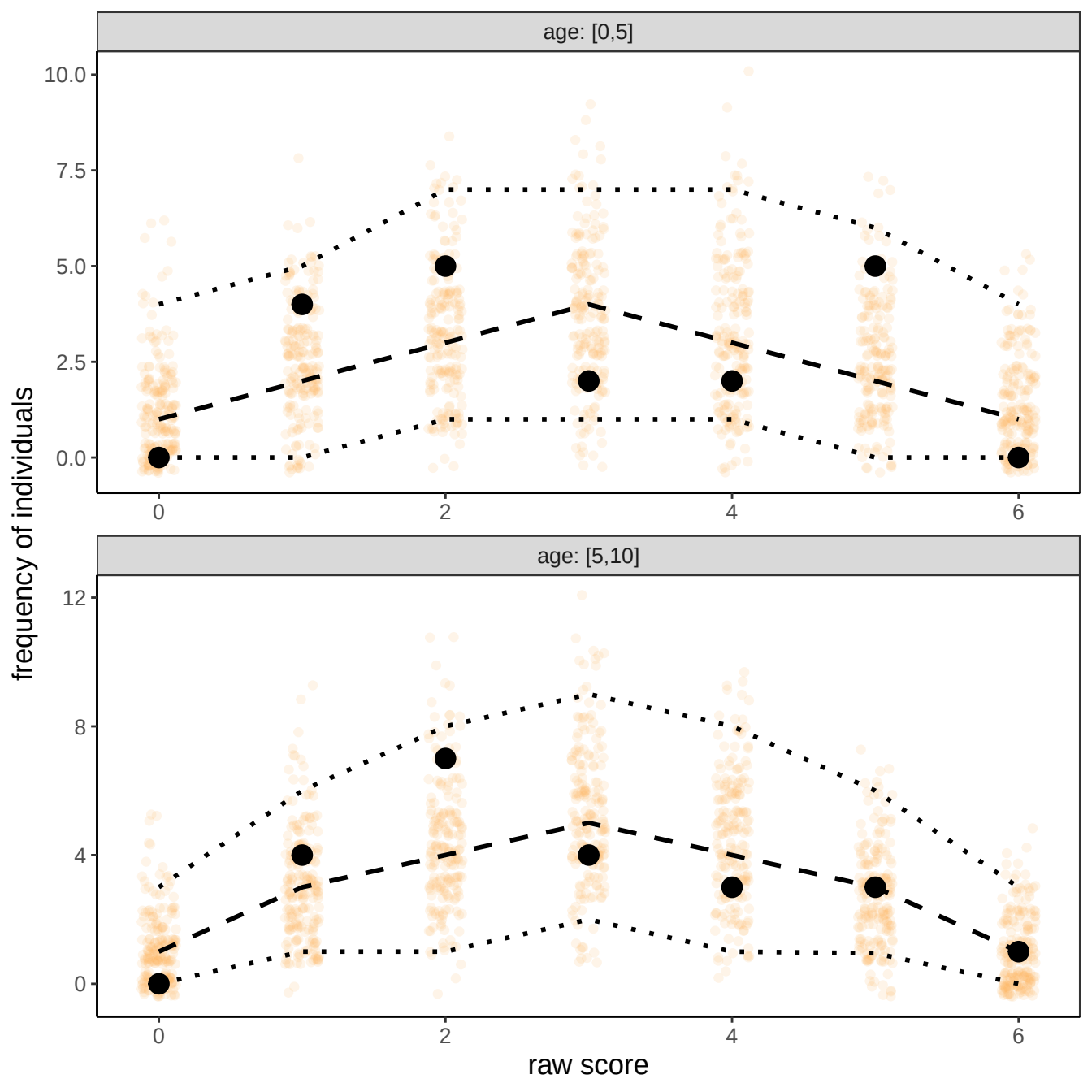

person fit: raw score
domain 1
domain 2
model−implied
observed
2.0
1.5
1.0
0
1
2
3

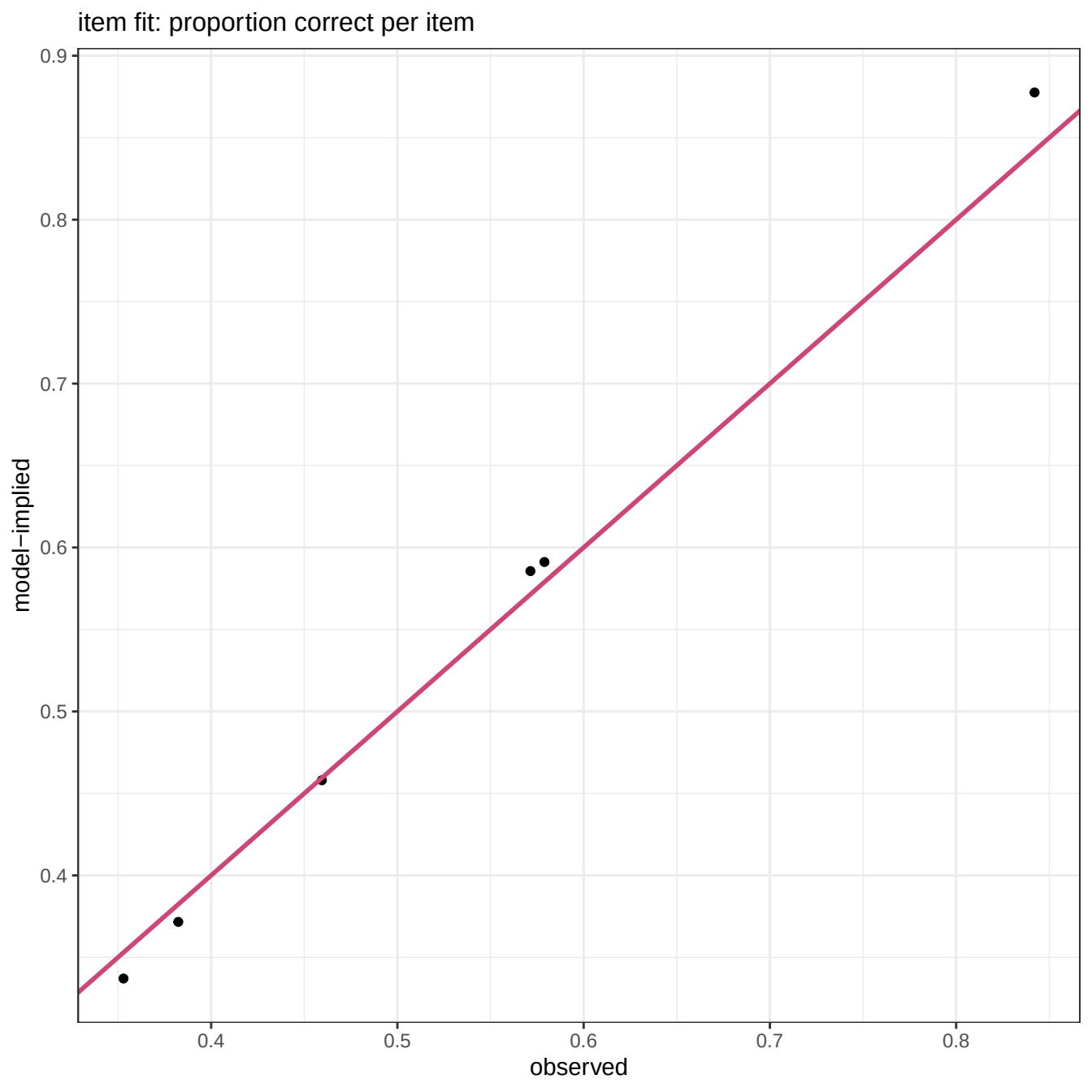

Corr
1.0
0.5
0.0
−0.5
−1.0

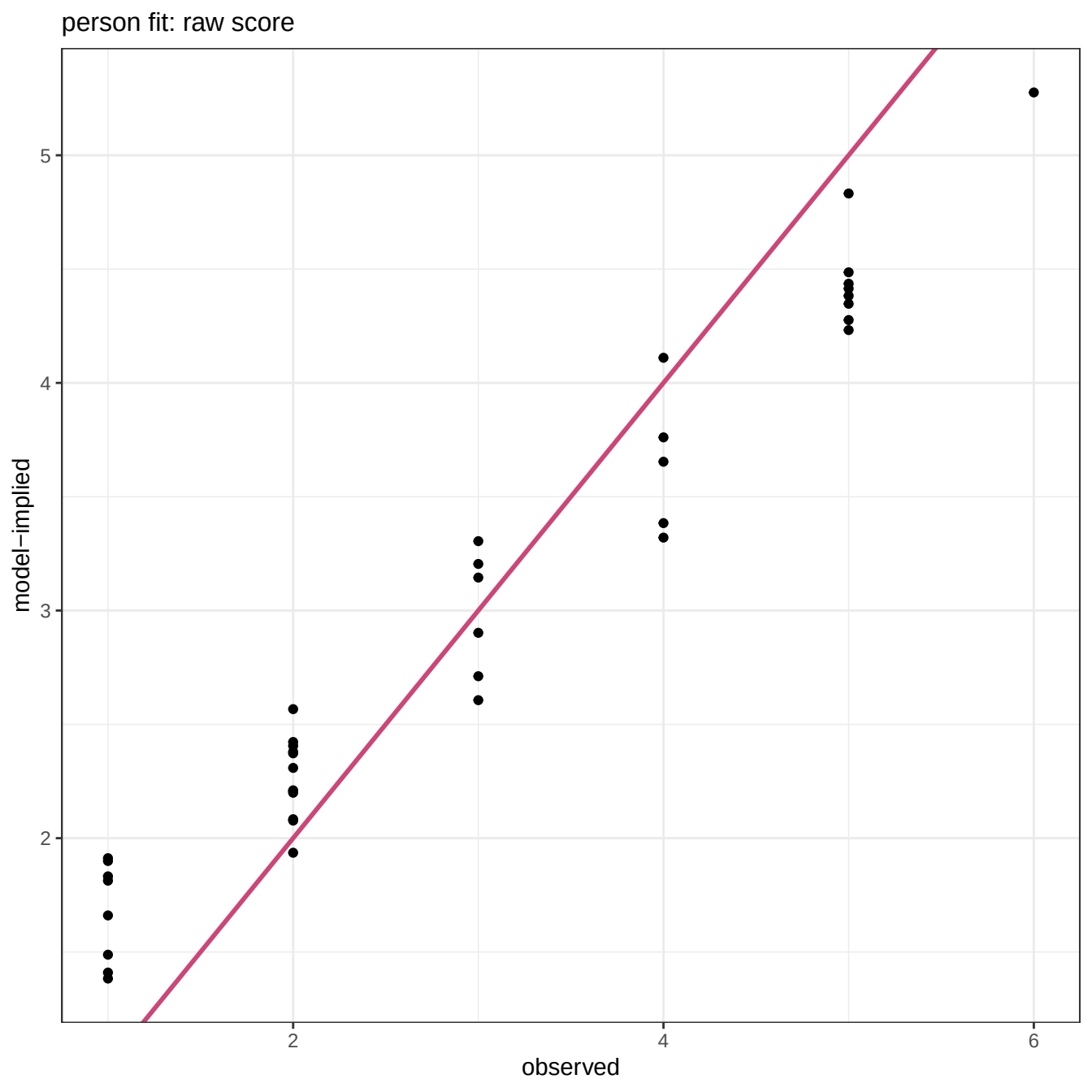

delta1
delta2
latent trait
age
2
1
0
−1
−2
2.5
5.0
7.5
10.0
p0.05
p0.1
p0.25
p0.5
p0.75
p0.9
p0.95

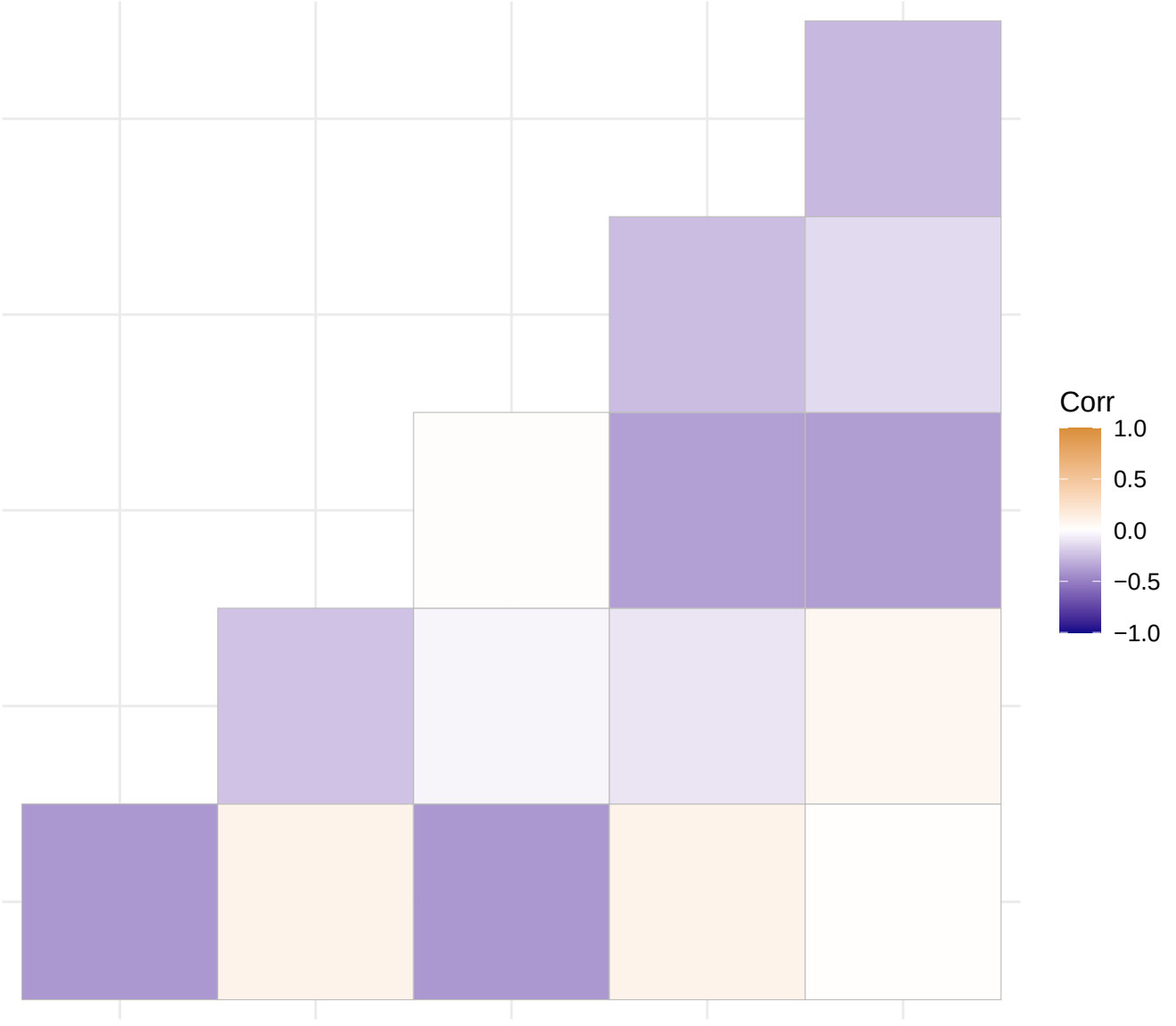

mean
delta1
mean
delta2
standard deviation
delta1
standard deviation
delta2
correlation
delta1
correlation
delta2
latent trait
age
0.0
0.4
0.8
0.00
−0.25
−0.50
−0.75
−1.00
0.5
0.6
0.7
0.8
0.2
0.4
0.6
0.8
0.200
0.225
0.250
0.18
0.20
0.22
0.24
0.26
2.5
5.0
7.5
10.0

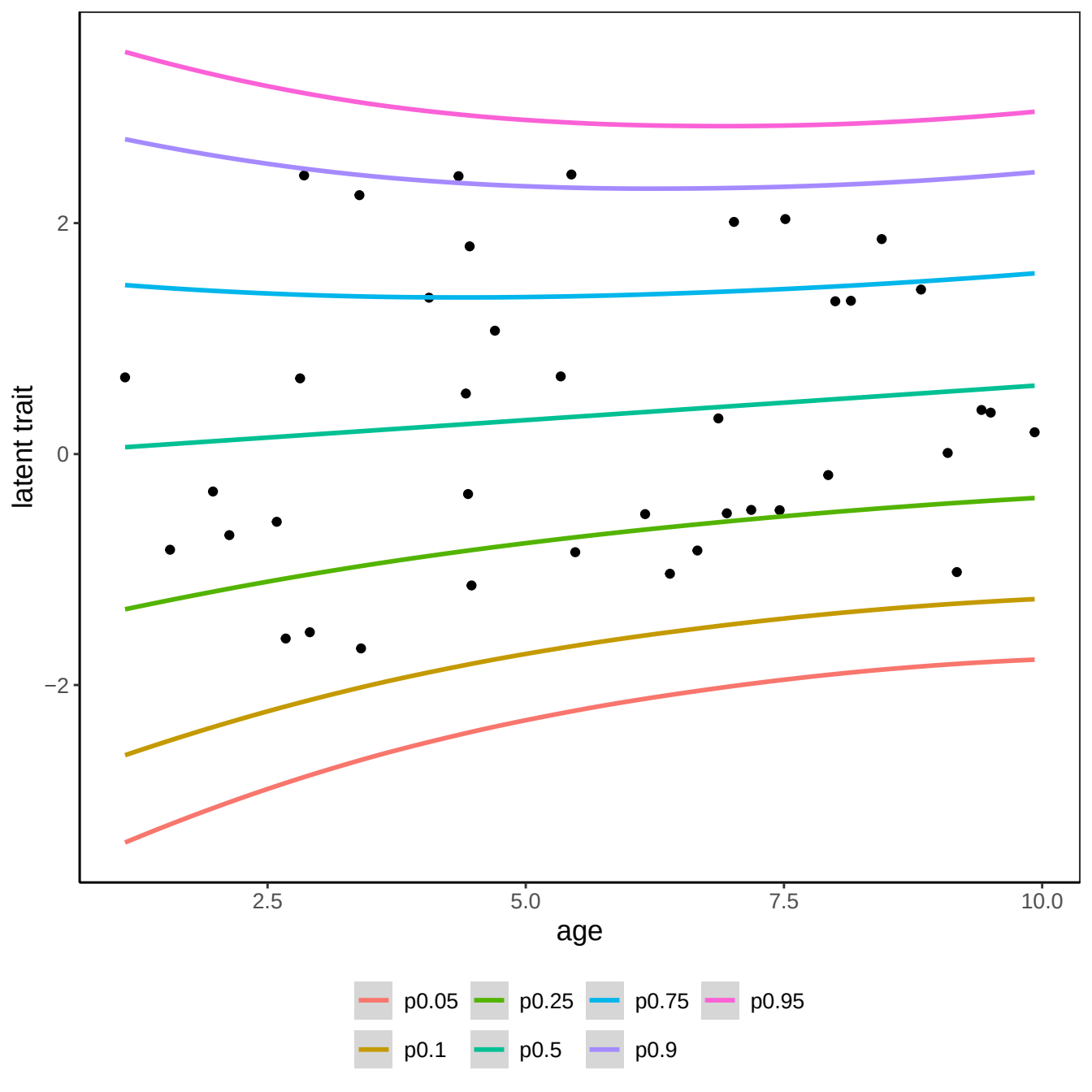

15
10
5
0
frequency of individuals
0
2
4
6
raw score

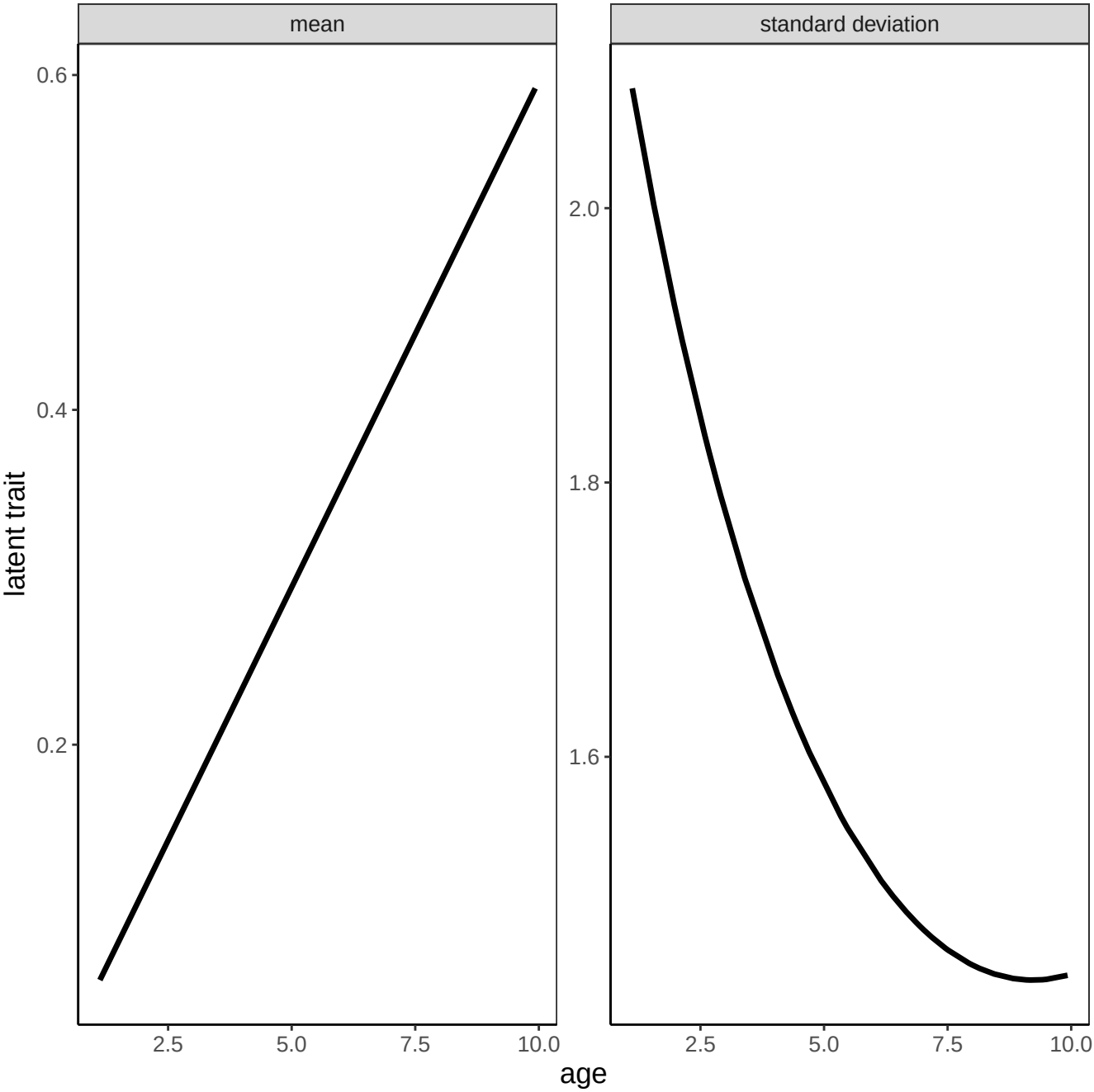

age: [0,5]
age: [5,10]
frequency of individuals
raw score
10.0
7.5
5.0
2.5
0.0
12
8
4
0
0
2
4
6

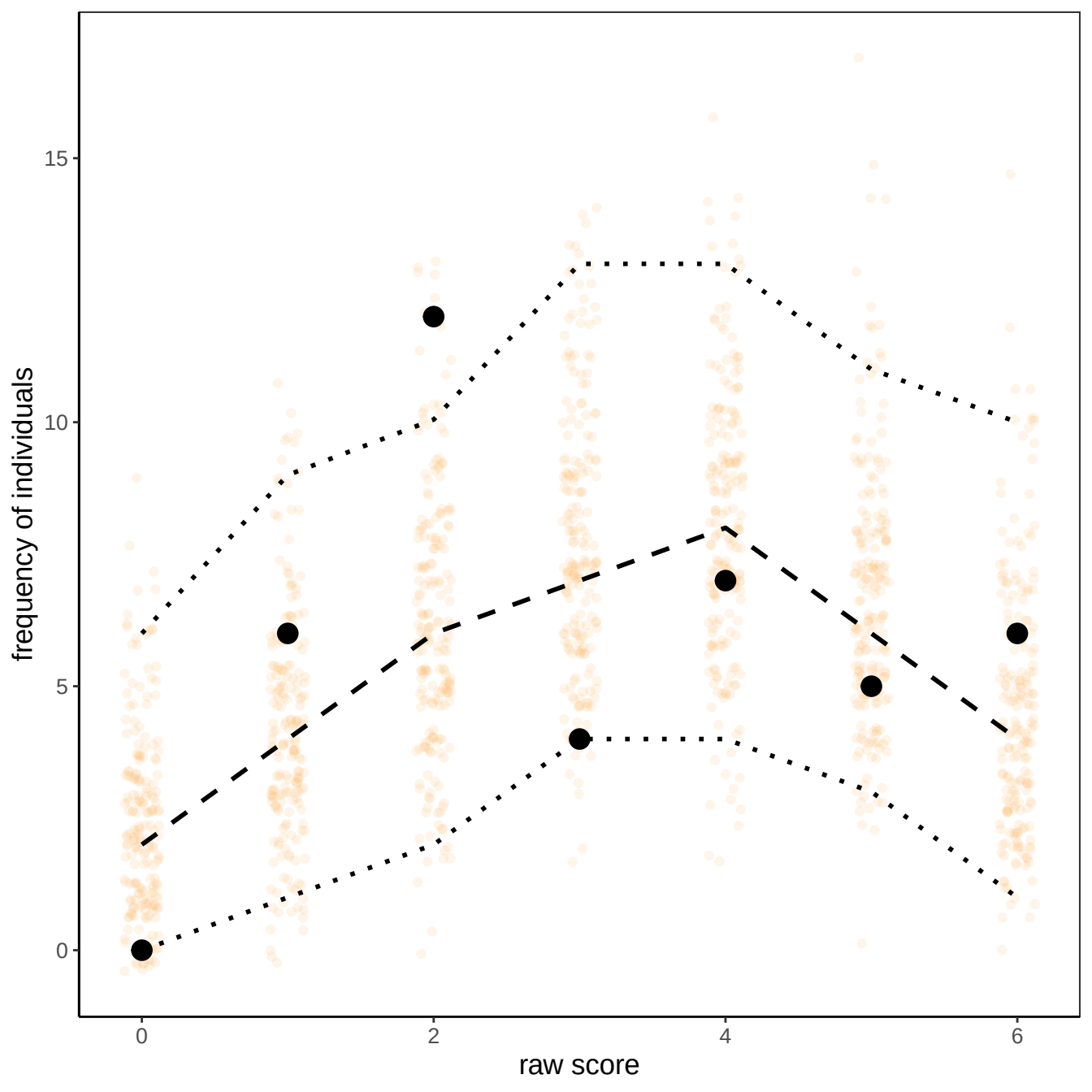

item fit: proportion correct per item
model−implied
observed
0.9
0.8
0.7
0.6
0.5
0.4
0.4
0.5
0.6
0.7
0.8

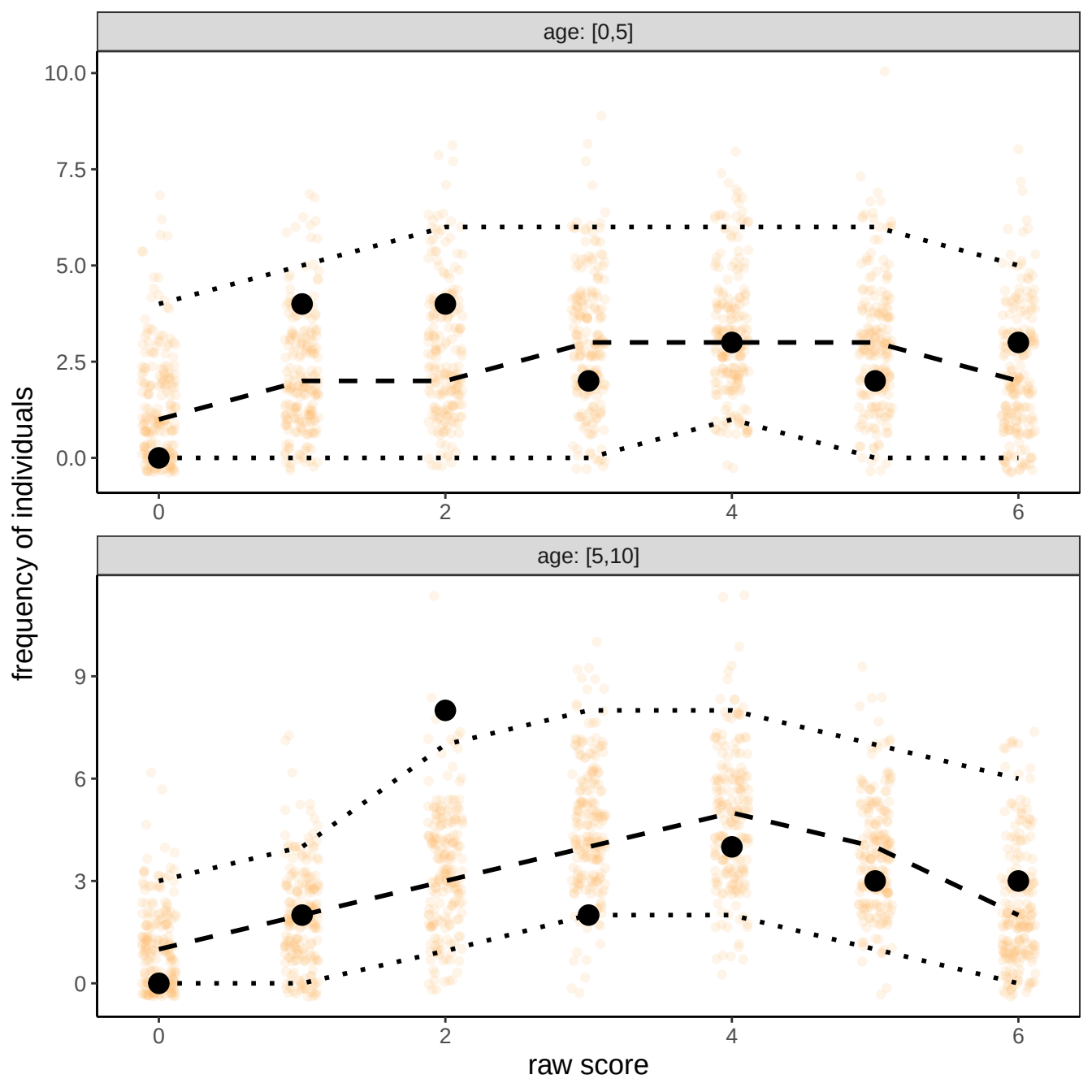

person fit: raw score
model−implied
observed
2
3
4
5
2
4
6

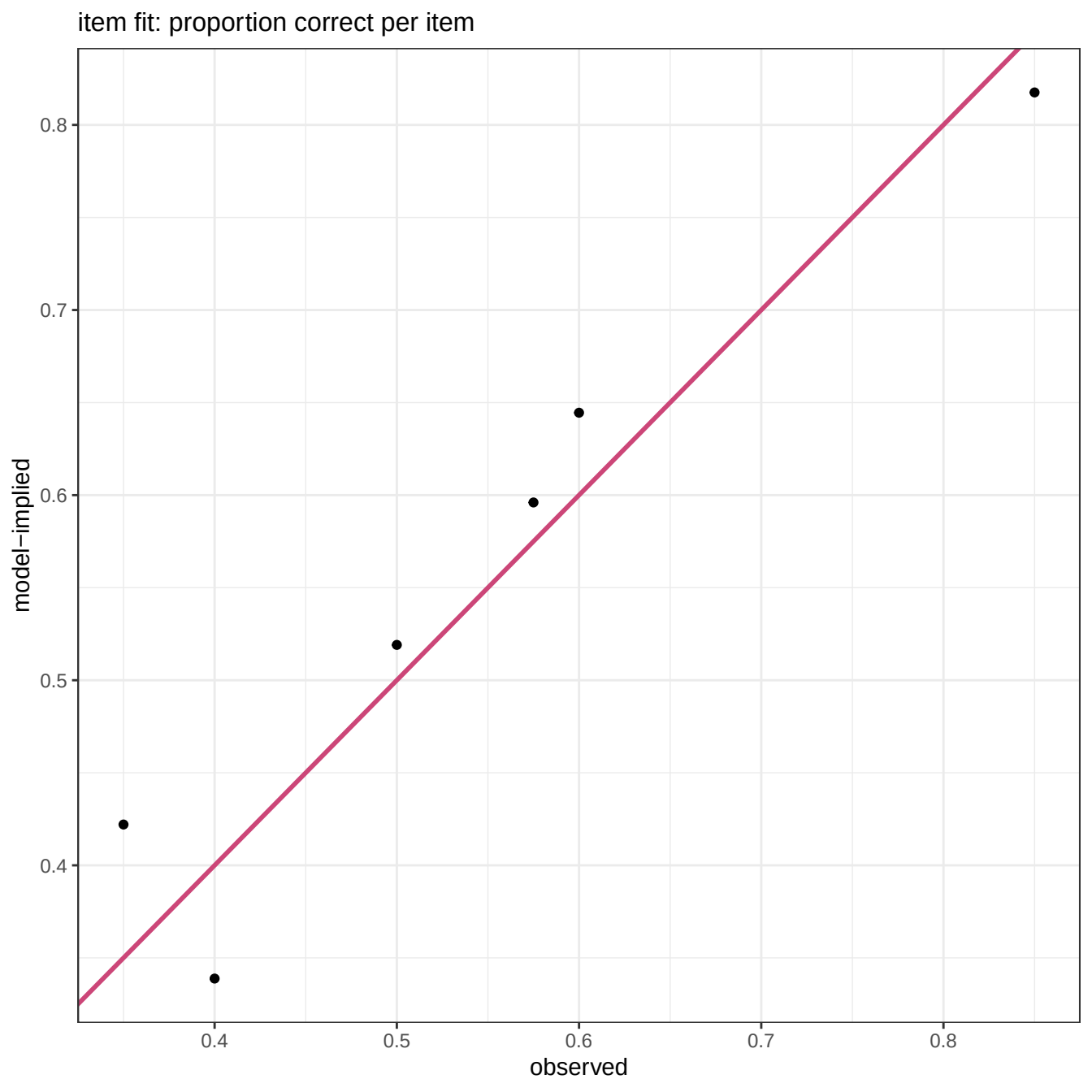

Corr
1.0
0.5
0.0
−0.5
−1.0

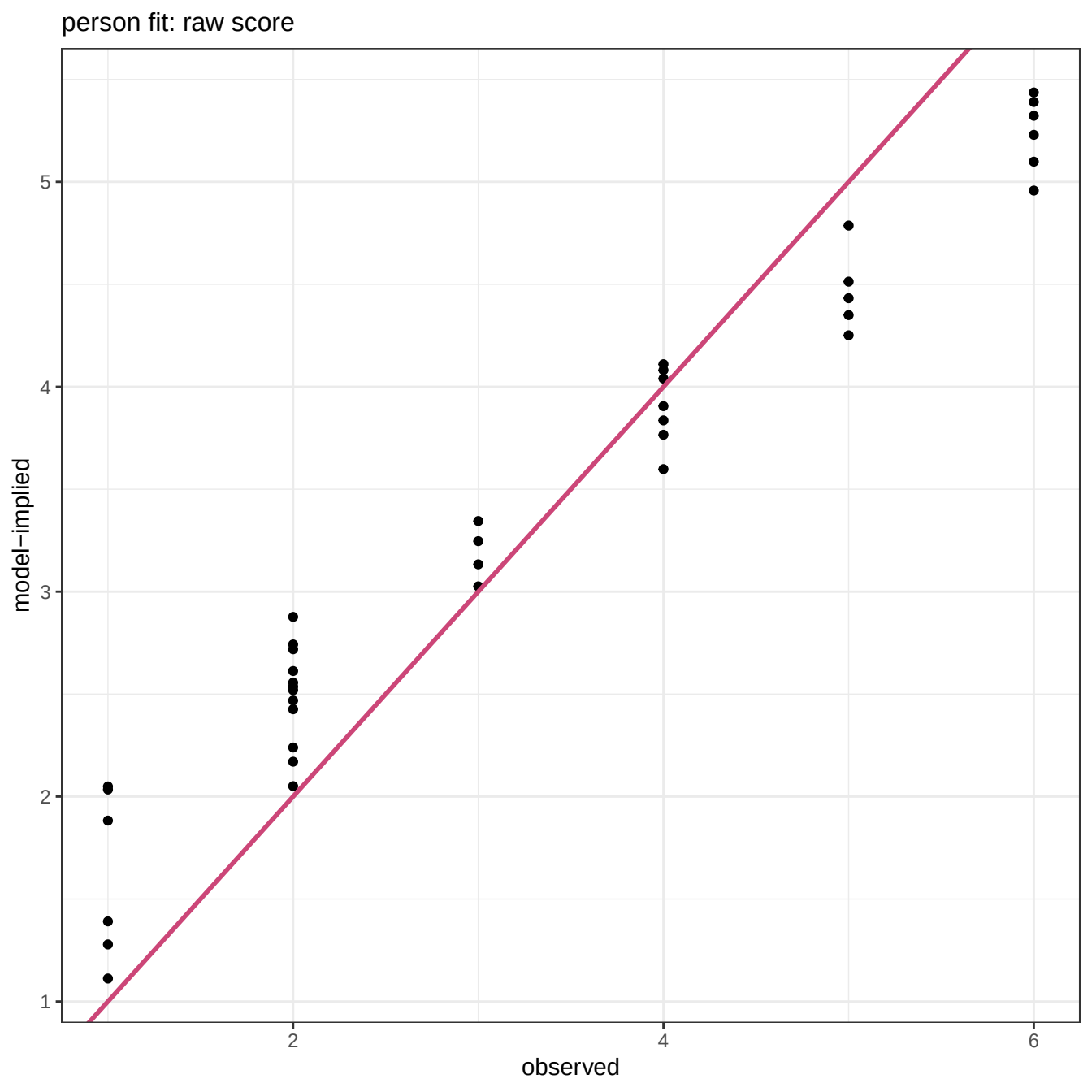

latent trait
2
0
−2
2.5
5.0
7.5
10.0
age
p0.05
p0.25
p0.75
p0.95
p0.1
p0.5
p0.9

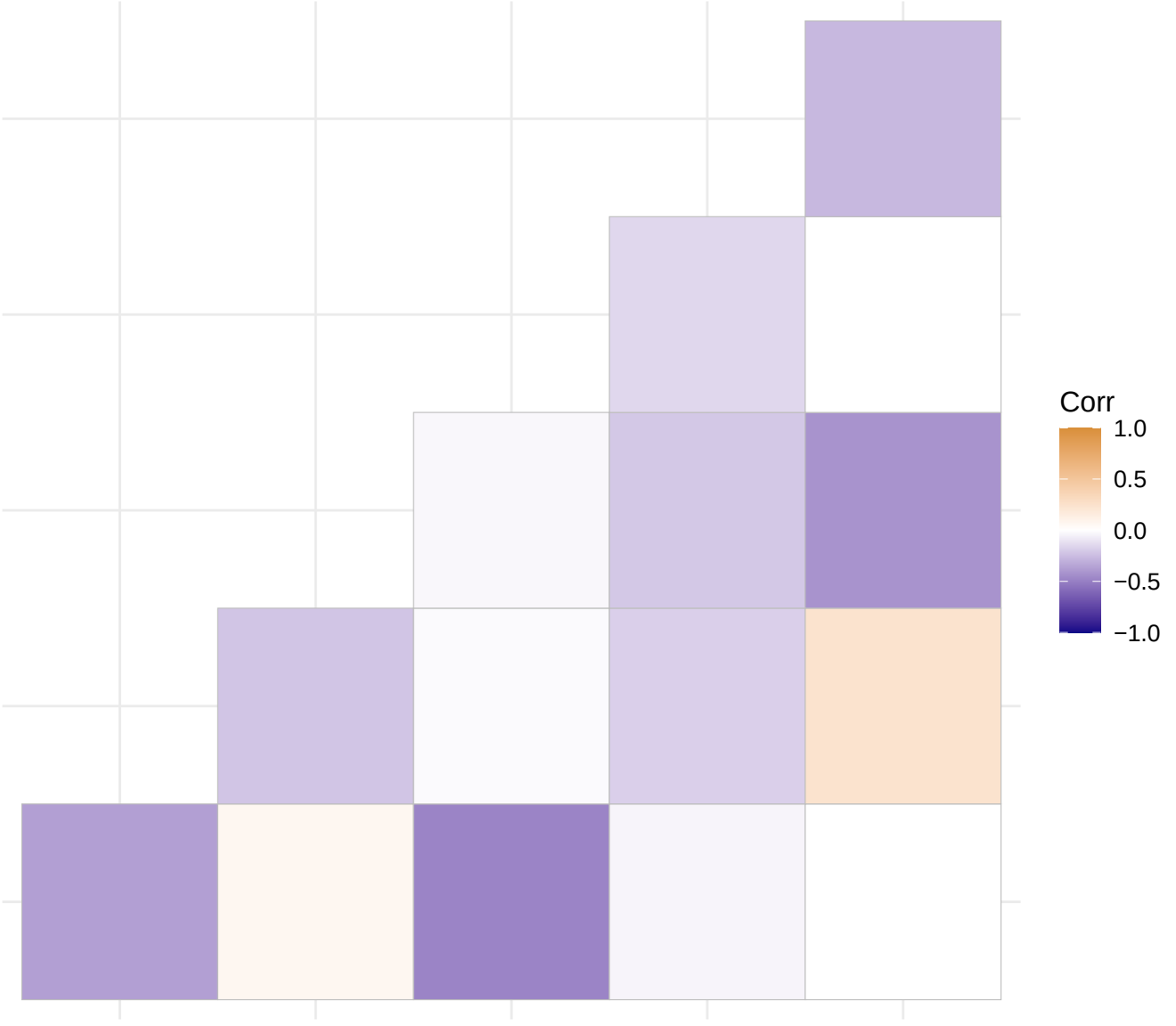

mean
standard deviation
0.6
0.4
0.2
2.0
1.8
1.6
latent trait
2.5
5.0
7.5
10.0
age

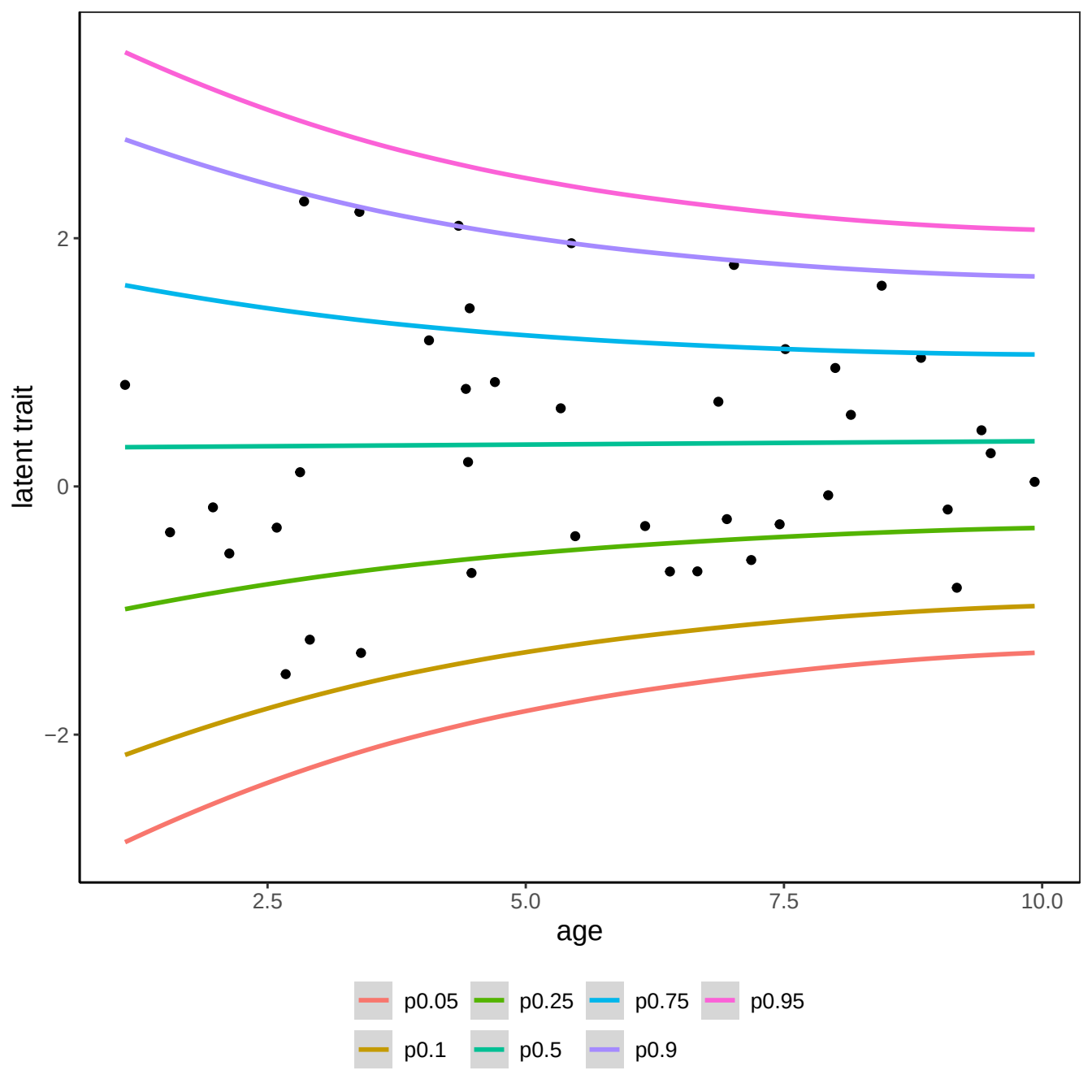

15
10
5
0
frequency of individuals
0
2
4
6
raw score

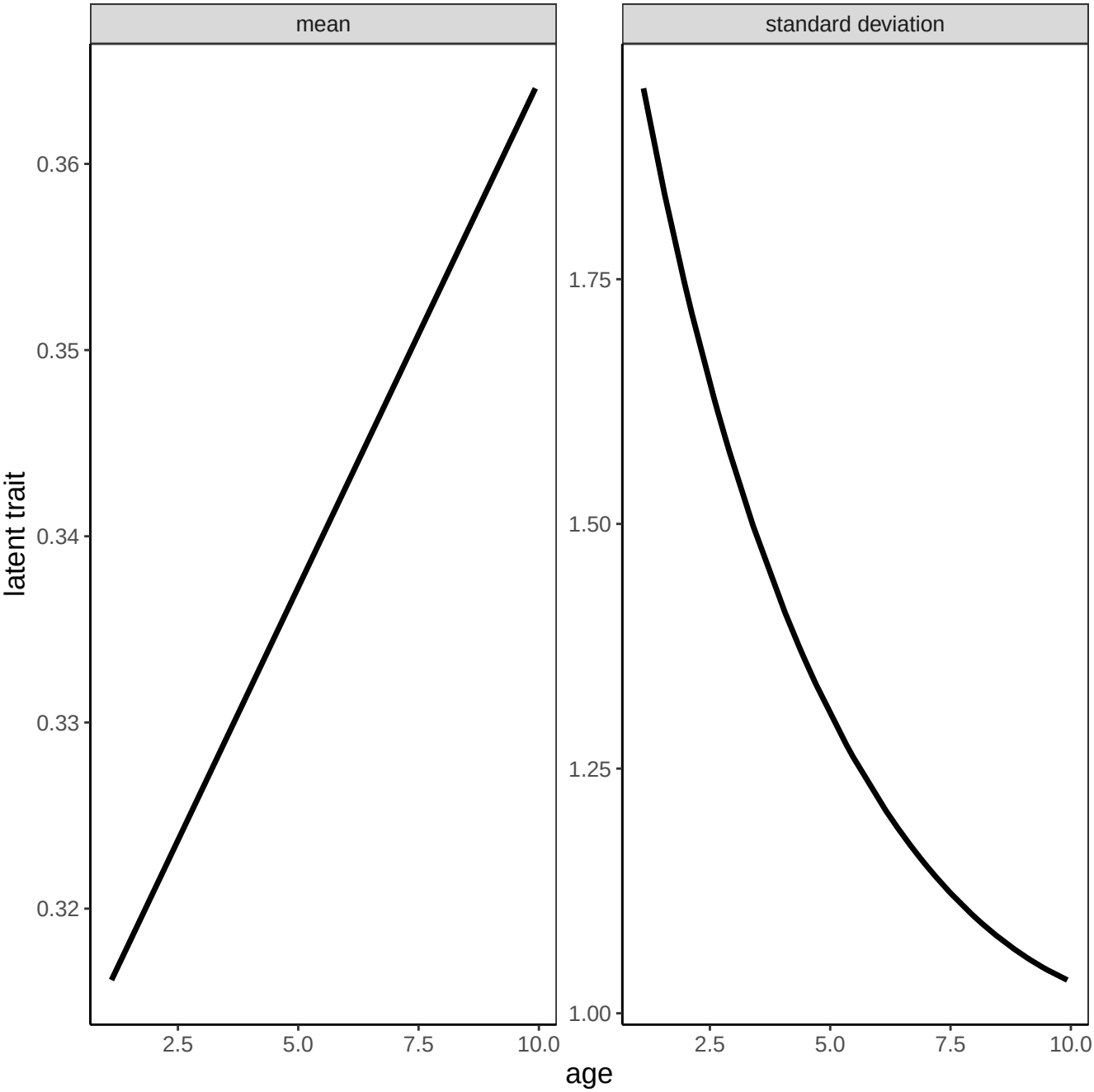

age: [0,5]
age: [5,10]
frequency of individuals
raw score
10.0
7.5
5.0
2.5
0.0
9
6
3
0
0
2
4
6

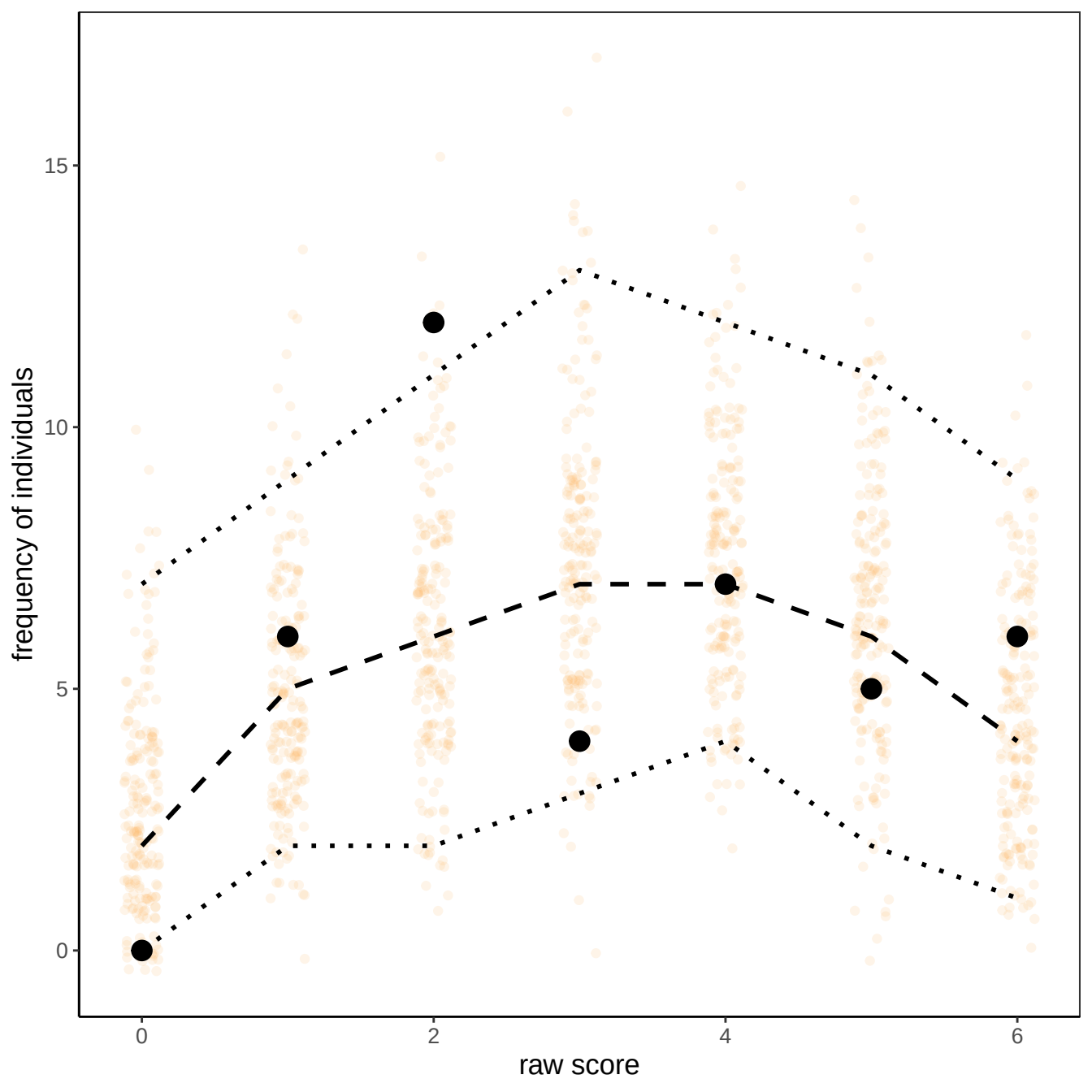

item fit: proportion correct per item
model−implied
observed
0.8
0.7
0.6
0.5
0.4
0.4
0.5
0.6
0.7
0.8

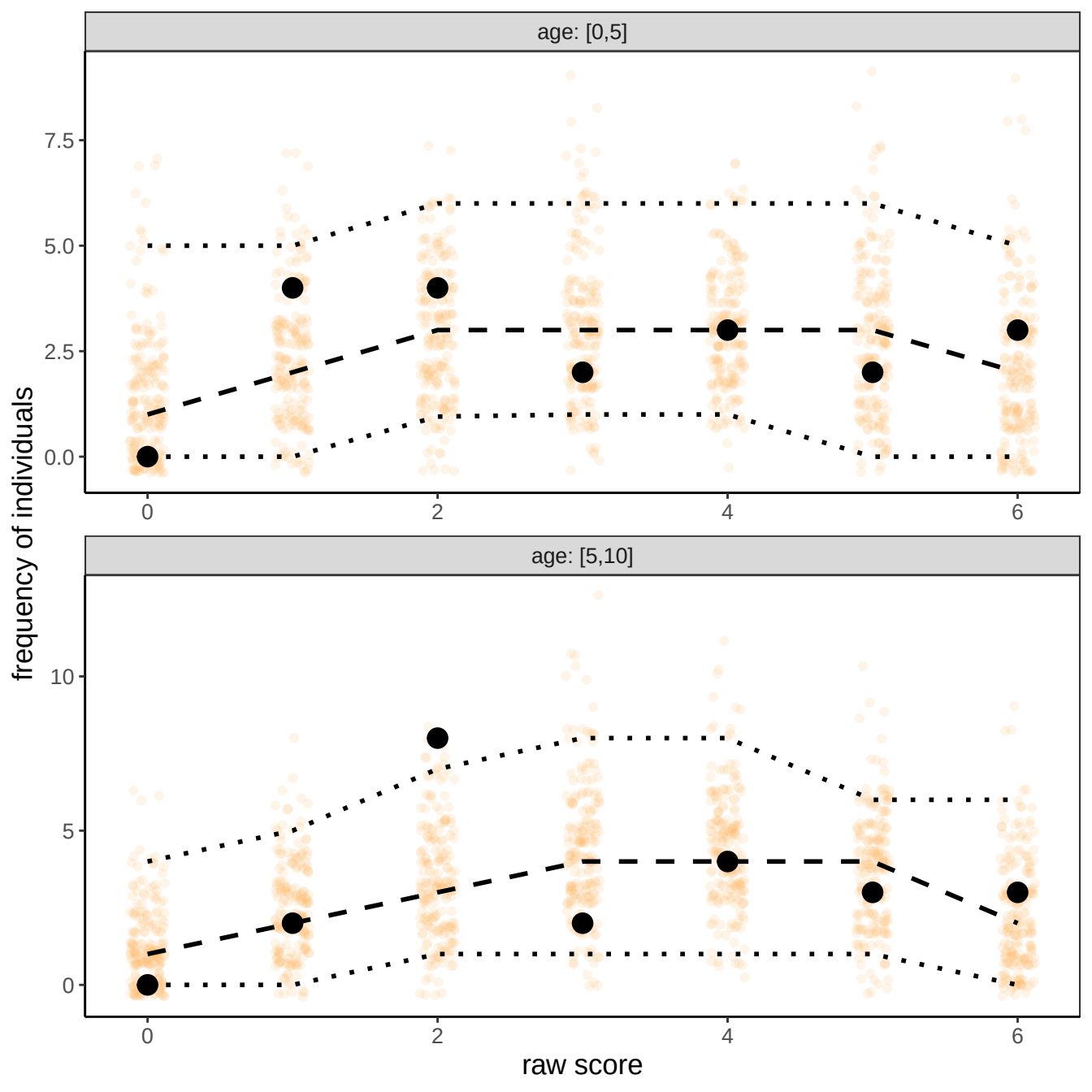

person fit: raw score
model−implied
observed
1
2
3
4
5
2
4
6

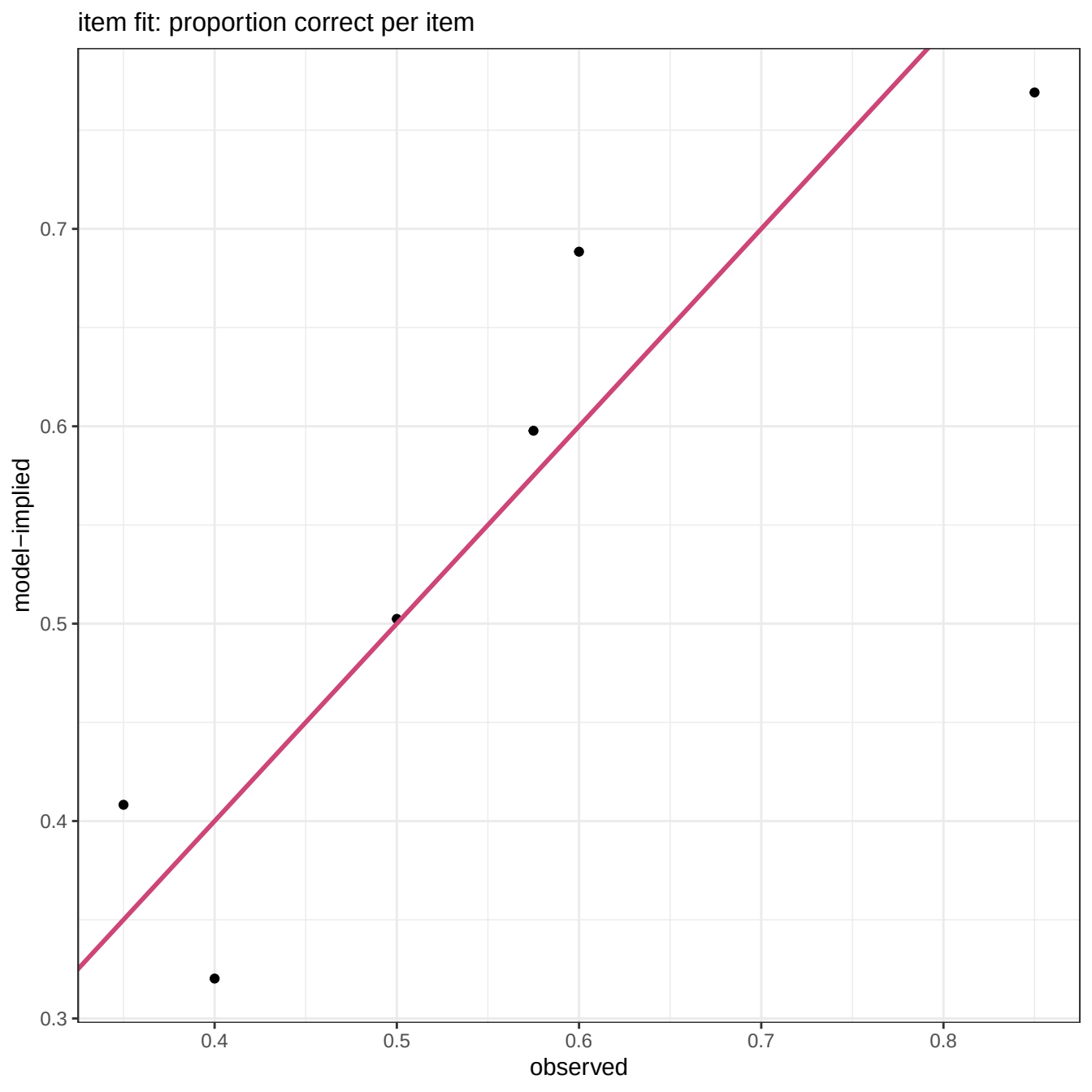

Corr
1.0
0.5
0.0
−0.5
−1.0

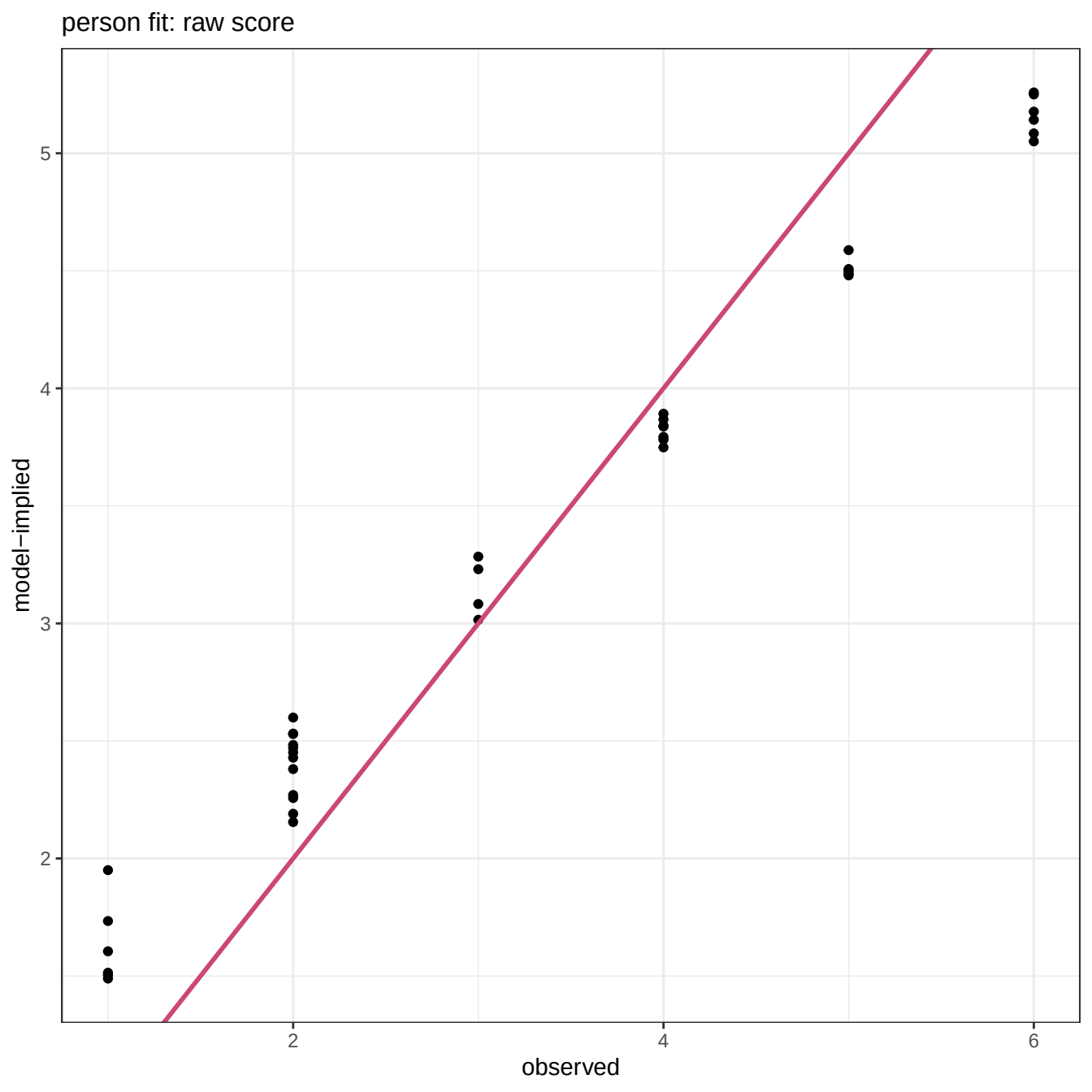

latent trait
2
0
−2
2.5
5.0
7.5
10.0
age
p0.05
p0.25
p0.75
p0.95
p0.1
p0.5
p0.9

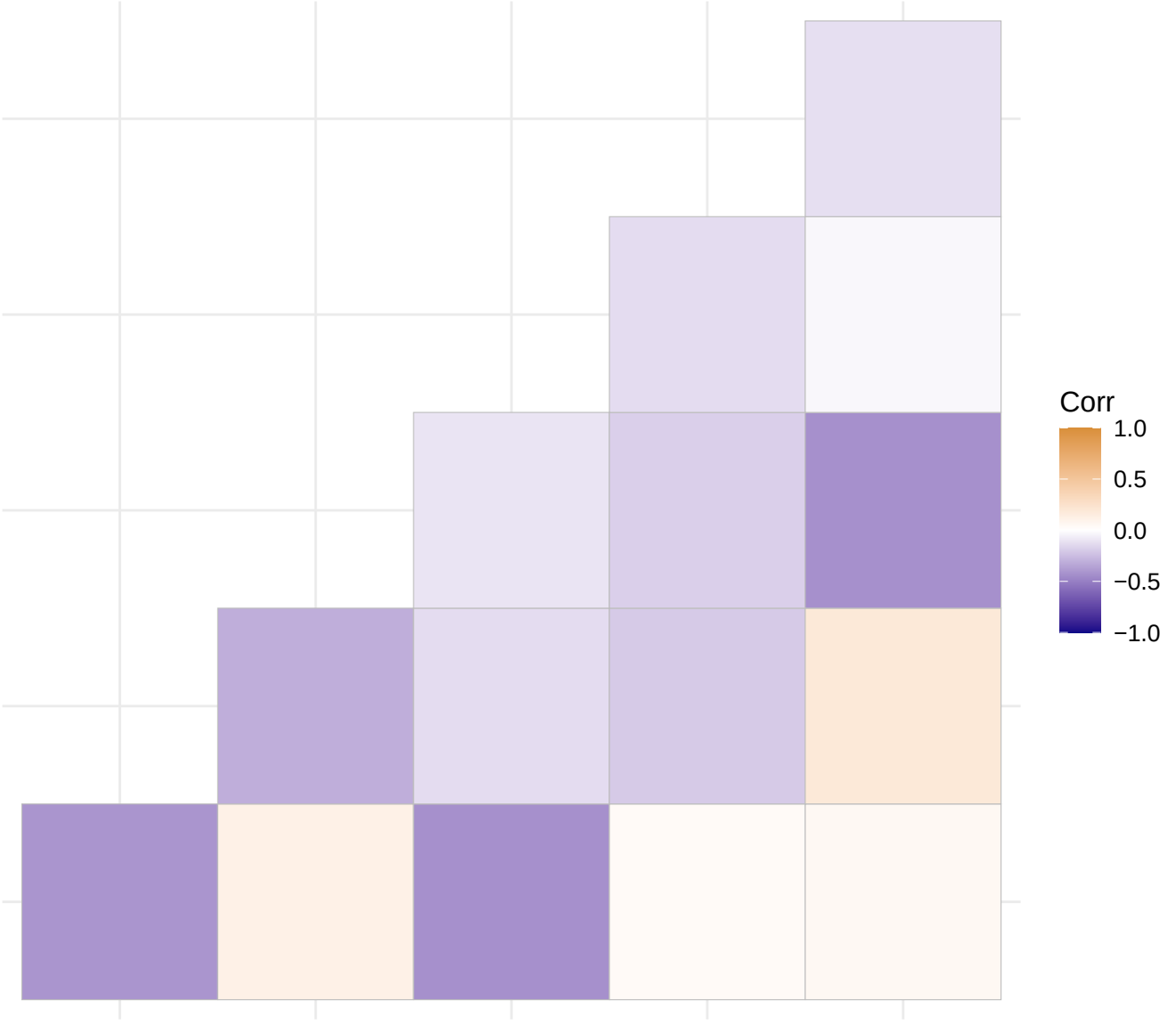

mean
standard deviation
latent trait
0.36
0.35
0.34
0.33
0.32
1.75
1.50
1.25
1.00
2.5
5.0
7.5
10.0
age

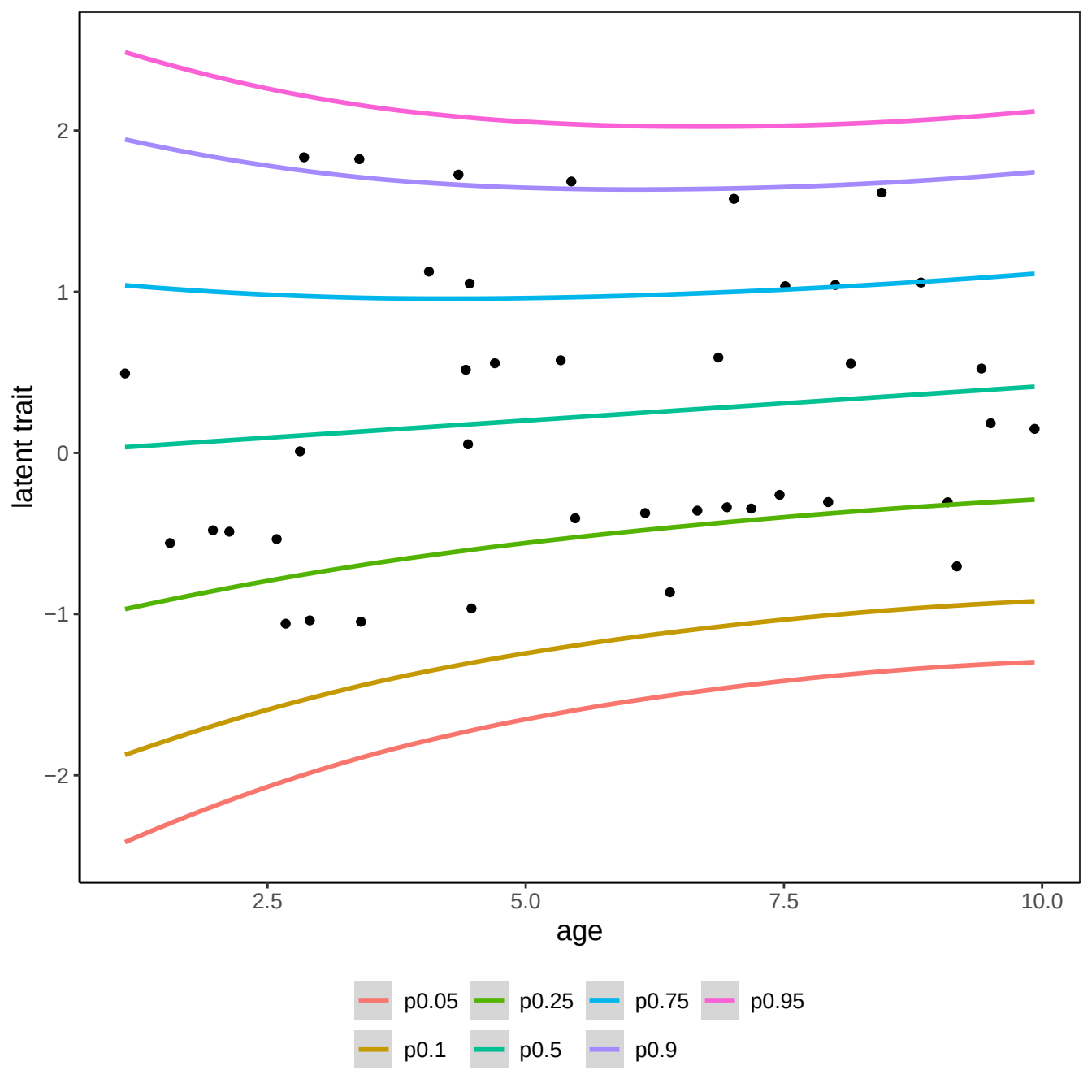

15
10
5
0
frequency of individuals
0
2
4
6
raw score

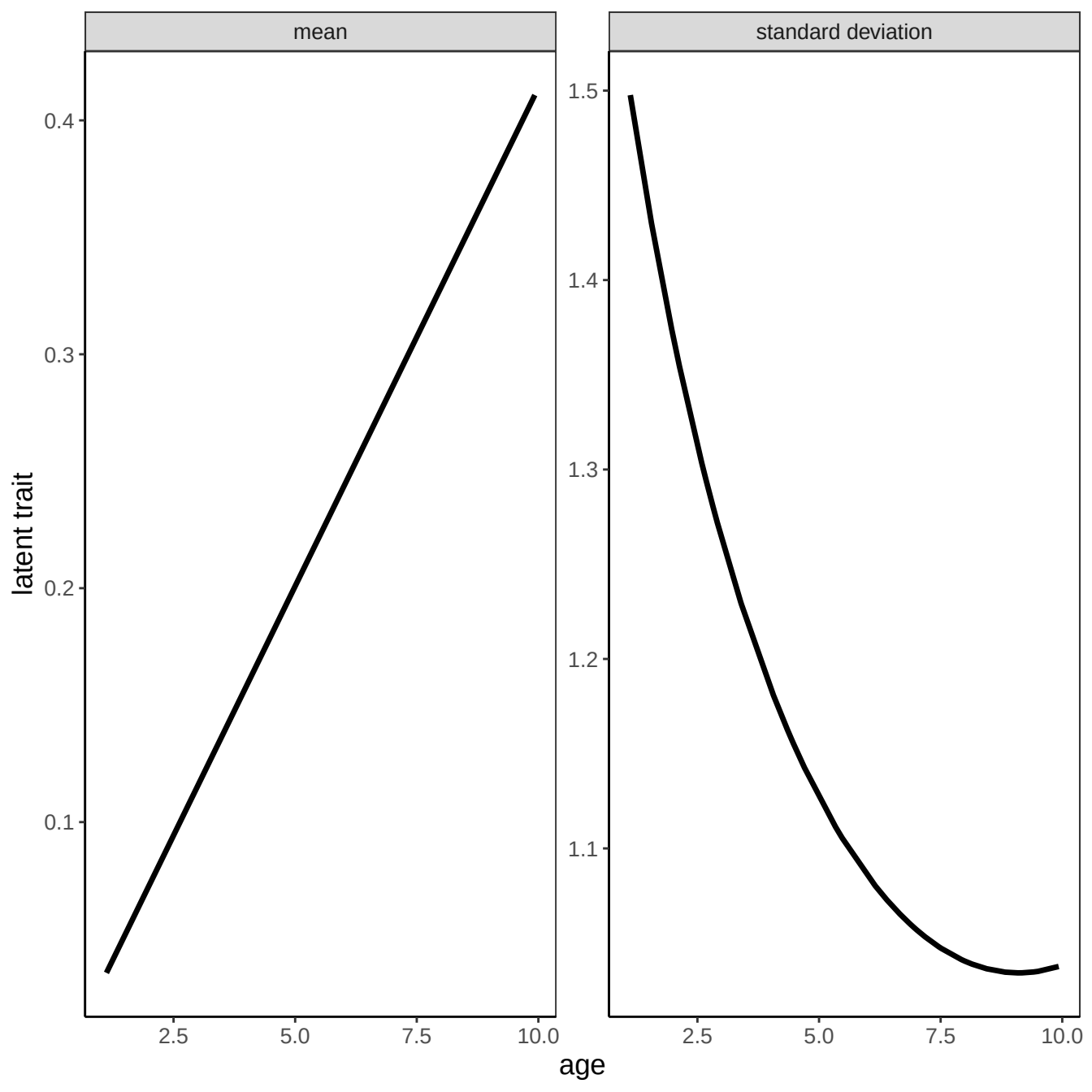

age: [0,5]
age: [5,10]
frequency of individuals
raw score
0.0
2.5
5.0
7.5
0
5
10
0
2
4
6

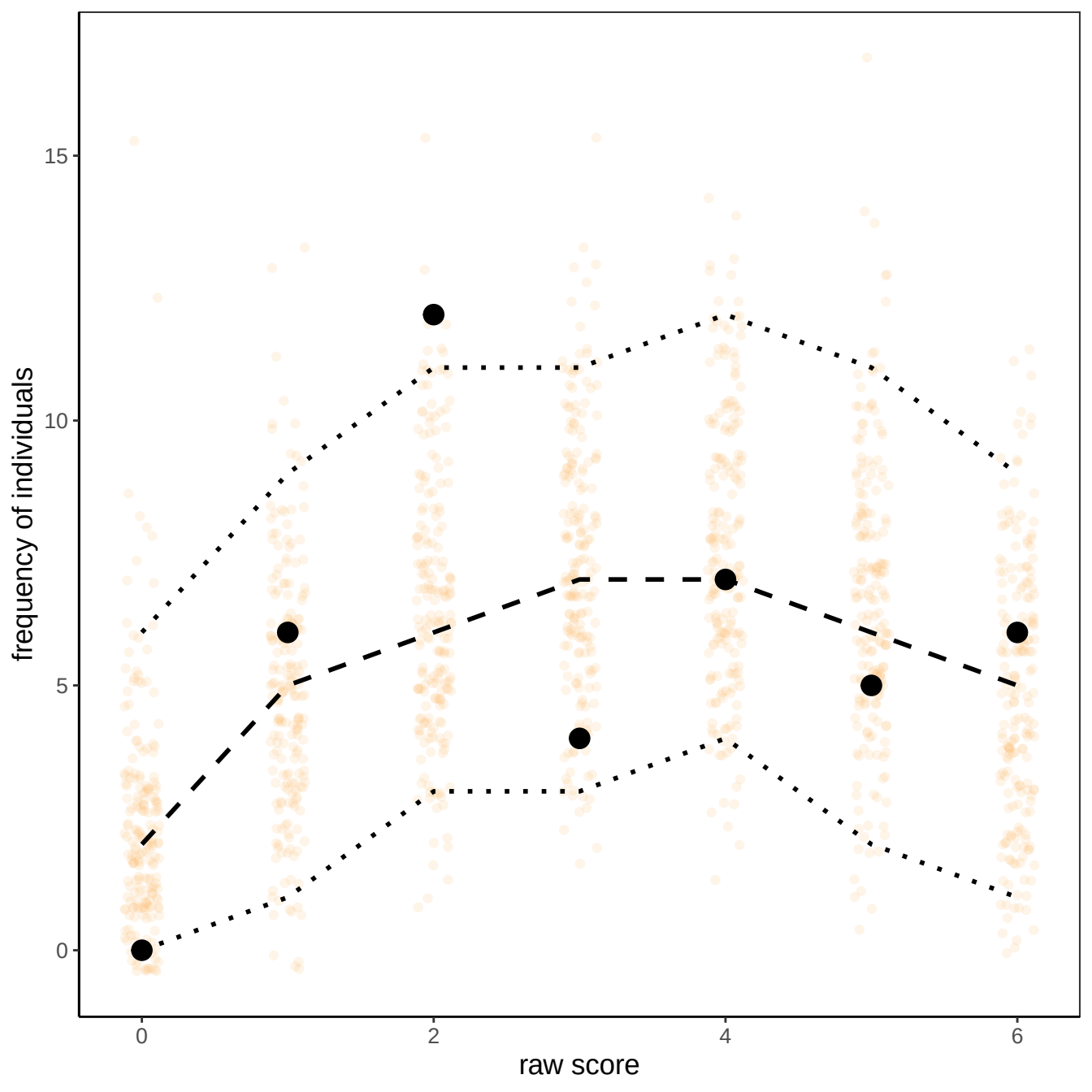

item fit: proportion correct per item
model−implied
observed
0.3
0.4
0.5
0.6
0.7
0.4
0.5
0.6
0.7
0.8

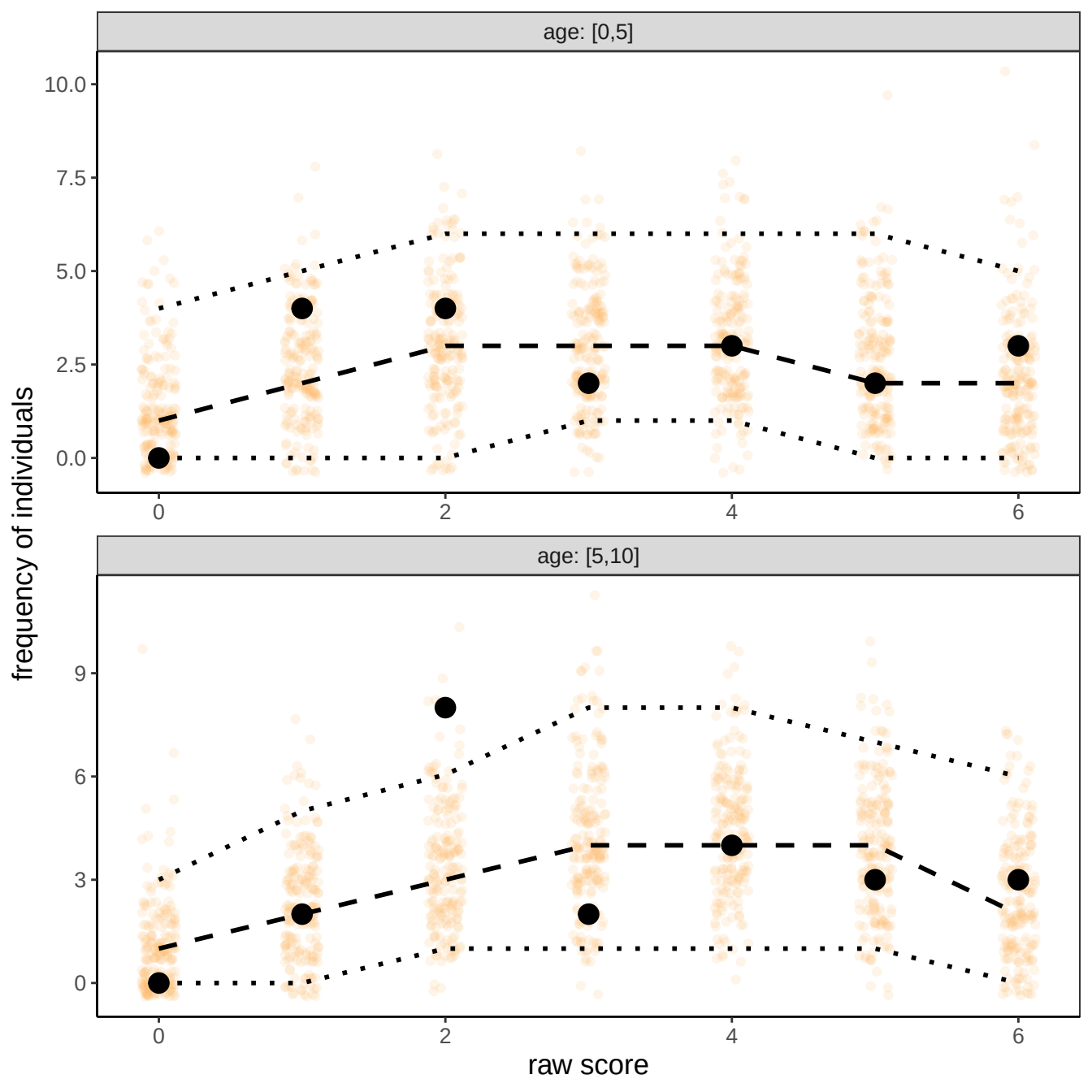

person fit: raw score
model−implied
observed
2
3
4
5
2
4
6

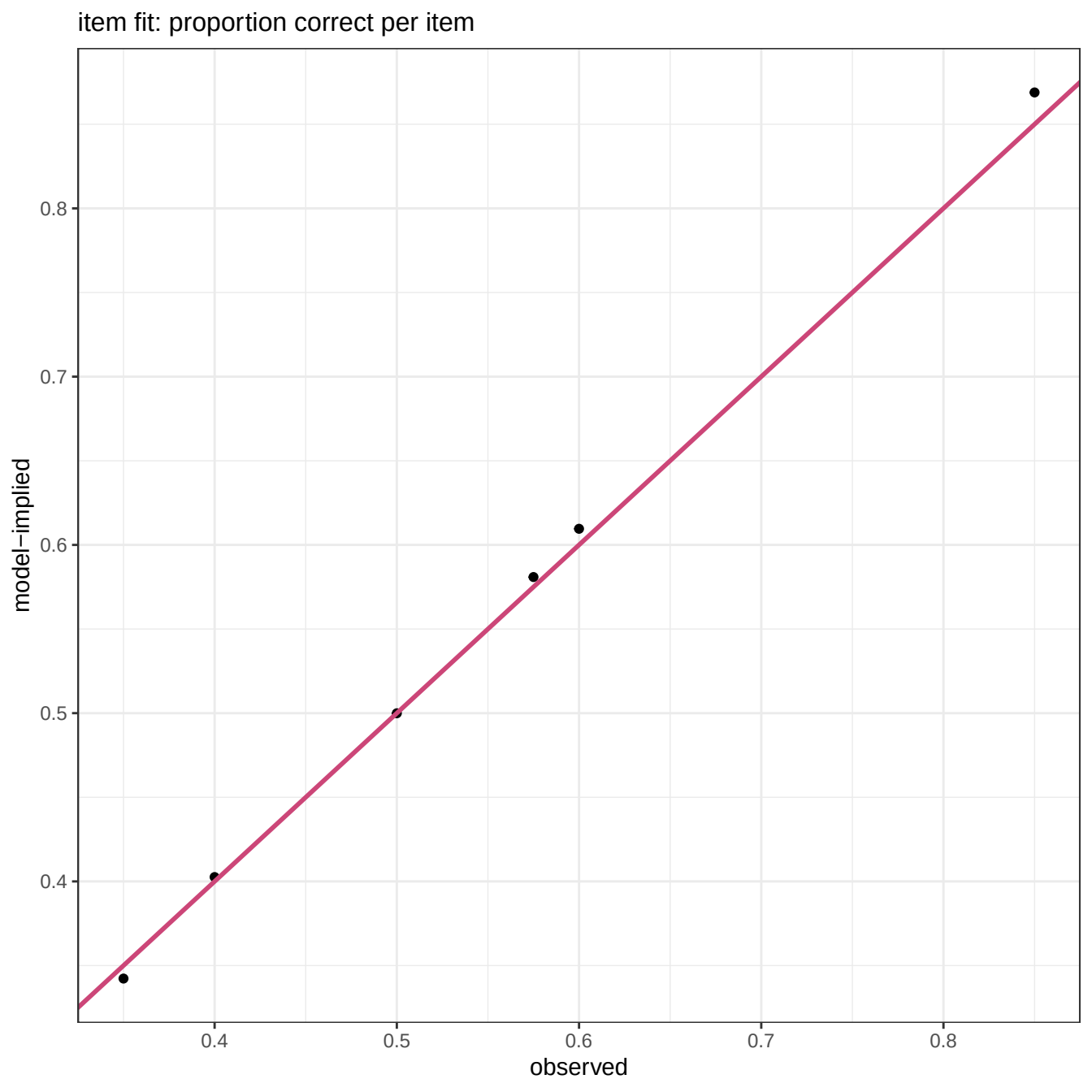

Corr
1.0
0.5
0.0
−0.5
−1.0

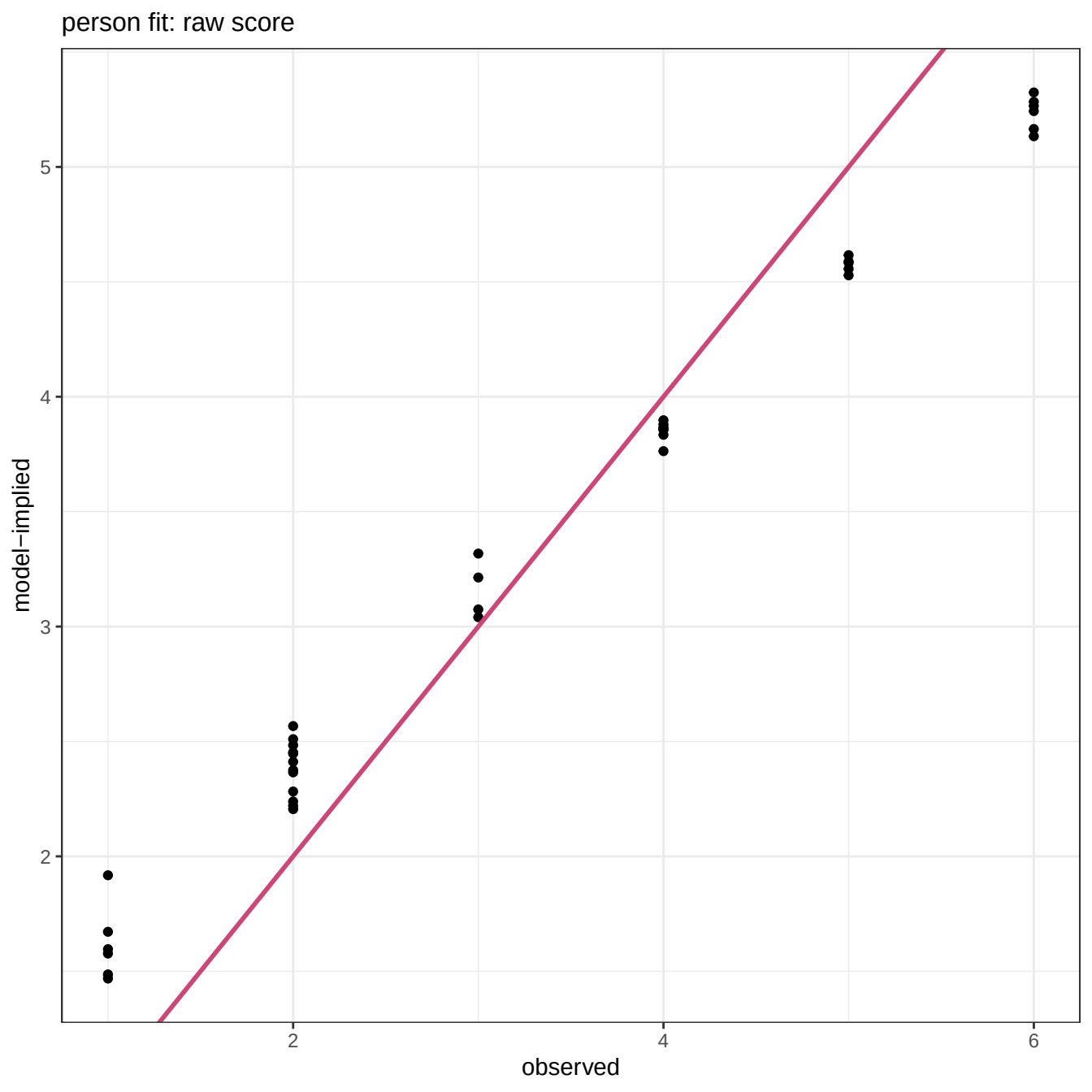

latent trait
2
1
0
−1
−2
2.5
5.0
7.5
10.0
age
p0.05
p0.25
p0.75
p0.95
p0.1
p0.5
p0.9

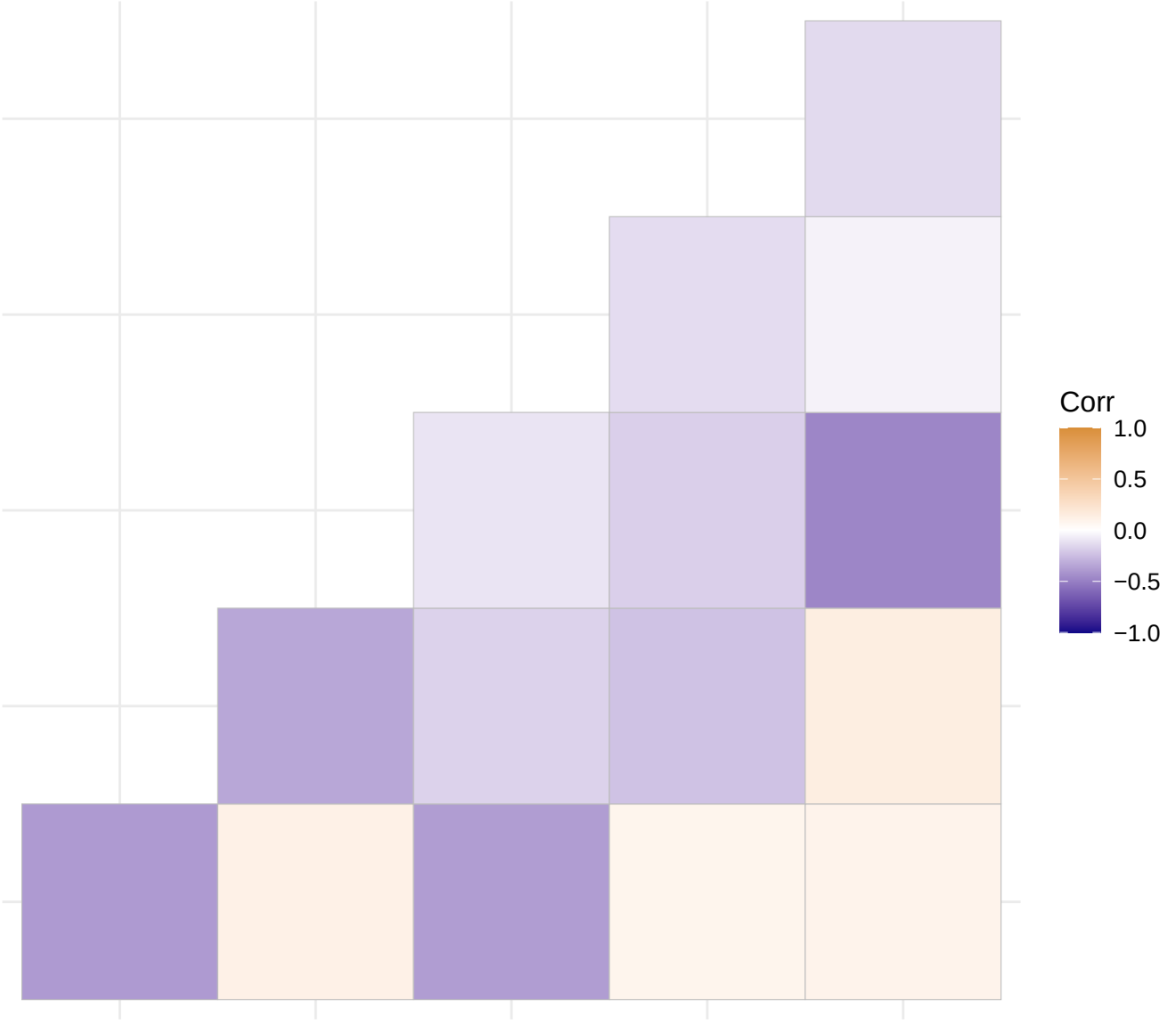

mean
standard deviation
latent trait
0.4
0.3
0.2
0.1
1.5
1.4
1.3
1.2
1.1
2.5
5.0
7.5
10.0
age

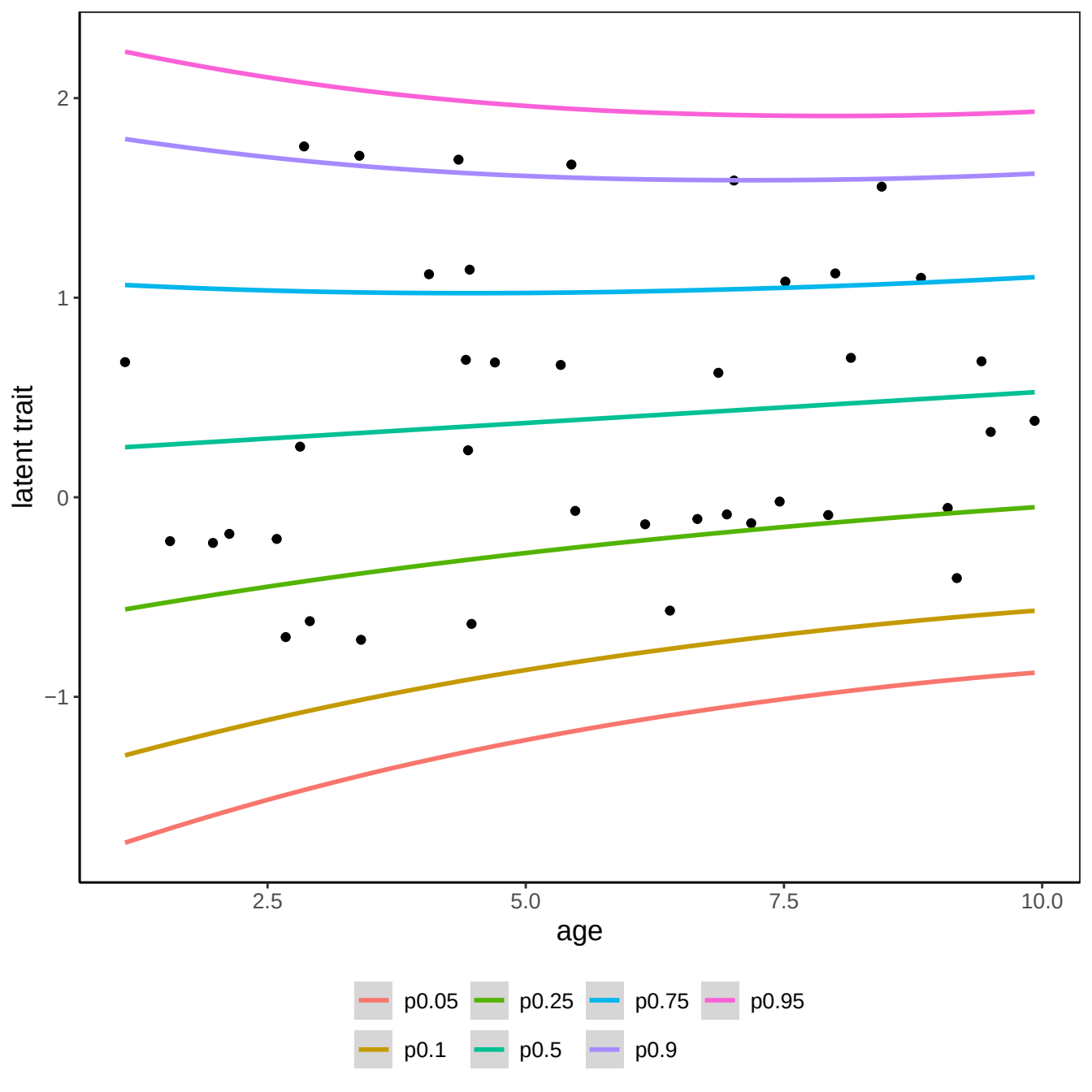

15
10
5
0
frequency of individuals
0
2
4
6
raw score

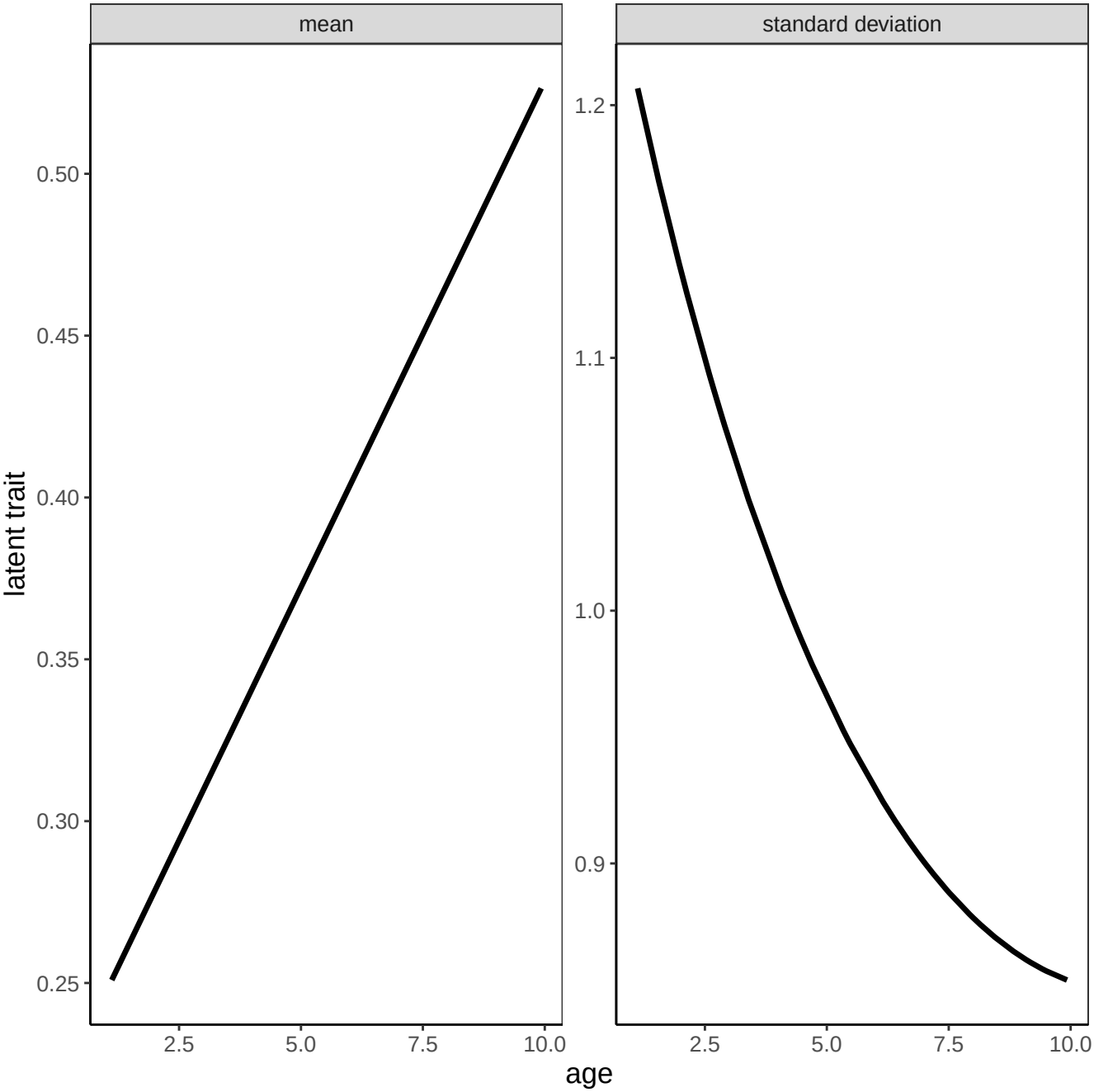

age: [0,5]
age: [5,10]
frequency of individuals
raw score
10.0
7.5
5.0
2.5
0.0
9
6
3
0
0
2
4
6

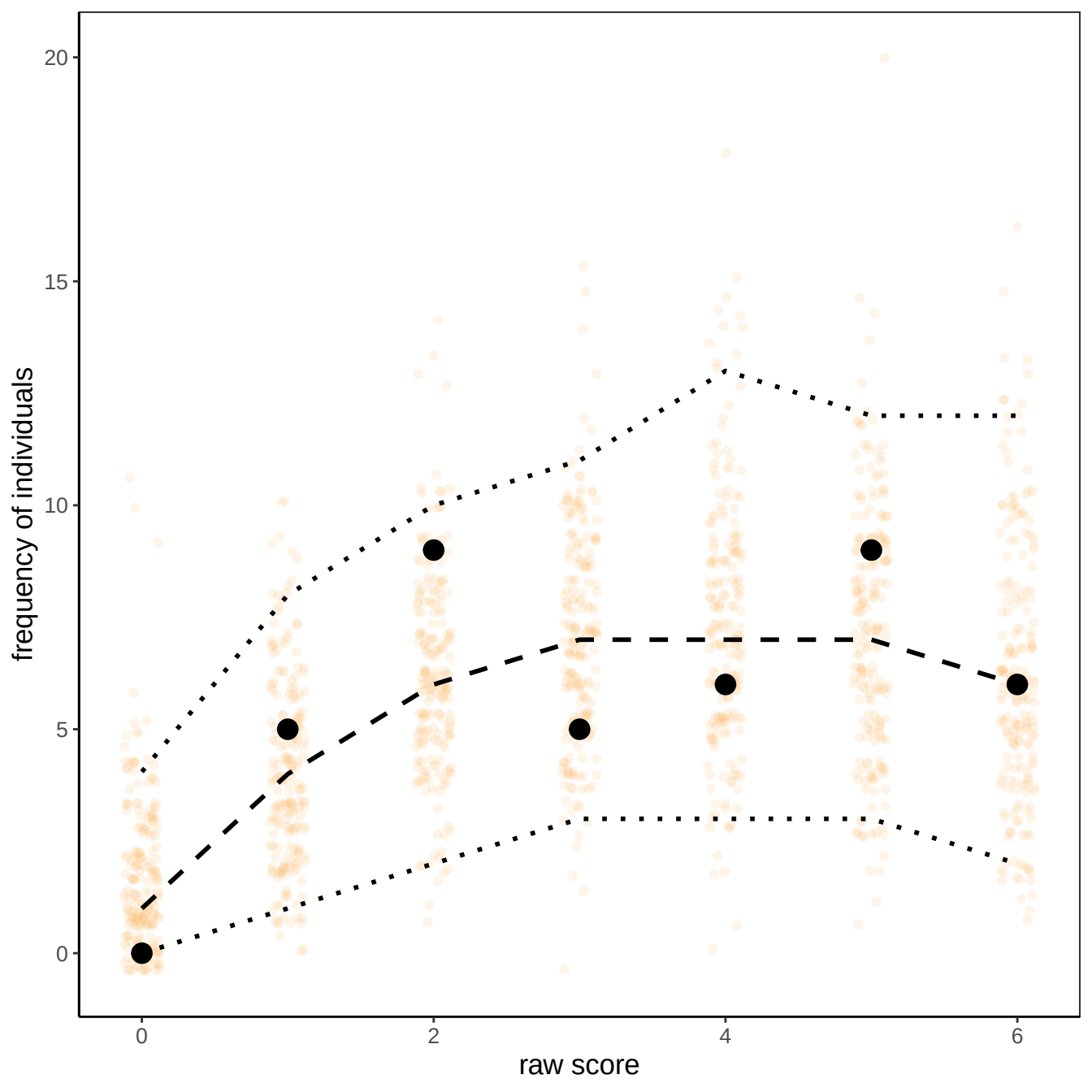

item fit: proportion correct per item
model−implied
observed
0.8
0.7
0.6
0.5
0.4
0.4
0.5
0.6
0.7
0.8

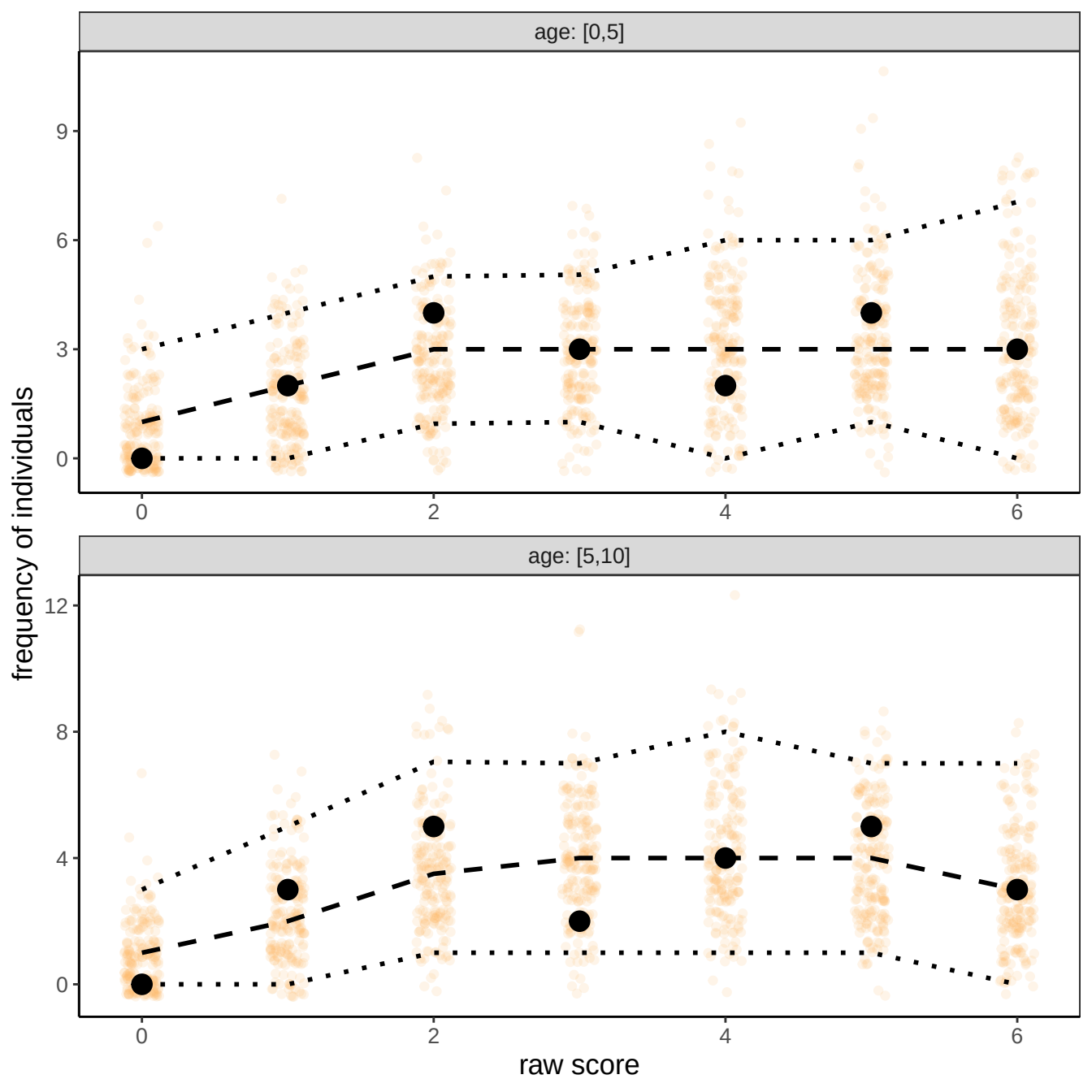

person fit: raw score
model−implied
observed
2
3
4
5
2
4
6

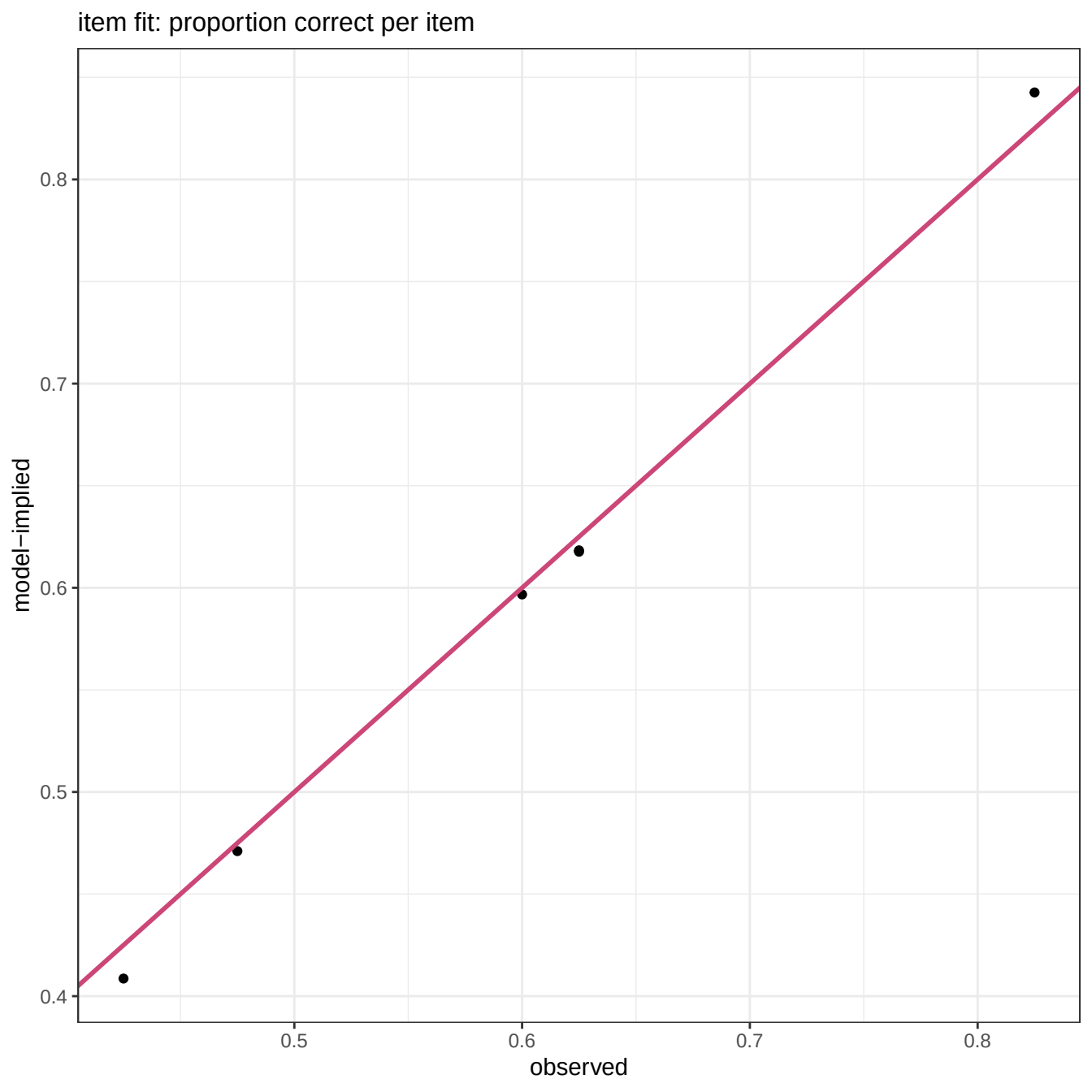

Corr
1.0
0.5
0.0
−0.5
−1.0

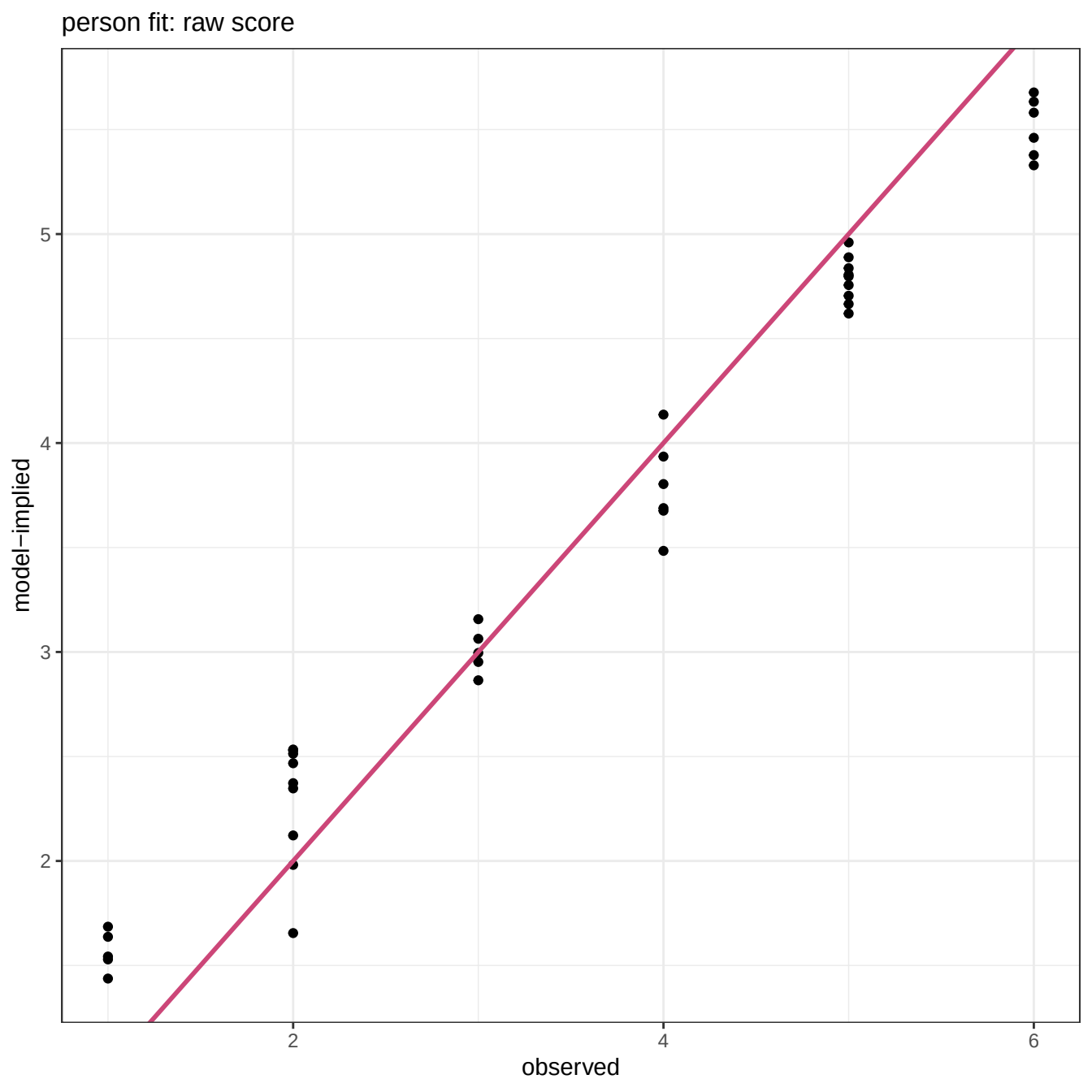

latent trait
2
1
0
−1
2.5
5.0
7.5
10.0
age
p0.05
p0.25
p0.75
p0.95
p0.1
p0.5
p0.9

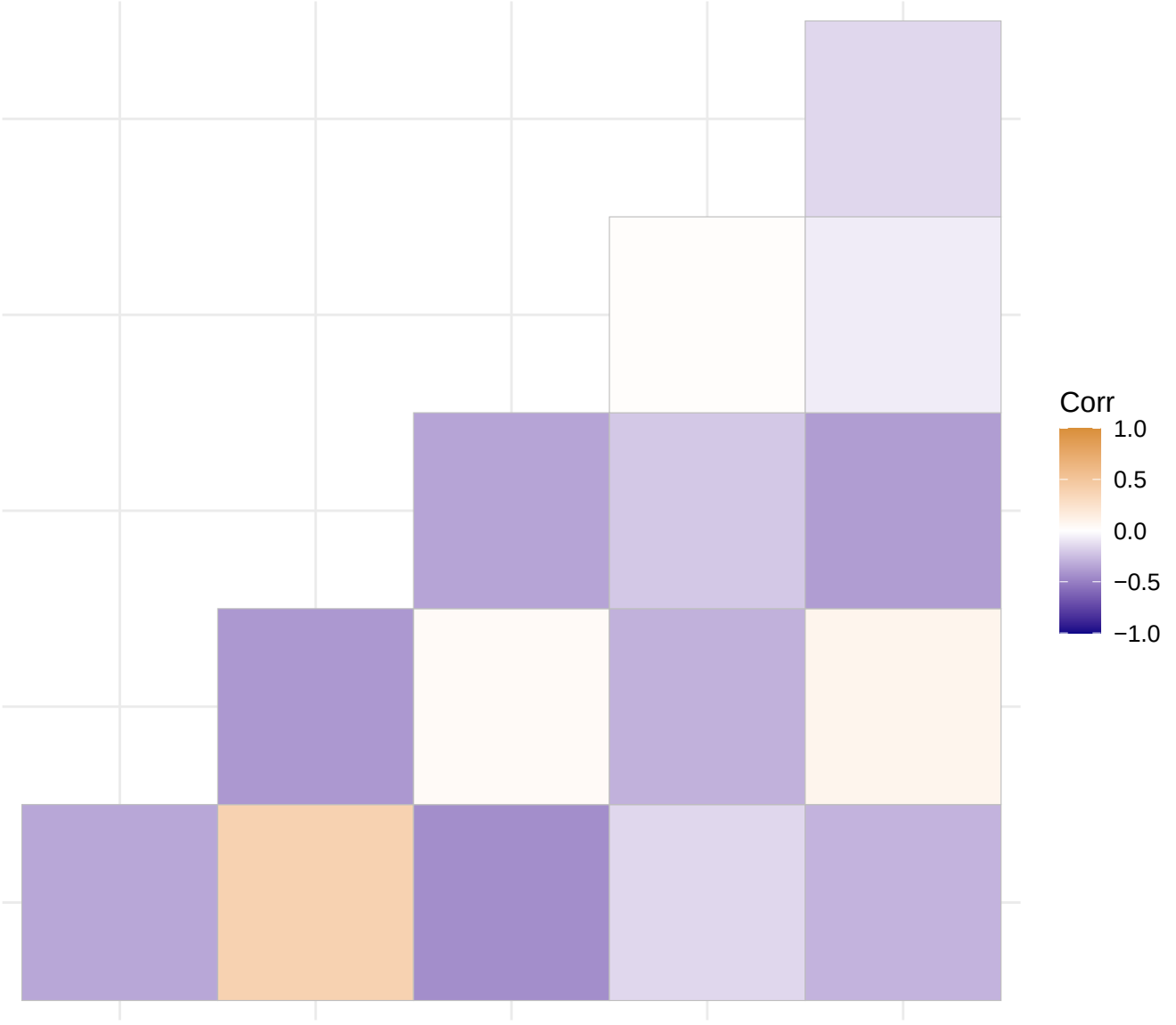

mean
standard deviation
latent trait
0.50
0.45
0.40
0.35
0.30
0.25
1.2
1.1
1.0
0.9
2.5
5.0
7.5
10.0
age

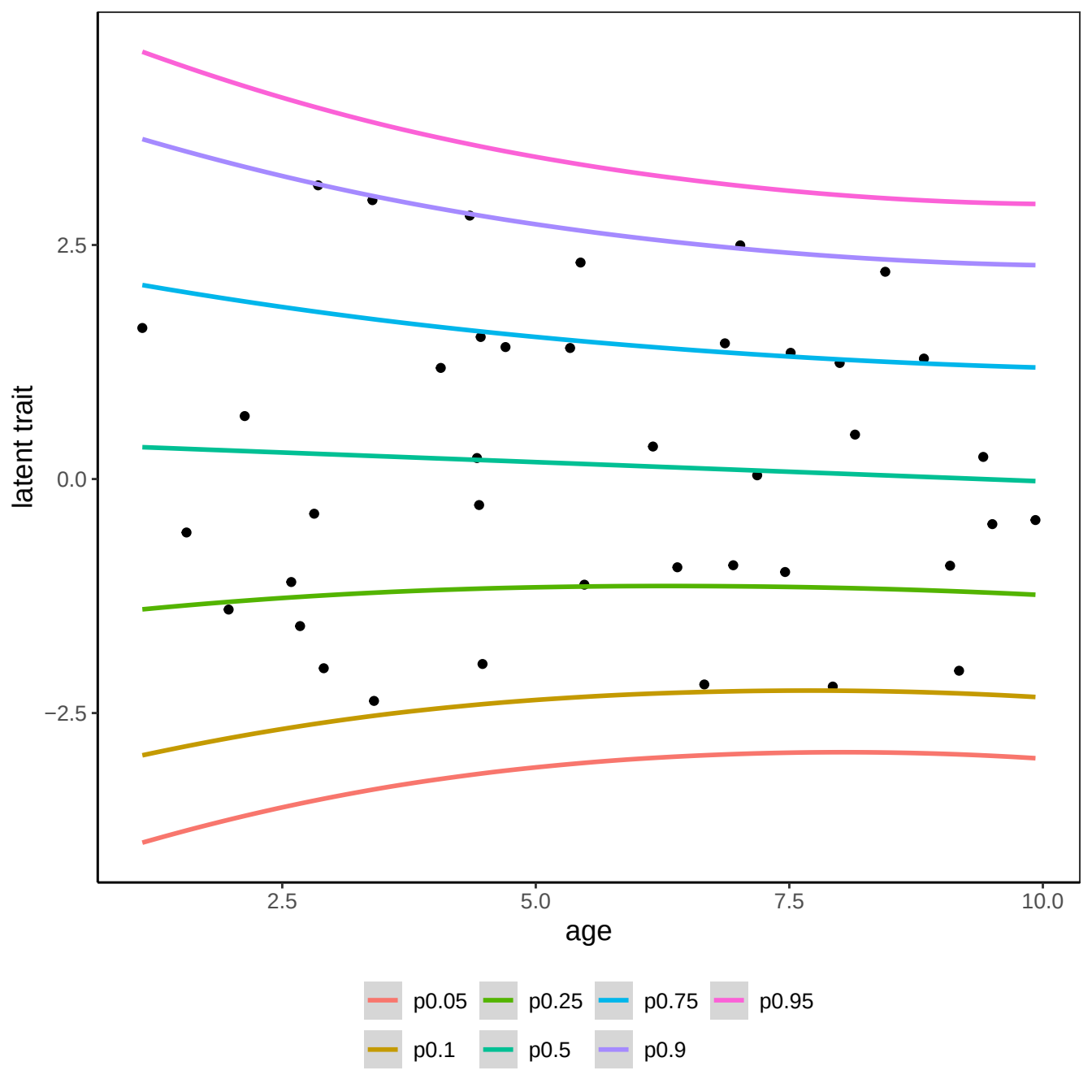

20
15
10
5
0
frequency of individuals
0
2
4
6
raw score

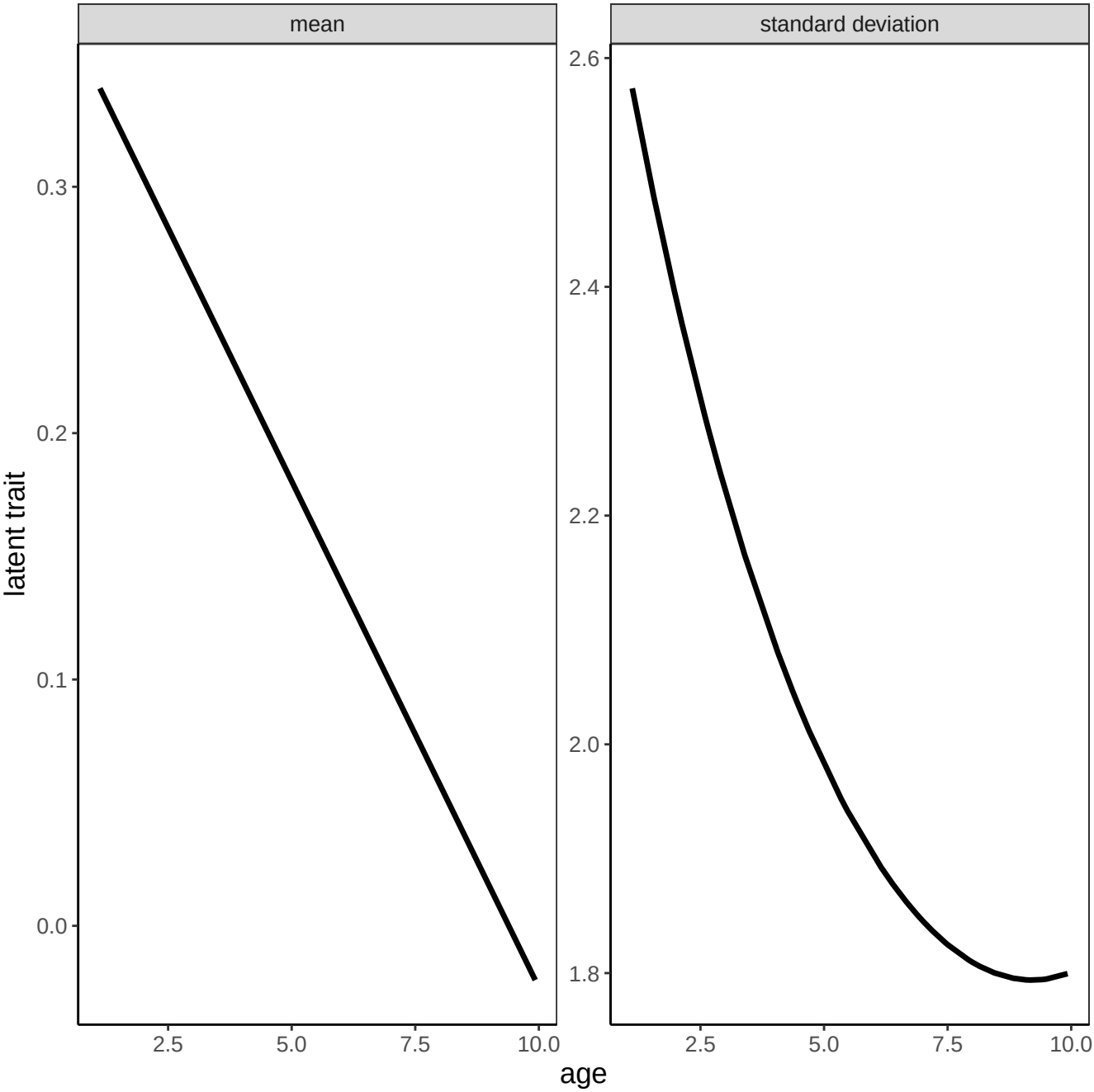

age: [0,5]
age: [5,10]
frequency of individuals
raw score
0
2
4
6
0
3
6
9
0
4
8
12

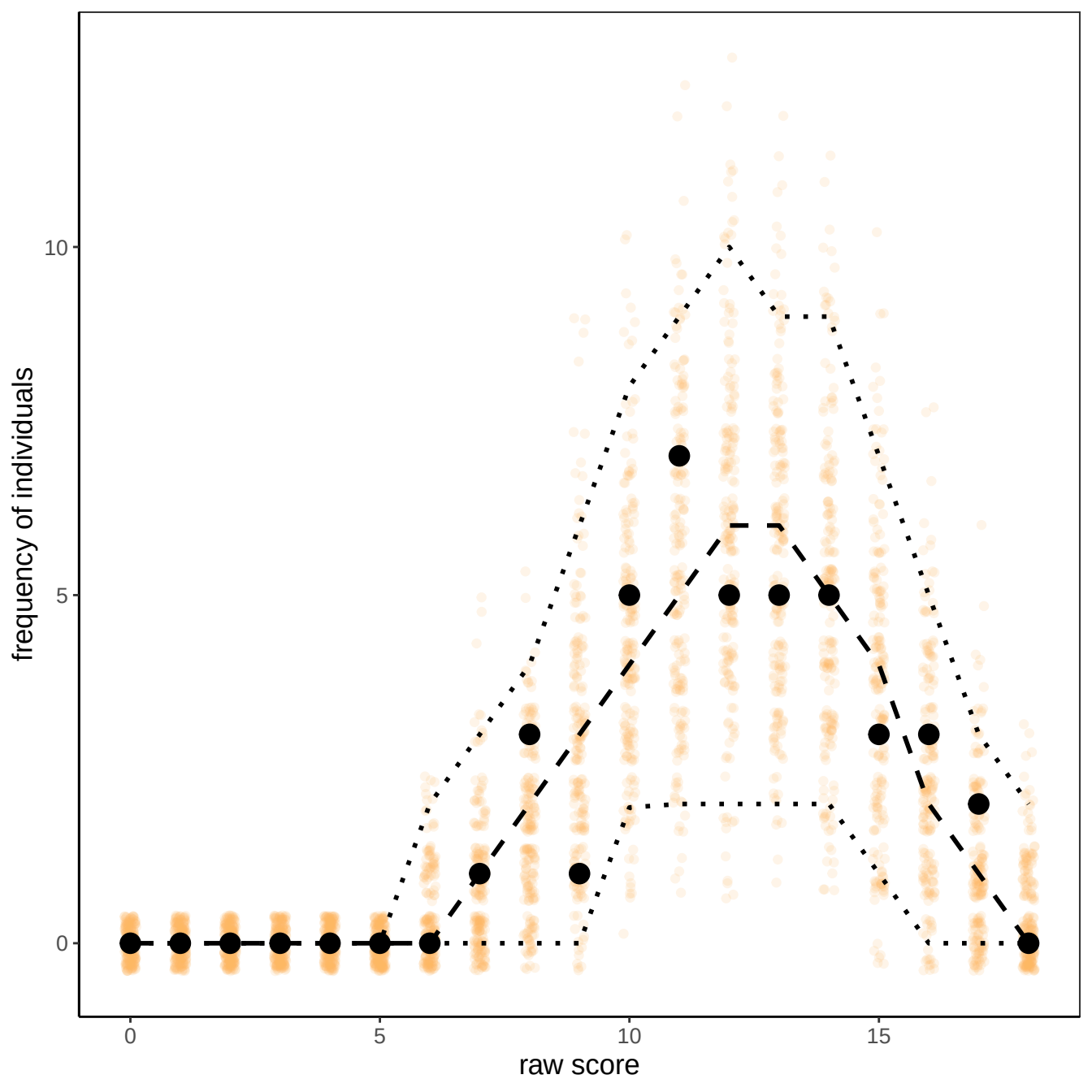

item fit: proportion correct per item
model−implied
observed
0.4
0.5
0.6
0.7
0.8
0.5
0.6
0.7
0.8

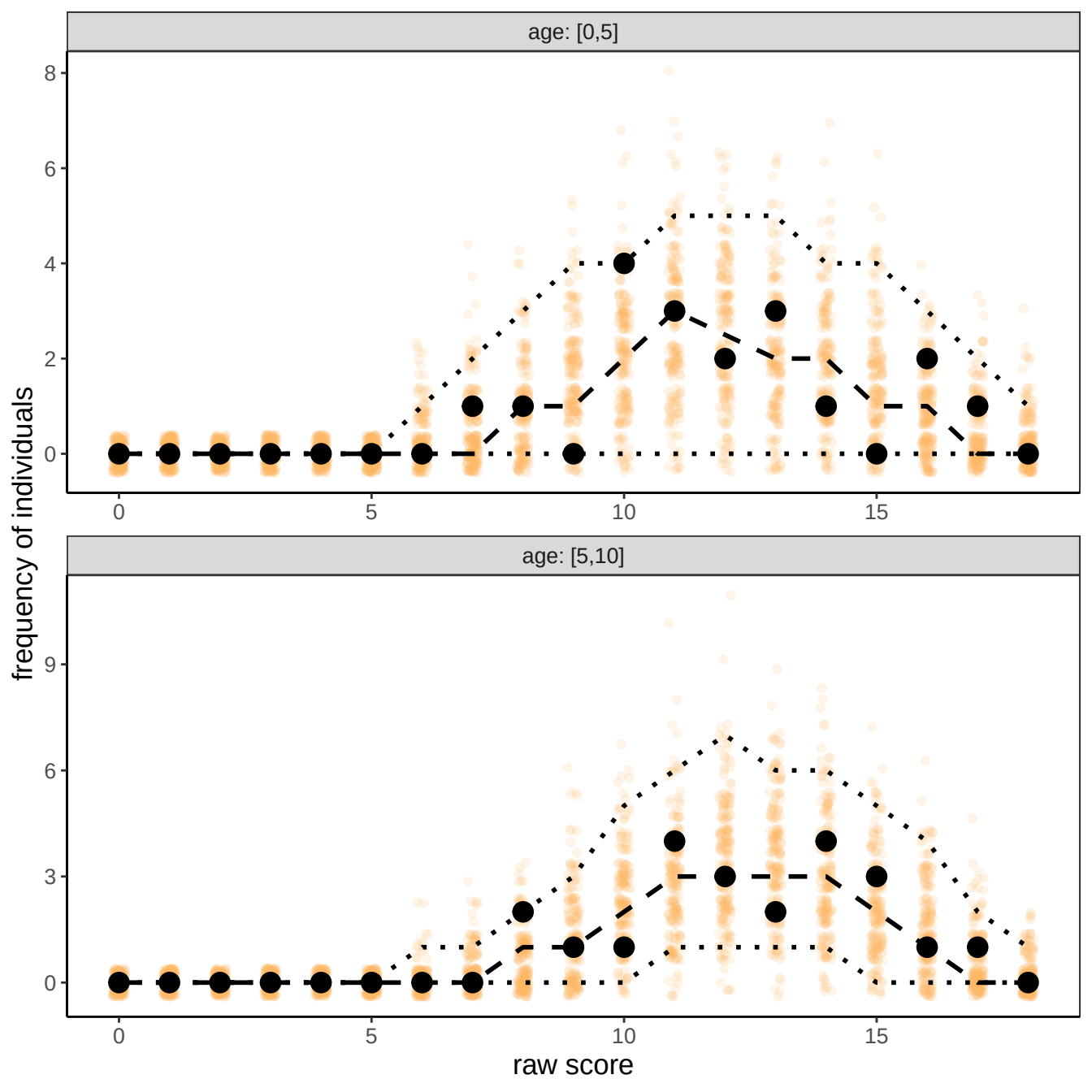

person fit: raw score
model−implied
observed
2
3
4
5
2
4
6

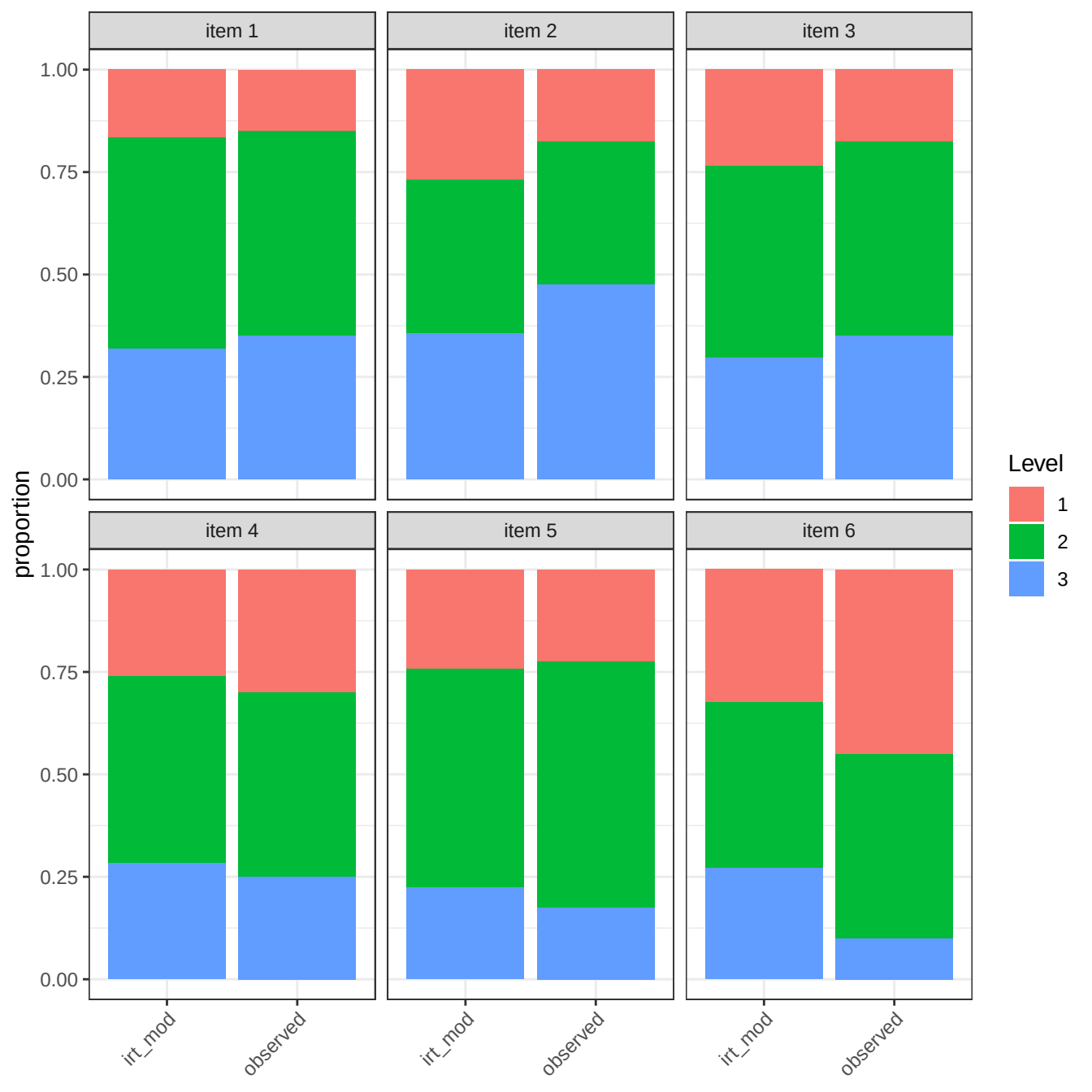

Corr
1.0
0.5
0.0
−0.5
−1.0

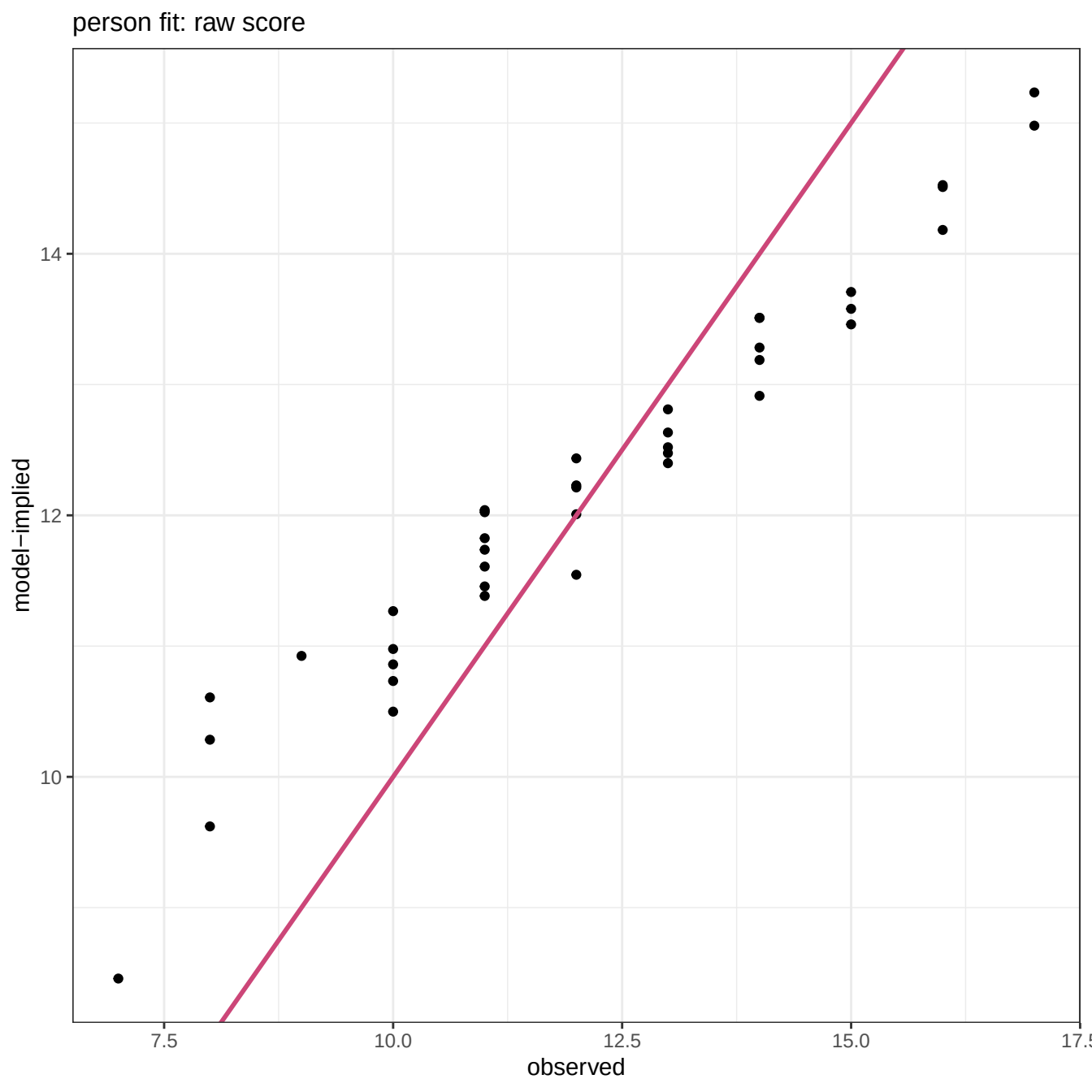

latent trait
2.5
0.0
−2.5
2.5
5.0
7.5
10.0
age
p0.05
p0.25
p0.75
p0.95
p0.1
p0.5
p0.9

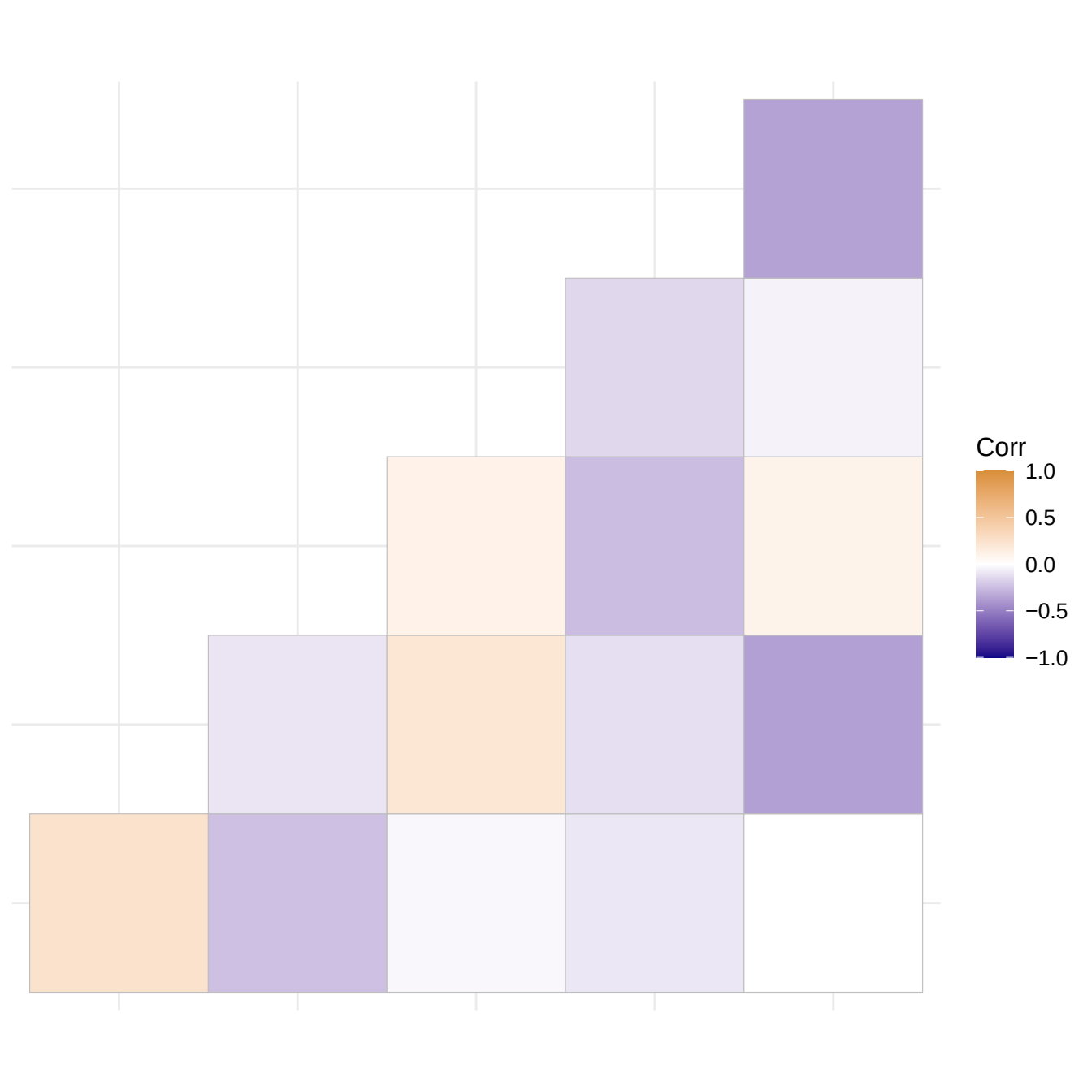

mean
standard deviation
latent trait
age
0.0
0.1
0.2
0.3
1.8
2.0
2.2
2.4
2.6
2.5
5.0
7.5
10.0

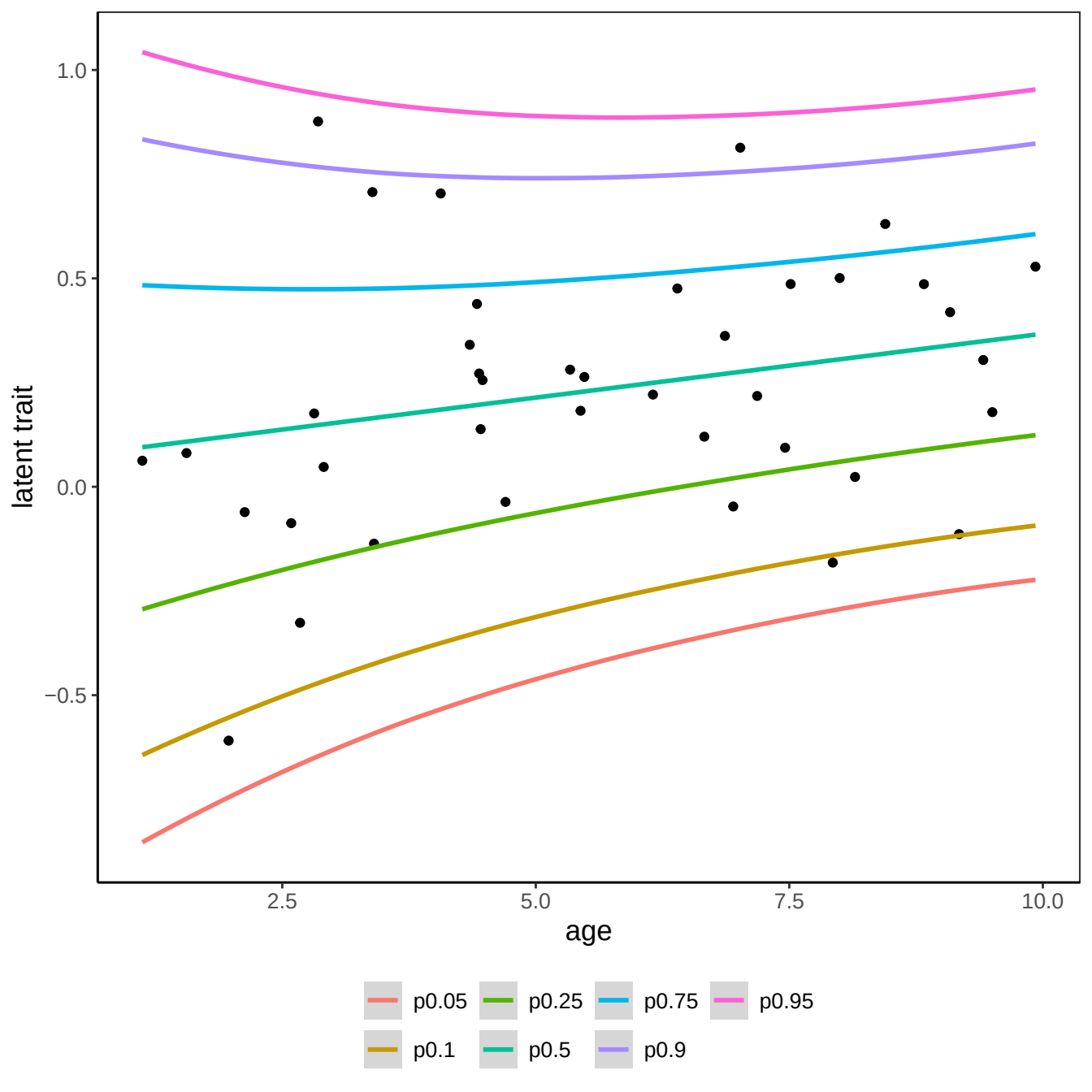

frequency of individuals
10
5
0
0
5
10
15
raw score

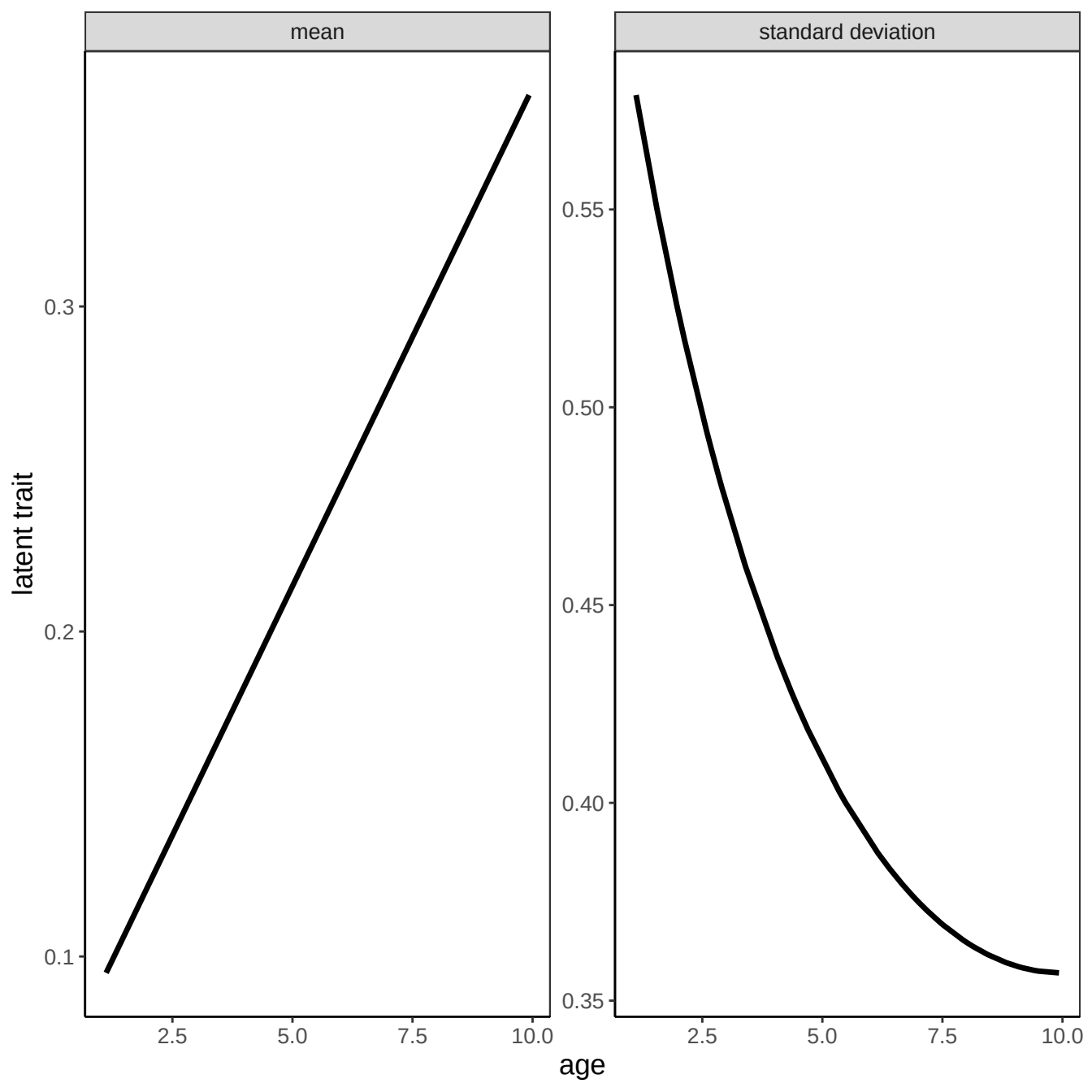

age: [0,5]
age: [5,10]
frequency of individuals
raw score
0
2
4
6
8
3
9
5
10
15

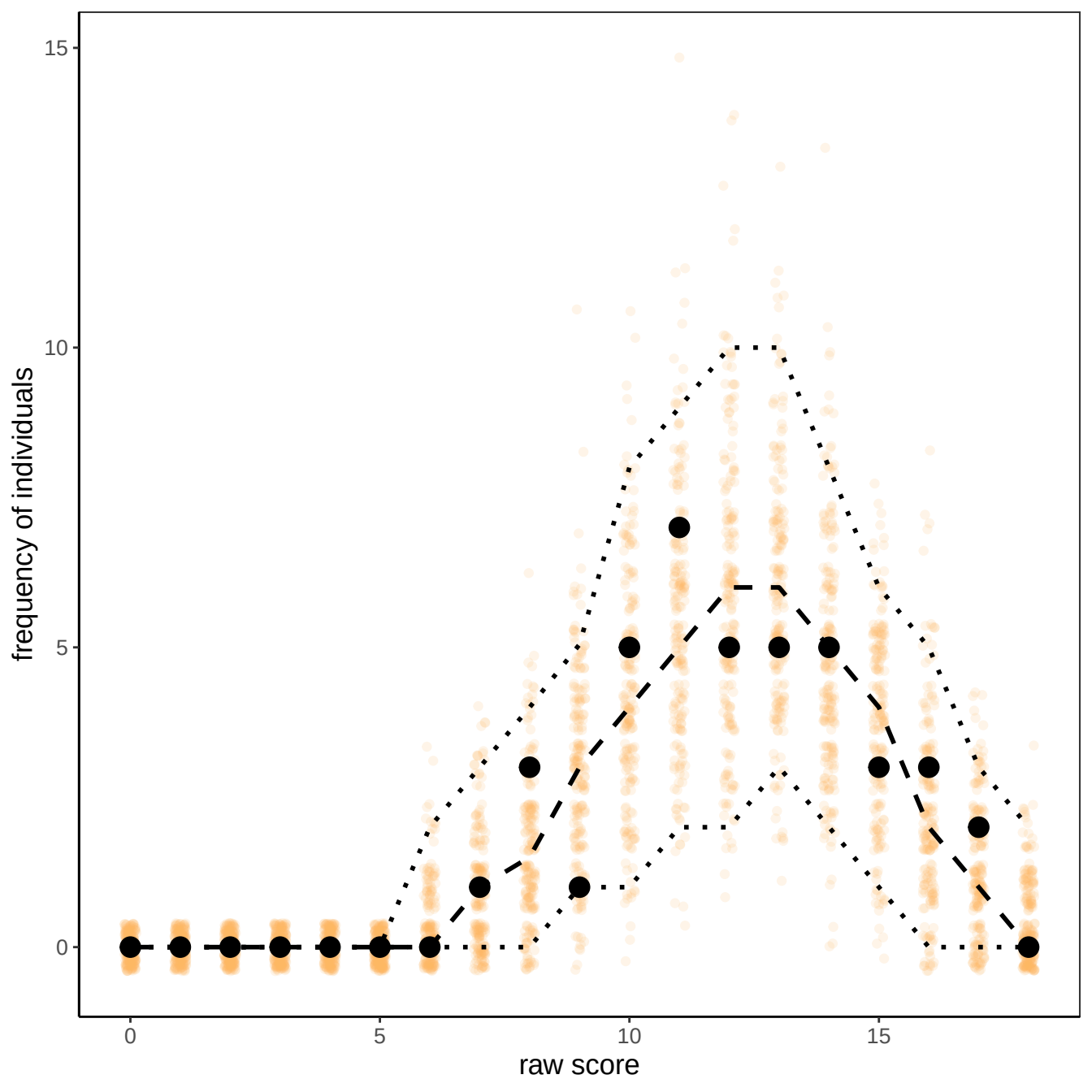

item 1
item 2
item 3
item 4
item 5
item 6
proportion
1.00
0.75
0.50
0.25
0.00
irt_mod
observed
Level
1
2
3

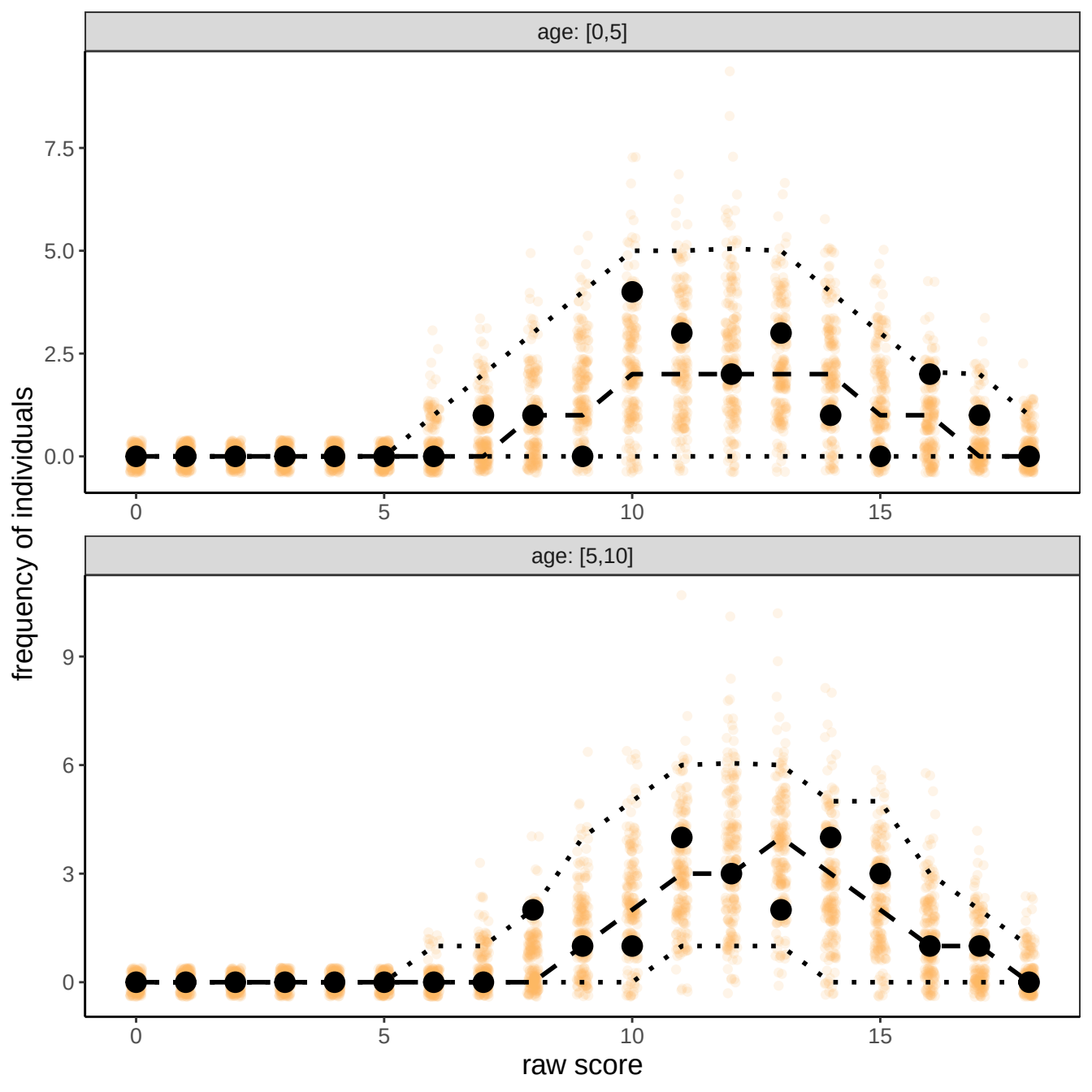

person fit: raw score
model−implied
observed
14
12
10
7.5
10.0
12.5
15.0
17.5

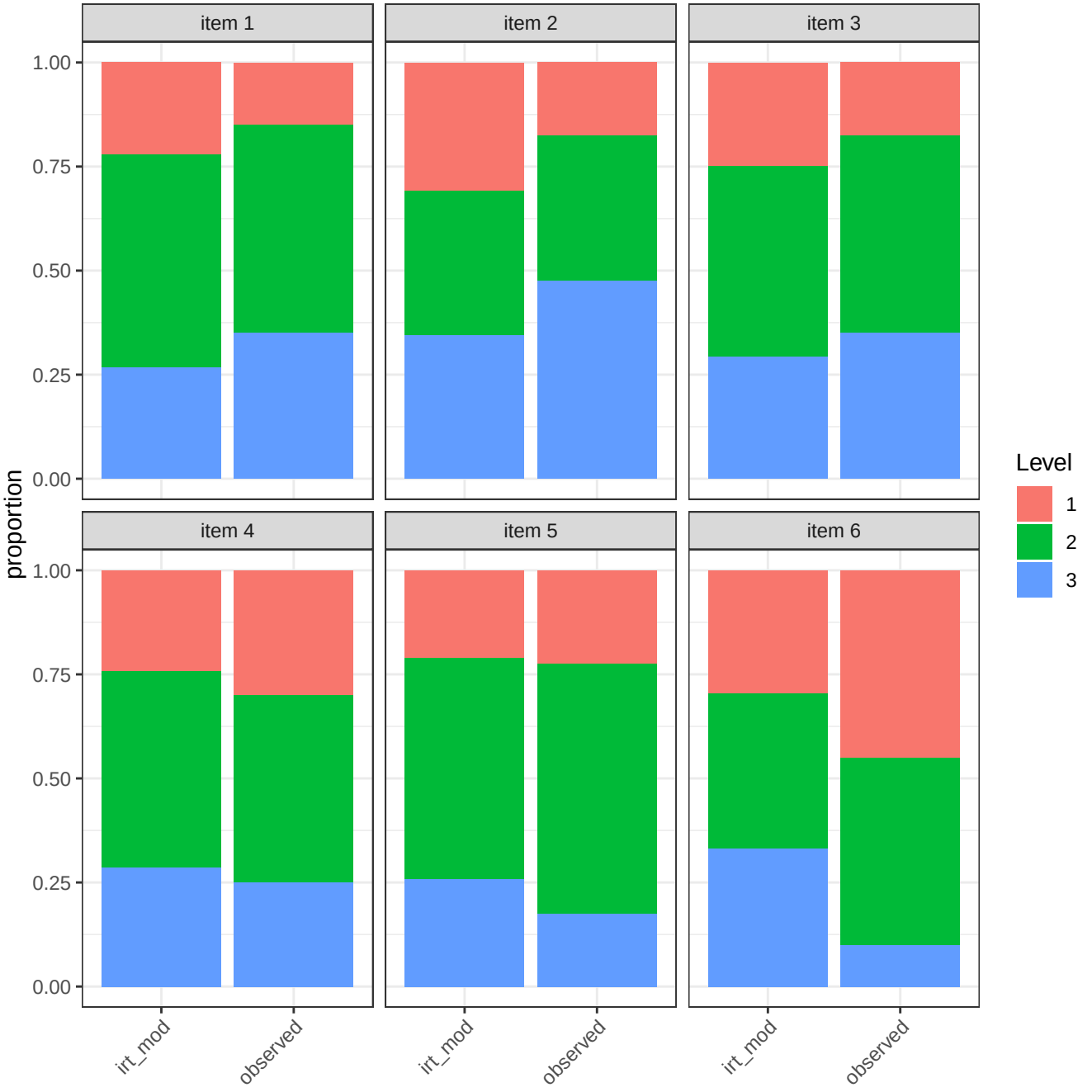

Corr
1.0
0.5
0.0
−0.5
−1.0

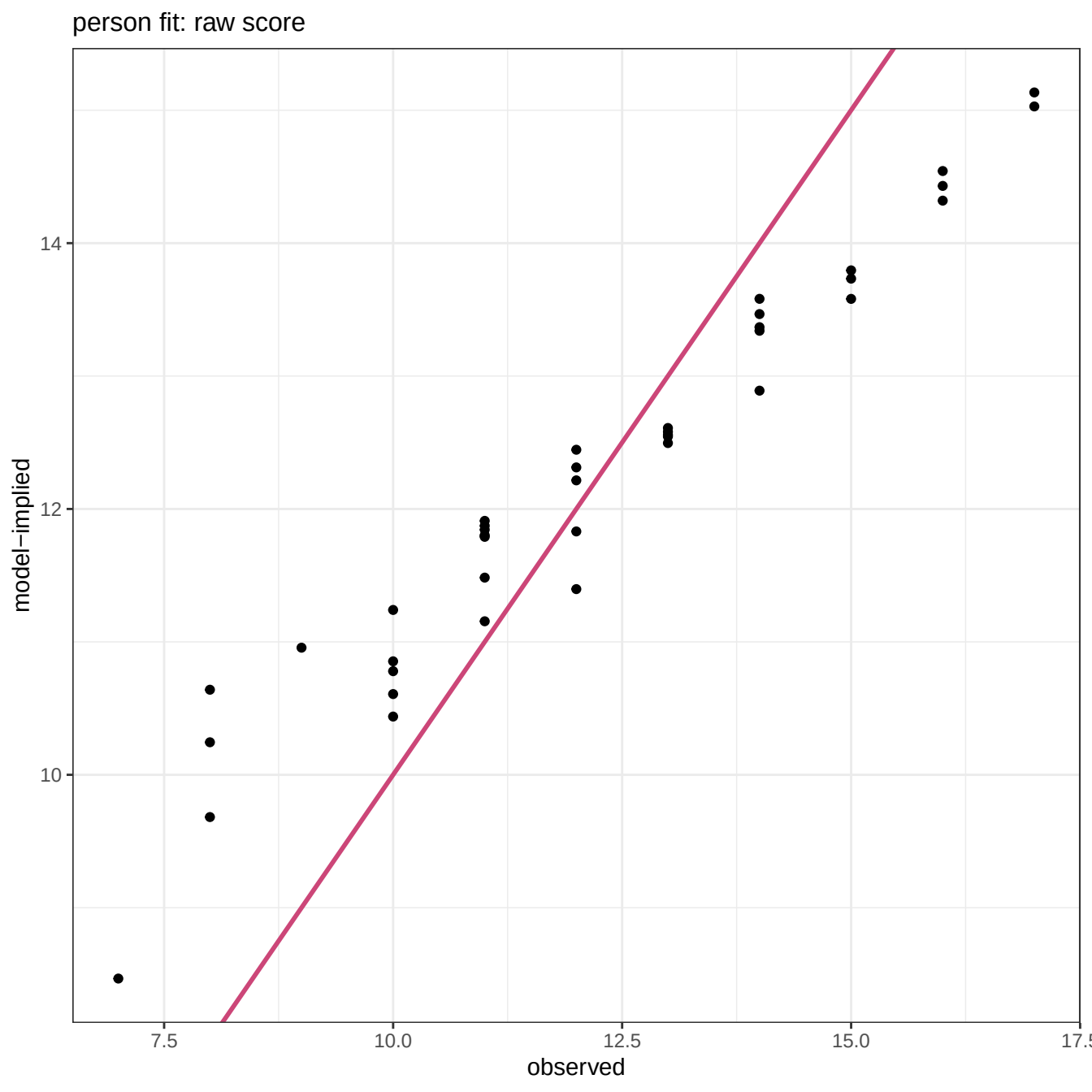

1.0
0.5
0.0
−0.5
latent trait
2.5
5.0
7.5
10.0
age
p0.05
p0.25
p0.75
p0.95
p0.1
p0.5
p0.9

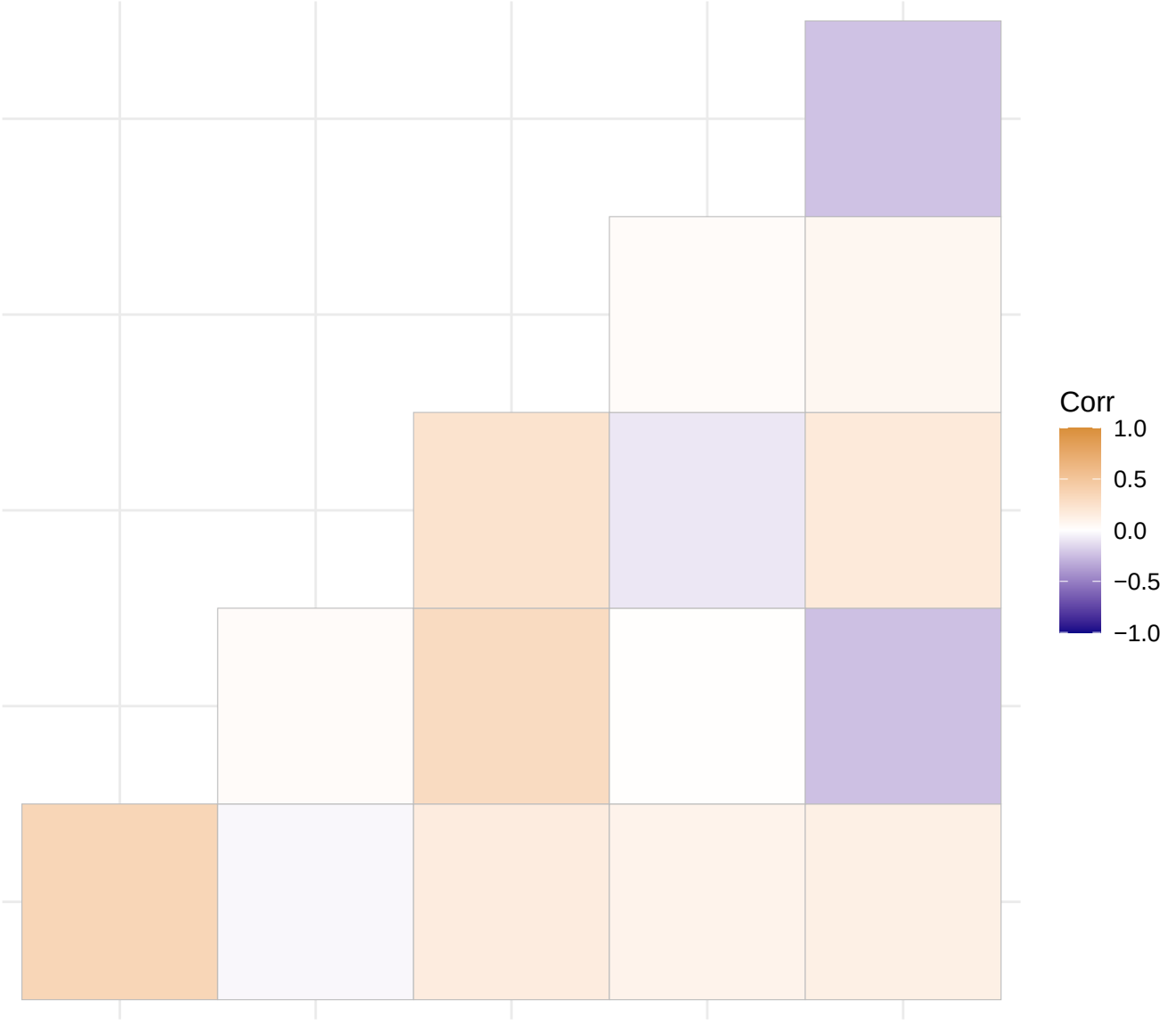

mean
standard deviation
latent trait
0.1
0.2
0.3
0.35
0.40
0.45
0.50
0.55
2.5
5.0
7.5
10.0
age

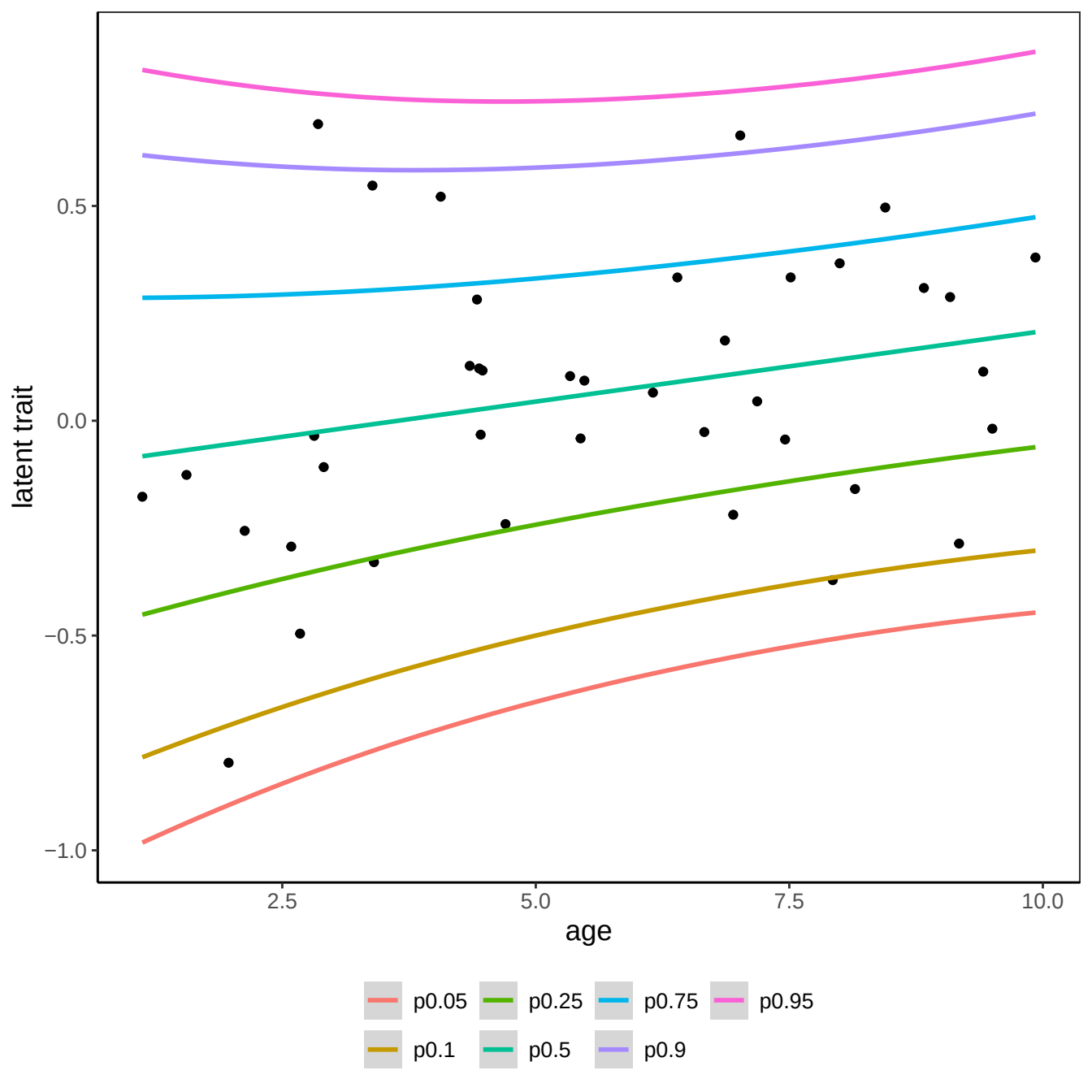

15
10
5
0
frequency of individuals
0
5
10
15
raw score

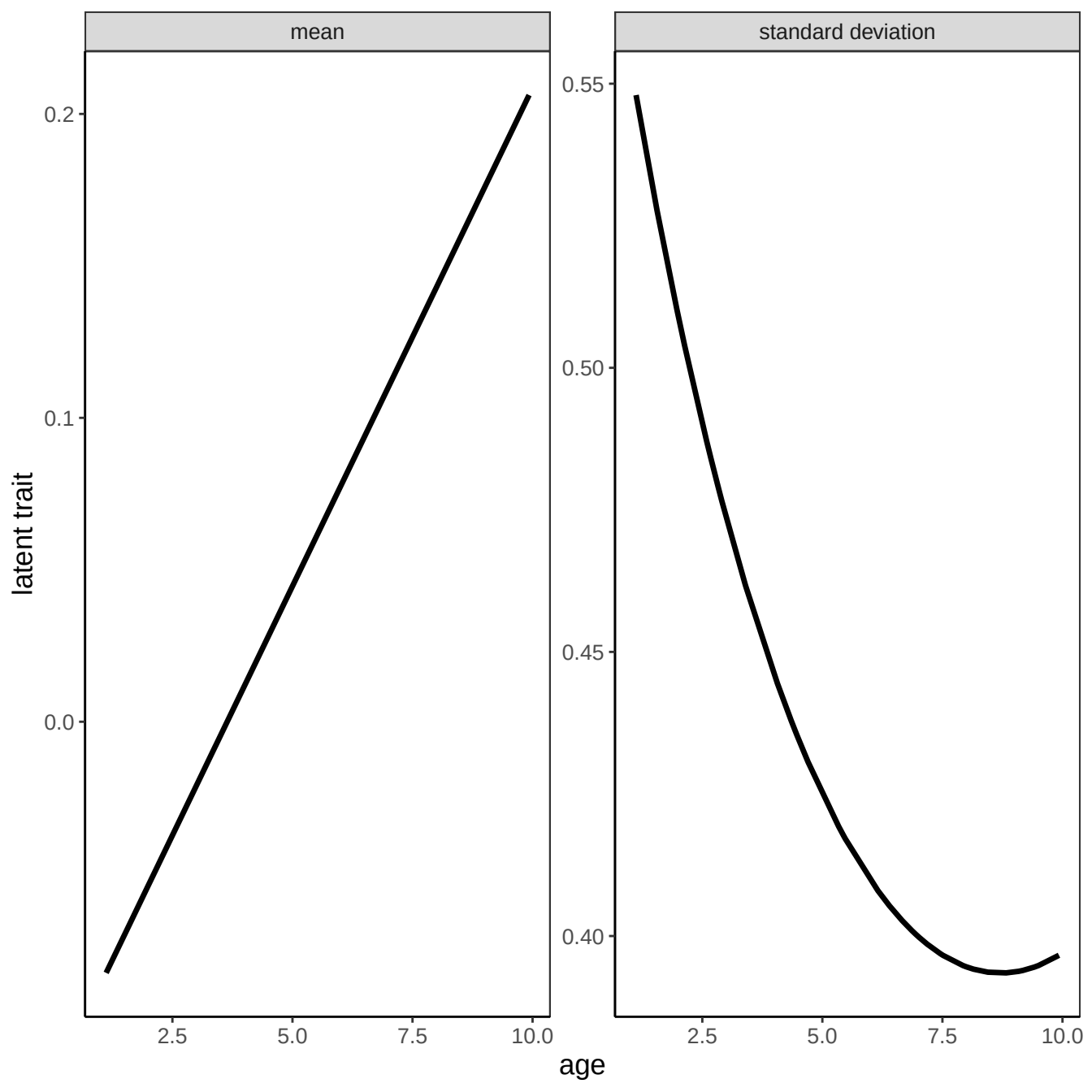

age: [0,5]
age: [5,10]
frequency of individuals
raw score
0
2.5
5.0
7.5
3
6
9
0
5
10
15

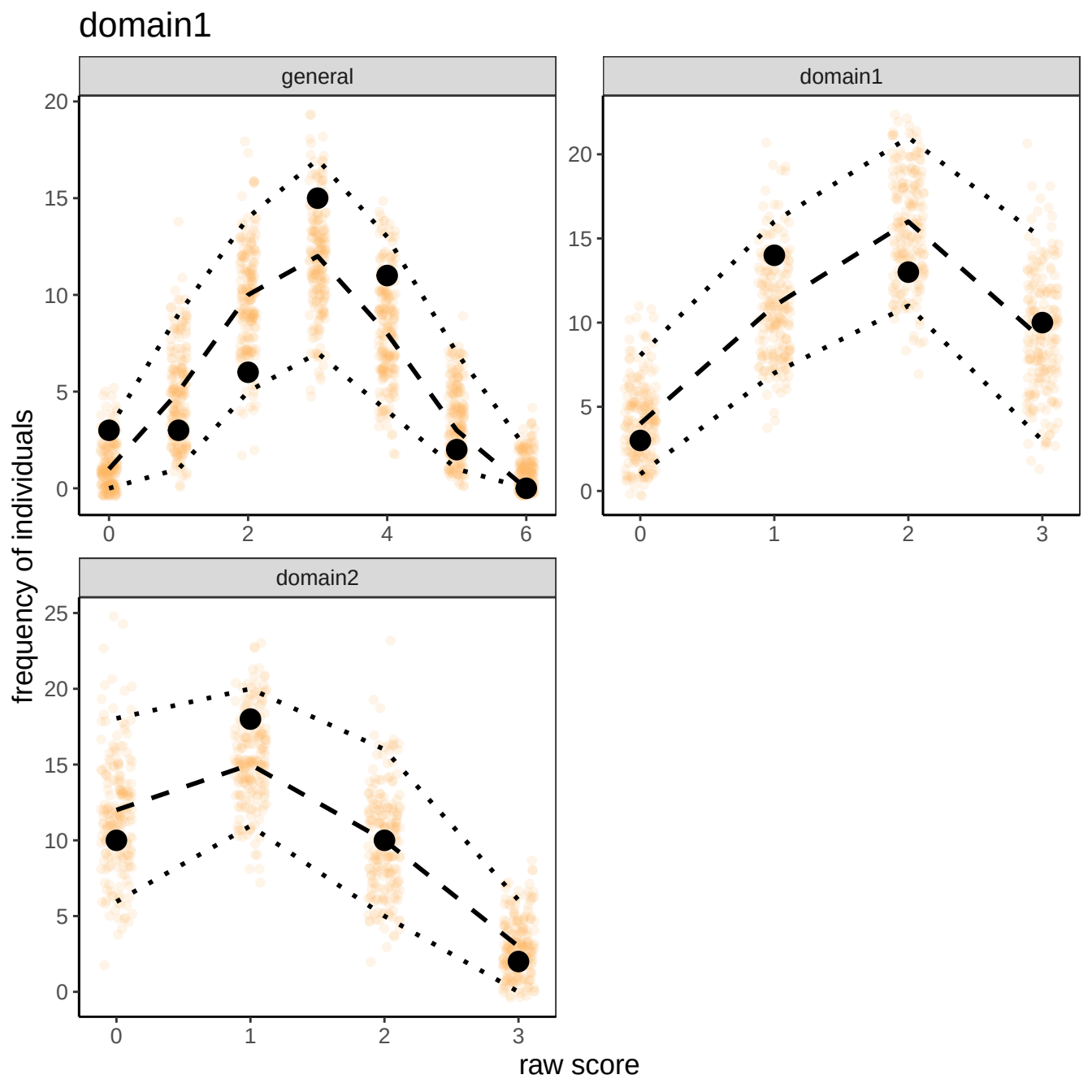

item 1
item 2
item 3
item 4
item 5
item 6
proportion
1.00
0.75
0.50
0.25
0.00
irt_mod
observed
Level
1
2
3

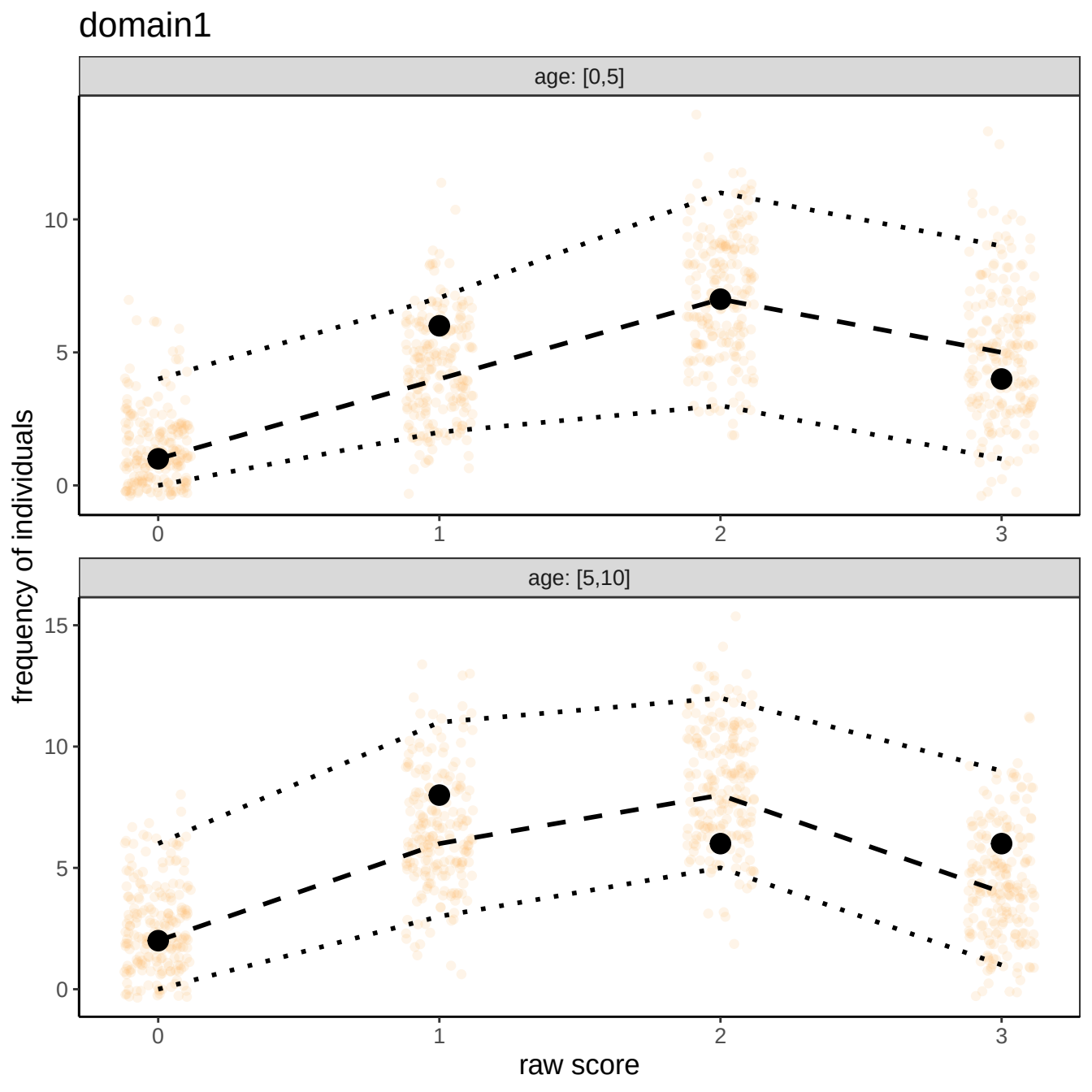

person fit: raw score
model−implied
observed
14
12
10
7.5
10.0
12.5
15.0
17.5

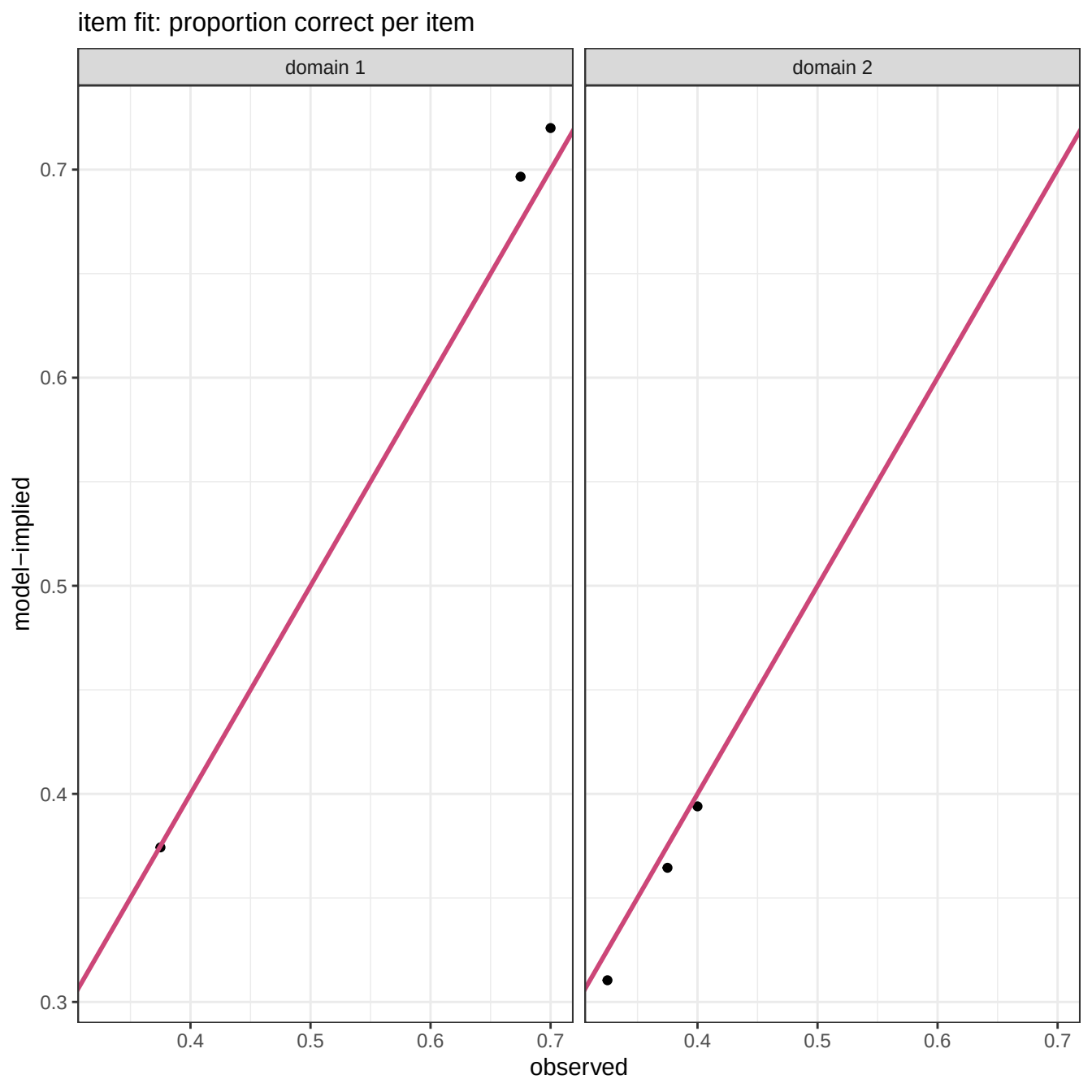

Corr
1.0
0.5
0.0
−0.5
−1.0

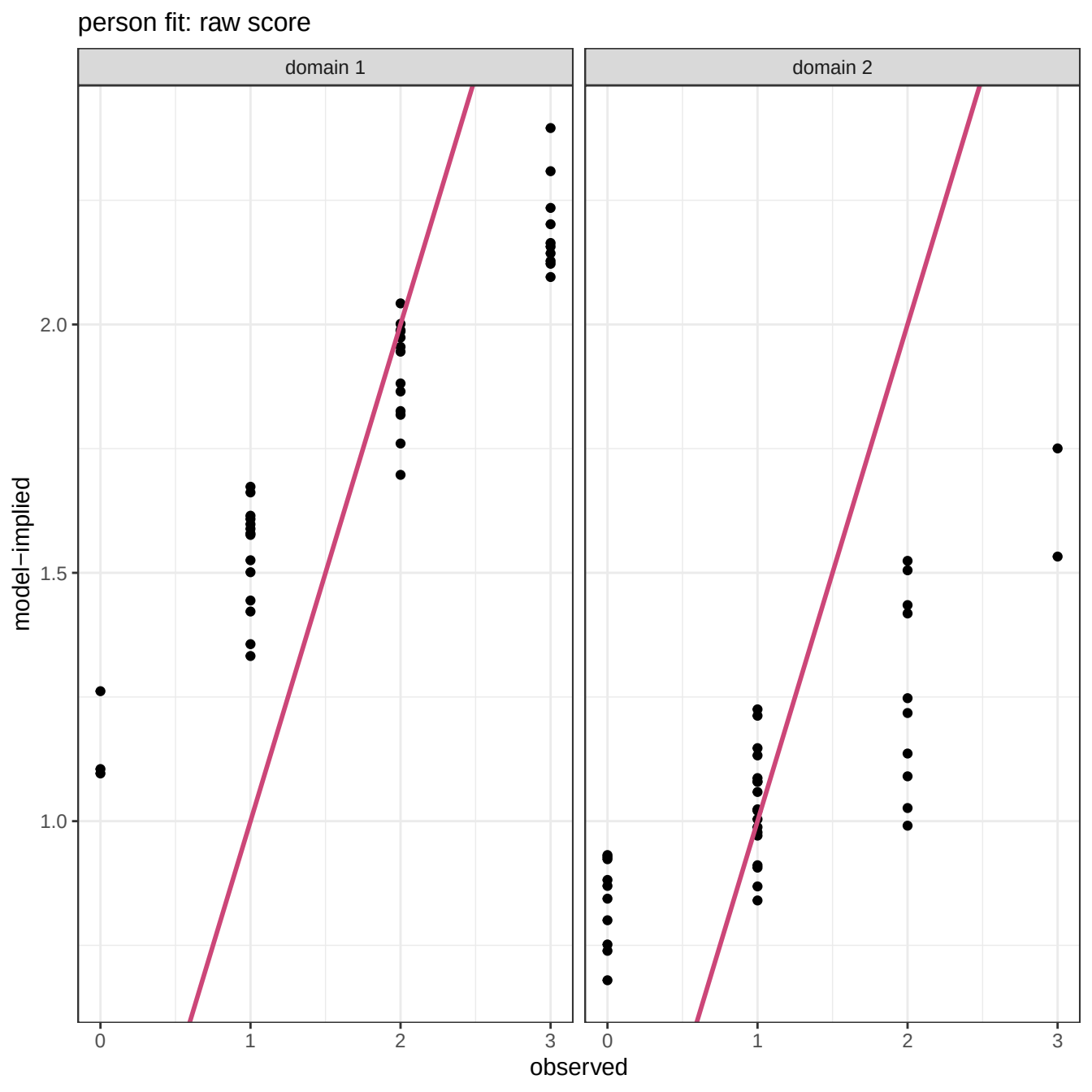

latent trait
0.5
0.0
−0.5
−1.0
2.5
5.0
7.5
10.0
age
p0.05
p0.25
p0.75
p0.95
p0.1
p0.5
p0.9

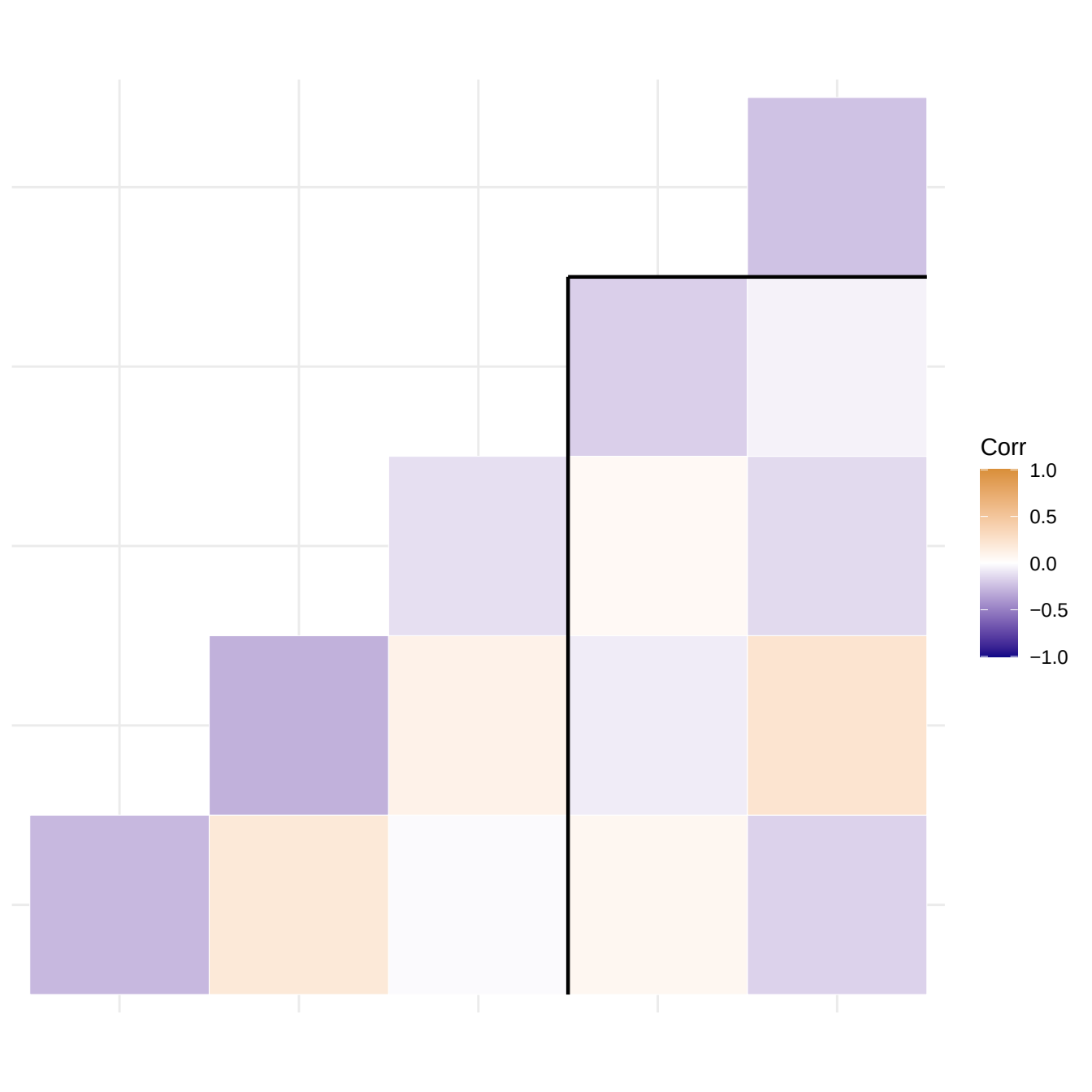

mean
standard deviation
latent trait
0.2
0.1
0.0
0.55
0.50
0.45
0.40
2.5
5.0
7.5
10.0
age

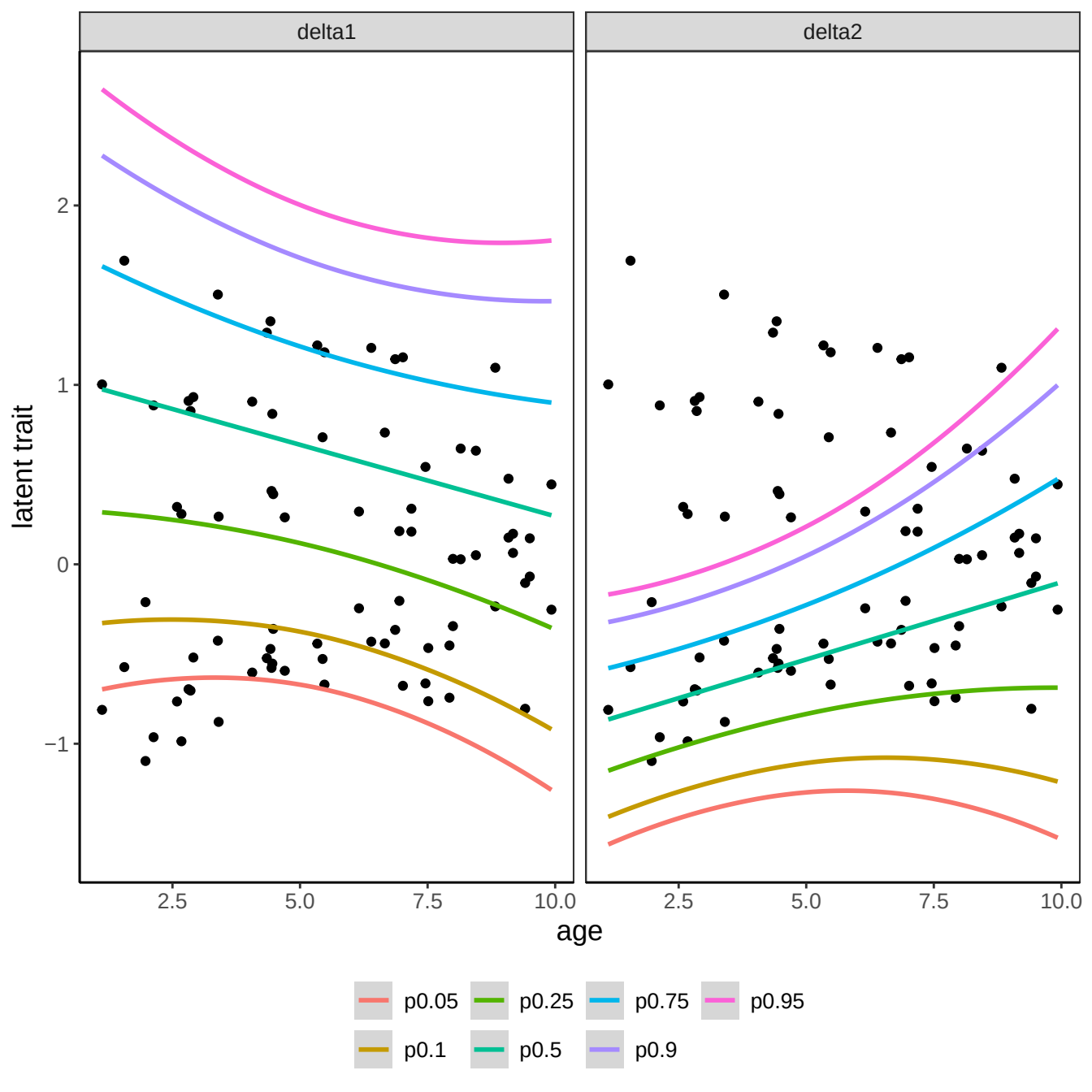

domain1
general
domain1
domain2
frequency of individuals
raw score
0
5
10
15
20
25
0
1
2
3
4
6

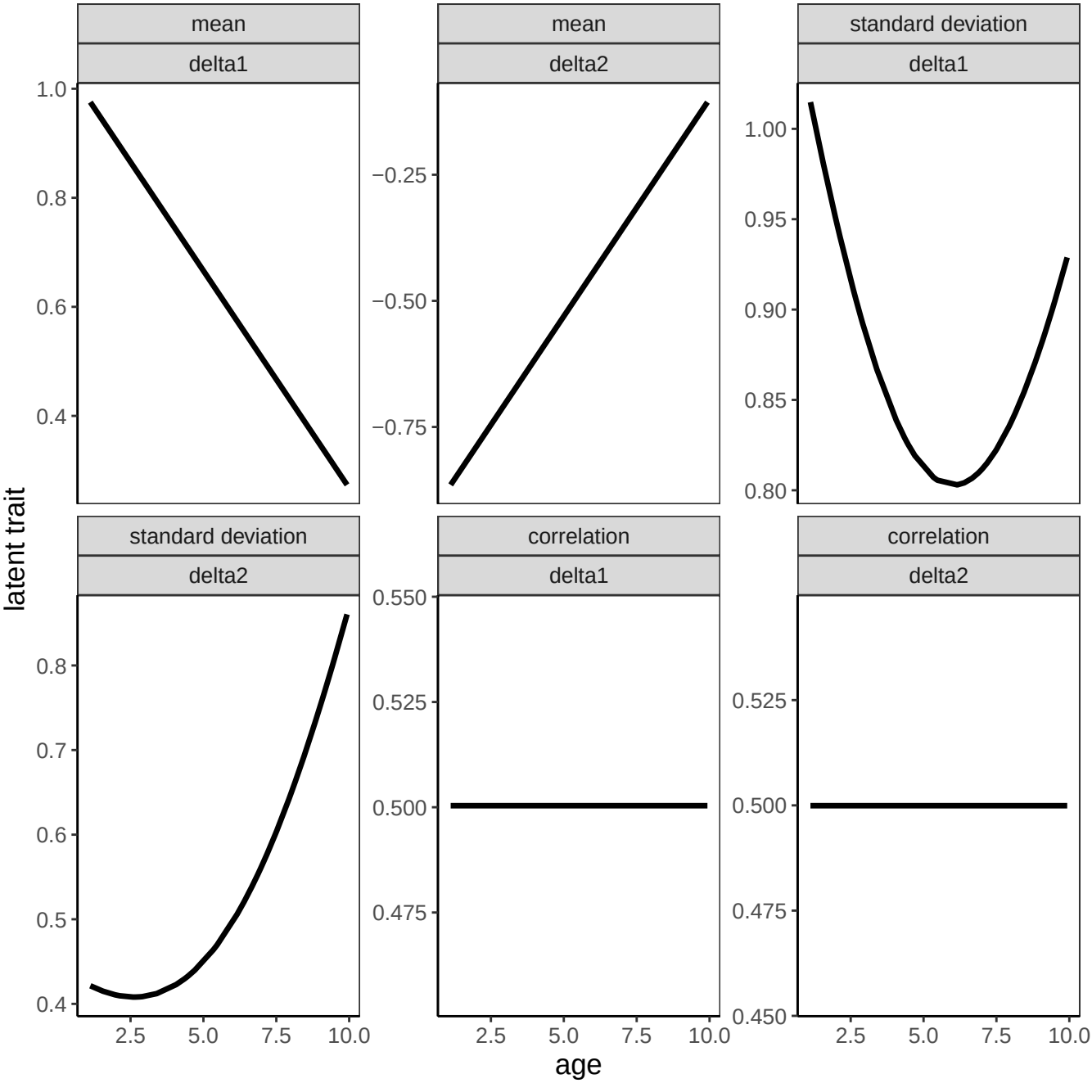

domain1
age: [0,5]
age: [5,10]
frequency of individuals
raw score
0
5
10
15
0
1
2
3

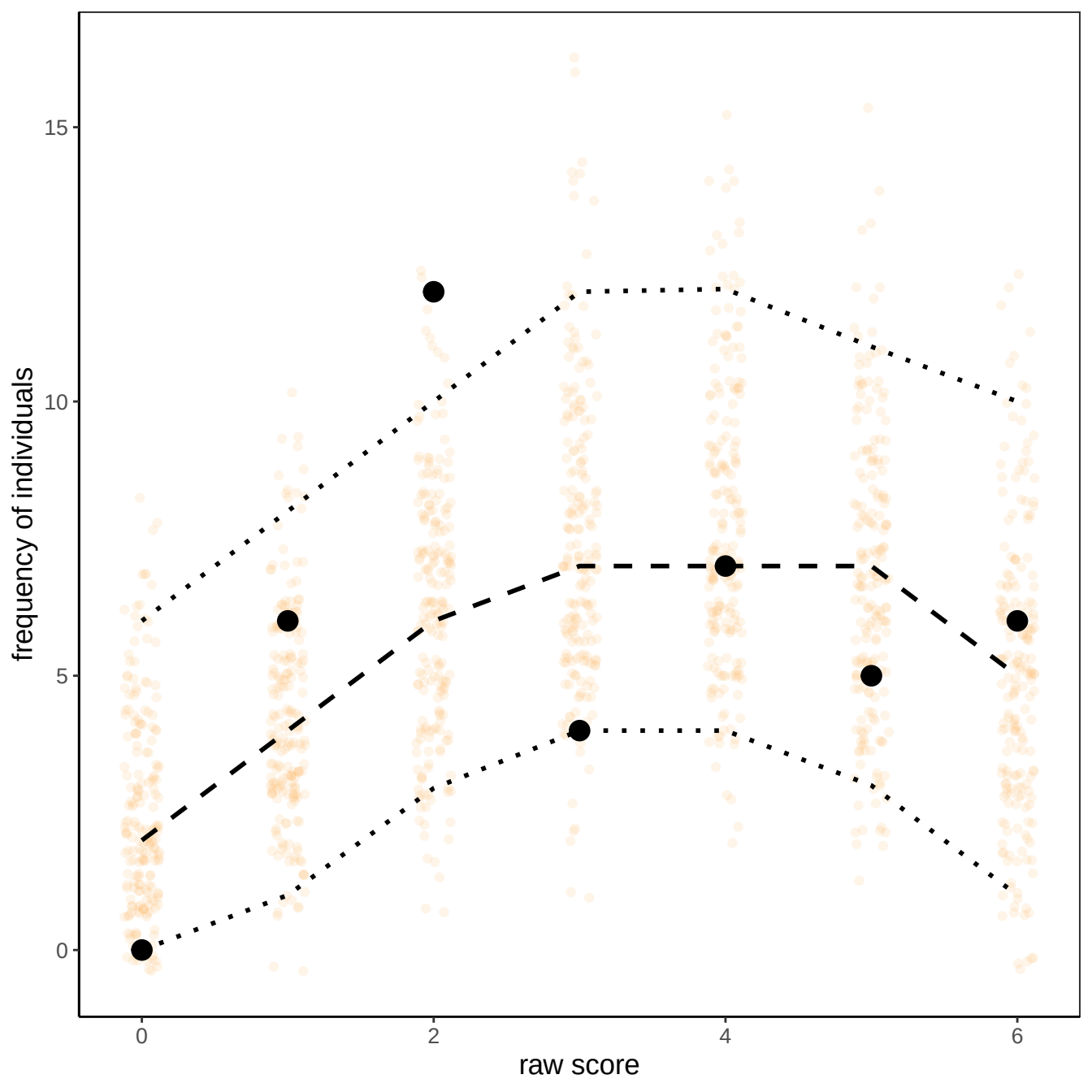

item fit: proportion correct per item
domain 1
domain 2
model−implied
observed
0.3
0.4
0.5
0.6
0.7
0.4
0.5
0.6
0.7

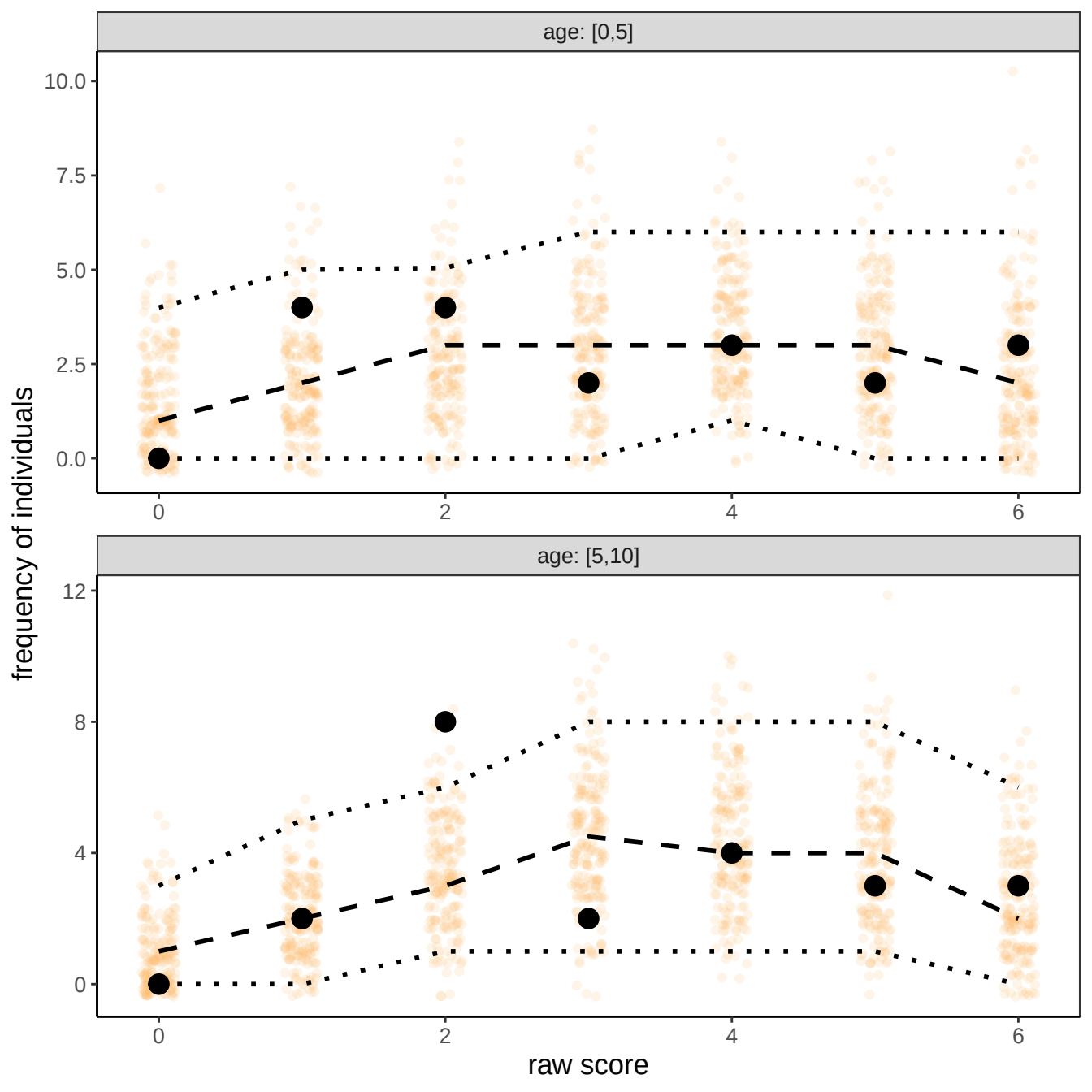

person fit: raw score
domain 1
domain 2
model−implied
observed
2.0
1.5
1.0
0
1
2
3

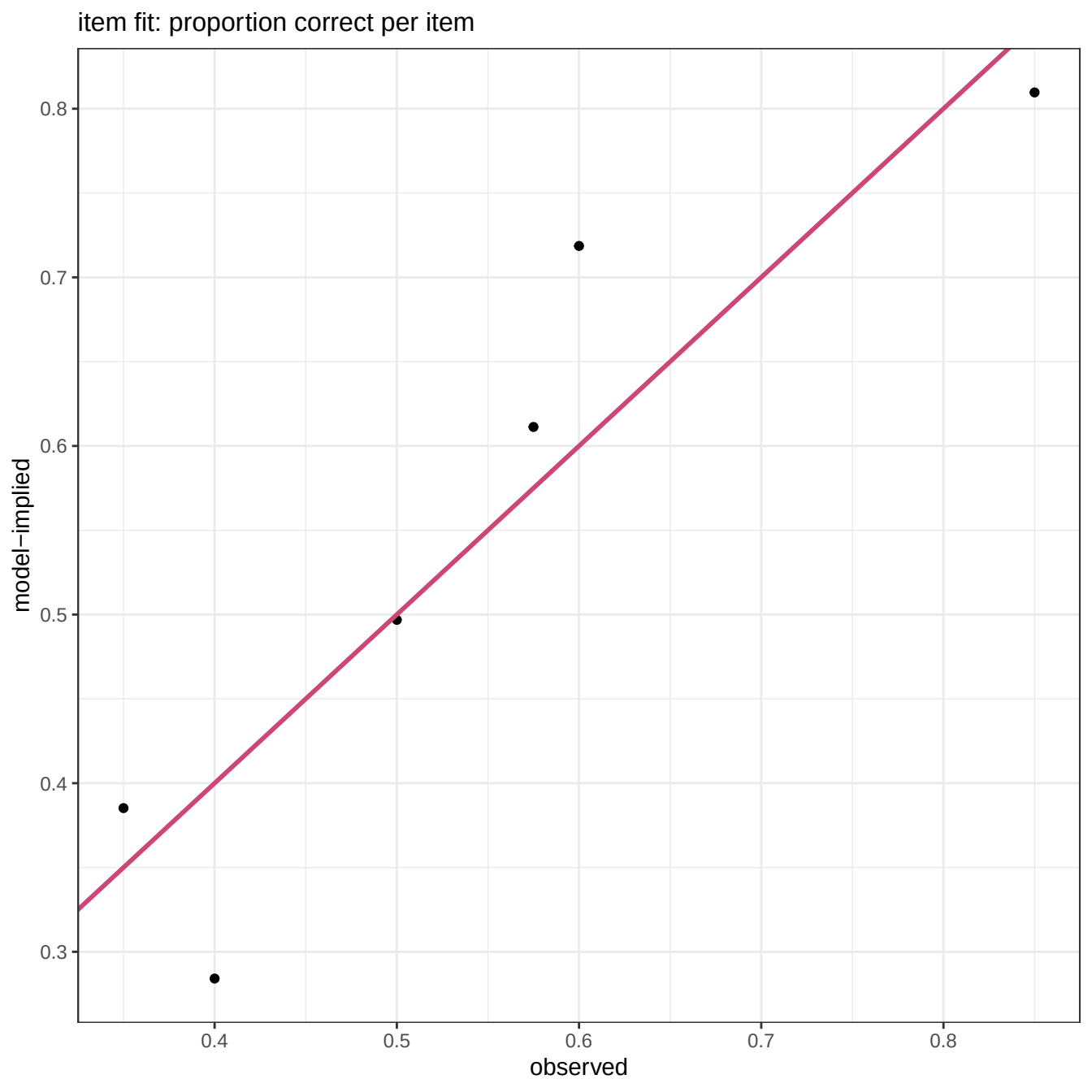

Corr
1.0
0.5
0.0
−0.5
−1.0

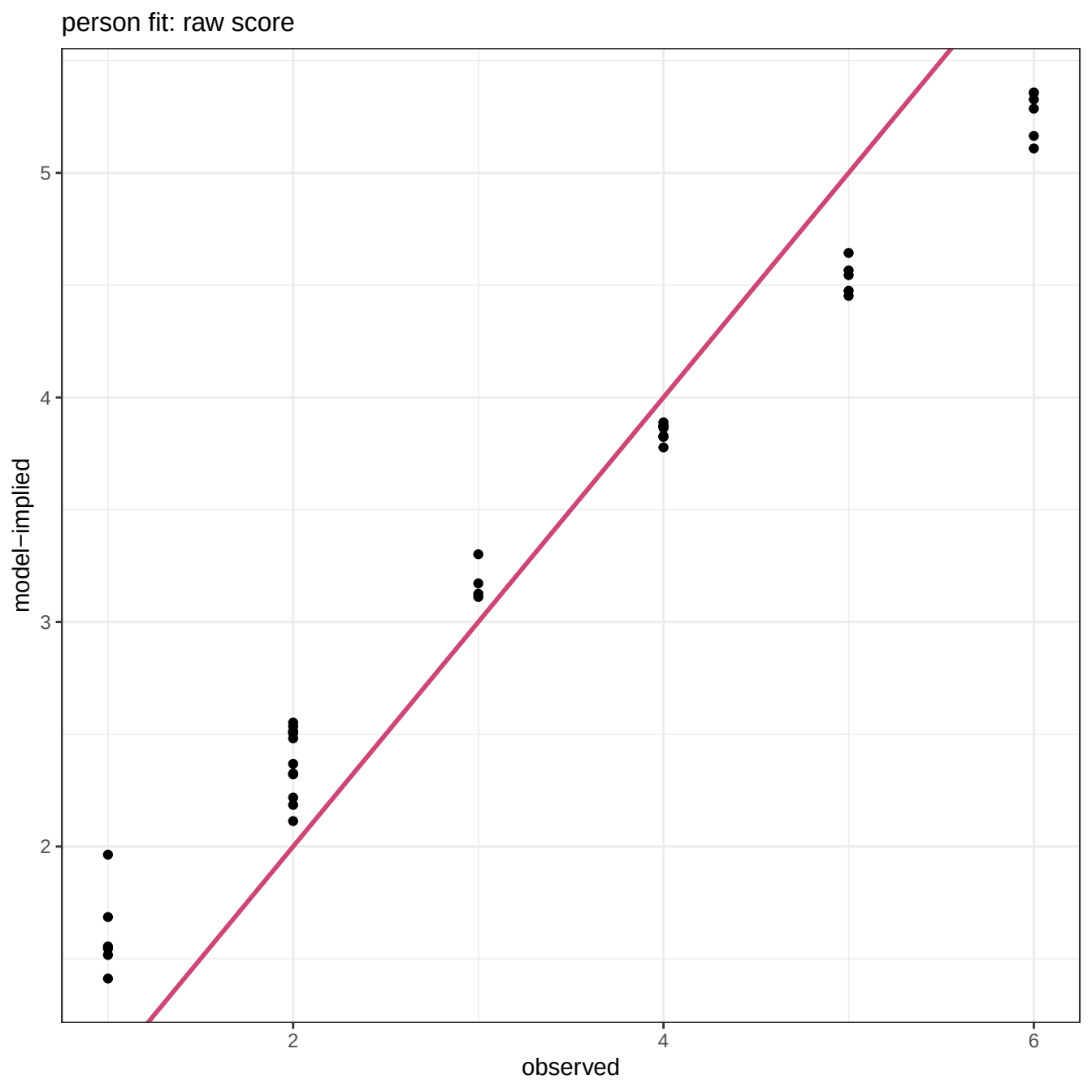

delta1
delta2
latent trait
2
1
0
−1
2.5
5.0
7.5
10.0
age
p0.05
p0.1
p0.25
p0.5
p0.75
p0.9
p0.95

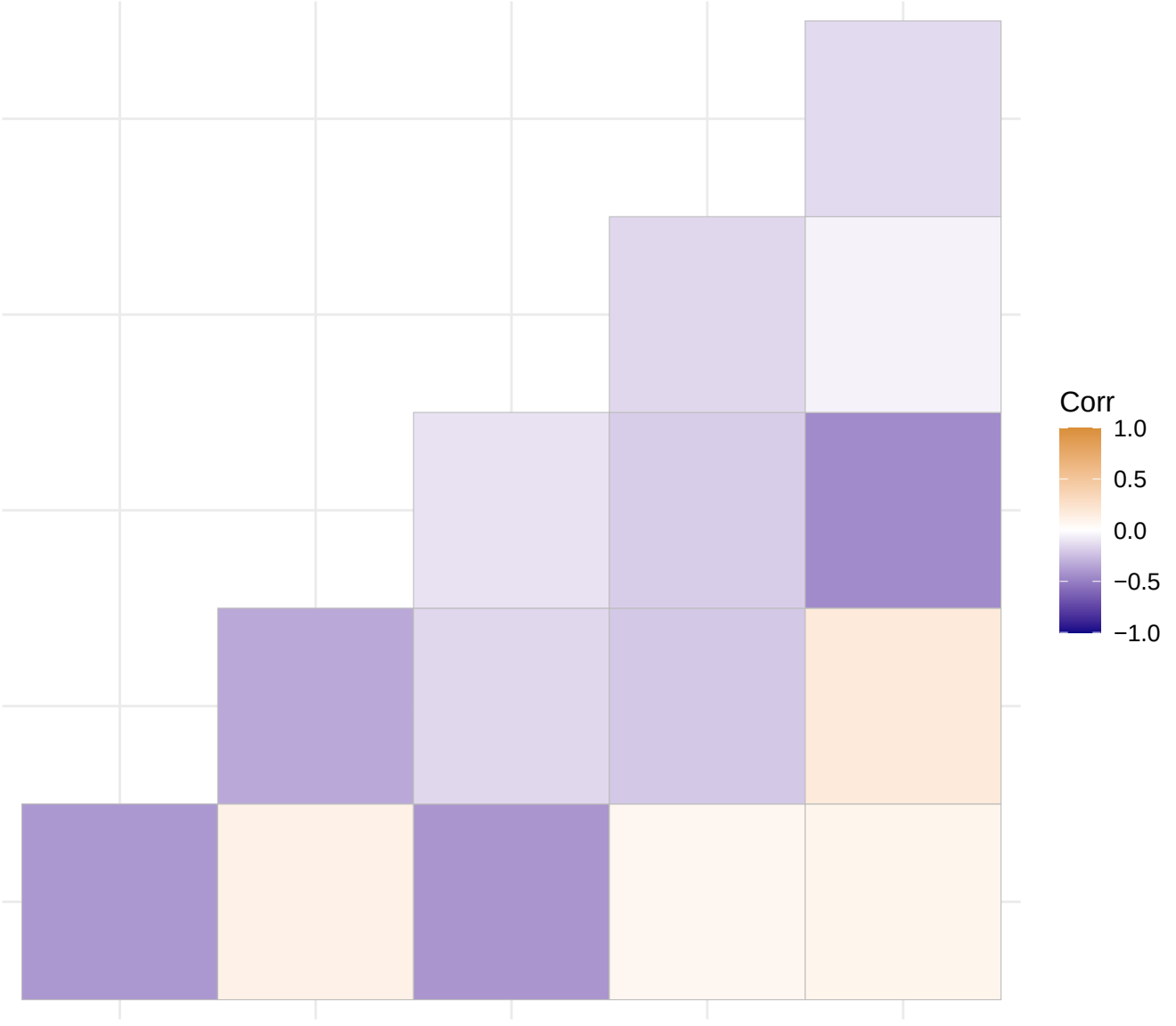

mean
delta1
mean
delta2
standard deviation
delta1
standard deviation
delta2
correlation
delta1
correlation
delta2
latent trait
age
1.0
0.8
0.6
0.4
−0.25
−0.50
−0.75
1.00
0.95
0.90
0.85
0.80
0.7
0.5
0.550
0.525
0.500
0.475
0.450
2.5
5.0
7.5
10.0

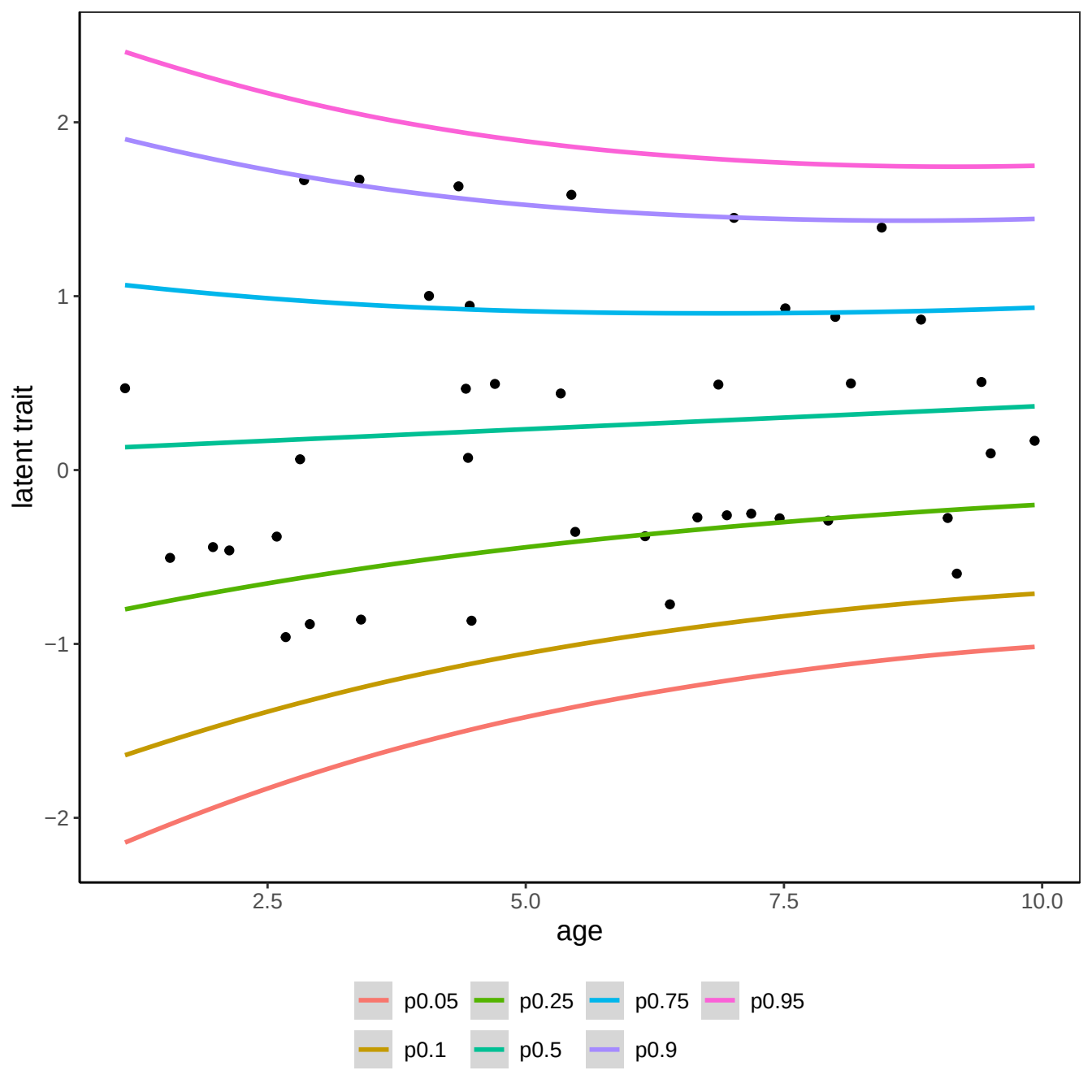

15
10
5
0
frequency of individuals
0
2
4
6
raw score

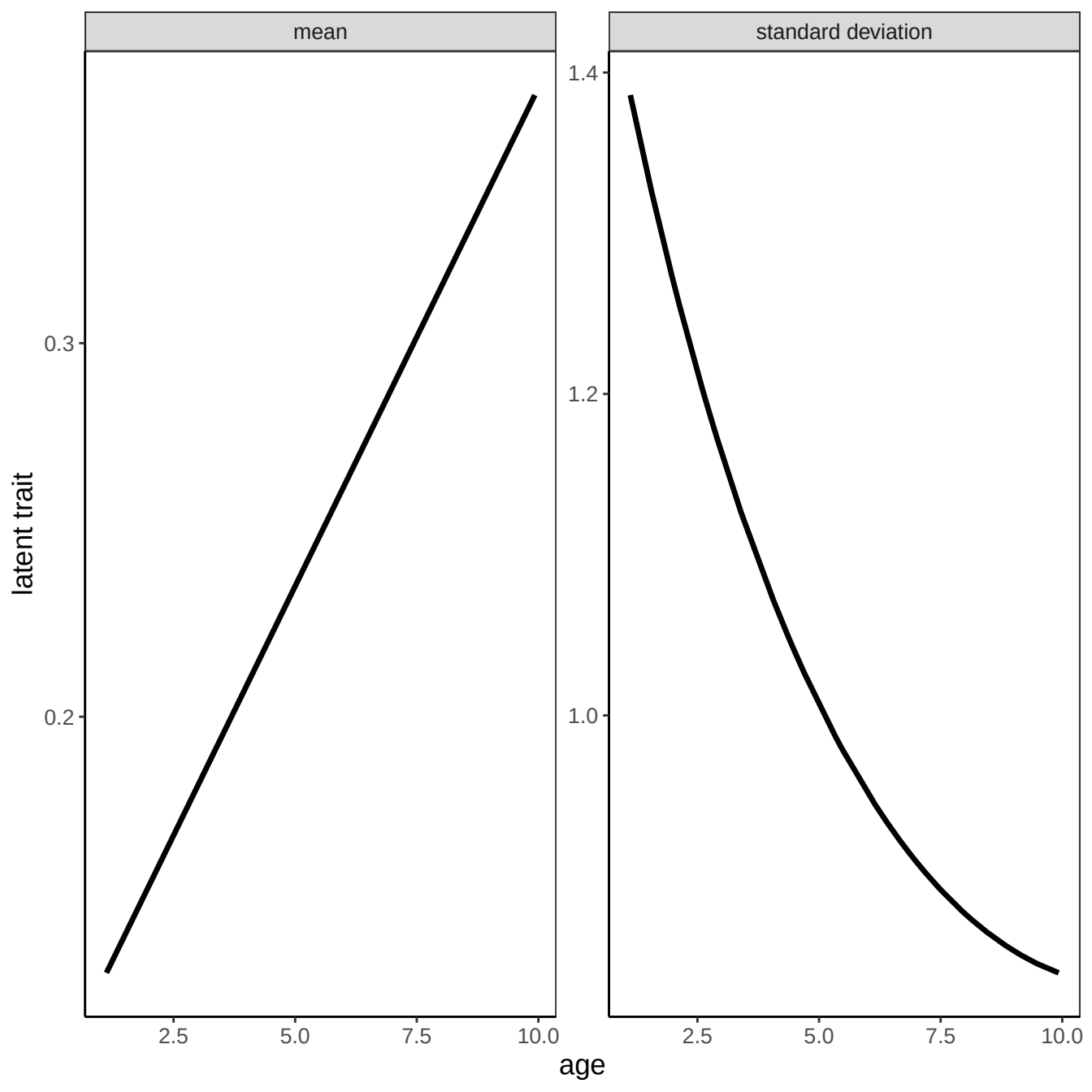

age: [0,5]
10.0
7.5
5.0
2.5
0.0
0
2
4
6
age: [5,10]
12
8
4
0
0
2
4
6
frequency of individuals
raw score


item fit: proportion correct per item
model−implied
observed
0.8
0.7
0.6
0.5
0.4
0.3
0.4
0.5
0.6
0.7
0.8


person fit: raw score
model−implied
observed
2
3
4
5
2
4
6


Corr
1.0
0.5
0.0
−0.5
−1.0


latent trait
2
1
0
−1
−2
2.5
5.0
7.5
10.0
age
p0.05
p0.25
p0.75
p0.95
p0.1
p0.5
p0.9


mean
standard deviation
latent trait
0.3
0.2
1.4
1.2
1.0
2.5
5.0
7.5
10.0
age